

Diseases of the Immune System

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The immune system is vital for survival because it protects us from infectious pathogens that abound in the environment and from the development of cancer. Predictably, immune deficiencies render individuals easy prey to infections and increase the incidence of certain cancers. But the immune system is itself capable of causing tissue injury and disease. Examples of disorders caused by immune responses include reactions to environmental substances that cause *allergies* and reactions against an individual's own tissues and cells (*autoimmunity*).

This chapter is devoted to diseases caused by too little or too much immunologic reactivity. We also consider amyloidosis, a disease in which an abnormal protein, derived in many cases from immunoglobulins, is deposited in tissues. First, we review some of the important features of normal immune responses, to provide a foundation for understanding the abnormalities that give rise to immunologic diseases.

NORMAL IMMUNE RESPONSE

The classic definition of immunity is protection from infectious pathogens, and the normal immune response is best understood in this context. However, immunity in its broader sense includes host reactions against cancers (tumor immunity), tissue transplants, and even self antigens (autoimmunity). The mechanisms of immunity fall into two broad categories (Fig. 6.1). *Innate immunity* (also called natural, or native, immunity) refers to intrinsic mechanisms that are poised to react immediately, and thus constitute the first line of defense. It is mediated by cells and molecules that recognize products of microbes and dead cells and induce rapid protective host reactions. *Adaptive immunity* (also called acquired, or specific, immunity) consists of mechanisms that are stimulated by (“adapt to”) exposure to microbes and other foreign substances. It develops more slowly than innate immunity but is even more powerful in combating infections. By convention, the term immune response usually refers to adaptive immunity.

Innate Immunity

Innate immunity is always present, ready to provide immediate defense against microbes and to eliminate damaged cells. The receptors

and components of innate immunity have evolved to serve these purposes.

Components of Innate Immunity

The major components of innate immunity are **epithelial barriers that block entry of microbes, phagocytic cells (mainly neutrophils and macrophages), dendritic cells, natural killer cells and other innate lymphoid cells, and several plasma proteins, including the proteins of the complement system.** Innate immunity has been divided into a cellular arm, which consists of cells (e.g., macrophages, NK cells) that eliminate microbes by phagocytosis and killing of infected cells (analogous to the cell-mediated adaptive immune response) and a humoral arm, which consists of plasma proteins, many of which are produced during the acute-phase response and eliminate extracellular microbes by opsonization and complement activation (similar to antibodies, the humoral arm of adaptive immunity).

- *Epithelia* of the skin and gastrointestinal and respiratory tracts act as mechanical barriers to the entry of microbes from the external environment. Epithelial cells also produce antimicrobial molecules such as defensins, and lymphocytes located in the epithelia combat microbes at these sites. If microbes breach epithelial boundaries, other defense mechanisms come into play.
- *Monocytes and neutrophils* are phagocytes in the blood that can be rapidly recruited to any site of infection; monocytes enter the tissues and mature into *macrophages*. Some tissue-resident macrophages (Kupffer cells in the liver, microglia in the brain, and alveolar macrophages in the lungs) develop from the yolk sac or fetal liver early in life and populate various tissues. Phagocytes sense the presence of microbes and other offending agents, ingest (phagocytose) these invaders, and destroy them. Because macrophages are the dominant cells of chronic inflammation, they are described in more detail in Chapter 3 in the discussion of chronic inflammation.
- *Dendritic cells (DCs)* are specialized cells present in epithelia, lymphoid organs, and most tissues. They capture protein antigens and display peptides for recognition by T lymphocytes. In addition to their antigen-presenting function, DCs are endowed with a rich

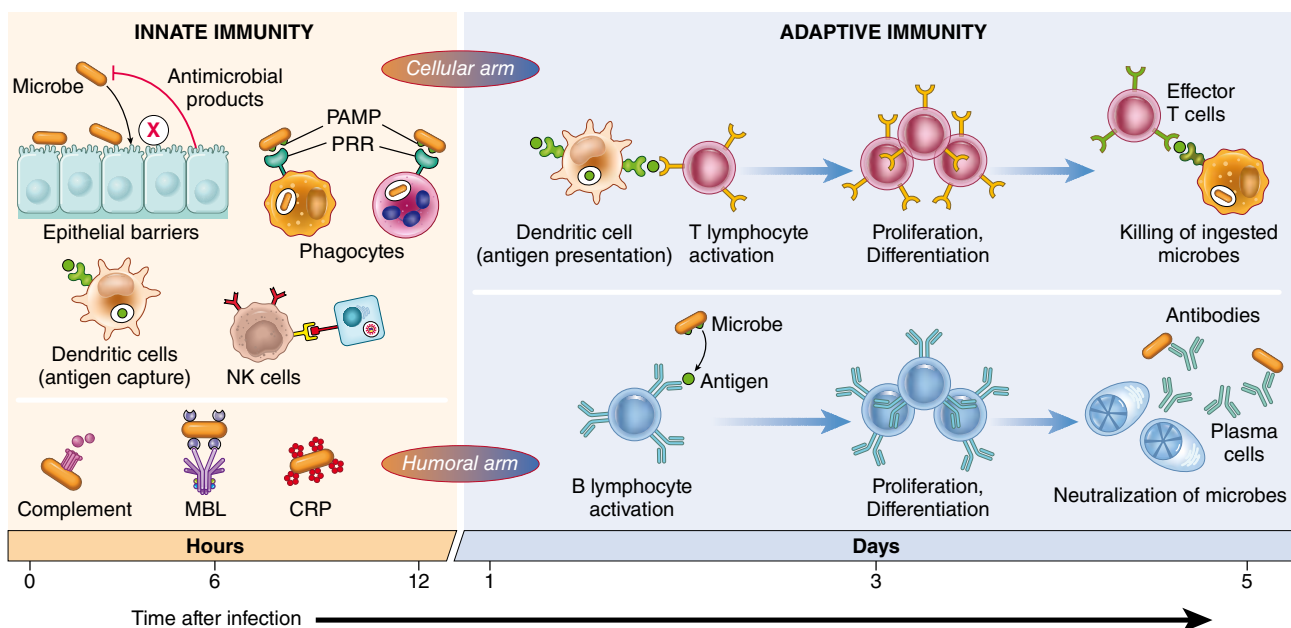


Fig. 6.1 The principal components of innate and adaptive immunity. Innate and adaptive immunity both consist of a cellular arm, composed of cells, and a humoral arm, composed of secreted proteins. *CRP*, C-reactive protein; *MBL*, mannose-binding lectin; *NK*, Natural killer; *PAMP*, pathogen-associated molecular pattern; *PRR*, pattern recognition receptor.

collection of receptors that sense microbes and cell damage and stimulate the secretion of cytokines, mediators that play critical roles in inflammation and antiviral defense. Thus, DCs serve as sentinels that detect danger and initiate innate immune responses, but, unlike macrophages, they are not directly involved in the destruction of microbes and other offending agents.

- **Innate lymphoid cells (ILCs)** are tissue-resident lymphocytes that lack T-cell antigen receptors and cannot respond to antigens, but instead are activated by cytokines and other mediators produced at sites of tissue damage. They are thought to be sources of inflammatory cytokines during early phases of immune reactions. ILCs are classified into groups based on the dominant cytokines they produce; groups 1, 2, and 3 ILCs produce many of the same cytokines as Th1, Th2, and Th17 subsets of CD4⁺ T cells, described later.
- **Natural killer (NK) cells** provide early protection against many viruses and intracellular bacteria; their properties and functions are described later.
- **Other cell types.** Several other cell types can sense and react to microbes. These include *mast cells*, which are capable of producing many mediators of inflammation (discussed later), and even epithelial and endothelial cells.
- **Plasma proteins.** Several soluble proteins play important roles in innate immunity. The *complement system* (described in Chapter 3) consists of plasma proteins that are activated by microbes. Complement activation may occur through the alternative and lectin pathways as part of innate immune responses or through the classical pathway, which involves antibody-antigen complexes, as part of adaptive immune responses (Chapter 3). Other circulating proteins of innate immunity are mannose-binding lectin and C-reactive protein, both of which coat microbes and promote phagocytosis. Lung surfactant is also a component of innate immunity, providing protection against inhaled microbes.

Cellular Receptors for Microbes, Products of Damaged Cells, and Foreign Substances

Cells that participate in innate immunity can recognize certain components that are shared among related microbes and that are often essential for infectivity (and thus cannot be mutated to allow the microbes to evade the defense mechanisms). These microbial structures are called pathogen-associated molecular patterns (PAMPs). Leukocytes also recognize molecules released by injured and necrotic cells, which are called damage-associated molecular patterns (DAMPs). The cellular receptors that recognize these molecules are called pattern recognition receptors.

Pattern recognition receptors are located in all cellular compartments where microbes may be present: plasma membrane receptors detect extracellular microbes, endosomal receptors detect ingested microbes, and cytosolic receptors detect microbes in the cytoplasm (Fig. 6.2). Several classes of these receptors have been identified.

Toll-like Receptors. The best known of the pattern recognition receptors are the Toll-like receptors (TLRs), whose founding member, *Toll*, was discovered in *Drosophila* as a gene involved in development of the fly. A family of related proteins was later shown to be essential for host defense against microbes. Mammals have 10 TLRs, each recognizing a different set of microbial molecules. The TLRs are present in the plasma membrane and endosomal vesicles (see Fig. 6.2). All TLRs signal by a common pathway that culminates in the activation of two sets of transcription factors: (1) NF- κ B, which stimulates the synthesis and secretion of cytokines and the expression of adhesion molecules, both of which are critical for the recruitment and activation of leukocytes (Chapter 3), and (2) interferon regulatory factors (IRFs), which stimulate the production of the antiviral cytokines, type I interferons (IFNs). Inherited loss-of-function mutations affecting TLRs and their signaling pathways are associated with rare but serious immunodeficiency

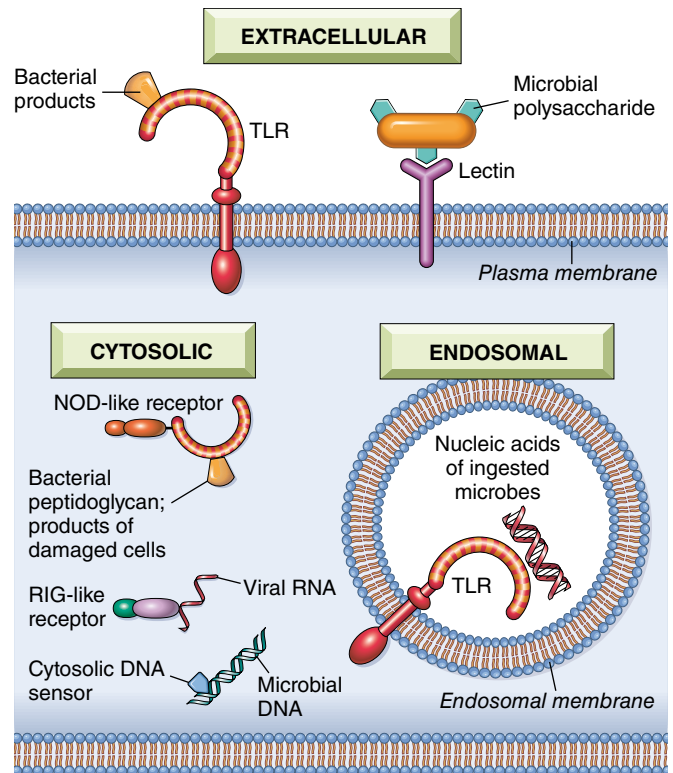


Fig. 6.2 Cellular receptors for microbes and products of cell injury. Phagocytes, dendritic cells, and many types of epithelial cells express different classes of receptors that sense the presence of microbes and dead cells. Toll-like receptors (TLRs) located in different membrane-bound cellular compartments, as well as other cytoplasmic and plasma membrane receptors, recognize products of different classes of microbes. Major classes of innate immune receptors are TLRs, NOD-like receptors in the cytosol, C-type lectin receptors, RIG-like receptors for viral nucleic acids, and cytosolic receptors for DNA. *RIG*, Retinoic acid-inducible gene.

syndromes, described later in the chapter. Inherited or acquired deficiencies in type I IFN production or function increase the risk of severe viral infection, including with SARS-CoV-2, the agent of COVID-19 disease.

NOD-like Receptors and the Inflammasome. NOD-like receptors (NLRs) are cytosolic receptors named after the founding member NOD-2 (nucleotide binding oligomerization domain containing protein 2). They recognize a wide variety of substances, including products released from necrotic or damaged cells (e.g., uric acid and adenosine triphosphate [ATP]) and some microbial products, and also “sense” the loss of intracellular K⁺ ions. How this family of sensors can detect so many diverse signs of danger or damage is not known. Several NLRs signal via a cytosolic multiprotein complex called the *inflammasome*, which activates an enzyme (caspase-1) that cleaves a precursor form of the cytokine interleukin-1 (IL-1) to generate biologically active IL-1 (Fig. 6.3). As discussed later, IL-1 is a mediator of inflammation that recruits leukocytes, which eliminate the damaged cells, and induces fever. Gain-of-function mutations in NLRs and related proteins, and loss-of-function mutations in regulators of the inflammasome, result in disorders that present with periodic fever and other symptoms called *autoinflammatory syndromes* (to be distinguished from autoimmune diseases, which result from T and B lymphocyte reactions against self antigens). The autoinflammatory syndromes respond very well to treatment with IL-1 antagonists. The inflammasome pathway may also play a role in many common disorders. For example, recognition of urate crystals by a class of NLRs

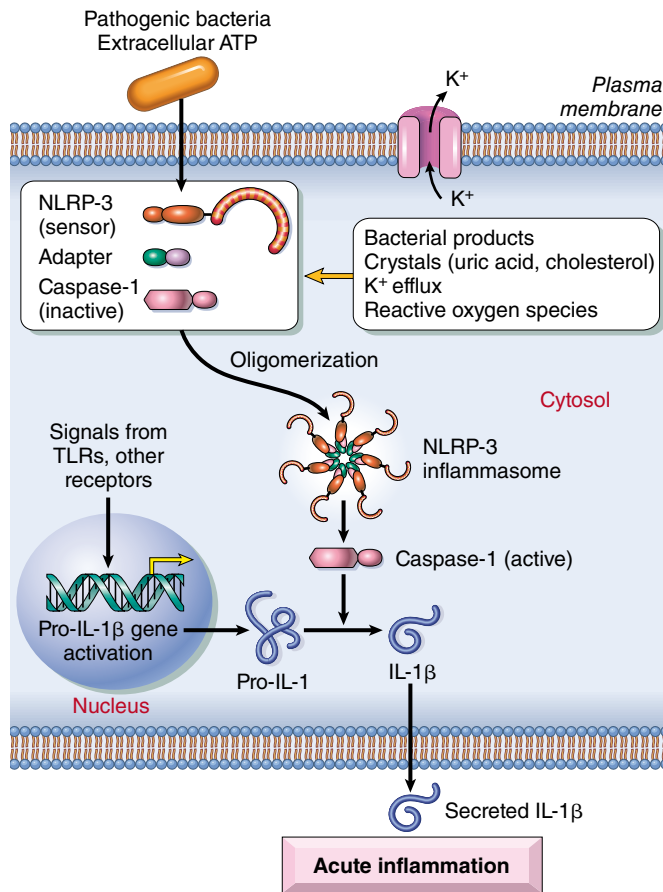


Fig. 6.3 The inflammasome. The inflammasome is a protein complex that recognizes products of dead cells and some microbes and foreign substances and induces the secretion of biologically active interleukin 1 (*IL-1*). The inflammasome consists of a sensor protein an example is the leucine-rich protein *NLRP3*, an adapter, and the enzyme caspase-1, which is converted from an inactive to an active form. *ATP*, Adenosine triphosphate; *TLR*, toll-like receptor.

underlies the inflammation associated with gout. These receptors can detect cholesterol crystals that are deposited in abnormally large amounts in tissues, and the resulting inflammation may contribute to atherosclerosis.

Cytosolic Receptors for Nucleic Acids. RIG-like receptors (RLRs), named after the founding member RIG-I (retinoic acid-inducible gene-1), are located in the cytosol of most cell types and detect RNA of viruses that replicate in the cytosol of infected cells. These receptors stimulate the production of the antiviral cytokine type I IFN. Cytosolic receptors for microbial DNA, often derived from viruses in the cell, and even self DNA that accumulates in the cytosol, activate a pathway called the STING (stimulator of interferon genes) pathway, which also leads to the production of type I IFN. Excessive activation of the STING pathway causes systemic inflammatory disorders collectively called *interferonopathies*.

Other Cellular Receptors for Microbial Products. C-type lectin receptors (CLRs, also known as Dectins) expressed on the plasma membrane of macrophages and DCs detect fungal glycans and elicit inflammatory reactions to fungi. Plasma membrane G protein-coupled receptors on neutrophils, macrophages, and most other types of leukocytes recognize short bacterial peptides containing *N*-formylmethionyl residues.

Because all bacterial proteins and few mammalian proteins (only those synthesized within mitochondria) are initiated by *N*-formylmethionine, this receptor enables neutrophils to detect bacterial proteins and move toward their source (chemotaxis). Mannose receptors recognize microbial sugars (which often contain terminal mannose residues, unlike mammalian glycoproteins) and induce phagocytosis of the microbes.

Natural Killer Cells

The function of natural killer (NK) cells is to recognize and destroy severely stressed or abnormal cells, such as virus-infected cells and tumor cells. NK cells make up 5% to 10% of peripheral blood lymphocytes. NK cells express CD16, a receptor for the Fc tails of IgG that confers on NK cells the ability to lyse IgG-coated target cells. This phenomenon is known as *antibody-dependent cellular cytotoxicity (ADCC)*.

Killing of target cells by NK cells is regulated by signals from activating and inhibitory receptors (Fig. 6.4). There are many types of activating receptors, which recognize surface molecules that are induced by various kinds of stress, such as infection and DNA damage. Thus, these receptors enable NK cells to recognize damaged or infected cells. NK cell inhibitory receptors recognize self major histocompatibility complex (MHC) class I molecules, which are expressed on all healthy cells. The inhibitory receptors prevent NK cells from killing normal cells. Virus infection or neoplastic transformation often enhances expression of ligands for activating receptors and at the same time reduces the expression of MHC class I molecules. As a result,

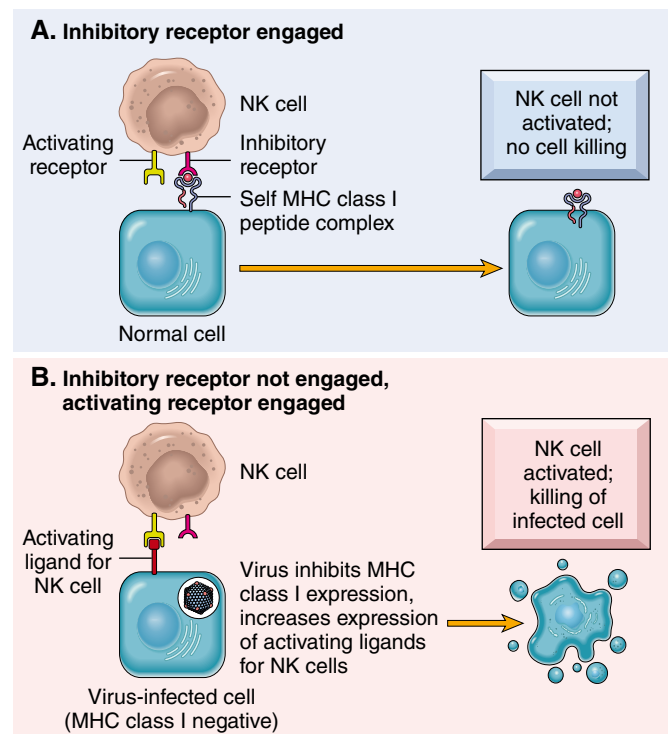


Fig. 6.4 Activating and inhibitory receptors of natural killer (NK) cells. (A) Healthy cells express self major histocompatibility complex (MHC) class I molecules, which are recognized by inhibitory receptors, thus ensuring that NK cells do not attack normal cells. Note that healthy cells may express ligands for activating receptors (not shown) or may not express such ligands (*as shown*), but they do not activate NK cells because they engage the inhibitory receptors. (B) In infected and stressed cells, MHC class I expression is reduced so that the inhibitory receptors are not engaged, and ligands for activating receptors are expressed. The result is that NK cells are activated, and the infected cells are killed.

when NK cells engage these abnormal cells, the balance is tilted toward activation, and the infected or tumor cell is killed.

NK cells also secrete cytokines such as interferon- γ (IFN- γ), which activates macrophages to destroy ingested microbes, and thus NK cells provide an early defense against intracellular microbial infections. The activity of NK cells is regulated by many cytokines, including the interleukins IL-2, IL-15, and IL-12. IL-2 and IL-15 stimulate proliferation of NK cells, whereas IL-12 activates the killing of target cells and the secretion of IFN- γ .

Reactions of Innate Immunity

The innate immune system provides host defense by two main reactions.

- **Inflammation.** Cytokines and products of complement activation, as well as other mediators, are produced during innate immune reactions and trigger the vascular and cellular components of inflammation. The recruited leukocytes destroy microbes and ingest and eliminate damaged cells. The innate immune response also triggers the repair of damaged tissues. These processes are described in Chapter 3.
- **Antiviral defense.** Type I IFNs produced in response to viruses act on infected and uninfected cells and activate enzymes that degrade viral nucleic acids and inhibit viral replication, inducing an antiviral state. NK cells recognize and kill virus-infected cells, as described earlier.

In addition to these direct defense functions, innate immunity produces the danger signals that stimulate the subsequent more powerful adaptive immune response. The nature of some of these signals is described later.

Innate immunity, unlike adaptive immunity, is not specific for particular microbial antigens and exhibits little or no memory. It is estimated that innate immunity uses about 100 different receptors to recognize 1000 molecular patterns. In contrast, adaptive immunity uses two types of receptors (antibodies and T-cell receptors [TCRs], described later), each with millions of variations, to recognize millions of antigens. Whereas the receptors of innate immunity are encoded in the germline (and are hence identical in all individuals and all cells in which they are expressed), the antigen receptors of adaptive immunity are encoded by receptor genes that undergo somatic recombination to generate tremendous diversity (and thus vary in different cells and individuals).

Adaptive Immunity

The adaptive immune system consists of lymphocytes and their products, including antibodies. The lymphocytes of adaptive immunity use highly diverse receptors to recognize a vast array of foreign substances. In the remainder of this introductory section, we focus on lymphocytes and the reactions of the adaptive immune system.

There are two types of adaptive immunity: humoral immunity, which protects against extracellular microbes and their toxins, and cell-mediated (or cellular) immunity, which is responsible for defense against intracellular microbes and against cancers. Humoral immunity is mediated by B (bone marrow–derived) lymphocytes and their secreted products, *antibodies* (also called *immunoglobulins*, Ig), and cellular immunity is mediated by T (thymus-derived) lymphocytes. Both classes of lymphocytes express highly specific receptors for a wide variety of substances, which are called *antigens*.

Cells of the Adaptive Immune System

The cells of the immune system are often divided into lymphoid cells (T and B lymphocytes) and myeloid cells (phagocytes, DCs, and others). The properties of lymphocytes and other cells and their roles in adaptive immune responses are described next.

Lymphocytes

Although T and B lymphocytes and their subsets are morphologically unimpressive and appear quite similar, they are remarkably heterogeneous and specialized in molecular properties and functions. The major classes of lymphocytes and their functions in adaptive immunity are illustrated in Fig. 6.5. Lymphocytes are not fixed in a particular tissue (as are cells in most of the organs of the body) but constantly circulate among lymphoid and other tissues via the blood and the lymphatic system. This property of lymphocytes allows the cells to home to any site of infection and thus promotes immune surveillance throughout the body. In lymphoid organs, different classes of lymphocytes are anatomically segregated and interact with one another only when stimulated to do so by encounters with antigens. Mature lymphocytes that have not encountered the antigen for which they are specific are said to be *naïve* (immunologically inexperienced). After they are activated by recognition of antigens and other signals described later, lymphocytes proliferate and differentiate into *effector cells*, which perform the function of eliminating microbes, and *memory cells*, which live in a state of heightened awareness and can react rapidly and strongly to combat the microbe in case it returns. The process of lymphocyte differentiation into effector and memory cells is summarized later. We start with a consideration of the diversity of T and B lymphocytes.

Lymphocyte Diversity. **Lymphocytes specific for many antigens exist before exposure to the antigen, and when an antigen appears, it selectively activates the antigen-specific cells.** This fundamental concept is called *clonal selection*. Lymphocytes of the same specificity are said to constitute a clone; all members of one clone express identical antigen receptors, which are different from the receptors in all other clones. There are about 10^{12} lymphocytes in a healthy adult, and it is estimated that there are 10^7 to 10^9 clones, each expressing receptors specific for a different antigen. It follows that the number of cells specific for any one antigen is very small, probably fewer than 1 in 100,000 lymphocytes. It is remarkable that so few cells with a particular specificity can accomplish the difficult task of combating various microbes; as discussed later, the immune system has developed many mechanisms for optimizing reactions to microbial antigens. It is also remarkable that the system can produce so many receptors, far more than can be individually encoded in the genome. The mechanisms by which this happens are now well understood and have many interesting clinical implications.

Antigen receptor diversity is generated by somatic recombination of the genes that encode antigen receptors. All cells of the body, including lymphocyte progenitors, contain antigen receptor genes in the germline (inherited) configuration, in which the genes encoding these receptors consist of spatially separated segments that cannot be transcribed into mRNAs. During lymphocyte maturation (in the thymus for T cells and the bone marrow for B cells), these gene segments are assembled by recombination, and sequence variation is introduced at the sites where the segments are joined. This creates many different genes that can be transcribed and translated into antigen receptors with diverse amino acid sequences, particularly in the regions of the receptors that recognize and bind antigen. The enzyme in developing lymphocytes that mediates recombination of these gene segments is the product of RAG-1 and RAG-2 (recombination-activating genes 1 and 2); inherited defects in RAG proteins result in a failure to generate mature lymphocytes. Although germline antigen receptor genes are present in all cells in the body, only T and B cells contain recombined (also called rearranged) antigen receptor genes (the TCR in T cells and Ig in B cells). Hence, the presence of recombined TCR or Ig genes, which can be demonstrated by molecular analysis, is a marker of T- or

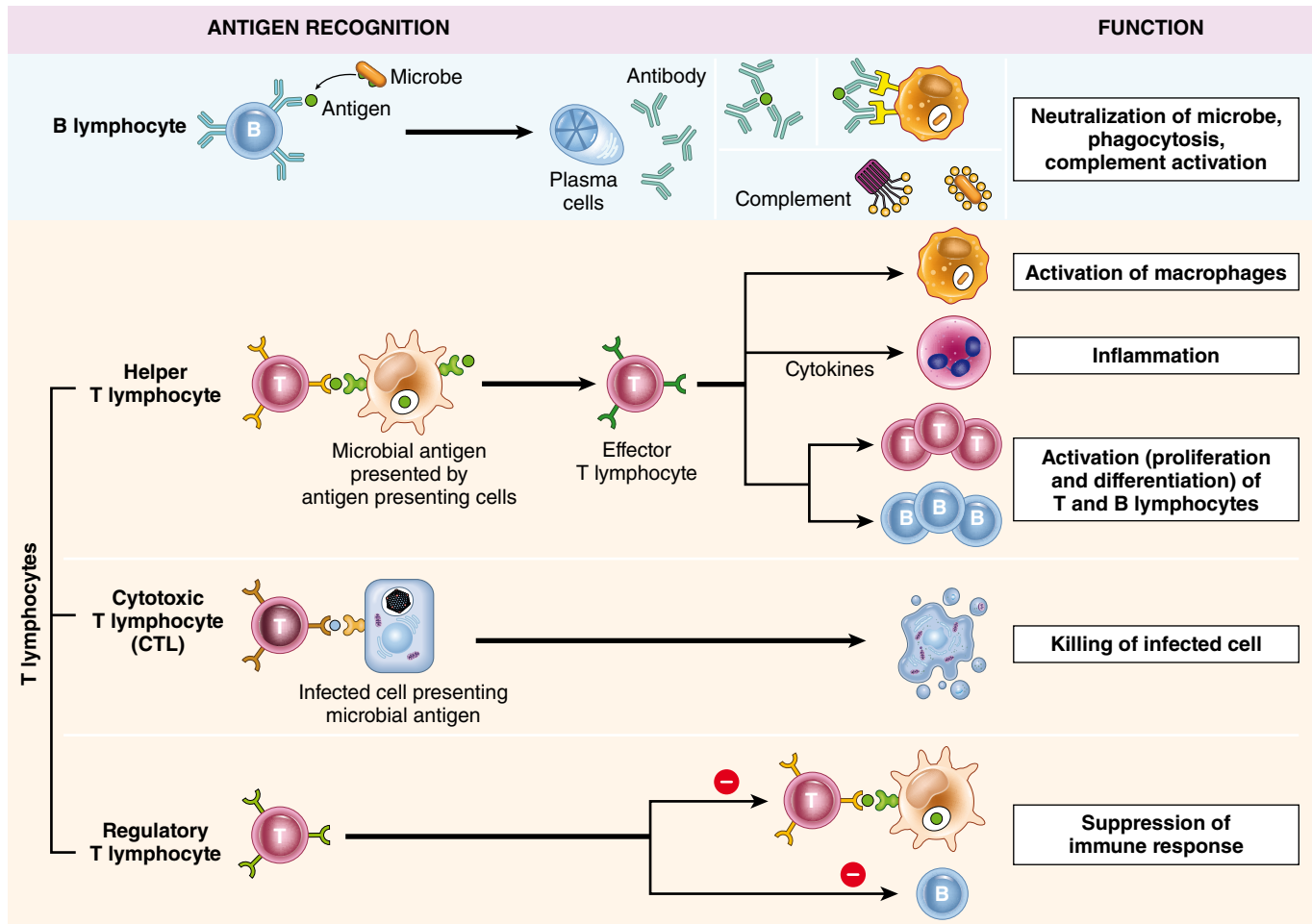


Fig. 6.5 The principal classes of lymphocytes and their functions. B and T lymphocytes are the cells of adaptive immunity. Several other classes of lymphocytes have been identified, including $\gamma\delta$ T cells, NK-T cells, and innate lymphoid cells (ILCs); the functions of these cells are not as well established as those of B and T lymphocytes and are not shown. Natural killer (NK) cells are discussed in Fig. 6.4.

B-lineage cells. Furthermore, because each T or B cell and its clonal progeny have a unique DNA rearrangement (that encodes a unique antigen receptor), it is possible to distinguish polyclonal (nonneoplastic) lymphocyte proliferations from monoclonal (neoplastic) lymphoid tumors by assessing the diversity of antigen receptor rearrangements within a population of lymphocytes. Thus, **assays that detect the clonality of antigen receptor gene rearrangements are useful in diagnosing lymphoid neoplasms** (Chapter 13).

T Lymphocytes. There are three major populations of T cells, which serve distinct functions (see Fig. 6.5). **Helper T lymphocytes stimulate B lymphocytes to make antibodies and activate other leukocytes (e.g., phagocytes) to destroy microbes; cytotoxic (killer) T lymphocytes (CTLs) kill infected cells; and regulatory T lymphocytes limit immune responses and prevent reactions against self antigens.** T cells recognize and respond to intracellular antigens (in contrast to B cells and their products, antibodies, which recognize extracellular antigens). To detect intracellular antigens, the antigen receptors of T cells recognize peptides that are derived from intracellular proteins and displayed by MHC molecules on the cell surface.

T lymphocytes develop in the thymus from precursors that arise from hematopoietic stem cells (HSCs). Mature T cells are found in the blood, where they constitute 60% to 70% of lymphocytes, and in T-cell zones of

secondary lymphoid organs (described later). In approximately 95% of T cells, the TCR consists of a disulfide-linked heterodimer made up of an α and a β polypeptide chain (Fig. 6.6), each having a variable (antigen-binding) region and a constant region. Each TCR is noncovalently linked to six polypeptide chains, which form the CD3 complex and the ζ chain dimer. The CD3 and ζ proteins are invariant (i.e., identical) in all T cells. They are involved in the transduction of signals into the T cell that are triggered by binding of antigen to the TCR. Together with the TCR, these proteins form the TCR complex.

A small population of mature T cells expresses another type of TCR composed of γ and δ polypeptide chains. The $\gamma\delta$ TCR recognizes peptides, lipids, and small molecules, without a requirement for display by MHC proteins. $\gamma\delta$ T cells tend to aggregate at epithelial surfaces, such as the skin and mucosa of the gastrointestinal and urogenital tracts, suggesting that these cells protect against microbes that try to enter through epithelia. However, the functions of $\gamma\delta$ T cells are not established. Another small subset of T cells expresses markers that are also found on NK cells; these cells are called NK-T cells. NK-T cells express a very limited diversity of TCRs, and they recognize glycolipids that are displayed by the MHC-like molecule CD1. The functions of NK-T cells are also not well defined.

In addition to CD3 and ζ proteins, T cells express several other proteins that assist the TCR complex in functional responses (see Fig. 6.6). These include CD4, CD8, CD28, and integrins. CD4 and

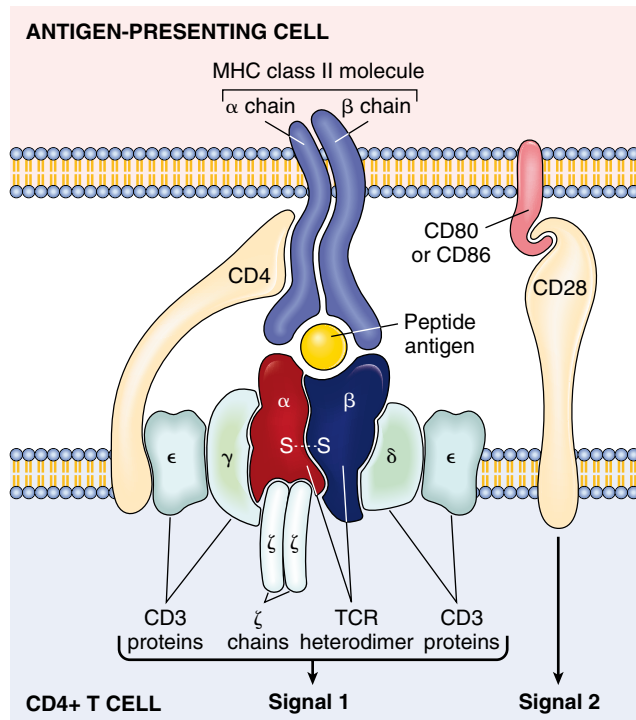


Fig. 6.6 The T-cell receptor (TCR) complex and other molecules involved in T-cell activation. The TCR heterodimer, consisting of an α and a β chain, recognizes antigen (in the form of peptide-MHC complexes expressed on antigen-presenting cells), and the linked CD3 complex and ζ chains initiate activating signals. CD4 and CD28 are also involved in T-cell activation. (Note that some T cells express CD8 and not CD4; these molecules serve analogous roles.) The sizes of the molecules are not drawn to scale. MHC, Major histocompatibility complex.

CD8 are expressed on two mutually exclusive subsets of $\alpha\beta$ T cells. Normally, approximately 60% of mature T cells are CD4+ and about 30% are CD8+. Most CD4+ T cells function as cytokine-secreting helper cells that assist macrophages and B lymphocytes to combat infections. Most CD8+ cells function as CTLs that destroy host cells

harboring microbes. CD4 and CD8 serve as coreceptors in T-cell activation. During antigen recognition, CD4 molecules bind to MHC class II molecules that are displaying antigen; CD8 molecules bind to MHC class I molecules; and the CD4 or CD8 coreceptor initiates signals that are necessary for activation of the T cells. Because of this requirement for coreceptors, CD4+ helper T cells can recognize and respond to antigen displayed only by MHC class II molecules, whereas CD8+ cytotoxic T cells recognize cell-bound antigens only in association with MHC class I molecules; this segregation is described later. Integrins are adhesion molecules that promote the attachment of T-cells to antigen-presenting cells (APCs).

To respond, T cells have to not only recognize antigen-MHC complexes, but also must receive additional signals provided by APCs. This process, in which CD28 plays an important role, is described later, when the steps in cell-mediated immune responses are summarized.

B Lymphocytes. B lymphocytes are the only cells in the body capable of producing antibodies, the mediators of humoral immunity. B lymphocytes develop from precursors in the bone marrow. Mature B cells constitute 10% to 20% of lymphocytes in the blood and are present in follicles in secondary lymphoid organs such as lymph nodes, spleen, and mucosa-associated lymphoid tissues. B cells recognize antigen by membrane-bound antibodies of the IgM and IgD isotypes, present on the surface of all mature, naive B cells, which serve as the B-cell antigen receptor (BCR) (Fig. 6.7). After stimulation by antigen and other signals (described later), B cells develop into *plasma cells*, veritable protein factories for producing antibodies, as well as long-lived memory cells. It is estimated that a single plasma cell can secrete hundreds to thousands of antibody molecules per second, a remarkable measure of the power of the immune response for combating pathogens.

In addition to membrane Ig, the B-cell antigen receptor complex contains a heterodimer of two invariant proteins called Ig α and Ig β . Like the CD3 and ζ proteins of the TCR complex, Ig α (CD79a) and Ig β (CD79b) are essential for signal transduction in response to antigen recognition. B cells also express several other molecules that serve important roles in their responses. These include the type 2 complement receptor (CR2, or CD21), which recognizes complement products generated during innate immune responses to microbes, and

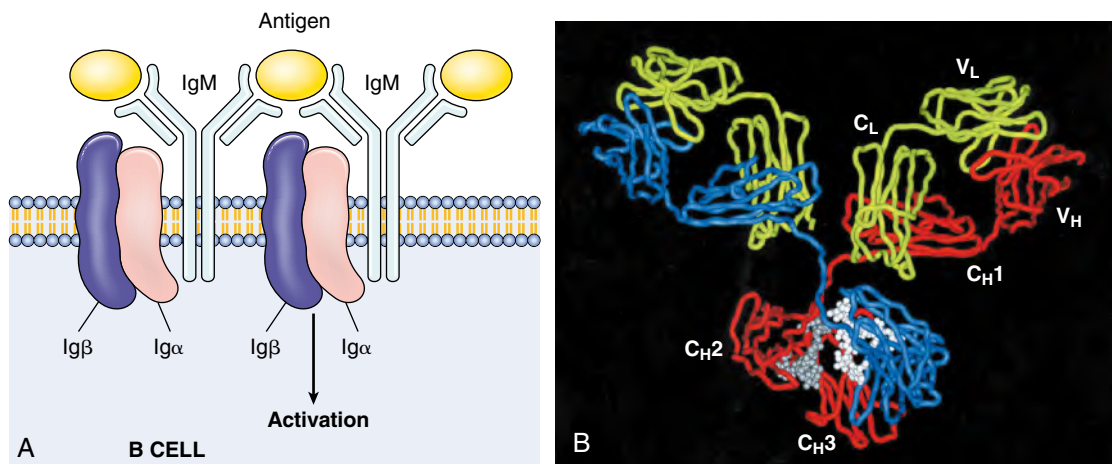


Fig. 6.7 Structure of antibodies and the B-cell antigen receptor. (A) The B-cell antigen receptor complex is composed of membrane immunoglobulin M (IgM; or IgD, not shown), which recognizes antigens, and the associated signaling proteins Ig α and Ig β . CD21 is a receptor for a complement component that also promotes B-cell activation. (B) Crystal structure of a secreted IgG molecule, showing the arrangement of the variable (V) and constant (C) regions of the heavy (H) and light (L) chains. Ig, Immunoglobulin. (Courtesy Dr. Alex McPherson, University of California, Irvine.)

CD40, which receives signals from helper T cells. CR2 is also used by Epstein-Barr virus (EBV) as a receptor to enter and infect B cells.

Myeloid Cells

The two types of myeloid cells that serve important functions in adaptive immune responses, DCs and macrophages, are described next.

Dendritic Cells. Dendritic cells (DCs; sometimes called interdigitating DCs) are the most important APCs for initiating T-cell responses against protein antigens. These cells have numerous fine cytoplasmic processes that resemble dendrites, from which they derive their name. Several features of DCs account for their key role in antigen presentation. (1) These cells are located at the right place to capture antigens—under epithelia, the common site of entry of microbes and foreign antigens, and in the interstitia of all tissues, where antigens may be produced. Immature DCs within the epidermis are called *Langerhans cells*. (2) DCs express many receptors for capturing and responding to microbes (and other antigens), including TLRs and lectins. (3) In response to microbes, DCs are recruited to the T-cell zones of lymphoid organs, where they are ideally located to present antigens to naive T cells. (4) DCs express high levels of MHC and other molecules needed for antigen presentation and T-cell activation.

A second type of cell with dendritic morphology is present in the germinal centers of lymphoid follicles in the spleen and lymph nodes and is called the *follicular dendritic cell* (FDC). These cells are specialized fibroblasts that bear Fc receptors for IgG and receptors for C3b and can trap antigens bound to antibodies or complement proteins. They play a role in humoral immune responses by presenting antigens to B cells in the germinal center, part of a process through which only B cells that express antibodies with high affinity for antigens survive and go on to mature into plasma cells or memory cells. FDCs do not participate in antigen presentation to T cells.

Macrophages. Macrophages are a part of the mononuclear phagocyte system; their origin, differentiation, and role in inflammation are discussed in Chapter 3. Here, their important functions in the induction and effector phases of adaptive immune responses are discussed.

- Macrophages that have phagocytosed microbes and protein antigens process the antigens and present peptide fragments to T cells. Thus, macrophages function as APCs in T-cell activation.
- Macrophages are key effector cells in certain forms of cell-mediated immunity, the reaction that serves to eliminate intracellular microbes. In this type of response, T cells activate macrophages and enhance their ability to kill ingested microbes (discussed later).
- Macrophages also participate in the effector phase of humoral immunity. Macrophages efficiently phagocytose and destroy microbes that are opsonized (coated) by IgG or C3b.

Tissues of the Immune System

The tissues of the immune system consist of the *primary* (also called *generative*, or *central*) lymphoid organs, in which T and B lymphocytes mature and become competent to respond to antigens, and the *secondary* (or *peripheral*) lymphoid organs, in which adaptive immune responses to microbes are initiated.

Primary Lymphoid Organs. The principal primary lymphoid organs are the thymus, where T cells develop, and the bone marrow, the site of production of all other blood cells, including naive B cells. These organs are described in Chapter 13.

Secondary Lymphoid Organs. The secondary lymphoid organs—lymph nodes, spleen, and the mucosal and cutaneous lymphoid

tissues—are the tissues where adaptive immune responses are initiated and develop. Several features of these organs promote the generation of adaptive immunity—antigens are concentrated in these organs, naive lymphocytes circulate through them searching for the antigens, and different lymphocyte populations (such as T and B cells) are brought together to coordinate cellular and humoral immune responses.

- *Lymph nodes* are nodular aggregates of lymphoid tissues located along lymphatic channels throughout the body (Fig. 6.8). As lymph slowly suffuses through lymph nodes, APCs are positioned to recognize antigens (e.g., derived from microbes that may enter through epithelia into tissues and are carried in the lymph). In addition, DCs pick up and transport antigens of microbes from epithelia and tissues via lymphatic vessels to the lymph nodes. Thus, the antigens of microbes that enter through epithelia or colonize tissues become concentrated in draining lymph nodes. Because most foreign antigens enter through epithelia or are produced in tissues, lymph nodes are the site of generation of most adaptive immune responses.
- The *spleen* is an abdominal organ that serves the same role in immune responses to blood-borne antigens as the lymph nodes do in responses to lymph-borne antigens. Blood entering the spleen flows through a network of sinusoids lined by macrophages and DCs. Blood-borne antigens are trapped in the spleen by these cells, which can then initiate adaptive immune responses to these antigens.
- The *cutaneous and mucosal lymphoid systems* are located under the epithelia of the skin and the gastrointestinal and respiratory tracts, respectively. They respond to antigens that enter through breaches in the epithelium. Pharyngeal tonsils and Peyer patches of the intestine are two anatomically defined mucosal lymphoid tissues. At any time, a large fraction of the body's lymphocytes are in the mucosal tissues (reflecting the large size of these tissues), and many of these are memory cells.

Within the secondary lymphoid organs, naive T and B lymphocytes are segregated into different regions (see Fig. 6.8C). In lymph nodes, the B cells are concentrated in discrete structures, called follicles, located around the periphery, or cortex, of each node. If the B cells in a follicle have recently responded to an antigen, this follicle may contain a central region called a germinal center. The T lymphocytes are concentrated in the paracortex, adjacent to the follicles. The follicles contain the FDCs that are involved in the activation of B cells, and the paracortex contains the DCs that present antigens to T lymphocytes. In the spleen, T lymphocytes are concentrated in periarteriolar lymphoid sheaths surrounding small arterioles, and B cells reside in follicles akin to those found in lymph nodes (the so-called splenic white pulp).

Lymphocyte Recirculation. Lymphocytes constantly recirculate through the blood to particular tissues; naive lymphocytes traverse the secondary lymphoid organs where immune responses are initiated, and effector lymphocytes migrate to sites of infection and inflammation. This process of lymphocyte recirculation is most important for T cells, because naive T cells have to circulate through the secondary lymphoid organs where antigens are concentrated and effector T cells have to migrate to sites of infection to eliminate microbes. In contrast, plasma cells remain in lymphoid organs and the bone marrow and do not need to travel to sites of infection because they secrete antibodies that are carried through the blood to distant tissues.

Major Histocompatibility Complex Molecules: The Peptide Display System of Adaptive Immunity

The function of MHC molecules is to display peptide fragments of protein antigens for recognition by antigen-specific T cells. Because MHC molecules are fundamental to antigen recognition by T cells and

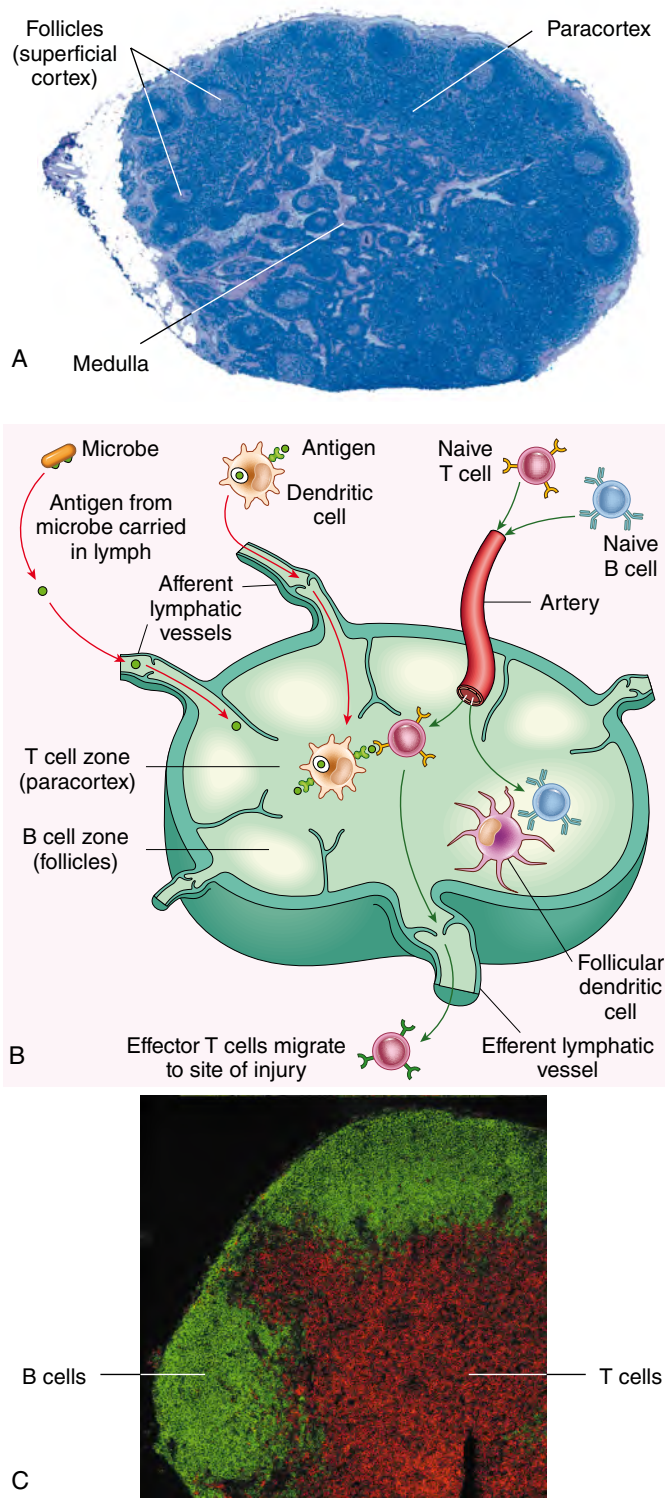


Fig. 6.8 Morphology of a lymph node. (A) The histology of a lymph node, with an outer cortex containing follicles and an inner medulla. (B) The migration of B cells and T cells in different regions of the lymph node, illustrated schematically. (C) The location of B cells (stained green, using the immunofluorescence technique) and T cells (stained red) in a lymph node. (Courtesy Drs. Kathryn Pape and Jennifer Walter, University of Minnesota School of Medicine, Minneapolis.)

are linked to many autoimmune diseases, it is important to briefly review their structure and function. MHC molecules were discovered as products of genes that evoke rejection of transplanted organs, and their name derives from their role in determining tissue compatibility between individuals. In humans, the MHC molecules are called *human leukocyte antigens* (HLAs) because they were initially detected on leukocytes. The genes encoding HLA molecules are clustered on a small segment of chromosome 6 (Fig. 6.9). The HLA system is highly polymorphic; there are thousands of distinct MHC gene alleles in humans, and as a result everyone's HLA alleles differ from those inherited by most other individuals in the population. This, as we see subsequently, constitutes a formidable barrier in organ transplantation.

Based on their structure, cellular distribution, and function, MHC proteins are divided into two major classes.

- **MHC class I molecules** are expressed on all nucleated cells and platelets. They are heterodimers consisting of a polymorphic α , or heavy, chain (44-kD) linked noncovalently to a smaller (12-kD) nonpolymorphic protein called β_2 -microglobulin. The α chains are encoded by three genes, designated *HLA-A*, *HLA-B*, and *HLA-C*, that lie close to one another in the MHC locus (see Fig. 6.9). The extracellular region of the α chain is divided into three domains: α_1 , α_2 , and α_3 . The α_1 and α_2 domains form a cleft, or groove, where peptides bind. The polymorphic amino acid residues line the sides and the base of the peptide-binding groove, explaining why different class I alleles bind different peptides.

MHC class I molecules display peptides that are derived from cytosolic proteins, including protein antigens of viruses and tumors, which are all recognized by CD8⁺ T cells. Cytosolic proteins are degraded in proteasomes, and peptides are transported into the endoplasmic reticulum (ER), where they bind to newly synthesized class I molecules. Peptide-loaded MHC molecules then associate with β_2 -microglobulin to form a stable complex, which is transported to the cell surface. The nonpolymorphic α_3 domain of MHC class I molecules has a binding site for CD8, and therefore the peptide-class I complexes are recognized by CD8⁺ T cells, which function as CTLs. In this interaction, the TCR recognizes the MHC-peptide complex, and the CD8 molecule, acting as a coreceptor, binds to the class I heavy chain. Since CD8⁺ T cells recognize peptides only if presented as a complex with MHC class I molecules, CD8⁺ T cells are said to be MHC class I-restricted. Because important functions of CD8⁺ CTLs include the elimination of viruses, which may infect any nucleated cell, and killing of tumor cells, which may arise from any nucleated cell, it makes good sense that all nucleated cells express MHC class I molecules and can be surveyed by CD8⁺ T cells.

- **MHC class II molecules** are encoded in a region called *HLA-D*, which has three subregions: *HLA-DP*, *HLA-DQ*, and *HLA-DR*. Each class II molecule is a heterodimer consisting of a noncovalently associated α chain and β chain, both of which are polymorphic. The extracellular portions of the α and β chains both have two domains, designated α_1 and α_2 , and β_1 and β_2 , respectively. The crystal structure of class II molecules has revealed that, like class I molecules, they have peptide-binding clefts facing outward (see Fig. 6.9). This cleft is formed by an interaction of the α_1 and β_1 domains, and it is in this portion that most class II alleles differ. Thus, as with class I molecules, polymorphism of class II molecules is associated with differential binding of antigenic peptides.

MHC class II molecules present antigens derived from extracellular microbes and proteins following their internalization into endosomes or lysosomes. Here, the internalized proteins are proteolytically digested, producing peptides that then associate with class II heterodimers in the vesicles, from which they are transported to the cell surface as stable peptide-MHC complexes. The class II β_2 domain has a binding site for CD4, and therefore, the class

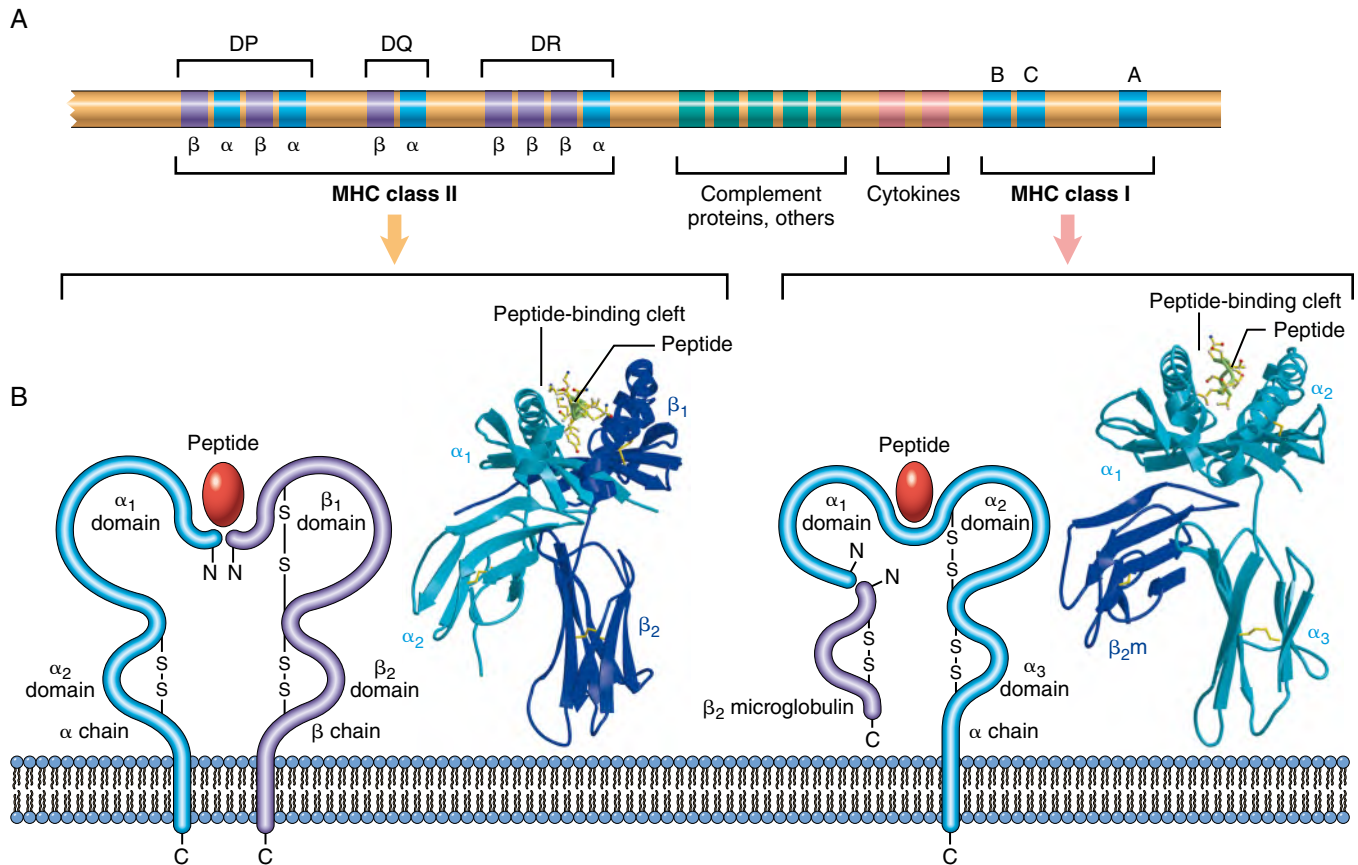


Fig. 6.9 The human leukocyte antigen (HLA) complex and the structure of HLA molecules. (A) The location of genes in the HLA complex. The relative locations, sizes, and distances between genes are not to scale. Genes that encode several proteins involved in antigen processing (the TAP transporter, components of the proteasome, and HLA-DM) are located in the class II region (not shown). (B) Schematic diagrams and crystal structures of HLA class I and class II molecules. *MHC*, Major histocompatibility complex. (Crystal structures courtesy Dr. P. Bjorkman, California Institute of Technology, Pasadena.)

II-peptide complex is recognized by CD4⁺ T cells, which function as helper cells. Because CD4⁺ T cells can recognize antigens only in the context of self class II molecules, they are referred to as MHC class II restricted. In contrast to class I molecules, class II molecules are mainly expressed on cells that present ingested antigens and respond to T-cell help (macrophages, B lymphocytes, and DCs).

The combination of HLA alleles found on a single chromosome is called the HLA haplotype. Any given individual inherits one set of HLA genes from each parent and (assuming the parents are unrelated) typically expresses two different molecules for every locus. Because of the polymorphism of the HLA genes, virtually innumerable combinations of molecules exist in the population, and everyone expresses an MHC profile on their cell surface that is different from the haplotypes of most other individuals. It is believed that this degree of polymorphism evolved to ensure that at least some individuals in a species would be able to display any microbial peptide and thus provide protection against any infection. This polymorphism also means that no two individuals (other than identical twins) are likely to express the same MHC molecules, and therefore grafts exchanged between these individuals are recognized as foreign and attacked by the immune system. Because each haplotype is inherited as a block, and there are two sets of genes from each parent, the chance that siblings will have the same MHC is 1 in 4. This is why siblings are screened first as potential donors for patients in need of a kidney or hematopoietic stem cell transplant when feasible.

MHC molecules play several key roles in T cell-mediated immune responses. They provide the mechanism for T cells to detect antigens that are inside cells (microbial and tumor antigens). Furthermore, by segregating cytosolic and internalized antigens, MHC molecules ensure that the correct immune response is mounted against different microbes—CTL-mediated killing of cells harboring cytosolic microbes and tumor antigens, and helper T cell-mediated antibody production and macrophage activation to combat extracellular and phagocytosed microbes.

Because different antigenic peptides bind to different MHC molecules, it follows that an individual mounts an immune response against a protein antigen only if that person inherits an MHC variant that can bind peptides derived from the antigen and present it to T cells. The consequences of inheriting a given MHC (e.g., class II) variant depend on the nature of the antigen bound by the class II molecule. For example, if the antigen is a peptide from ragweed pollen, the individual who expresses class II molecules capable of binding the antigen would be genetically prone to allergic reactions against ragweed. In contrast, an inherited capacity to bind a bacterial peptide may provide resistance to the infection by evoking a protective antibody response. This role of MHC molecules in determining T cell responses to antigens may explain why several autoimmune and other diseases are associated with the inheritance of particular HLA alleles. The mechanisms underlying these associations are discussed when the pathogenesis of autoimmune diseases is considered later in the chapter.

Cytokines: Messenger Molecules of the Immune System

The induction and regulation of immune responses involve multiple interactions among lymphocytes, DCs, macrophages, other inflammatory cells (e.g., neutrophils), and endothelial cells. Some of these interactions depend on cell-cell interactions; however, **many functions of leukocytes are stimulated and regulated by secreted proteins called cytokines**. Molecularly defined cytokines are called interleukins because they mediate communication between leukocytes (although many also act on cells other than leukocytes). Most cytokines have a wide spectrum of effects, and some are produced by several different cell types. Most of these cytokines act on the cells that produce them (autocrine actions), on neighboring cells (paracrine), and rarely at a distance (endocrine).

Cytokines contribute to different types of immune responses.

- In *innate immune responses*, cytokines are produced rapidly after an encounter with microbes and other stimuli, and they function to induce inflammation and inhibit virus replication. These cytokines include TNF, IL-1, IL-12, type I IFNs, IFN- γ , and chemokines (Chapter 3). Their major sources are macrophages, DCs, ILCs, and NK cells, but endothelial, epithelial, and various stromal cells can also produce them.
- In *adaptive immune responses*, cytokines are produced principally by CD4⁺ T lymphocytes activated by antigen and other signals, and they function to promote lymphocyte proliferation and differentiation and to activate effector cells. The main ones in this group are IL-2, IL-4, IL-5, IL-17, and IFN- γ ; their roles in immune responses are described later. Some cytokines serve mainly to limit and terminate immune responses; these include TGF- β and IL-10.
- Some cytokines stimulate *hematopoiesis* and are called colony-stimulating factors (CSFs) because they are assayed by their ability to stimulate formation of blood cell colonies from bone marrow progenitors (Chapter 13). Their functions are to increase leukocyte production during immune and inflammatory responses, both to increase their numbers and to replace leukocytes that die during such responses. They are produced by marrow stromal cells, T lymphocytes, macrophages, and other cells. Examples include G-CSF, GM-CSF, and IL-3.

The knowledge gained about cytokines has numerous therapeutic applications. Inhibiting cytokine production or actions is an approach for controlling the harmful effects of inflammation and tissue-damaging immune reactions. Patients with rheumatoid arthritis often show dramatic responses to antagonists of the cytokine tumor necrosis factor (TNF), an example of molecularly targeted therapy. Many other cytokine antagonists are now approved for the treatment of various inflammatory disorders. Conversely, administration of cytokines is used to boost processes that are normally dependent on these proteins, such as hematopoiesis (erythropoietin, GM-CSF).

Overview of Lymphocyte Activation and Immune Responses

All adaptive immune responses develop in steps, which include antigen recognition; activation of specific lymphocytes to proliferate and then differentiate into effector and memory cells; elimination of the antigen; and decline of the response, with memory cells being the long-lived survivors. The major events in each step are summarized next; these general principles apply to protective responses against microbes as well as pathologic responses that injure the host.

Display and Recognition of Antigens

Microbes and other foreign antigens can enter the body anywhere. It is impossible for lymphocytes to effectively patrol every portal of antigen entry, because there are not enough antigen-specific lymphocytes to

constantly cover all of this “terrain.” To overcome this problem, antigens are captured and concentrated in secondary lymphoid organs through which naive lymphocytes circulate, thus increasing the likelihood of a lymphocyte finding the antigen it can recognize. Microbes and their protein antigens are captured by DCs that are resident in epithelia and tissues. These cells carry their antigenic cargo to draining lymph nodes (Fig. 6.10). Here the antigens are processed and displayed complexed with MHC molecules on the cell surface, where the antigens are recognized by T cells.

B lymphocytes use their antigen receptors (membrane-bound antibody molecules) to recognize antigens of many different chemical types, including proteins, polysaccharides, and lipids.

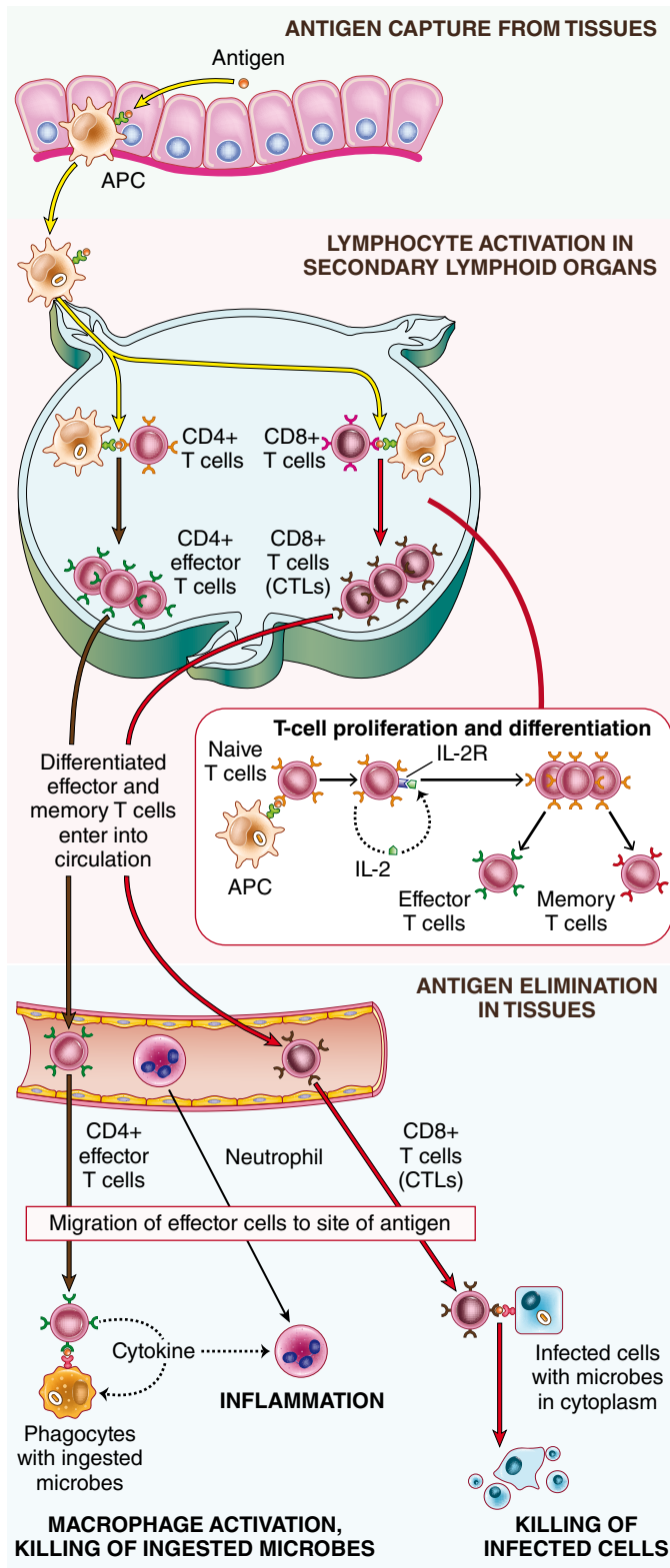
Even before the antigens of a microbe are recognized by T and B lymphocytes, the microbe elicits an innate immune response through pattern recognition receptors expressed on innate immune cells; this is the first line of defense that also serves to activate adaptive immunity. In the case of immunization with a protein antigen, microbial mimics, called adjuvants, are given with the antigen, and these stimulate innate immune responses. During the innate response, the microbe or adjuvant activates APCs to express molecules called *co-stimulators* and to secrete cytokines that stimulate the proliferation and differentiation of T lymphocytes. The principal costimulators for T cells are the B7 proteins (CD80 and CD86) that are expressed on APCs and are recognized by the CD28 receptor on naive T cells. Thus, antigen (“signal 1”) and costimulatory molecules produced during innate immune responses to microbes (“signal 2”) function cooperatively to activate antigen-specific lymphocytes (see Fig. 6.6). The requirement for microbe-triggered signal 2 ensures that the adaptive immune response is induced by microbes and not by harmless substances. In immune responses to tumors and transplants, “signal 2” may be provided by substances released from necrotic cells (the “damage-associated molecular patterns” mentioned earlier). Note, however, that simply increased expression of costimulators, e.g., in an infection, is not sufficient to induce immune responses against self antigens that may be presented by APCs, probably because powerful control mechanisms prevent such anti-self responses (discussed later).

The reactions and functions of T and B lymphocytes differ in important ways and are best considered separately even though both may be activated concurrently in an immune response.

Cell-Mediated Immunity: Activation of T Lymphocytes and Elimination of Intracellular Microbes

Naive T lymphocytes are activated by antigen and costimulators in secondary lymphoid organs and proliferate and differentiate into effector cells that migrate to any site where microbial antigens are present (see Fig. 6.10). One of the earliest responses of CD4⁺ helper T cells is secretion of the cytokine IL-2 and expression of high-affinity receptors for IL-2. This creates an autocrine loop wherein IL-2 acts as a growth factor that stimulates T-cell proliferation, leading to an increase in the number of antigen-specific lymphocytes. The functions of helper T cells are mediated by the combined actions of CD40-ligand (CD40L) and cytokines. When CD4⁺ helper T cells recognize antigens being displayed by macrophages or B lymphocytes, the T cells express CD40L, which engages CD40 on the macrophages or B cells and activates these cells.

Some of the activated CD4⁺ T cells differentiate into effector cells that secrete distinct sets of cytokines and perform different functions (Fig. 6.11). Cells of the Th1 (type 1 T helper) subset secrete the cytokine IFN- γ , which is a potent macrophage activator. The combination of CD40- and IFN- γ -mediated activation results in “classical” macrophage activation (Chapter 3), leading to the production of microbicidal substances in macrophages that destroy ingested microbes.



Th2 cells produce IL-5, which activates eosinophils that kill helminthic parasites. IL-4 and IL-13 produced by Th2 cells induce the “alternative” pathway of macrophage activation, which is associated with tissue repair and fibrosis (Chapter 3). Th17 cells, so called because the signature cytokine of these cells is IL-17, recruit neutrophils and monocytes,

Fig. 6.10 Cell-mediated immunity. Dendritic cells capture microbial antigens from epithelia and tissues and transport them to lymph nodes. During this process, the dendritic cells mature and express high levels of major histocompatibility complex (MHC) molecules and costimulators. Naive T cells recognize MHC-associated peptide antigens displayed on dendritic cells. The T cells are activated to proliferate and to differentiate into effector and memory cells, which migrate to sites of infection and serve various functions in cell-mediated immunity. CD4+ effector T cells of the Th1 subset recognize the antigens of microbes ingested by phagocytes and activate the phagocytes to kill the microbes; other subsets of effector cells enhance leukocyte recruitment and stimulate different types of immune responses. CD8+ cytotoxic T lymphocytes (CTLs) kill infected cells harboring microbes in the cytoplasm. Some activated T cells remain in the lymphoid organs and help B cells to produce antibodies, and some T cells differentiate into long-lived memory cells (not shown). APC, Antigen-presenting cell; IL, interleukin.

which destroy extracellular bacteria and fungi and are involved in some inflammatory diseases. Each type of helper cell is induced by the type of microbe it is best at combating. Th1 cells are induced by the cytokine IL-12, which is produced by macrophages and DCs that encounter intracellular microbes and viruses. Th2 cells are induced by cytokines produced by damaged or stressed epithelial cells (IL-25, IL-33, TSLP [thymic stromal lymphopoietin]), accounting for the abundance of this subset at epithelial barriers and in response to microbes encountered at these barriers. Th17 cells are induced by cytokines produced by macrophages and DCs in response to fungi and some bacteria. Antibodies that block the inducing and effector cytokines of each subset are used to treat diseases caused by that subset.

The Th1, Th2, and Th17 effector cells are generated in lymphoid organs and migrate to sites of infection, where they perform their defensive functions. Some helper T cells remain in the lymphoid organs and are concentrated in follicles where B cells reside, and are hence called T follicular helper (Tfh) cells. Their function is to help B cells to produce high-affinity antibodies.

The early steps in the activation of CD8+ T cells are similar to those of CD4+ cells. CD8+ cells proliferate even more than CD4+ cells, and this proliferation is driven by costimulators and IL-2. Activated CD8+ T lymphocytes differentiate into CTLs that kill cells harboring microbes in the cytosol. By destroying the infected cells, CTLs eliminate the reservoirs of infection. CTLs also kill tumor cells by recognizing tumor-specific antigens derived from mutated or abnormal cytosolic proteins.

Humoral Immunity: Activation of B Lymphocytes and Elimination of Extracellular Microbes

Upon activation, B lymphocytes proliferate and then differentiate into plasma cells that secrete different classes of antibodies with distinct functions (Fig. 6.12). Antibody responses to most protein antigens require T cell help and are said to be *T-dependent*. In these responses, B cells that recognize protein antigens by their Ig receptors endocytose these antigens into vesicles, degrade them, and display peptides bound to MHC class II molecules for recognition by helper T cells. The helper T cells are activated and express CD40L and secrete cytokines, which work together to stimulate the B cells. Many polysaccharide and lipid antigens cannot be recognized by T cells (because they cannot bind to MHC molecules) but have multiple identical antigenic determinants (epitopes) that are able to engage many antigen receptor molecules on each B cell and initiate the process of B-cell activation; these responses are said to be *T-independent*. T-independent

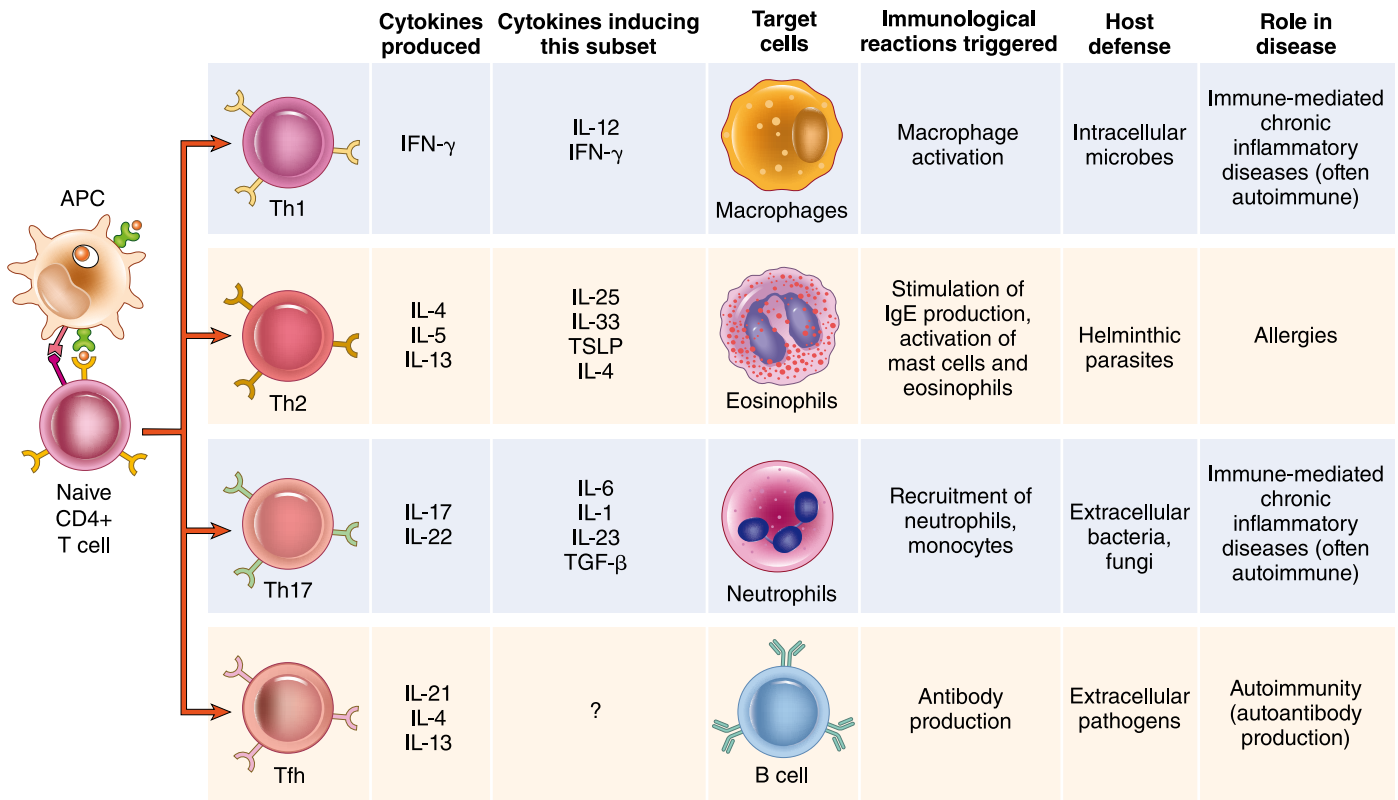


Fig. 6.11 Subsets of helper T (*Th*) cells. In response to stimuli (mainly cytokines) present at the time of antigen recognition, naive CD4⁺ T cells may differentiate into populations of effector cells that produce distinct sets of cytokines and perform different functions. The dominant immune reactions elicited by each subset, and its role in host defense and immunologic diseases, are summarized. Some activated T cells produce multiple cytokines or only a limited number of cytokines and do not fall into a distinct subset. APC, Antigen-presenting cell; IFN, interferon; IL, interleukin; TGF, transforming growth factor; TSLP, thymic stromal lymphopoietin.

responses are relatively simple, whereas T-dependent responses show features such as Ig isotype switching and affinity maturation (described later), which require T cell help and lead to responses that are more varied and effective.

Each plasma cell is derived from an antigen-stimulated B cell and secretes antibodies that recognize the same antigen that was bound by the BCR to initiate the response. Polysaccharides and lipids stimulate secretion mainly of IgM antibody. Protein antigens, by virtue of CD40L- and cytokine-mediated helper actions of Tfh cells, induce the production of antibodies of different classes, or isotypes (IgG, IgA, IgE), a process called *isotype* (or *class*) *switching*. Tfh cells also stimulate the production of antibodies with high affinities for the antigen. This process, called *affinity maturation*, improves the quality of the humoral immune response, and stimulating affinity maturation is an important goal of vaccination. These two processes are initiated when B cells activated by recognition of protein antigens migrate into follicles, receive signals from Tfh cells, and begin to proliferate to form germinal centers, which are sites of isotype switching and affinity maturation.

The humoral immune response combats microbes in many ways (see Fig. 6.12). Antibodies bind to microbes and prevent them from infecting cells, thus neutralizing the microbes. IgG antibodies coat (opsonize) microbes and target them for phagocytosis, since phagocytes (neutrophils and macrophages) express receptors for the Fc tails of IgG. IgG and IgM activate the complement system by the classic pathway, and complement products promote phagocytosis

and destruction of microbes. Some antibodies serve special roles at particular anatomic sites. IgA is secreted from mucosal epithelia and neutralizes microbes in the lumens of the respiratory and gastrointestinal tracts (and other mucosal tissues). IgG is actively transported across the placenta and protects the newborn until the immune system becomes mature. IgE binds to mast cells and induces their degranulation.

Some antibody-secreting plasma cells, particularly those that are generated in germinal centers, migrate to the bone marrow, and take up residence there for months or even years, continuously producing antibodies during this time.

Decline of Immune Responses and Immunologic Memory

Most effector lymphocytes induced by an infectious pathogen die by apoptosis after the microbe is eliminated and the stimuli for lymphocyte activation wane, thus returning the immune system to its resting state. The initial activation of lymphocytes also generates long-lived memory cells, which may survive for many years after the infection in secondary lymphoid organs, the circulation, and peripheral tissues. Memory cells are an expanded pool of antigen-specific lymphocytes (more numerous than the naive cells specific for any antigen that are present before encounter with that antigen), and they respond faster and more effectively when reexposed to the antigen than do naive cells. Generation of memory cells underlies the effectiveness of vaccination.

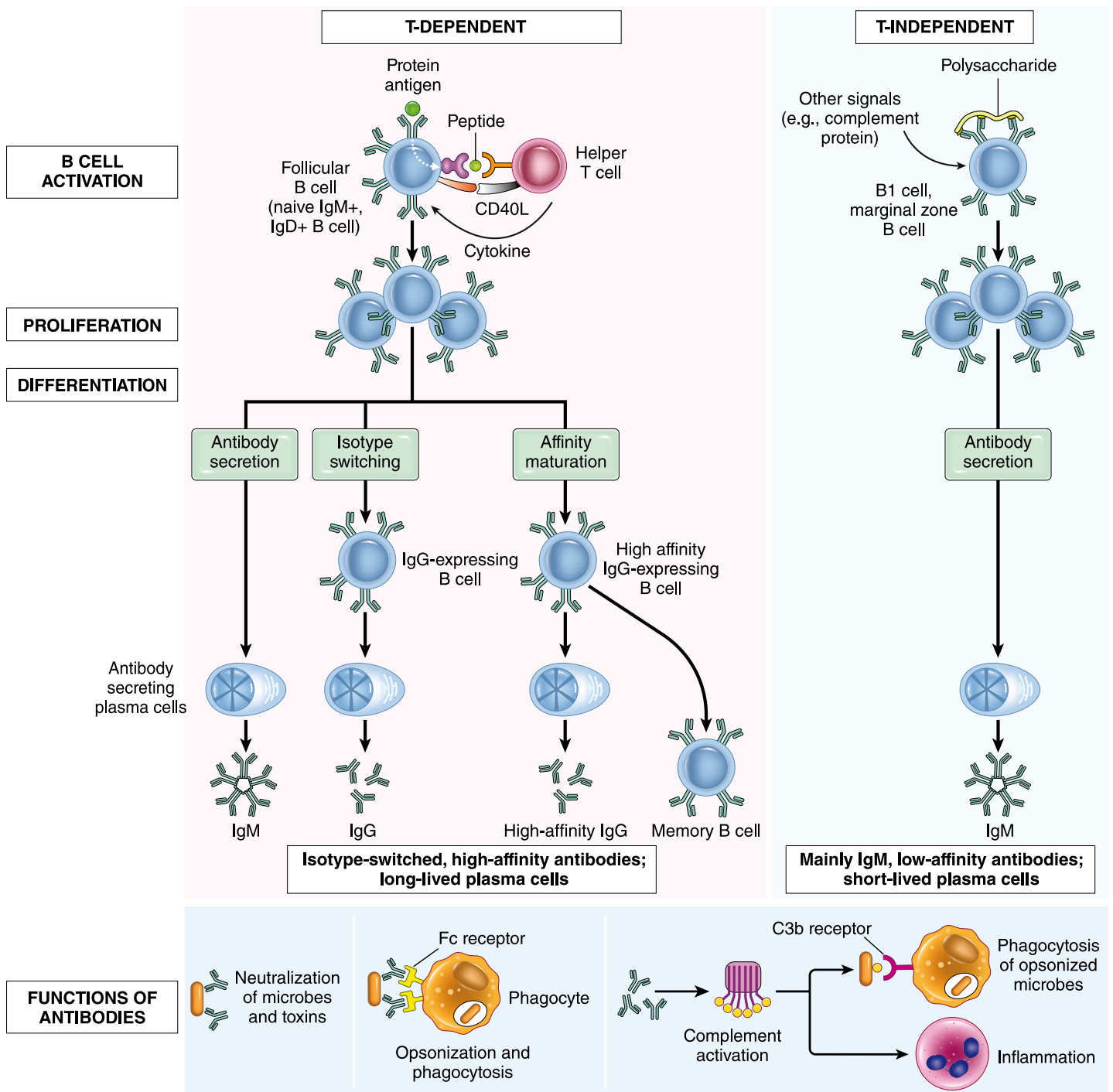


Fig. 6.12 Humoral immunity. In T-dependent responses, naive B lymphocytes recognize antigens, and under the influence of helper T cells and other stimuli (not shown), the B cells are activated to proliferate and to differentiate into antibody-secreting plasma cells. Some of the activated B cells undergo heavy-chain class switching and affinity maturation, and some become long-lived memory cells. Antibodies of different heavy-chain classes (isotypes) perform different effector functions, shown on the right. In T-independent responses, the B cells respond to multivalent antigens in the absence of T cell help and produce mainly IgM of low affinity. The bottom panel shows the principal functions of antibodies in host defense. *Ig*, Immunoglobulin.

KEY CONCEPTS

The Normal Immune Response: Overview of Cells, Tissues, Receptors, and Mediators

- Innate immunity provides early rapid defense against microbes and clears damaged and dead cells; adaptive immunity provides the later and more effective defense.
- Components of the innate immune system include epithelial barriers, phagocytes, NK cells, and plasma proteins (e.g., complement). Innate immune reactions are often manifested as inflammation.
- The innate immune system uses several families of receptors (e.g., toll-like receptors) to recognize molecules present in and shared by various types of microbes and produced by damaged cells.
- Lymphocytes are the mediators of adaptive immunity and the only cells that produce specific and diverse receptors for antigens.
- The antigen receptors of T (thymus-derived) lymphocytes, called T-cell receptors (TCRs), recognize peptide fragments of protein antigens that are displayed by MHC molecules on the surface of APCs.
- B (bone marrow–derived) lymphocytes express membrane-bound antibodies that recognize a wide variety of antigens. B cells are activated to become plasma cells, which secrete antibodies.
- NK cells destroy cells that are infected by some microbes, or are stressed and damaged beyond repair. NK cells express inhibitory receptors that recognize MHC molecules that are normally expressed on healthy cells and are thus prevented from killing normal cells.
- The cells of the immune system are organized in tissues, some of which are the sites of production of mature lymphocytes (the primary, or generative, lymphoid organs, the bone marrow and thymus), and others are the sites of immune responses (the secondary, or peripheral, lymphoid organs, including lymph nodes, spleen, and mucosal lymphoid tissues).
- Microbes and other foreign antigens are captured by dendritic cells (DCs) and transported to lymph nodes, where the antigens are recognized by naive T lymphocytes. The lymphocytes are activated to proliferate and differentiate into effector and memory cells.
- Cell-mediated immunity is the reaction of T lymphocytes, designed to combat cell-associated microbes (e.g., phagocytosed microbes and microbes in the cytoplasm of infected cells). Humoral immunity is mediated by antibodies and is effective against extracellular microbes (in the circulation and mucosal lumens).
- CD4+ helper T cells help B cells to produce antibodies, activate macrophages to destroy ingested microbes, stimulate recruitment of leukocytes, and regulate immune responses to protein antigens. The functions of CD4+ T cells are mediated by secreted proteins called cytokines.
- CD8+ cytotoxic T lymphocytes kill cells that express antigens in the cytoplasm that are seen as foreign (e.g., virus-infected and tumor cells) and can also produce cytokines.
- Antibodies secreted by plasma cells neutralize microbes and block their infectivity, and they promote the phagocytosis and destruction of pathogens. Antibodies also confer passive immunity to neonates.

The brief outline of basic immunology presented here provides a foundation for considering the diseases of the immune system. We first discuss immune reactions that cause injury, called hypersensitivity reactions, and then disorders caused by the failure of tolerance to self antigens, called autoimmune disorders, and the rejection of transplants. This is followed by diseases caused by a defective immune system, called immunodeficiency diseases. We close with a consideration of amyloidosis, a disorder that is often associated with immune and inflammatory diseases.

HYPERSENSITIVITY: IMMUNOLOGICALLY MEDIATED TISSUE INJURY

Injurious immune reactions, called hypersensitivity, are responsible for the pathology associated with immunologic diseases. This term arose from the idea that individuals who have been previously exposed to an antigen will manifest detectable reactions to that antigen and are therefore said to be sensitized. Hypersensitivity implies an excessive or harmful reaction to an antigen. There are several important general features of hypersensitivity disorders.

- *Hypersensitivity reactions can be elicited by exogenous environmental antigens (microbial and nonmicrobial) or endogenous self antigens.* Exogenous antigens include those in dust, pollen, food, drugs, microbes, and various chemicals. The immune responses against such exogenous antigens may take several forms, ranging from annoying but trivial discomforts, such as itching of the skin, to potentially fatal reactions, such as anaphylaxis. Some of the most common reactions to environmental antigens cause the group of diseases known as allergy. Immune responses against self, or autologous, antigens cause autoimmune diseases.
- *Hypersensitivity usually results from an imbalance between the effector mechanisms of immune responses and the control mechanisms that serve to limit such responses.* In fact, in many hypersensitivity diseases, it is suspected that the underlying cause is a failure of normal regulation. We will return to this concept when we consider autoimmunity.
- *The development of hypersensitivity diseases (both allergic and autoimmune) is often associated with the inheritance of particular susceptibility genes.* HLA genes and many non-HLA genes have been implicated in different diseases; specific examples will be described in the context of the diseases.
- *The mechanisms of tissue injury in hypersensitivity reactions are the same as the effector mechanisms of defense against infectious pathogens.* The problem in hypersensitivity is that these reactions are poorly controlled, excessive, or misdirected (e.g., against normally harmless environmental and self antigens).

Classification of Hypersensitivity Reactions

Hypersensitivity reactions can be classified based on the underlying immunologic mechanism (Table 6.1). This classification is of value in distinguishing how different types of immune response may cause tissue injury and disease, and the accompanying pathologic and clinical manifestations. However, it is now increasingly recognized that multiple mechanisms may be operative in any one disease. The main types of hypersensitivity reactions are:

- *Immediate hypersensitivity (type I hypersensitivity):* injury is caused by Th2 cells, IgE antibodies, and mast cells and other leukocytes. Mast cells release mediators that act on vessels and smooth muscle and proinflammatory cytokines that recruit inflammatory cells.
- *Antibody-mediated disorders (type II hypersensitivity):* antibodies, usually IgG and rarely IgM, injure cells by promoting their phagocytosis or lysis and injure tissues by inducing inflammation. The leukocytes that are recruited (neutrophils and monocytes) produce tissue damage by release of lysosomal enzymes and generation of toxic free radicals. Antibodies may also interfere with cellular functions and cause disease without tissue injury.
- *Immune complex–mediated disorders (type III hypersensitivity):* antibodies, typically IgG, bind antigens in the circulation or in tissues. The antigen-antibody complexes deposit in tissues and induce inflammation.
- *Cell-mediated immune disorders (type IV hypersensitivity):* T lymphocytes (Th1 and Th17 cells and CD8+ CTLs) are the cause of the tissue injury in these disorders.

TABLE 6.1 Mechanisms of Hypersensitivity Reactions

Type	Immune Mechanisms	Histopathologic Lesions	Prototypical Disorders
Immediate (type I) hypersensitivity	Production of IgE antibody → immediate release of vasoactive amines and other mediators from mast cells; later recruitment of inflammatory cells	Vascular dilation, edema, smooth muscle contraction, mucus production, tissue injury, inflammation	Anaphylaxis; allergies; bronchial asthma (atopic forms)
Antibody-mediated (type II) hypersensitivity	Production of IgG, IgM → binds to antigen on target cell or tissue → phagocytosis or lysis of target cell by activated complement or Fc receptors; recruitment of leukocytes	Phagocytosis and lysis of cells; inflammation; in some diseases, functional derangements without cell or tissue injury	Autoimmune hemolytic anemia; Goodpasture syndrome
Immune complex-mediated (type III) hypersensitivity	Deposition of antigen-antibody complexes → complement activation → recruitment of leukocytes by complement products and Fc receptors → release of enzymes and other toxic molecules	Inflammation, necrotizing vasculitis (fibrinoid necrosis)	Systemic lupus erythematosus; some forms of glomerulonephritis; serum sickness; Arthus reaction
Cell-mediated (type IV) hypersensitivity	Activated T lymphocytes → (1) release of cytokines, inflammation, and macrophage activation; (2) T cell-mediated cytotoxicity	Perivascular cellular infiltrates; edema; granuloma formation; cell destruction	Contact dermatitis; multiple sclerosis; type 1 diabetes; tuberculosis

Ig, Immunoglobulin.

Immediate (Type I) Hypersensitivity

Immediate, or type I, hypersensitivity is a rapid immunologic reaction occurring in a previously sensitized individual that is triggered by the binding of an antigen to IgE antibody on the surface of mast cells. These reactions are often called *allergy*, and the antigens that elicit them are allergens. Immediate hypersensitivity may occur as a systemic disorder or as a local reaction. The systemic reaction most often follows injection of an antigen into a sensitized individual (e.g., by a bee sting), but can also follow antigen ingestion (e.g., peanut allergens). Sometimes, within minutes the patient goes into a state of shock, which may be fatal. Local reactions are diverse and vary depending on the portal of entry of the allergen. They may take the form of localized cutaneous rash or blisters (skin allergy, hives), nasal and conjunctival discharge (allergic rhinitis and conjunctivitis, hay fever), airway obstruction (asthma), or gastroenteritis (food allergy).

Many local type I hypersensitivity reactions have two well-defined phases (Fig. 6.13). The *immediate reaction* is characterized by vasodilation,

vascular leakage, and, depending on the tissue, smooth muscle spasm or glandular secretions. These changes usually become evident within minutes after exposure to an allergen and tend to subside in a few hours. In many instances (e.g., allergic rhinitis and asthma), a second, *late-phase reaction* sets in 2 to 24 hours later without additional exposure to the antigen and may last for several days. This late-phase reaction is characterized by infiltration of tissues with eosinophils, neutrophils, basophils, monocytes, and CD4⁺ T cells, as well as tissue destruction, typically in the form of mucosal epithelial cell damage. Although immediate hypersensitivity was named because of the rapid and often transient reaction to antigen challenge in previously sensitized individuals, many allergic diseases in humans are, in fact, chronic and characterized by repeated episodes.

Allergic diseases are the prototypic type 2 inflammatory disorders, mediated by the cytokines IL-4, IL-5, and IL-13, different combinations of which are secreted by Th2 cells, Tfh cells, group 2 ILCs, and a few other cell types. The cytokine responses of these cells are

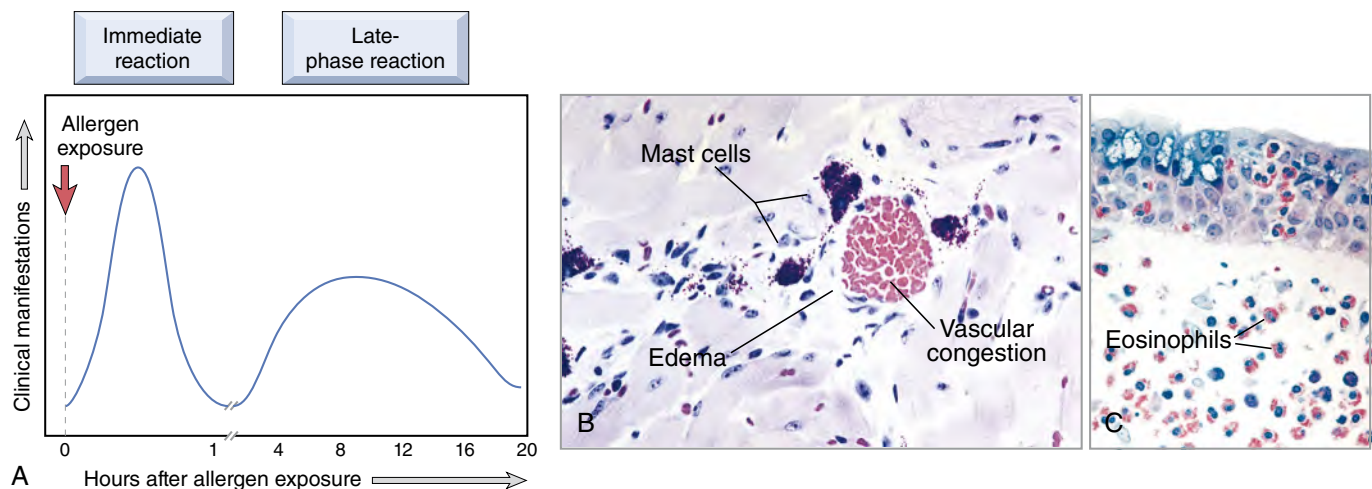


Fig. 6.13 Phases of immediate hypersensitivity reactions. (A) Kinetics of the immediate and late-phase reactions. The immediate vascular and smooth muscle reaction to allergen develops within minutes after challenge (allergen exposure in a previously sensitized individual), and the late-phase reaction develops 2 to 24 hours later. The immediate reaction (B) is characterized by vasodilation, congestion, and edema, and the late-phase reaction (C) is characterized by an inflammatory infiltrate rich in eosinophils, neutrophils, and T cells. (Courtesy Dr. Daniel Friend, Department of Pathology, Brigham and Women's Hospital, Boston.)

often collectively called type 2 immune responses (contrasted with type 1 responses, which are mediated by Th1 cells and group 1 ILCs). Many of the early events and pathologic features of the reaction are triggered by these cytokines, which may be produced by Tfh cells in lymphoid organs and by Th2 cells and possibly ILC2s in tissues. Since the dominant cell type in type 2 responses are Th2 cells, we next describe the characteristic sequence of events in Th2 responses (Fig. 6.14).

Activation of Th2 Cells and Production of IgE Antibody

The first step in the generation of Th2 cells is the presentation of the antigen to naive CD4⁺ helper T cells, probably by DCs that capture the antigen from its site of entry and take it to draining lymph nodes. Only some environmental antigens elicit strong Th2 responses and thus serve as allergens. Following injury by certain environmental substances, epithelial cells produce cytokines (including IL-25, IL-33, and TSLP). In response to antigen and these cytokines, naive T cells in draining lymph nodes differentiate into Th2 cells, which migrate to peripheral tissues, and related Tfh cells that stay in the lymph nodes. The newly minted Th2 and Tfh cells produce several cytokines on subsequent encounter with the antigen; as mentioned earlier, the signature cytokines of this subset are IL-4, IL-5, and IL-13. IL-4 and IL-13 secreted by Tfh cells in lymph nodes act on B cells to stimulate isotype switching to IgE. At sites of allergic reactions, IL-5 from Th2 cells activates eosinophils, which are important effectors of type 1 hypersensitivity (discussed later), and IL-13 acts on epithelial cells to stimulate mucus secretion. In addition, Th2 cells (as well as mast cells and epithelial cells) produce chemokines that attract more Th2 cells, as well as other leukocytes, notably eosinophils, to the reaction site. Patients with chronic atopic diseases such as asthma and atopic dermatitis are sometimes classified into Th2-high and Th2-low based on biomarkers that reflect the intensity of the pathologic T-cell response in individual patients. This separation may serve as a guide to therapy, as antagonists of Th2 cytokines (IL-4, IL-5) are predictably most effective in the Th2-high group.

Before Th2 responses develop, group 2 ILCs in tissues may respond to the cytokines that are produced by injured epithelia. These ILCs secrete IL-5 and IL-13 and are thus able to induce the same tissue reactions as the classic Th2 cells. Over time, the Th2 cells become the dominant contributors to local cytokine production.

Sensitization and Activation of Mast Cells

Because mast cells are central to the development of immediate hypersensitivity, we first review some of their salient characteristics. Mast cells are bone marrow–derived cells that are widely distributed in the tissues. They are abundant near small blood vessels and nerves and in subepithelial tissues, which explains why local immediate hypersensitivity reactions often occur at these sites. Mast cells have cytoplasmic membrane-bound granules that contain a variety of biologically active mediators, described later. The granules also contain acidic proteoglycans that bind basic dyes such as toluidine blue. (*Mast* in German refers to fattening of animals, and the name of these cells came from the erroneous belief that the granules of these cells had a nutritional function.) As detailed next, mast cells (and their circulating counterpart, basophils) are activated by the cross-linking of high-affinity IgE Fc receptors; in addition, mast cells may also be triggered by several other stimuli, such as complement components C5a and C3a (called *anaphylatoxins* because they elicit reactions that mimic anaphylaxis), both of which act by binding to receptors on the mast cell membrane. Other mast cell activators include some chemokines (e.g., IL-8), drugs such as codeine and morphine, adenosine, melittin (present in bee venom), and physical stimuli (e.g., heat, cold, sunlight). Basophils are similar to mast cells in many respects, including the presence of cell surface IgE Fc receptors as well as cytoplasmic granules. In contrast to mast cells, however, basophils are not normally present in tissues but rather circulate in the blood in small numbers. Like other granulocytes, basophils can be recruited to inflammatory sites.

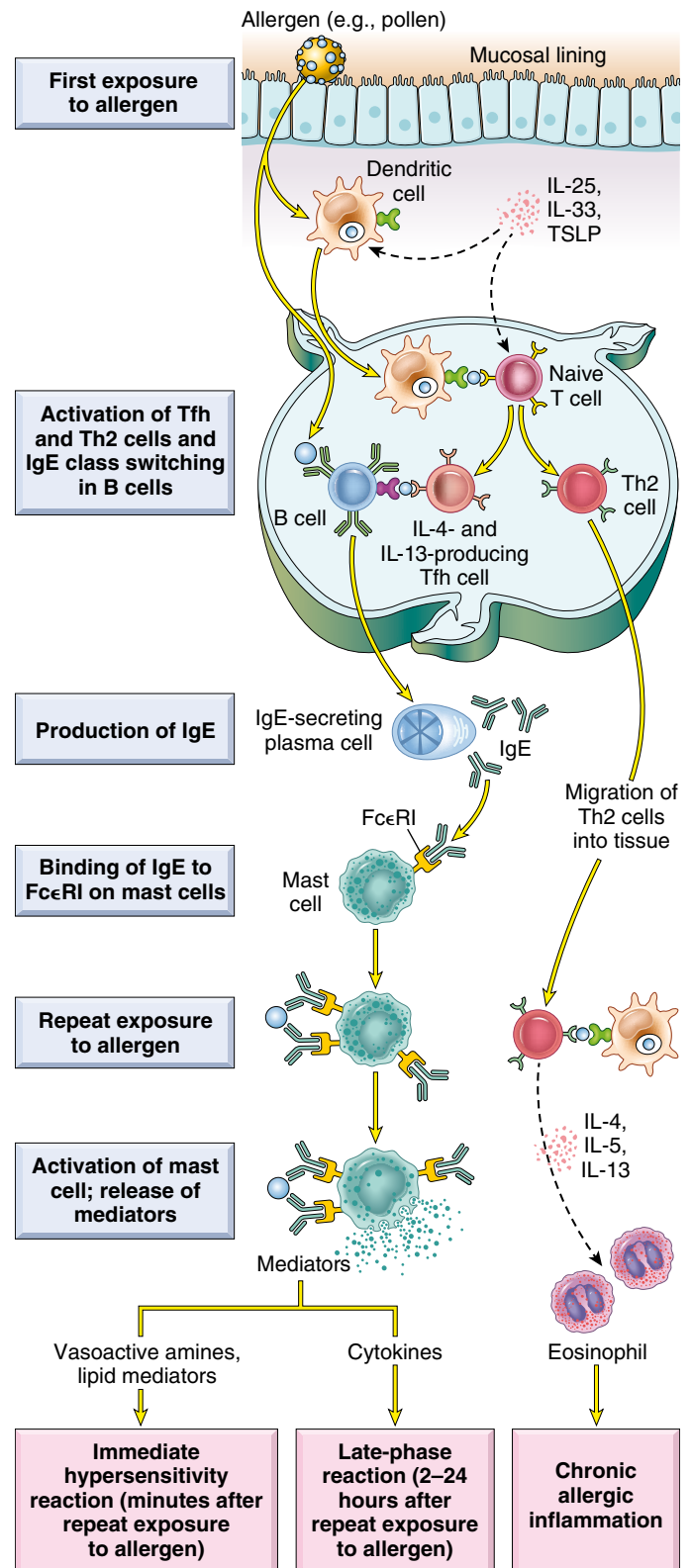


Fig. 6.14 Sequence of events in allergy. Allergic reactions are initiated by the introduction of an allergen, which stimulates type 2 immune responses and IgE production in genetically susceptible individuals. (Innate lymphoid cells are not shown.) IgE binds to Fc receptors (*FcεRI*) on mast cells, and subsequent exposure to the allergen activates the mast cells to secrete the mediators that are responsible for the pathologic manifestations of immediate hypersensitivity. *Ig*, Immunoglobulin; *IL*, interleukin; *Tfh*, T follicular helper cell.

When a mast cell armed with IgE antibodies previously produced in response to an antigen is exposed to the same antigen, the cell is activated, leading to the release of an arsenal of powerful mediators that are responsible for immediate hypersensitivity reactions. Mast cells and basophils express a high-affinity receptor called FcεRI that is specific for the Fc portion of IgE and avidly binds IgE antibodies. IgE-coated mast cells are said to be sensitized because they react to subsequent encounters with antigen. In the first step of activation, the antigen binds to the IgE antibodies on the surface of the sensitized mast cell. If the antigen is multivalent, it binds to and cross-links adjacent IgE antibodies, bringing the underlying Fcε receptors together. This triggers signal transduction pathways from the cytoplasmic portion of the receptors that lead to the release of preformed mediators and production of mediators that are responsible for the initial symptoms of immediate hypersensitivity. It also sets into motion the events that lead to the late-phase reaction.

Mediators of Immediate Hypersensitivity

Mast cell activation leads to degranulation, with the discharge of preformed mediators that are stored in the granules, and *de novo* synthesis and release of additional mediators including lipid products and cytokines (Fig. 6.15).

Preformed Mediators (Granule Contents). Mediators contained within mast cell granules are the first to be released and can be divided into three categories:

- **Vasoactive amines.** The most important mast cell–derived amine is *histamine* (Chapter 3). Histamine causes intense smooth muscle contraction, increases vascular permeability, and stimulates mucus secretion by nasal, bronchial, and gastric glands.
- **Enzymes.** These are contained in the granule matrix and include neutral proteases (chymase, tryptase) and several acid hydrolases. The enzymes cause tissue damage and lead to the generation of kinins and activated components of complement (e.g., C3a) by acting on their precursor proteins.
- **Proteoglycans.** These include heparin, a well-known anticoagulant, and chondroitin sulfate. The proteoglycans serve to package and store the amines in the granules.

Lipid Mediators. The major lipid mediators are arachidonic acid–derived products (Chapter 3). Mast cell activation is associated with activation of phospholipase A₂, an enzyme that converts membrane phospholipids to arachidonic acid. This is the parent compound from which leukotrienes and prostaglandins are produced by the 5-lipoxygenase and the cyclooxygenase pathways, respectively.

- **Leukotrienes.** Leukotrienes C₄ and D₄ are the most potent vasoactive and spasmogenic agents known. On a molar basis, they are several thousand times more active than histamine in increasing vascular permeability and causing bronchial smooth muscle contraction. Leukotriene B₄ is highly chemotactic for neutrophils, eosinophils, and monocytes.
- **Prostaglandin D₂.** This is the most abundant mediator produced in mast cells by the cyclooxygenase pathway. It causes intense bronchospasm and increases mucus secretion.

Cytokines. Mast cells are sources of many cytokines, which may play an important role at several stages of immediate hypersensitivity reactions. The cytokines include TNF, IL-1, and chemokines, which promote leukocyte recruitment (typical of the late-phase reaction); IL-4, which amplifies the Th2 response; and numerous others. The inflammatory cells that are recruited by mast cell–derived TNF and chemokines are additional sources of cytokines.

The mediators produced by mast cells are responsible for most of the manifestations of immediate hypersensitivity reactions. Some, such as histamine and leukotrienes, are released rapidly from sensitized mast cells and trigger the intense immediate reactions characterized by edema, mucus secretion, and smooth muscle spasm; others, exemplified by cytokines, including chemokines, set the stage for the late-phase reaction by recruiting more leukocytes. Not only do these inflammatory cells release additional waves of mediators (including cytokines), but they also cause epithelial cell damage. Epithelial cells themselves are not passive bystanders in this reaction; they can also produce soluble mediators, such as chemokines and cytokines that promote Th2 differentiation.

Late-Phase Reaction

In the late-phase reaction, leukocytes are recruited that amplify and sustain the inflammatory response without additional exposure to the triggering antigen. *Eosinophils* are often abundant in these reactions (see Fig. 6.13C). They are recruited to sites of immediate hypersensitivity by chemokines, such as eotaxin and others, that may be produced by epithelial cells, Th2 cells, and mast cells. The Th2 cytokine IL-5 is the most potent eosinophil-activating cytokine known. Upon activation, eosinophils liberate proteolytic enzymes as well as two proteins called major basic protein and eosinophil cationic protein, which damage tissues. Eosinophils also contain large amounts of the protein galectin-10, which undergoes crystallization after eosinophil activation to form Charcot-Leyden crystals. Upon lysis of the eosinophils, the crystals are released into the extracellular space and can be detected in the sputum of patients with asthma.

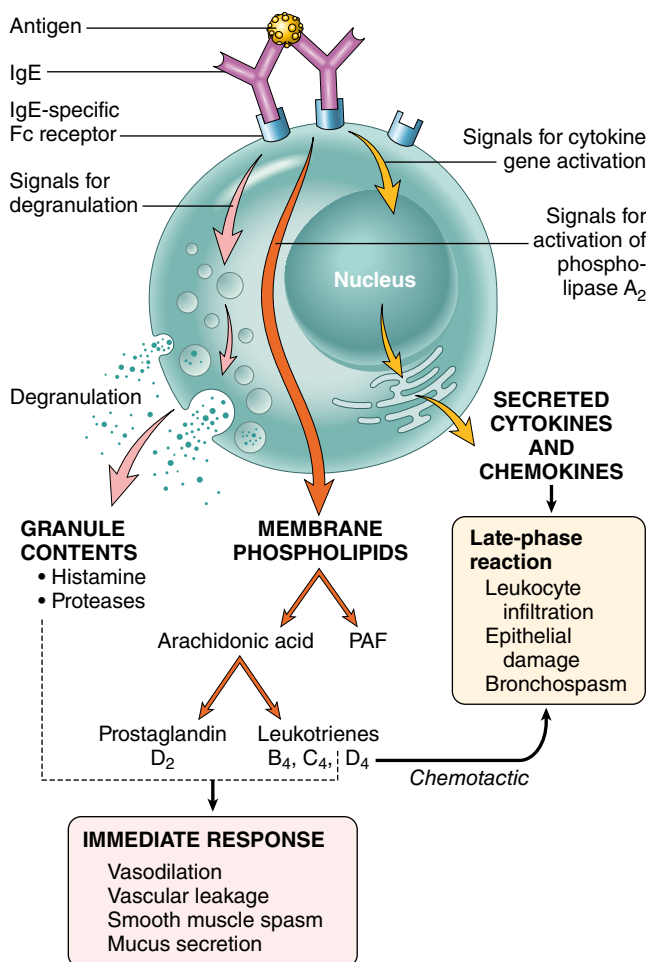


Fig. 6.15 Mast cell mediators. On activation, mast cells release various classes of mediators that are responsible for the immediate and late-phase reactions. *Ig*, Immunoglobulin; *PAF*, platelet-activating factor.

Charcot-Leyden crystals promote inflammation and enhance Th2 responses, so they may contribute to allergic reactions. It is now believed that the late-phase reaction is a major cause of symptoms in some type I hypersensitivity disorders, such as allergic asthma. Therefore, treatment of these diseases requires the use of broad-spectrum antiinflammatory drugs, such as steroids, rather than antihistamine drugs, which are of benefit only in the immediate reaction as occurs in allergic rhinitis (hay fever).

Development of Allergies

Susceptibility to allergic disease is genetically determined. A propensity to develop allergy is called *atopy*. Atopic individuals may have one or more types of allergy, the most common forms being allergic rhinitis, asthma, atopic dermatitis, and food allergies. Frequently, an individual will develop more than one atopic disorder. The clinical presentation of atopic dermatitis in babies followed later in childhood by allergic rhinitis and asthma is known as the atopic march, and these three conditions are together called the atopic triad. Atopic individuals tend to have higher serum IgE levels and more IL-4–producing Th2 cells than does the general population. A positive family history of allergy is found in almost 50% of atopic individuals. The basis of familial predisposition is not clear, but studies in patients with asthma reveal linkage to polymorphisms in several genes encoding cytokines with important roles in allergic reactions, including the genes for IL-3, IL-4, IL-5, IL-9, IL-13, and GM-CSF. How the disease-associated polymorphisms influence the development of allergies is not known. Linkage has also been noted to polymorphisms lying within the HLA genes located on chromosome 6, suggesting that the inheritance of certain HLA alleles permits reactivity to certain allergens.

Environmental factors are important in the development of allergic diseases. Exposure to environmental pollutants, which is common in industrialized societies, is a predisposing factor for allergy. Viral infections of the airways are triggers for asthma (Chapter 15).

It is estimated that 20% to 30% of immediate hypersensitivity reactions are induced by nonantigenic stimuli such as temperature extremes and exercise, and do not involve Th2 cells or IgE; such reactions are sometimes called *nonatopic allergy*. It is believed that in these cases, mast cells are abnormally sensitive to activation by various nonimmune stimuli.

The incidence of many allergic diseases has increased in developed countries, as populations have urbanized. These observations have led to an idea, sometimes called the *hygiene hypothesis*, that early childhood and even prenatal exposure to environmental, including microbial, antigens “educates” the immune system in such a way that subsequent pathologic responses against common allergens are prevented. Thus, paradoxically, improved hygiene in early childhood may increase allergies later in life. An example of the protective effect of early exposure to allergens is seen when infants and young children are given small amounts of peanut-containing foods; this reduces the incidence of peanut allergy later in life.

With this consideration of the basic mechanisms of type I hypersensitivity, we turn to some clinically important examples of allergic disease. These reactions can lead to a wide spectrum of injury and clinical manifestations (Table 6.2).

Systemic Anaphylaxis

Systemic anaphylaxis is characterized by vascular shock, widespread edema, and bronchospasm. It may occur in sensitized individuals in hospital settings after administration of foreign proteins (e.g., antisera), hormones, enzymes, polysaccharides, and drugs (e.g., the antibiotic penicillin), or in the community setting following exposure to food allergens (e.g., peanuts, shellfish) or insect toxins (e.g., those in bee venom). Extremely small doses of antigen may trigger anaphylaxis, for example, the tiny amounts used in skin testing for allergies. Because of the risk of severe allergic reactions to minute quantities of peanuts in confined quarters, many commercial airlines have stopped serving peanuts or foods

TABLE 6.2 Examples of Disorders Caused by Immediate Hypersensitivity

Clinical Syndrome	Clinical and Pathologic Manifestations
Anaphylaxis (may be caused by drugs, bee sting, food)	Fall in blood pressure (shock) caused by vascular dilation; airway obstruction due to laryngeal edema
Bronchial asthma	Airway obstruction caused by bronchial smooth muscle hyperactivity; inflammation and tissue injury caused by late-phase reaction
Allergic rhinitis, sinusitis (hay fever)	Increased mucus secretion; inflammation of upper airways, sinuses
Food allergies	Increased peristalsis due to contraction of intestinal muscles

containing peanuts. Within minutes after exposure to allergens, itching, hives, and skin erythema appear, followed shortly thereafter by a striking contraction of pulmonary bronchioles and respiratory distress. Laryngeal edema results in hoarseness and further compromises breathing. Vomiting, abdominal cramps, diarrhea, and laryngeal obstruction follow, and the patient may go into shock and even die within the hour. The risk of anaphylaxis must be borne in mind when certain therapeutic agents are administered. Although patients at risk often have a previous history of some form of allergy, the absence of such a history does not preclude the possibility of an anaphylactic reaction.

Local Immediate Hypersensitivity Reactions

About 10% to 20% of the population suffers from allergies involving localized reactions to common environmental allergens, such as pollen, animal dander, house dust, foods, and the like. Specific diseases include urticaria, allergic rhinitis (hay fever), asthma, atopic dermatitis, and food allergies; these are discussed in the context of diseases of particular organ systems.

KEY CONCEPTS

Immediate (Type I) Hypersensitivity (Allergy)

- Induced by environmental antigens (allergens) that stimulate strong Th2 responses and IgE production in genetically susceptible individuals.
- IgE attaches to mast cells by binding to Fcε receptors; reexposure to the allergen leads to cross-linking of the IgE and FcεRI, activation of mast cells, and release of mediators.
- The principal mediators are histamine, proteases, and other granule contents; prostaglandins and leukotrienes; and cytokines.
- The mediators are responsible for the immediate vascular and smooth muscle reactions and the late-phase reaction (inflammation).
- The clinical manifestations may be local or systemic and range from mildly annoying rhinitis to fatal anaphylaxis.

Antibody-Mediated (Type II) Hypersensitivity

Antibodies, most often IgG, that react with antigens present on cell surfaces or in the extracellular matrix cause disease by destroying these cells, triggering inflammation, or interfering with normal functions. The antibodies may be specific for normal fixed cell or tissue antigens or for exogenous antigens, such as chemical or microbial proteins, that bind to a cell surface or tissue matrix. The antibody-dependent mechanisms that cause tissue injury and disease are illustrated in Fig. 6.16 and described next. These reactions are the cause of several important diseases (Table 6.3).

Opsonization and Phagocytosis

Phagocytosis is largely responsible for depletion of cells coated with antibodies. Cells opsonized by IgG antibodies are recognized by phagocyte Fc receptors, which are specific for the Fc portions of some IgG subclasses. In addition, when IgM or IgG antibodies are deposited on the surfaces of cells, they may activate the complement system by the classical pathway. Complement activation generates cleavage products of C3, mainly C3b, which are deposited on the surfaces of the cells and recognized by receptors for these proteins expressed on macrophages and neutrophils. The net result is phagocytosis of the opsonized cells and their destruction inside the phagocytes (see Fig. 6.16A). Complement activation on cells also leads to the formation of the membrane attack complex, which disrupts membrane integrity by “drilling holes” through the lipid bilayer, thereby causing osmotic lysis of the cells. This mechanism is most effective in destroying cells and microbes with thin cell walls.

Antibody-mediated destruction of cells may also occur by ADCC. Cells that are coated with IgG antibody are killed by effector cells, mainly NK cells and macrophages, which bind to the target by their receptors for the Fc fragment of IgG, and cell lysis proceeds without phagocytosis. The contribution of ADCC to common hypersensitivity diseases is uncertain.

Antibody-mediated cell destruction and phagocytosis contribute to several clinical disorders: (1) transfusion reactions, in which cells from a blood group–mismatched donor react with and are opsonized by preformed antibody in the host (Chapter 14); (2) hemolytic disease of the fetus and newborn (erythroblastosis fetalis), in which there is an antigenic difference between the mother and the fetus, and IgG anti-erythrocyte antibodies from the mother cross the placenta and cause destruction of fetal red cells (Chapter 10); (3) autoimmune hemolytic anemia, thrombocytopenia, and agranulocytosis, in which individuals produce antibodies to their own blood cells, which are then destroyed (Chapter 14); and (4) certain drug reactions. In some cases of drug-induced antibody-mediated destruction of blood cells, the drug binds to plasma membrane proteins on host cells, and antibodies are produced against the drug-protein complex. In other instances, the offending drug modifies the conformation of an antigen, generating new antigenic epitopes against which the individual reacts.

Inflammation

When antibodies deposit in fixed tissues, such as basement membranes and extracellular matrix, the resultant injury is due to inflammation. The deposited antibodies activate complement, generating cleavage products, including chemotactic agents (mainly C5a), which direct the

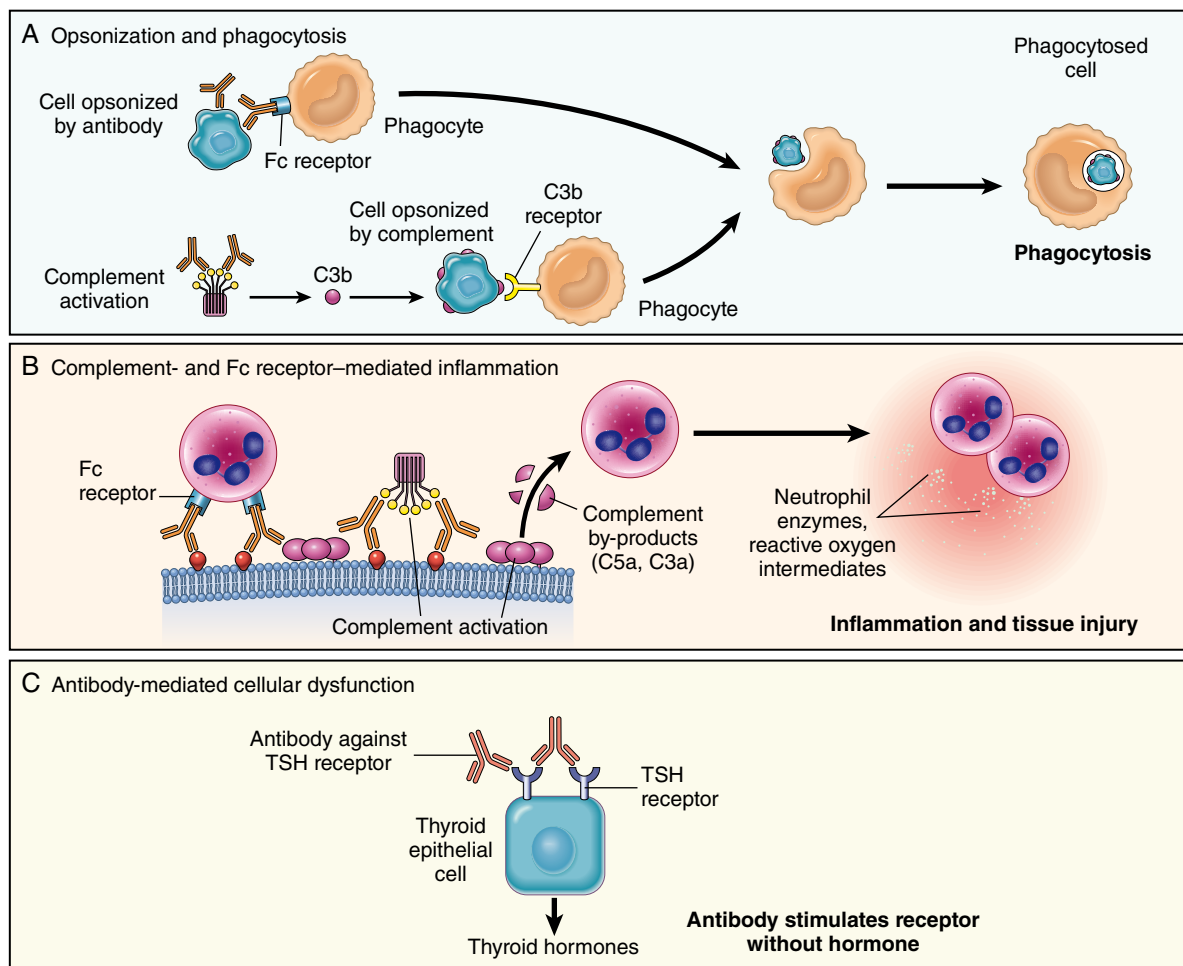


Fig. 6.16 Mechanisms of antibody-mediated injury. (A) Opsonization of cells by antibodies and complement components and ingestion by phagocytes. (B) Inflammation induced by antibody binding to Fc receptors of leukocytes and by complement breakdown products. (C) Antireceptor antibodies disturb the normal function of receptors. In the example shown, antibodies against the thyroid-stimulating hormone (TSH) receptor activate thyroid cells in Graves disease.

TABLE 6.3 Examples of Antibody-Mediated Diseases (Type II Hypersensitivity)

Disease	Target Antigen	Mechanisms of Disease	Clinicopathologic Manifestations
Autoimmune hemolytic anemia	Red cell membrane proteins (Rh blood group antigens, I antigen)	Opsonization and phagocytosis of red cells	Hemolysis, anemia
Autoimmune thrombocytopenic purpura	Platelet membrane proteins (GpIIb/IIIa integrin)	Opsonization and phagocytosis of platelets	Bleeding
Pemphigus vulgaris	Proteins in intercellular junctions of epidermal cells (desmogleins)	Antibody-mediated activation of proteases, disruption of intercellular adhesions	Skin vesicles (bullae)
Vasculitis caused by ANCA	Neutrophil granule proteins, presumably released from activated neutrophils	Neutrophil degranulation and inflammation	Vasculitis
Goodpasture syndrome	Noncollagenous protein in basement membranes of kidney glomeruli and lung alveoli	Complement- and Fc receptor-mediated inflammation	Nephritis, lung hemorrhage
Acute rheumatic fever	Streptococcal cell wall antigen; antibody cross-reacts with myocardial antigen	Inflammation, macrophage activation	Myocarditis, arthritis
Myasthenia gravis	Acetylcholine receptor	Antibody inhibits acetylcholine binding, complement-mediated injury	Muscle weakness, paralysis
Graves disease (hyperthyroidism)	TSH receptor	Antibody-mediated stimulation of TSH receptors	Hyperthyroidism
Pernicious anemia	Intrinsic factor of gastric parietal cells	Neutralization of intrinsic factor and injury to parietal epithelial cells, decreased absorption of vitamin B ₁₂	Abnormal hematopoiesis, anemia, spinal cord injury

ANCA, Antineutrophil cytoplasmic antibodies; TSH, thyroid-stimulating hormone.

migration of granulocytes and monocytes, and anaphylatoxins (C3a and C5a), which increase vascular permeability (see Fig. 6.16B). The leukocytes are activated by engagement of their C3b and Fc receptors. This results in the release of substances from the leukocytes that damage tissues, such as lysosomal enzymes, including proteases capable of digesting basement membrane, collagen, elastin, and cartilage, and by generation of reactive oxygen species.

Antibody-mediated inflammation is the mechanism responsible for tissue injury in some forms of glomerulonephritis (Chapter 20), vascular rejection in organ grafts (discussed later), and numerous other disorders (see Table 6.3).

Cellular Dysfunction

In some cases, antibodies directed against cell surface receptors impair or dysregulate function without causing cell injury or inflammation. For example, in myasthenia gravis, antibodies reactive with acetylcholine receptors in the motor end plates of skeletal muscles block neuromuscular transmission and therefore cause muscle weakness (Chapter 27). The success of complement inhibition therapy in generalized myasthenia suggests that complement-mediated injury to neuromuscular junctions may also contribute to the disease. In pernicious anemia, antibodies block intrinsic factor, and antibodies (and probably T cells) damage parietal epithelial cells, both resulting in reduced absorption of vitamin B₁₂ and anemia (Chapter 14). The converse (i.e., antibody-mediated stimulation of cell function) is the basis of Graves disease (Chapter 24). In this disorder, antibodies against the thyroid-stimulating hormone receptor on thyroid epithelial cells stimulate the cells, resulting in hyperthyroidism (see Fig. 6.16C).

Immune Complex–Mediated (Type III) Hypersensitivity

Complexes of antigens and IgG antibodies produce tissue damage mainly by eliciting inflammation at the sites of deposition. The pathologic reaction is usually initiated when antigen combines with antibody in the circulation, creating immune complexes that typically

deposit in vessel walls. Less frequently, the complexes may be formed at sites where antigen has been “planted” previously (called *in situ* immune complexes). The antigens that form immune complexes may be exogenous, such as a foreign protein that is injected or produced by an infectious microbe, or endogenous, if the individual produces antibody against self antigens (autoimmunity). Examples of immune complex disorders and the antigens involved are listed in Table 6.4. Immune complex–mediated diseases tend to be systemic, but often preferentially involve the kidney (glomerulonephritis), joints (arthritis), and small blood vessels (vasculitis), all of which are common sites of immune complex deposition for reasons mentioned below.

Systemic Immune Complex Disease

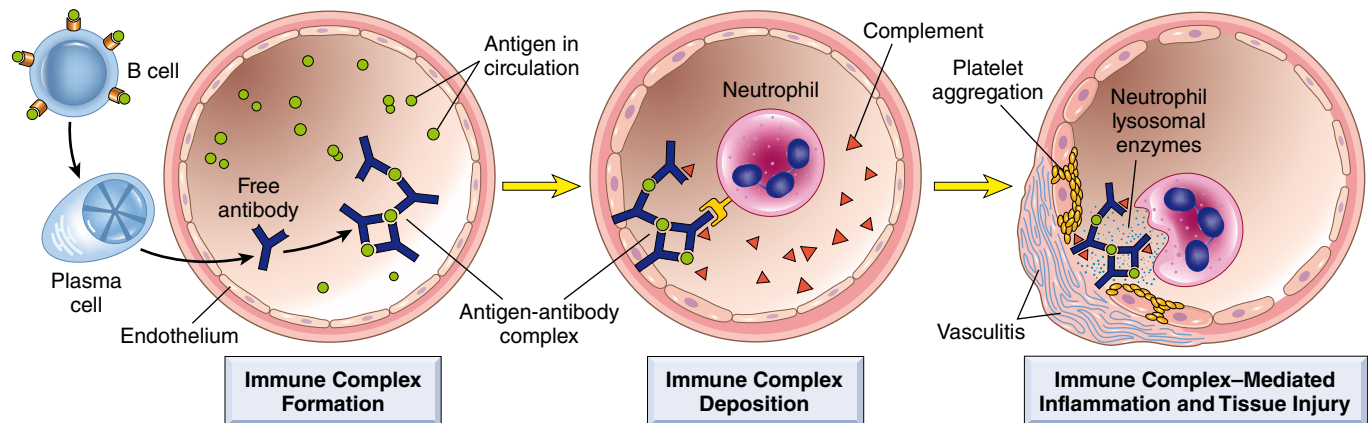
Serum sickness is the prototype of a systemic immune complex disease; it was once a frequent sequela to the administration of large amounts of foreign serum (e.g., serum from immunized horses used for protection against diphtheria). In modern times, the disease is infrequent and usually seen in individuals who receive antibodies from other individuals or species. Nevertheless, it is an informative model that has taught us a great deal about systemic immune complex disorders.

The pathogenesis of systemic immune complex disease can be divided into three phases (Fig. 6.17).

1. **Formation of immune complexes.** The introduction of a protein antigen triggers an immune response that results in the formation of antibodies, typically about 1 week after the injection of the protein. These antibodies are secreted into the blood, where they react with the antigen still present in the circulation and form antigen-antibody complexes.
2. **Deposition of immune complexes.** In the next phase, the circulating antigen-antibody complexes are deposited in vessels. The factors that determine whether immune complex formation will lead to tissue deposition and disease are not fully understood, but the major influences seem to be the characteristics of the complexes and local vascular alterations. In general, complexes that are of medium

TABLE 6.4 Examples of Immune Complex–Mediated Diseases

Disease	Antigen Involved	Clinicopathologic Manifestations
Systemic lupus erythematosus	Nuclear antigens	Nephritis, skin lesions, arthritis, others
Poststreptococcal glomerulonephritis	Streptococcal cell wall antigen(s); may be “planted” in glomerular basement membrane	Nephritis
Polyarteritis nodosa	Hepatitis B virus antigens in some cases	Systemic vasculitis
Reactive arthritis	Bacterial antigens (e.g., <i>Yersinia</i>)	Acute arthritis
Serum sickness	Various proteins, e.g., foreign serum protein (horse antithymocyte globulin)	Arthritis, vasculitis, nephritis
Arthus reaction (experimental)	Various foreign proteins	Cutaneous vasculitis

**Fig. 6.17** Immune complex disease. The sequential phases in the induction of systemic immune complex–mediated diseases (type III hypersensitivity).

size, formed under conditions of slight antigen excess, are the most pathogenic. Organs where blood is filtered at high pressure to form other fluids, like urine and synovial fluid, are sites where immune complexes become concentrated and tend to deposit; hence, immune complex disease often involves glomeruli and joints.

3. **Inflammation and tissue injury.** Once immune complexes are deposited in the tissues, they initiate an acute inflammatory reaction. During this phase (approximately 10 days after antigen administration), clinical features such as fever, urticaria, joint pain, lymph node enlargement, and proteinuria appear. Wherever complexes deposit, inflammation and tissue injury occur through the antibody-mediated mechanisms that were discussed earlier. The important role of complement in the pathogenesis of the tissue injury is supported by the observations that complement proteins can be detected at the site of injury and, during the active phase of the disease, consumption of complement leads to a decrease in serum levels of C3. In fact, serum C3 levels can, in some cases, be used to monitor disease activity.

MORPHOLOGY

The principal morphologic manifestation of immune complex injury is acute *vasculitis*, associated with necrosis of the vessel wall and intense neutrophilic infiltration. The necrotic tissue and deposits of immune complexes, complement, and plasma protein appear as a smudgy eosinophilic area of tissue destruction, an appearance termed *fibrinoid necrosis* (see Fig. 2.15). When deposited in the kidney, the complexes can be seen on immunofluorescence microscopy as granular deposits of immunoglobulin and complement and on electron microscopy as electron-dense deposits along the glomerular basement membrane (Chapter 20).

In *acute serum sickness*, caused by a single exposure to a large amount of antigen, the lesions tend to resolve because of catabolism of the immune complexes. A form of *chronic serum sickness* results from repeated or prolonged exposure to an antigen. This occurs in several diseases, such as systemic lupus erythematosus (SLE), which is associated with persistent antibody responses to autoantigens. In many diseases, the morphologic changes and other findings suggest immune complex deposition, but the inciting antigens are known in only some of these disorders. Included in this category are membranous glomerulonephritis and several vasculitides.

Local Immune Complex Disease

The *Arthus reaction* is a localized area of tissue necrosis resulting from acute immune complex vasculitis, usually elicited in the skin. The reaction can be produced experimentally by intracutaneous injection of antigen in a previously immunized animal that contains circulating antibodies against the antigen. As the antigen diffuses into the vascular wall, it binds the preformed antibody, and large immune complexes are formed locally. These complexes precipitate in the vessel walls and cause fibrinoid necrosis, and superimposed thrombosis worsens the ischemic injury.

KEY CONCEPTS

Diseases Caused by Antibodies and Immune Complexes (Types II and III Hypersensitivity)

- Antibodies can coat (opsonize) cells, with or without complement proteins, and target these cells for phagocytosis by phagocytes (macrophages), which express receptors for the Fc tails of IgG and for complement proteins. The result is depletion of the opsonized cells.

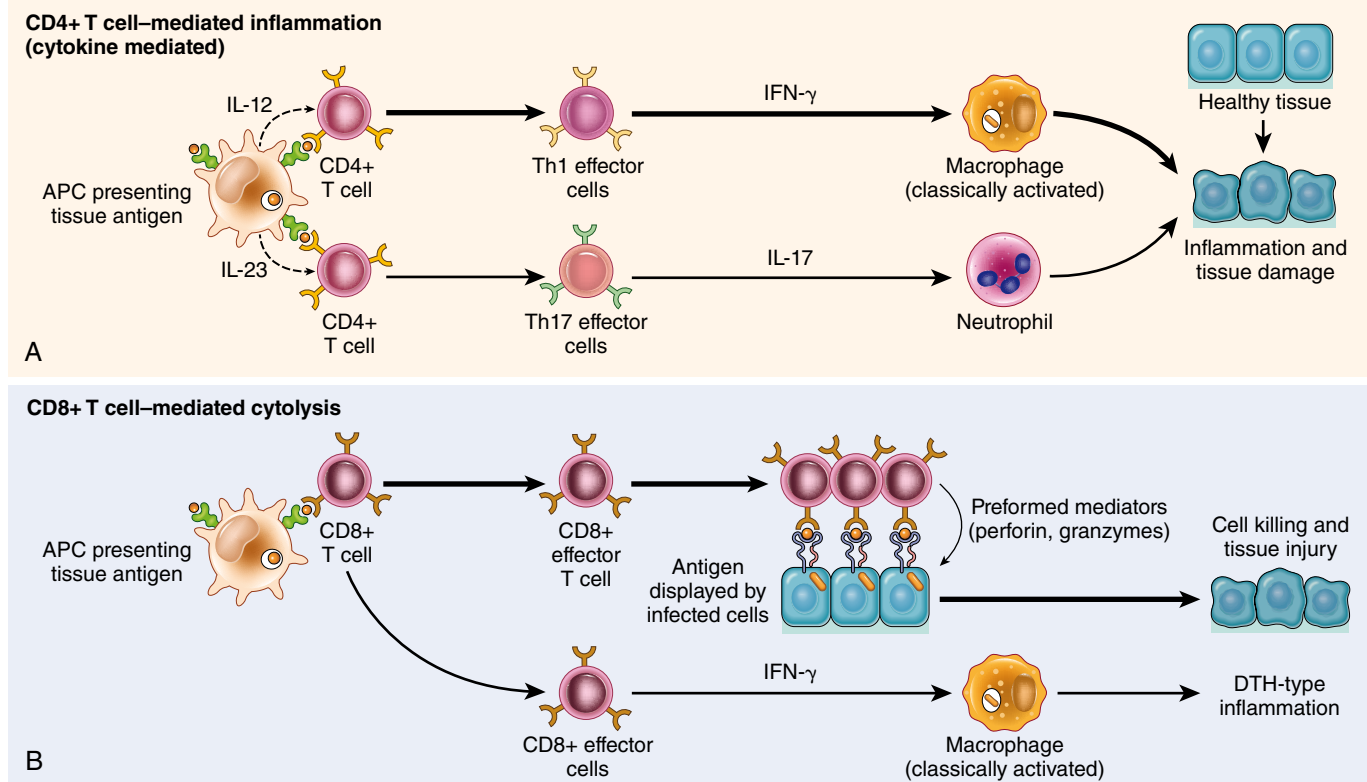


Fig. 6.18 Mechanisms of T cell-mediated (type IV) hypersensitivity reactions. (A) CD4+ Th1 and Th17 cells respond to tissue antigens by secreting cytokines that recruit leukocytes and activate phagocytes, leading to tissue injury. (B) In some diseases, CD8+ cytotoxic T lymphocytes directly kill tissue cells. APC, Antigen-presenting cell; DTH, delayed-type hypersensitivity; IFN, interferon; IL, interleukin.

- Antibodies and immune complexes may deposit in tissues or blood vessels and elicit an acute inflammatory reaction by activating complement, with release of breakdown products, or by engaging Fc receptors of leukocytes. The inflammatory reaction causes tissue injury.
- Antibodies can bind to cell surface receptors or other essential molecules and cause functional derangements (either inhibition or unregulated activation) without cell injury or inflammation.

T Cell-Mediated (Type IV) Hypersensitivity

Cell-mediated hypersensitivity is caused mainly by inflammation resulting from cytokines produced by CD4+ T cells (Fig. 6.18). CD4+ T cell-mediated hypersensitivity induced by environmental and self antigens is the cause of many autoimmune and other chronic inflammatory diseases (Table 6.5). Cell killing by CD8+ cells may also be involved in some autoimmune diseases and may be the dominant mechanism of tissue injury in certain reactions, especially those that follow viral infections.

CD4+ T Cell-Mediated Inflammation

In CD4+ T cell-mediated hypersensitivity reactions, cytokines produced by T cells induce inflammation that may be chronic and destructive. The prototype of T cell-mediated inflammation is *delayed-type hypersensitivity* (DTH), a tissue reaction to antigens given to immune individuals. In this reaction, an antigen administered into the skin of a previously immunized individual results in a detectable cutaneous reaction within 24 to 48 hours (hence the term delayed, in

contrast to immediate hypersensitivity). Both Th1 and Th17 cells contribute to organ-specific diseases in which chronic inflammation is a prominent aspect of the pathology. The inflammatory reaction associated with Th1 cells is dominated by activated macrophages, whereas that triggered by Th17 cells has a greater neutrophil component.

The inflammatory reactions stimulated by CD4+ T cells can be divided into sequential stages.

Activation of CD4+ T Cells. As described earlier, naive CD4+ T cells recognize peptides displayed by DCs and secrete IL-2, which functions as an autocrine growth factor to stimulate proliferation of the antigen-responsive T cells. The subsequent differentiation of antigen-stimulated T cells to Th1 or Th17 cells is driven by the cytokines produced by APCs and other cells at the time of T-cell activation. In some situations, the APCs (DCs and macrophages) produce IL-12, which induces differentiation of CD4+ T cells to the Th1 subset. IFN- γ produced by these effector cells promotes further Th1 development, thus amplifying the reaction. If the APCs produce the inflammatory cytokines IL-1, IL-6, and a close relative of IL-12 called IL-23, the T cells are induced to differentiate to the Th17 subset. Some of the differentiated effector cells enter the circulation and join the pool of memory T cells, where they persist for long periods, sometimes years.

Responses of Differentiated Effector T Cells. On repeat exposure to an antigen, Th1 cells secrete cytokines, mainly IFN- γ , which are responsible for many of the manifestations of delayed-type hypersensitivity. IFN- γ -activated (“classically activated”) macrophages are altered in several ways: their ability to phagocytose and kill microorganisms is markedly augmented; they express more MHC class II molecules on the

TABLE 6.5 T Cell–Mediated Diseases

Disease	Specificity of Pathogenic T Cells	Principal Mechanisms of Tissue Injury	Clinicopathologic Manifestations
Rheumatoid arthritis	Collagen? Citruinated self proteins?	Inflammation mediated by Th17 (and Th1?) cytokines; role of antibodies and immune complexes?	Chronic arthritis with inflammation, destruction of articular cartilage
Multiple sclerosis	Protein antigens in myelin (e.g., myelin basic protein)	Inflammation mediated by Th1 and Th17 cytokines, myelin destruction by activated macrophages	Demyelination in CNS with inflammation; paralysis, optic neuritis
Type 1 diabetes	Antigens of pancreatic islet β cells (insulin, glutamic acid decarboxylase, others)	T-cell–mediated inflammation, destruction of islet cells by CTLs	Insulinitis (chronic inflammation in islets), destruction of β cells; diabetes
Inflammatory bowel disease	Enteric bacteria; self antigens?	Inflammation mediated by Th1 and Th17 cytokines	Chronic intestinal inflammation, obstruction
Psoriasis	Unknown	Inflammation mediated mainly by Th17 cytokines	Destructive plaques in the skin
Contact sensitivity	Various environmental chemicals (e.g., urushiol from poison ivy or poison oak)	Inflammation mediated by Th1 (and Th17?) cytokines	Epidermal necrosis, dermal inflammation, causing skin rash and blisters

Examples of human T cell–mediated diseases are listed. In many cases, the specificity of the T cells and the mechanisms of tissue injury are inferred based on the similarity with experimental animal models of the diseases.

CTLs, Cytotoxic T lymphocytes.

surface, enhancing antigen presentation; they secrete TNF, IL-1, and chemokines, which promote inflammation (Chapter 3); and they produce more IL-12, amplifying the Th1 response. Thus, activated macrophages serve to eliminate the offending antigen; if the activation is sustained, continued inflammation and tissue injury result.

Activated Th17 cells secrete IL-17, IL-22, chemokines, and several other cytokines. Collectively, these cytokines recruit neutrophils and monocytes to the reaction, thus promoting inflammation.

Clinical Examples of CD4+ T Cell–Mediated Inflammatory Reactions.

The classic example of DTH is the tuberculin reaction, which is produced by the intracutaneous injection of purified protein derivative (PPD, also called tuberculin), a protein-containing antigen of the tubercle bacillus. In a previously sensitized individual, reddening and induration of the site appear in 8 to 12 hours, reach a peak in 24 to 72 hours, and thereafter slowly subside. Morphologically, delayed-type hypersensitivity is characterized by the accumulation of mononuclear cells, mainly CD4+ T cells and macrophages, around venules, producing perivascular “cuffing” (Fig. 6.19). In fully developed lesions, the venules show marked endothelial hypertrophy, reflecting cytokine-mediated endothelial activation.

With certain persistent or nondegradable antigens, such as tubercle bacilli colonizing the lungs or other tissues, the infiltrate is dominated by macrophages over a period of 2 or 3 weeks. With sustained activation, macrophages often undergo a morphologic transformation into epithelioid cells, large cells with abundant cytoplasm. Aggregates of epithelioid cells, usually surrounded by lymphocytes, form grossly visible small, granule-like nodules called *granulomas* (see Fig. 3.22). This pattern of chronic inflammation, called *granulomatous inflammation* (Chapter 3), is commonly associated with strong Th1 cell activation and production of cytokines such as IFN- γ . It can also be caused by indigestible foreign bodies, which activate macrophages without eliciting an adaptive immune response. In some helminthic infections, such as schistosomiasis, the worms lay eggs that elicit granulomatous reactions. These reactions are usually rich in eosinophils and are elicited by strong Th2 responses, which are typical of many helminthic infections.

Contact dermatitis is a common example of tissue injury resulting from DTH reactions. It may be evoked by contact with urushiol, the

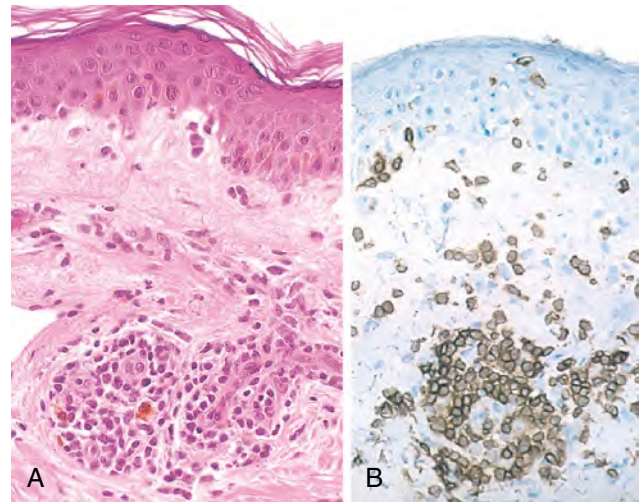


Fig. 6.19 Delayed-type hypersensitivity reaction in the skin. (A) Perivascular accumulation (“cuffing”) of mononuclear inflammatory cells (lymphocytes and macrophages), with associated dermal edema and fibrin deposition. (B) Immunoperoxidase staining reveals a predominantly perivascular cellular infiltrate that marks positively with anti-CD4 antibody. (Courtesy Dr. Louis Picker, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

antigenic component of poison ivy or poison oak, and presents as an itchy, vesicular (blistering) dermatitis. It is thought that in these reactions, the environmental chemical binds to and structurally modifies self proteins, and peptides derived from these modified proteins are recognized by T cells and elicit the reaction. Chemicals may also modify HLA molecules, making them appear foreign to T cells. The same mechanism is responsible for most *drug reactions*, among the most common immunologic reactions. These also often manifest as skin rashes.

CD4+ T cell–mediated inflammation is the basis of tissue injury in many organ-specific and systemic autoimmune diseases, such as rheumatoid arthritis and multiple sclerosis, as well as diseases linked to

uncontrolled reactions to bacterial commensals, such as inflammatory bowel disease (see Table 6.5).

CD8+ T Cell–Mediated Cytotoxicity

In this type of T cell–mediated reaction, CD8+ CTLs kill antigen-expressing target cells. Tissue destruction by CTLs may be a component of some T cell–mediated diseases, such as type 1 diabetes. CTLs directed against cell surface histocompatibility antigens play an important role in graft rejection, discussed later. They also play a role in reactions against viruses. In a virus-infected cell, viral peptides are displayed by MHC class I molecules, and the complex is recognized by the TCRs of CD8+ T lymphocytes. The killing of infected cells leads to the elimination of the infection, but in some cases, it is responsible for cell damage that accompanies the infection (e.g., in viral hepatitis). Tumor antigens are also presented on the surface of tumor cells, and CTLs are involved in the host response to transformed cells (Chapter 7).

The principal mechanism of T cell–mediated killing of targets involves perforins and granzymes, preformed mediators contained in the lysosome-like granules of CTLs. CTLs that recognize the target cells secrete a complex consisting of perforin, granzymes, and other proteins that enters target cells by endocytosis. In the target cell cytoplasm, perforin facilitates the release of the granzymes from the complex. Granzymes are proteases that cleave and activate caspases, which induce apoptosis of the target cells (Chapter 2). A less important mechanism of cell killing by CTLs involves expression on activated CTLs of Fas ligand, a molecule with homology to TNF, which can induce apoptosis by binding and activating Fas receptor expressed on target cells.

CD8+ T cells also produce cytokines, notably IFN- γ , and are involved in inflammatory reactions resembling DTH, especially following viral infections and exposure to some contact sensitizing agents.

KEY CONCEPTS

T Cell–Mediated Hypersensitivity Reactions (Type IV Hypersensitivity)

- Cytokine-mediated inflammation: CD4+ T cells are activated by exposure to a protein antigen and differentiate into Th1 and Th17 effector cells. Subsequent exposure to the antigen results in the secretion of cytokines. IFN- γ activates macrophages to produce substances that cause tissue damage and promote fibrosis, and IL-17 and other cytokines recruit leukocytes, thus promoting inflammation.
- The classic T cell–mediated inflammatory reaction is *delayed-type hypersensitivity* (DTH). Chronic DTH reactions associated with macrophage activation often lead to granuloma formation.
- T cell–mediated cytotoxicity: CD8+ cytotoxic T lymphocytes (CTLs) specific for an antigen recognize cells expressing the target antigen and kill these cells. CD8+ T cells also secrete IFN- γ .

Now that we have described how the immune system can cause tissue damage, we turn to diseases in which normal mechanisms of immune regulation fail. The prototypes of such diseases are autoimmune disorders, which are the result of failure of tolerance to self antigens.

AUTOIMMUNE DISEASES

Immune reactions against self antigens—autoimmunity—are an important cause of numerous diseases in humans, estimated to affect 5%

TABLE 6.6 Autoimmune Diseases

Organ-Specific	Systemic
Diseases Mediated by Antibodies	
Autoimmune hemolytic anemia	Systemic lupus erythematosus
Autoimmune thrombocytopenia	
Autoimmune atrophic gastritis of pernicious anemia	
Myasthenia gravis	
Graves disease	
Goodpasture syndrome	
Diseases Mediated by T Cells^a	
Type 1 diabetes mellitus	Rheumatoid arthritis
Multiple sclerosis	Systemic sclerosis (scleroderma) ^b
	Sjögren syndrome ^b
Diseases Postulated to Be Autoimmune	
Inflammatory bowel diseases (Crohn's disease, ulcerative colitis) ^c	Polyarteritis nodosa ^b
Primary biliary cholangitis ^b	Inflammatory myopathies ^b
Autoimmune (chronic active) hepatitis	

^aA role for T cells has been demonstrated in these disorders, but antibodies may also be involved in tissue injury.

^bAn autoimmune basis of these disorders is suspected, but the supporting evidence is not strong.

^cThese disorders may result from excessive immune responses to commensal enteric microbes, autoimmunity, or a combination of the two.

to 8% of the US population, and the incidence is increasing, especially in highly resourced countries. A growing number of diseases have been attributed to autoimmunity (Table 6.6). Similarity to experimental models driven by autoimmunity is sometimes used to support this mechanism in human disease. The detection of autoantibodies is the most widely used clinical test for autoimmune diseases.

The clinical manifestations of autoimmune disorders are extremely varied. On one end are conditions in which the immune responses are directed against a single organ or tissue, resulting in *organ-specific disease*, and on the other end are diseases in which the autoimmune reactions are against widespread antigens, resulting in *systemic disease*. Examples of organ-specific autoimmune diseases are type 1 diabetes, in which the autoreactive T cells and antibodies are specific for β cells of the pancreatic islets, and multiple sclerosis, in which autoreactive T cells react against central nervous system (CNS) myelin. The best example of systemic autoimmune disease is SLE, in which a diversity of antibodies directed against DNA, platelets, red cells, and protein-phospholipid complexes result in widespread lesions throughout the body. In the middle of the spectrum falls Goodpasture syndrome, in which antibodies to basement membranes of lung and kidney induce lesions in these organs.

It is obvious that autoimmunity results from the loss of self tolerance, and the question arises as to how this happens. Before we look for answers to this question, we review the mechanisms of immunologic tolerance to self antigens.

Immunologic Tolerance

Immunologic tolerance is the phenomenon of unresponsiveness to an antigen induced by exposure of lymphocytes to that antigen. *Self-tolerance* refers to lack of responsiveness to an individual's own antigens, and it underlies our ability to live in harmony with our cells and tissues.

The mechanisms of self-tolerance can be broadly classified into two groups: central tolerance and peripheral tolerance (Fig. 6.20). Each of these is considered briefly.

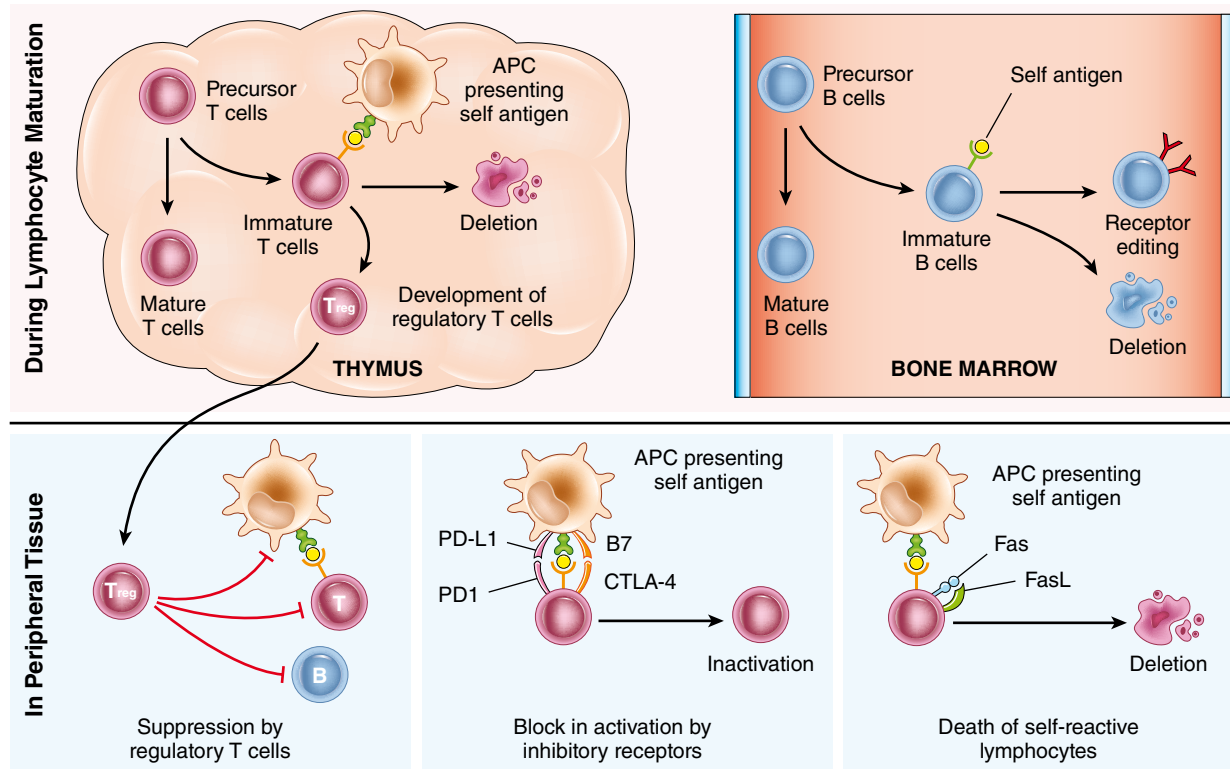


Fig. 6.20 Mechanisms of immunologic tolerance to self antigens. The principal mechanisms of central and peripheral self tolerance in T and B cells are illustrated. APC, Antigen-presenting cell.

Central Tolerance

In this process, immature self-reactive T and B lymphocyte clones that recognize self antigens during their maturation in the central (primary, or generative) lymphoid organs (the thymus for T cells and the bone marrow for B cells) are killed or rendered harmless. In developing lymphocytes, random somatic antigen receptor gene rearrangements generate diverse antigen receptors, many of which by chance may have high affinity for self antigens. The mechanisms of central tolerance eliminate these potentially dangerous lymphocytes.

- When immature T cells expressing TCRs specific for self antigens encounter these antigens in the thymus, signals are produced that result in killing of the cells by apoptosis. This process is called *negative selection* or *clonal deletion*. A wide variety of autologous protein antigens, including antigens thought to be restricted to peripheral tissues, are processed and presented by thymic APCs in association with self MHC molecules and can, therefore, be recognized by potentially self-reactive T cells. A protein called AIRE (autoimmune regulator) stimulates expression of some peripheral tissue–restricted self antigens in the thymus and promotes the differentiation of thymic epithelial cells into cells that mimic peripheral tissue cells, and is thus critical for deletion of immature T cells specific for these antigens. The importance of this mechanism is emphasized by rare patients with germline loss-of-function mutations in the AIRE gene, who develop an autoimmune disorder called *autoimmune polyglandular syndrome* that leads to destruction of multiple endocrine organs (Chapter 24). In the CD4⁺ T-cell lineage, some of the cells that see self antigens in the thymus do not die but develop into regulatory T cells (described later). What determines the choice between deletion and development of regulatory T cells in the thymus is not established; it may be partly related to the affinity of the antigen receptor on immature T cells for self antigens present in the thymus.

- When developing B cells strongly recognize self antigens in the bone marrow, many of the cells reactivate the machinery of antigen receptor gene rearrangement and begin to express new antigen receptors, not specific for self antigens. This process is called *receptor editing*; it is estimated that one-fourth to one-half of all B cells in the body may have undergone receptor editing during their maturation. If receptor editing does not occur, the self-reactive cells undergo apoptosis, thus purging potentially dangerous lymphocytes from the mature pool.

Central tolerance, however, is imperfect. Not all self antigens may be present in the thymus and bone marrow, and hence lymphocytes bearing receptors for such autoantigens escape into the periphery. Self-reactive lymphocytes that escape negative selection can inflict tissue injury unless they are deleted or muzzled in the peripheral tissues.

Peripheral Tolerance

Several mechanisms silence potentially autoreactive T and B cells in peripheral tissues; these are best defined for T cells. These mechanisms include:

- Anergy.** Lymphocytes that recognize self antigens may be rendered functionally unresponsive, a phenomenon called anergy. We discussed earlier that activation of antigen-specific T cells requires two signals: recognition of peptide antigen in association with self MHC molecules on the surface of APCs and a set of costimulatory signals (“second signals”) from APCs. These second signals are provided by certain T cell–associated molecules, such as CD28, that bind to their ligands (the costimulators B7-1 and B7-2) on APCs. If the antigen is presented to T cells without adequate levels of costimulators, the cells become anergic. Because costimulatory molecules are not expressed or are weakly expressed on resting DCs in

normal tissues, the encounter between autoreactive T cells and their specific self antigens displayed by these DCs may lead to anergy. The coinhibitory receptor CTLA-4, which (like CD28) binds to B7 molecules, is expressed mostly on regulatory T cells. It removes B7 costimulators from APCs and thus promotes the development of anergy. The second coinhibitory receptor, PD-1, binds to two ligands, PD-L1 and PD-L2, that are expressed on a wide variety of cells. It delivers signals that block activation of T cells. The importance of these inhibitory mechanisms has been established by study of CTLA-4 and PD-1 “knockout” mice and humans with inherited mutations in *CTLA4*, all of which lead to systemic inflammatory diseases. Interestingly, some tumors and viruses use the same pathways of immune regulation to evade immune attack. This realization has led to the development of antibodies that block CTLA-4, PD-1, and its ligand PD-L1 for tumor immunotherapy; by removing the brakes on the immune response, these antibodies promote responses against tumors (Chapter 7).

Anergy also affects mature B cells in peripheral tissues. It is believed that if B cells encounter self antigen in peripheral tissues, especially in the absence of specific helper T cells, the B cells become unable to respond to subsequent antigenic stimulation and may be excluded from lymphoid follicles, resulting in their death.

- **Suppression by regulatory T cells.** A population of T cells called *regulatory T cells* functions to prevent immune reactions against self antigens. Regulatory T cells develop mainly in the thymus, because of recognition of self antigens (see Fig. 6.20), but they may also be induced in peripheral lymphoid tissues. The best defined regulatory T cells are CD4⁺ cells that express high levels of CD25, the α chain of the IL-2 receptor, and FOXP3, a transcription factor of the forkhead family. Both IL-2 and FOXP3 are required for the development and maintenance of functional CD4⁺ regulatory T cells. Mutations in *FOXP3* result in severe autoimmunity in humans and mice; in humans these mutations are the cause of a systemic autoimmune disease called *IPEX* (an acronym for *immune dysregulation, polyendocrinopathy, enteropathy, X-linked*). Mutation of the gene encoding IL-2 or the IL-2 receptor α or β chain also results in a rare multiorgan autoimmune disease because IL-2 is essential for the maintenance of regulatory T cells. Recent genome-wide association studies have revealed that polymorphisms in the promoter of the *CD25* gene are associated with type 1 diabetes, multiple sclerosis, and other autoimmune diseases, raising the possibility of a regulatory T-cell defect contributing to these diseases. As mentioned earlier, regulatory T cells express CTLA-4 and use this receptor to prevent costimulation of T cells. Their inhibitory activity may also be mediated in part by the secretion of immunosuppressive cytokines such as IL-10 and TGF- β , which inhibit lymphocyte activation and effector functions.

Regulatory T cells prevent immune responses not only against self antigens but also against the fetus and commensal microbes. Placental mammals face a unique challenge because the developing fetus expresses paternal antigens that are foreign to the mother yet must be tolerated. There is emerging evidence that regulatory T cells prevent immune reactions against fetal antigens that are inherited from the father and therefore foreign to the mother. In line with this idea, during evolution, placentation appeared simultaneously with the ability to stably express the FOXP3 transcription factor. Experiments in mice have shown that fetal antigens induce long-lived FOXP3⁺ regulatory T cells, and depletion of these cells results in fetal loss. There is great interest in determining the contribution of regulatory T cells in human pregnancy and whether defects in these cells serve as a basis for recurrent spontaneous abortions.

- **Deletion by apoptosis.** T cells that recognize self antigens may receive signals that promote their death by apoptosis. Depletion of T cells occurs not only in the thymus, discussed earlier, but also in the periphery. Two mechanisms of deletion of mature T cells in the periphery have been proposed, based mainly on studies in mice. It is postulated that when T cells recognize self antigens, they increase expression of a pro-apoptotic member of the BCL family, called Bim, that is unopposed by anti-apoptotic members of the family like BCL-2 and BCL-X (whose induction requires the full set of signals for lymphocyte activation). Unopposed Bim triggers apoptosis by the mitochondrial pathway (Chapter 2). A second mechanism involves the Fas-Fas ligand system. Upon recognition of self antigens, lymphocytes express the death receptor Fas (CD95), a member of the TNF-receptor family. Fas ligand (FasL), a membrane protein that is structurally homologous to the cytokine TNF, is expressed mainly on activated T lymphocytes. The engagement of Fas by FasL induces apoptosis by the death receptor pathway (Chapter 2). If self antigens engage antigen receptors of self-reactive T cells, Fas and FasL are co-expressed, leading to elimination of the cells via Fas-mediated apoptosis. Mutations in Fas or FasL in mice and humans result in an autoimmune disease resembling lupus called the *autoimmune lymphoproliferative syndrome (ALPS)*.
- Some antigens are hidden (sequestered) from the immune system because the tissues in which these antigens are located do not communicate with the blood and lymph. As a result, self antigens in these tissues fail to elicit immune responses and are essentially ignored by the immune system. This is believed to be the case for the testis, eye, and brain, all of which are called *immune-privileged sites*. If the antigens of these tissues are released, for example, because of trauma or infection, the result may be an immune response that leads to prolonged tissue inflammation and injury. This is the postulated mechanism for posttraumatic orchitis and uveitis.

Mechanisms of Autoimmunity: General Principles

The immune system normally exists in a state of equilibrium in which lymphocyte activation, which is required for defense against pathogens, is balanced by the mechanisms of tolerance, which prevent reactions against self antigens. The underlying cause of autoimmune diseases is the failure of tolerance, which allows responses to develop against self antigens. Understanding why tolerance fails in these diseases is an important goal of immunologists.

Autoimmunity arises from a combination of the inheritance of susceptibility genes, which may contribute to the breakdown of self tolerance, and environmental triggers, such as infections and tissue damage, which promote the activation of self-reactive lymphocytes (Fig. 6.21). Although much remains unknown about the enigma of autoimmunity, the following abnormalities appear to contribute to its development:

- **Defective tolerance or regulation.** Some clues about how tolerance mechanisms are disrupted have come from the analysis of patients with rare inherited (Mendelian) autoimmune disorders and from gene knockout mice that develop autoimmune lesions, mentioned earlier, and discussed in more detail later. However, despite these advances, it is still not known why self-tolerance fails in most common autoimmune diseases.
- **Abnormal display of self antigens.** Abnormalities may include increased expression and persistence of self antigens that are normally cleared, or structural changes in these antigens resulting from posttranslational enzymatic modifications or from cellular stress or injury. If these changes lead to the display of “neoantigens” (new epitopes that are not expressed normally), the immune

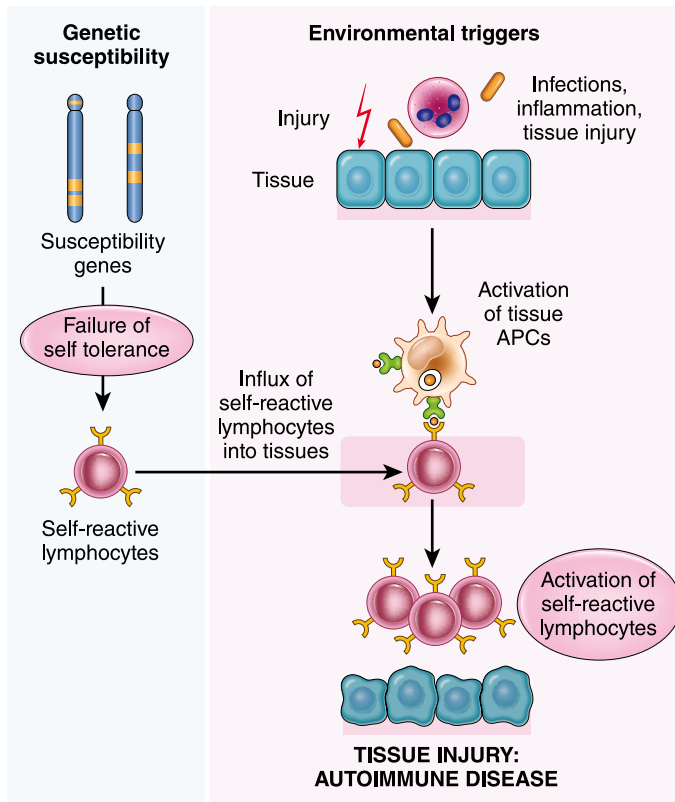


Fig. 6.21 Pathogenesis of autoimmunity. Autoimmunity results from multiple factors, including susceptibility genes that may interfere with self tolerance and environmental triggers (such as infections, tissue injury, and inflammation) that promote lymphocyte entry into tissues, activation of self-reactive lymphocytes, and tissue damage. APC, Antigen-presenting cell.

system may not be tolerant to these epitopes, and anti-self responses may develop.

- **Inflammation or an initial innate immune response.** As discussed earlier, the innate immune response is a strong stimulus for the subsequent activation of lymphocytes and the generation of adaptive immune responses. Microbes or cell injury may elicit local inflammatory reactions that may be triggers for subsequent autoimmunity.

Although these are appealing hypotheses, which of these abnormalities actually play a role in specific autoimmune diseases in humans remains largely a matter of speculation.

Role of Susceptibility Genes

Most autoimmune diseases are complex multigenic disorders. It has been known for decades that autoimmunity has a genetic component. The incidence of many autoimmune diseases is greater in twins of affected individuals than in the general population, and greater in monozygotic than in dizygotic twins, proof that genetics contributes to the development of these disorders.

Association of HLA Alleles With Autoimmune Diseases. Among the genes known to be associated with autoimmunity, the greatest contribution is that of HLA genes (Table 6.7). The most striking of these associations is between ankylosing spondylitis and HLA-B27; individuals who inherit this HLA class I allele have a 100- to 200-fold greater chance (odds ratio, or relative risk) of developing the disease compared

TABLE 6.7 Association of HLA Alleles and Inflammatory Diseases

Disease	HLA Allele	Odds Ratio ^a
Rheumatoid arthritis (anti-CCP Ab positive) ^b	DRB1, 1 SE allele ^c	4
	DRB1, 2 SE alleles	12
Type 1 diabetes	DRB1*0301-DQA1*0501-DQB1*0201 haplotype	4
	DRB1*0401-DQA1*0301-DQB1*0302 haplotype	8
	DRB1*0301/0401 haplotype heterozygotes	35
Multiple sclerosis	DRB1*1501	3
Systemic lupus erythematosus	DRB1*0301	2
	DRB1*1501	1.3
Ankylosing spondylitis	B*27 (mainly B*2705 and B*2702)	100–200
Celiac disease	DQA1*0501-DQB1*0201 haplotype	7

^aThe odds ratio reflects approximate values of increased risk of the disease associated with the inheritance of particular HLA alleles. The data are from European-derived populations.

^bAnti-CCP Ab = antibodies directed against cyclic citrullinated peptides. Data are from patients who test positive for these antibodies in the serum.

^cSE refers to *shared epitope*, so called because the susceptibility alleles map to one region of the DRB1 protein (positions 70–74).

Courtesy Dr. Michelle Fernando, Imperial College London.

with those who do not carry HLA-B27. Many autoimmune diseases are associated with various HLA class II alleles. It is reasonable to postulate that these associations reflect the ability of some HLA molecules to preferentially display natural or modified self peptides, which has been demonstrated in some diseases, such as rheumatoid arthritis. It is also important to understand that although different HLA alleles may contribute to a disease, their presence is not, by itself, the cause of any disease. Thus, in the example of HLA-B27, most individuals who inherit this allele never develop ankylosing spondylitis.

Association of Non-MHC Genes With Autoimmune Diseases. Genome-wide association studies and family studies have shown that polymorphisms in multiple non-MHC genes are associated with autoimmune diseases (Table 6.8). Some of these genes are disease-specific, but many of the associations are seen in multiple disorders, suggesting that the products of these genes affect general mechanisms of immune regulation and self tolerance. Most of the polymorphisms are not in the coding sequences of the genes but in regulatory regions (promoters and enhancers), suggesting that they affect gene expression rather than the functions of the expressed proteins. Three recently described genetic associations are especially interesting.

- Polymorphisms in a gene called *PTPN22*, which encodes a protein tyrosine phosphatase, are associated with rheumatoid arthritis, type 1 diabetes, and several other autoimmune diseases. Because these disorders have a fairly high prevalence (especially rheumatoid arthritis), *PTPN22* is a non-HLA gene that is frequently implicated in autoimmunity. It is postulated that the disease-associated variants are unable to dampen the activity of tyrosine kinases that are involved in many responses of lymphocytes. The net result is excessive lymphocyte activation.
- Polymorphisms in the gene for NOD2 are associated with Crohn disease, a form of inflammatory bowel disease, especially in certain ethnic populations. NOD2, a member of the NLR family (discussed

TABLE 6.8 Selected Non-HLA Genes Associated With Autoimmune Diseases

Putative Gene Involved	Diseases	Postulated Function of Encoded Protein and Role of Mutation/Polymorphism in Disease
Genes Involved in Immune Regulation		
<i>PTPN22</i>	RA, T1D, IBD	Protein tyrosine phosphatase, may affect signaling in lymphocytes and may alter negative selection or activation of self-reactive T cells
<i>IL23R</i>	IBD, PS, AS	Receptor for the Th17-inducing cytokine IL-23; may alter differentiation of CD4+ T cells into pathogenic Th17 effector cells
<i>CTLA4</i>	T1D, RA	Inhibits T-cell responses by terminating activation and promoting activity of regulatory T cells; may interfere with self tolerance
<i>IL2RA</i>	MS, T1D	α chain of the receptor for IL-2, which is a growth and survival factor for activated and regulatory T cells; may affect development of effector cells and/or regulation of immune responses
Genes Involved in Immune Responses to Microbes		
<i>NOD2</i>	IBD	Cytoplasmic sensor of bacteria expressed in Paneth and other intestinal epithelial cells; may control resistance to gut commensal bacteria
<i>ATG16L1</i>	IBD	Involved in autophagy; possible role in defense against microbes and maintenance of epithelial barrier function
<i>IRF5, IFIH1</i>	SLE	Role in type I interferon production; type I interferon is involved in the pathogenesis of SLE (see text)

The probable linkage of these genes with various autoimmune diseases has been defined by genome-wide association studies and other methods for studying disease-associated polymorphisms.

AS, Ankylosing spondylitis; IBD, inflammatory bowel disease; MS, multiple sclerosis; PS, psoriasis; RA, rheumatoid arthritis; SLE, systemic lupus erythematosus; T1D, type 1 diabetes mellitus.

Modified from Zenewicz LA, Abraham C, Flavell RA, Cho JH. Unraveling the genetics of autoimmunity. *Cell*. 2010;140:791-797.

earlier), is a cytoplasmic sensor of microbes that is expressed in intestinal epithelial and other cells. According to one hypothesis, the disease-associated variant is ineffective at sensing gut microbes, including commensal bacteria, resulting in entry of and chronic inflammatory responses against these normally well-tolerated organisms.

- Polymorphisms in the gene encoding the IL-2 receptor (CD25) are associated with type 1 diabetes, multiple sclerosis, and other autoimmune diseases. These variants may influence the expression and activity of this key receptor and thereby affect the balance between regulatory and effector T cells.

Many other polymorphisms have been described in different autoimmune diseases, and we will mention some of these when we describe specific disorders. We have previously mentioned that gene knockouts in mice and sporadic mutations in humans affecting several individual genes result in autoimmunity. These genes include *AIRE*, *CTLA4*, *PD1*, *FAS*, *FASL*, and *IL2* and its receptor *CD25*. In addition, B cells express an Fc receptor that recognizes IgG antibodies bound to antigens and switches off further antibody production (a normal negative-feedback mechanism). Knockout of this receptor results in autoimmunity, presumably because the B cells can no longer be controlled. These examples provide valuable information about pathways of self tolerance and immune regulation, but the diseases caused by these single-gene mutations are rare, and mutations in these genes are not the cause of most common autoimmune disorders.

Role of Infections and Other Environmental Factors

Autoimmune reactions may be triggered by infections. Two mechanisms have been postulated to explain the link between infections and autoimmunity (Fig. 6.22). First, infections may upregulate the expression of costimulators on APCs. If these cells are presenting self antigens, the result may be a breakdown of anergy and activation of T cells specific for the self antigens, especially if regulatory mechanisms are defective. Second, some microbes may express antigens that share amino acid sequences with self antigens. Immune responses against microbial antigens may result in the activation of self-reactive

lymphocytes. This phenomenon is called *molecular mimicry*. An example of such mimicry is rheumatic heart disease, in which antibodies against streptococcal proteins cross-react with myocardial proteins and cause myocarditis (Chapter 12). More subtle molecular mimicry may be involved in classic autoimmune diseases as well.

Microbes may induce other abnormalities that promote autoimmune reactions. Some viruses, such as EBV and human immunodeficiency virus (HIV), cause polyclonal B-cell activation, which may result in production of autoantibodies. The tissue injury that is common in infections may release and structurally modify self antigens, creating neoantigens that are able to activate T cells. Infections may induce the production of cytokines that recruit lymphocytes, including potentially self-reactive lymphocytes, to sites of self antigens.

Infections may protect against some autoimmune diseases. Although the role of infections in triggering autoimmunity has received a great deal of attention, epidemiologic studies suggest that the incidence of autoimmune diseases is increasing in high-resource countries in parallel with better infection control. In some animal models (e.g., of type 1 diabetes), infections greatly reduce the incidence of disease. The underlying mechanisms are unclear; one possibility is that infections promote low-level IL-2 production, and this is essential for maintaining regulatory T cells.

Other Factors. Recently, there has been great interest in the idea that the normal gut and skin microbiome influences the development of autoimmunity. It is possible that different commensal microbes affect the relative proportions of effector and regulatory T cells and shape the host response toward or away from aberrant activation. However, it is still not clear which microbes actually contribute to specific diseases in humans, or if the microbiome can be manipulated to prevent or treat these disorders.

In addition to infections, the display of tissue antigens also may be altered by a variety of environmental insults. As discussed later, ultraviolet (UV) radiation causes cell death and may lead to the exposure of nuclear antigens, which elicit pathologic immune responses in lupus; this mechanism is the proposed explanation for the association of lupus flares with exposure to sunlight. Smoking is a risk factor for rheumatoid

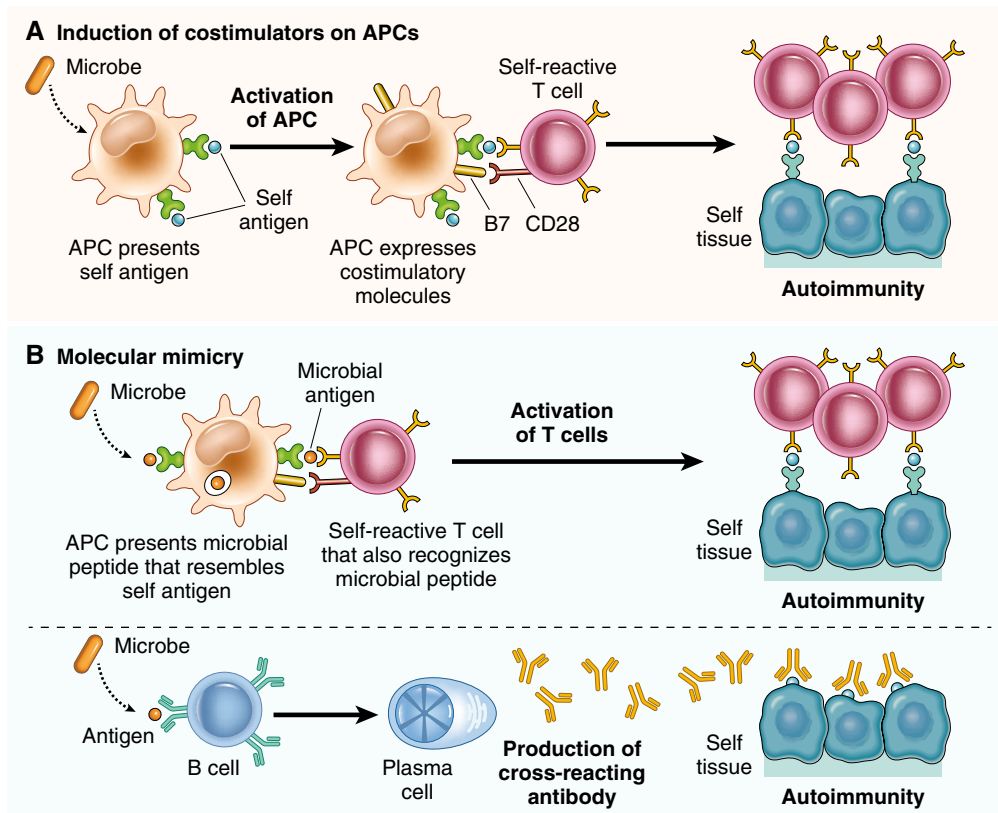


Fig. 6.22 Postulated role of infections in autoimmunity. Infections may promote activation of self-reactive lymphocytes by inducing the expression of costimulators (A), or microbial antigens may mimic self antigens and activate self-reactive lymphocytes as a cross-reaction (B). APC, Antigen-presenting cell.

arthritis, perhaps because it leads to chemical modification of self antigens. Local tissue injury for any reason may lead to the release of self antigens and autoimmune responses.

Finally, autoimmunity has a strong sex bias, with many of these diseases being more common in females than in males (Fig. 6.23). The underlying mechanisms are not understood but may involve the effects of sex hormones and genes on the X chromosome.

KEY CONCEPTS

Immunologic Tolerance and Autoimmunity

- Tolerance (unresponsiveness) to self antigens is a fundamental property of the immune system, and breakdown of tolerance is the basis of autoimmune diseases.
- Central tolerance: immature lymphocytes that recognize self antigens in the central (generative) lymphoid organs are eliminated by apoptosis; in the B-cell lineage, some of the self-reactive lymphocytes switch to new antigen receptors (receptor editing) that are not self-reactive.
- Peripheral tolerance: mature lymphocytes that recognize self antigens in peripheral tissues are suppressed by regulatory T lymphocytes, engage inhibitory receptors (such as CTLA-4 and PD-1) that block activation, or die by apoptosis.
- The factors that lead to a failure of self tolerance and the development of autoimmunity include (1) inheritance of susceptibility genes that may disrupt different tolerance pathways and (2) infections and tissue injury that may expose self antigens and activate APCs and lymphocytes in the tissues.

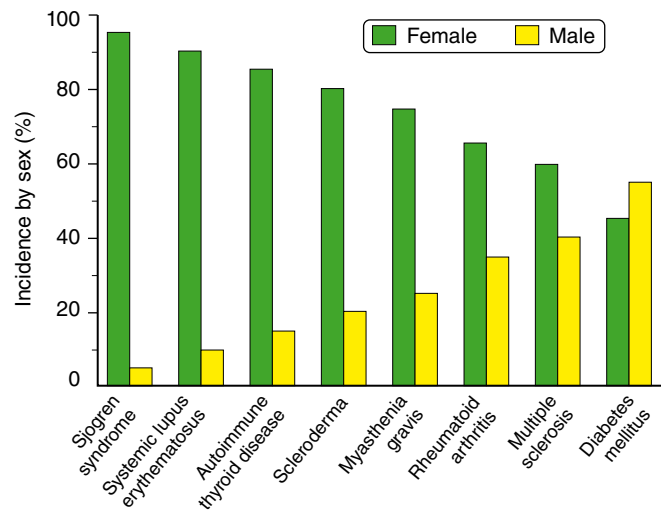


Fig. 6.23 Sex distribution of major autoimmune diseases. (From Whitacre C. Sex differences in autoimmune disease. *Nat Immunol.* 2001;2:777-780.)

General Features of Autoimmune Diseases

Diseases caused by autoimmunity have some important general features.

- Autoimmune diseases tend to be chronic, sometimes with relapses and remissions, and the damage is often progressive. An obvious reason for the chronicity is that self antigens are always present, capable of eliciting repetitive autoimmune reactions. Another

reason may be the phenomenon of *epitope spreading*, in which an immune response against one self antigen causes tissue damage, releasing other antigens, and resulting in the activation of lymphocytes that recognize these newly encountered epitopes.

- The clinical and pathologic manifestations of an autoimmune disease are determined by the nature of the underlying immune response. Some of these diseases are caused by autoantibodies, whose formation may be associated with dysregulated germinal center reactions. Most chronic inflammatory diseases are caused by abnormal and excessive Th1 and Th17 responses; examples of these diseases include psoriasis, multiple sclerosis, and some types of inflammatory bowel disease. In some autoimmune diseases, such as rheumatoid arthritis, both antibodies and T cell–mediated inflammation may be involved.

With this background, we can proceed to a discussion of specific autoimmune diseases. Table 6.6 lists major systemic and organ-specific autoimmune disorders. The systemic diseases tend to involve blood vessels and connective tissues, and therefore, they are often called *collagen vascular diseases* or *connective tissue diseases*. Our focus here is on selected systemic autoimmune diseases; organ-specific disorders are covered in relevant chapters.

Systemic Lupus Erythematosus

Systemic lupus erythematosus (SLE) is an autoimmune disease involving multiple organs, characterized by a vast array of autoantibodies,

particularly antinuclear antibodies (ANAs), in which injury is caused mainly by deposition of immune complexes and binding of antibodies to various cells and tissues. The disease may be acute or insidious in its onset and is typically a chronic, remitting and relapsing, often febrile, illness. Injury to the skin, joints, kidneys, and serosal membranes is prominent, but virtually any organ in the body may be affected. Because of this, the disease is very heterogeneous, and patients may present with any number of clinical features. In recognition of this, the American College of Rheumatology has established a complex set of criteria for this disorder, which is helpful for clinicians and for the assessment of patients in clinical trials (Table 6.9).

SLE is a fairly common disease, with a prevalence that may be as high as 1 in 2500 in certain populations. Similar to many autoimmune diseases, SLE predominantly affects females, with a frequency of 1 in 700 among females of childbearing age and a female-to-male ratio of 9:1 in the reproductive age group of 17 to 55 years. By comparison, the female-to-male ratio is only 2:1 for disease developing during childhood or after 65 years of age. For unclear reasons, in the United States the disease is more prevalent and severe in individuals of African, Hispanic, and Asian descent than in Americans of European ancestry. Although SLE most often presents in people aged in their twenties and thirties, it may develop at any age, even in early childhood.

Spectrum of Autoantibodies in Systemic Lupus Erythematosus

The hallmark of SLE is the production of autoantibodies, some of which (antibodies to double-stranded DNA and the so-called Smith

TABLE 6.9 1997 Revised Criteria for Classification of Systemic Lupus Erythematosus^a

Criterion	Definition
Clinical Criteria	
Acute cutaneous lupus	Malar rash (fixed erythema, flat or raised, over the malar eminences), photosensitivity
Chronic cutaneous lupus	Discoid rash: erythematous raised patches with adherent keratotic scaling and follicular plugging
Nonscarring alopecia	Diffuse thinning or hair fragility in the absence of other causes
Oral or nasal ulcers	Oral or nasopharyngeal ulceration, usually painless
Joint disease	Nonerosive synovitis involving two or more peripheral joints, characterized by tenderness, swelling, or effusion
Serositis	Pleuritis (pleuritic pain or rub or evidence of pleural effusion), pericarditis
Renal disorder	Persistent proteinuria >0.5 g/24 hours, or red cell casts
Neurologic disorder	Seizures, psychosis, myelitis, or neuropathy, in the absence of offending drugs or other known causes
Hemolytic anemia	Hemolytic anemia
Leukopenia or lymphopenia	Leukopenia: $<4.0 \times 10^9$ cells/L (4000 cells/mm ³) total on two or more occasions, or Lymphopenia: $<1.5 \times 10^9$ cells/L (1500 cells/mm ³) on two or more occasions
Thrombocytopenia	Thrombocytopenia: $<100 \times 10^9$ cells/L (100×10^3 cells/mm ³) in the absence of offending drugs and other conditions
Immunologic Criteria	
Antinuclear antibody (ANA)	Abnormal titer of antinuclear antibody by immunofluorescence
Anti-dsDNA antibody	Abnormal titer
Anti-Sm antibody	Presence of antibody to Smith (Sm) nuclear antigen
Antiphospholipid antibody	Positive finding of antiphospholipid antibodies based on (1) an abnormal serum level of IgG or IgM anticardiolipin antibodies, (2) a positive test for lupus anticoagulant using a standard test, or (3) a false-positive serologic test for syphilis known to be positive for at least 6 months and confirmed by negative <i>Treponema pallidum</i> immobilization or fluorescent treponemal antibody absorption test
Low complement	Low C3, C4, or CH50
Direct Coombs test	Assay for anti–red cell antibody, in the absence of clinically evident hemolytic anemia

^aThis classification was initially proposed in 1997 by the American College of Rheumatology for the purpose of identifying patients in clinical studies. It was updated in 2012 by the Systemic Lupus International Collaborating Clinics. A patient is classified as having SLE if four of the clinical and immunologic criteria are present at any time (not necessarily concurrently), including at least one clinical and one immunologic criterion. Some details have been omitted from the table.

Modified from Petri M, Orbai AM, Alarcón GS, et al. Derivation and validation of the Systemic Lupus International Collaborating Clinics classification criteria for systemic lupus erythematosus. *Arthritis Rheum.* 2012;64:2677-2686.

TABLE 6.10 Autoantibodies in Systemic Autoimmune Diseases

Disease	Specificity of Autoantibody	% Positive	Association With Specific Disease Features
Systemic lupus erythematosus (SLE)	Double-stranded DNA	40–60	Nephritis; specific for SLE
	U1-RNP	30–40	
	Smith (Sm) antigen (core protein of small RNP particles)	20–30	Specific for SLE
	Ro (SS-A)/La (SS-B) nucleoproteins	30–50	Congenital heart block; neonatal lupus
	Phospholipid-protein complexes (anti-PL)	30–40	Antiphospholipid syndrome (in ~10% of SLE patients)
	Multiple nuclear antigens (“generic ANAs”)	95–100	Found in other autoimmune diseases, not specific
Systemic sclerosis	DNA topoisomerase 1	30–70	Diffuse skin disease, lung disease; specific for systemic sclerosis
	Centromeric proteins (CENPs) A, B, C	20–40	Limited skin disease, ischemic digital loss, pulmonary hypertension
	RNA polymerase III	15–20	Acute onset, scleroderma renal crisis, cancer
Sjögren syndrome	Ro/SS-A La/SS-B	70–95	
Autoimmune myositis	Histidyl aminoacyl-tRNA synthetase, Jo1	25	Interstitial lung disease, Raynaud phenomenon
	Mi-2 nuclear antigen	5–10	Dermatomyositis, skin rash
	MDA5 (melanoma differentiation associated protein 5; cytoplasmic receptor for viral RNA)	20–35 (Japanese ancestry)	Vascular skin lesions, interstitial lung disease
	TIF1 γ (transcriptional intermediary factor 1 γ) nuclear protein	15–20	Dermatomyositis, cancer
Rheumatoid arthritis	CCP (cyclic citrullinated peptides); various citrullinated proteins	60–80	Specific for rheumatoid arthritis
	Rheumatoid factor (not specific)	60–70	

“Generic” antinuclear antibodies (ANAs), which may react against many nuclear antigens, are positive in a large fraction of patients with SLE but are also positive in other autoimmune diseases. % Positive refers to the approximate % of patients who test positive for each antibody. Table compiled with the help of Dr. Antony Rosen, Johns Hopkins University, Baltimore.

[Sm] antigen) are virtually diagnostic. These and other autoantibodies are pathogenic, either by forming immune complexes or by attacking their target cells. The levels of these autoantibodies in the blood are also helpful for the diagnosis and management of patients with SLE. Autoantibodies are found in many diseases in addition to SLE, and antibodies of different specificities tend to be associated with different autoimmune disorders (Table 6.10).

Antinuclear Antibodies. The most widely used method for detecting ANAs is an indirect immunofluorescence assay (IFA), in which patient’s serum is layered on cultured HEP-2 cells and binding is detected with an anti-Ig antibody labeled with a fluorochrome. This test has identified antibodies that bind to over 200 antigens in the nucleus, nucleolus, cytoplasm, and mitotic apparatus. Because of the diversity of antibodies detected, some experts are replacing the term “ANA” with the terms “anti-cell antibodies” and “Hep-2 IFA.”

As each autoantigen has a peculiar topographic distribution across the cell, fluorescence patterns suggest the autoantibody specificities in the serum. The International Consensus on ANA Patterns (ICAP) has defined over 30 patterns organized into a classification tree at www.anapatterns.org. Some common patterns are depicted in Fig. 6.24:

- **Homogeneous nuclear pattern** usually reflects antibodies to chromatin, double-stranded DNA, and, occasionally, histones.
- **Rim or peripheral nuclear patterns** indicate antibodies to nuclear envelope proteins, including anti-gp210, anti-p68, and anti-lamin antibodies. Some of these are useful biomarkers for the diagnosis of primary sclerosing cholangitis (Chapter 18).
- **Coarse speckled nuclear pattern** refers to variable-sized speckles that may represent RNA spliceosomes. This pattern usually occurs at high titer and is strongly associated with autoantibodies to ribonucleoproteins (e.g., Sm and U1-RNP).

- **Fine speckled nuclear pattern** refers to small speckles evenly distributed across the nucleus. This pattern is very common but has low specificity.
- **Nucleolar patterns** can be subdivided into homogeneous (associated with antibodies against RNA processing enzymes Th/To), clumpy (strongly associated with anti-fibrillarin/U3-RNP antibodies), and punctate patterns (associated with antibodies to RNA polymerase I and NOR-90). Antibodies to Th/To, fibrillarin, NOR-90, and RNA polymerase I are associated with systemic sclerosis. Therefore, high titer nucleolar patterns suggest the diagnosis of systemic sclerosis.
- **Centromeric pattern** is caused by autoantibodies against the centromeric proteins CENP-A, CENP-B, or CENP-C, which occur in systemic sclerosis, primary biliary cholangitis, Sjögren syndrome, and, occasionally, SLE. Patients with isolated Raynaud phenomenon and anticentromere antibodies are at increased risk for developing systemic sclerosis.
- **Cytoplasmic reticular pattern**, a reticulum of discrete dots that represent cytoplasmic mitochondria, is caused by antibodies against the mitochondrial enzyme pyruvate dehydrogenase E2, which are specific biomarkers for primary biliary cholangitis.
- **Cytoplasmic dense fine speckled pattern** can be caused by antibodies against the ribosomal proteins P0, P1, and P2 (specific for SLE) or antibodies against tRNA synthetases (especially PL-7 and PL-12), which are biomarkers of the antisynthetase syndrome.

The immunofluorescence assay persists as the recommended method for autoantibody screening, despite attempts to develop alternative, more quantitative tests based on solid-phase immunoassays with recombinant/purified antigens or whole-cell extracts. Although not specific, immunofluorescence patterns provide valuable preliminary indications of the putative autoantibody specificities in the sample. Antigen-specific immunoassays are necessary to confirm the autoantibody specificities suggested by the fluorescence patterns. A

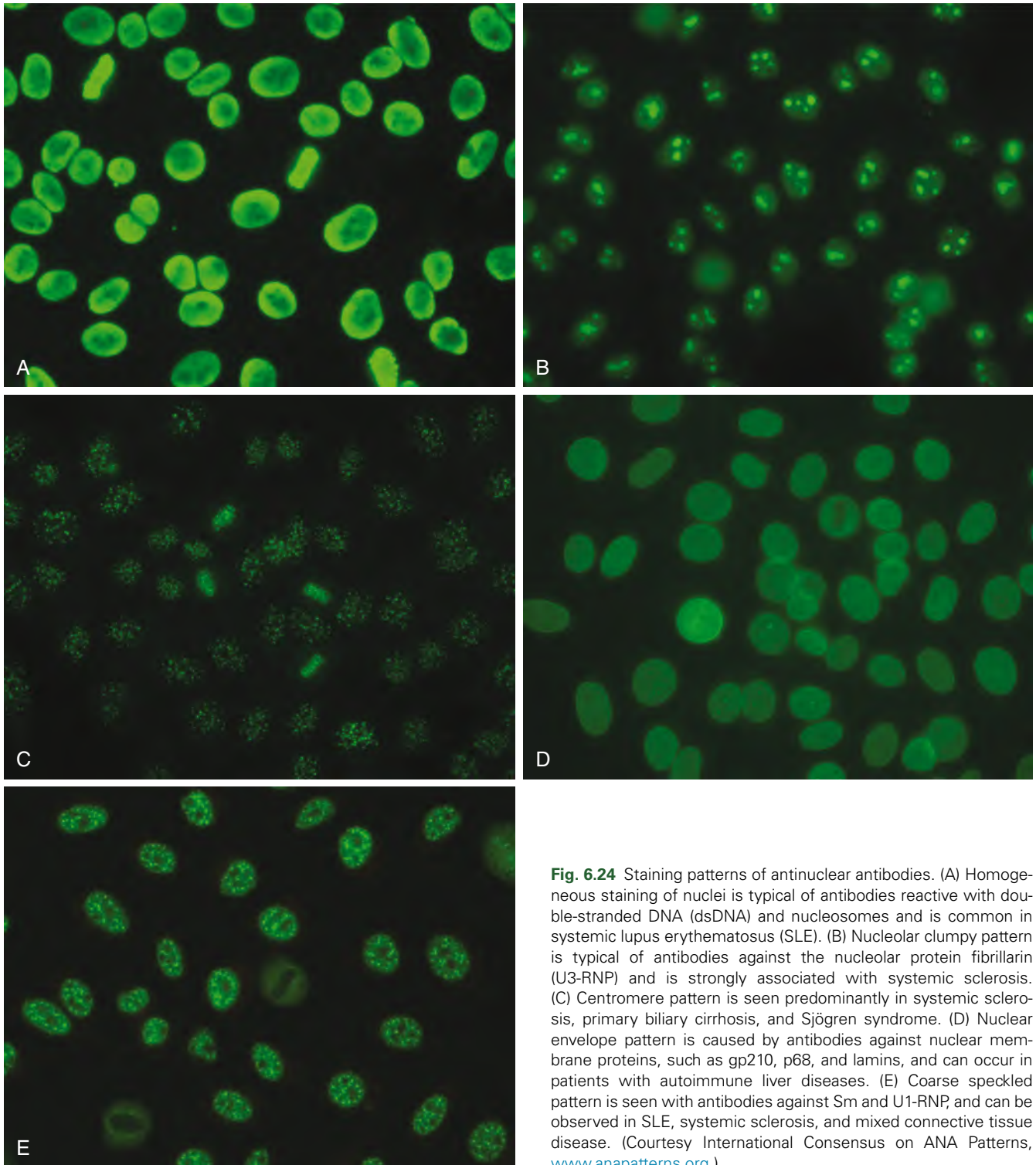


Fig. 6.24 Staining patterns of antinuclear antibodies. (A) Homogeneous staining of nuclei is typical of antibodies reactive with double-stranded DNA (dsDNA) and nucleosomes and is common in systemic lupus erythematosus (SLE). (B) Nucleolar clumpy pattern is typical of antibodies against the nucleolar protein fibrillarin (U3-RNP) and is strongly associated with systemic sclerosis. (C) Centromere pattern is seen predominantly in systemic sclerosis, primary biliary cirrhosis, and Sjögren syndrome. (D) Nuclear envelope pattern is caused by antibodies against nuclear membrane proteins, such as gp210, p68, and lamins, and can occur in patients with autoimmune liver diseases. (E) Coarse speckled pattern is seen with antibodies against Sm and U1-RNP, and can be observed in SLE, systemic sclerosis, and mixed connective tissue disease. (Courtesy International Consensus on ANA Patterns, www.anapatterns.org.)

variety of immunoassay platforms is available for the determination of the most clinically relevant autoantibodies. These include multiplex assays covering the most important autoantibodies in a specific clinical scenario (e.g., a 14-antigen line blot for autoantibodies associated with or specific to autoimmune inflammatory myopathies).

Other Autoantibodies. In addition to ANAs, lupus patients have a host of other autoantibodies. Some are directed against blood cells, such as

red cells, platelets, and lymphocytes; others react with proteins in complex with phospholipids. *Antiphospholipid antibodies* are present in 30% to 40% of lupus patients. They are specific for epitopes of plasma proteins that are revealed when the proteins are in complex with phospholipids. Included among these proteins are prothrombin, annexin V, β_2 -glycoprotein I, protein S, and protein C. Antibodies against the phospholipid- β_2 -glycoprotein complex also bind to cardiolipin antigen, used in syphilis serology, and therefore lupus patients may have a

false-positive test result for syphilis. Some of these antibodies interfere with in vitro clotting tests, such as partial thromboplastin time (PTT). Therefore, these antibodies are sometimes referred to as *lupus anticoagulant*. Despite the observed clotting delays in vitro, however, patients with antiphospholipid antibodies have complications related to excessive clotting (a hypercoagulable state), such as thrombosis (Chapter 4).

Pathogenesis

The fundamental defect in SLE is a failure of the mechanisms that maintain self-tolerance. Although what causes this failure of self-tolerance remains unknown, as is true of most autoimmune diseases, both genetic and environmental factors may play a role.

Genetic Factors. SLE is a genetically complex disease with contributions from MHC and multiple non-MHC genes. Many lines of evidence support a genetic predisposition.

- Family members of patients have an increased risk of developing SLE. In addition, laboratory testing has revealed that as many as 20% of clinically unaffected first-degree relatives of SLE patients have autoantibodies and other immune abnormalities.
- There is a higher rate of concordance (>20%) in monozygotic twins than dizygotic twins (1% to 3%).
- Studies of HLA associations support the concept that MHC genes regulate production of particular autoantibodies. Specific alleles of the *HLA-DQ* locus have been linked to the production of anti-double-stranded DNA, anti-Sm, and antiphospholipid antibodies, although the relative risk is small.
- Some lupus patients have inherited deficiencies of early complement components, such as C2, C4, or C1q. Knockout mice lacking C4 or certain complement receptors are also prone to develop lupus-like autoimmunity. Various mechanisms have been invoked, including failure to clear immune complexes and loss of B-cell self tolerance. It has also been proposed that deficiency of C1q results in defective phagocytic clearance of apoptotic cells. Many cells normally undergo apoptosis, and if they are not cleared, their nuclear components may elicit immune responses.
- Genome-wide association studies have identified several genetic loci that are associated with the disease. Many of these loci encode proteins involved in lymphocyte signaling and interferon responses, both of which may play a role in lupus pathogenesis, as discussed later. The relative risk for each locus is small, and even taken together these loci account for 20% or less of the disease predisposition, suggesting an important role for environmental factors, discussed later.

Immunologic Factors. Recent studies in animal models and patients have revealed several immunologic aberrations that collectively may result in the persistence and uncontrolled activation of self-reactive lymphocytes.

- Failure of self tolerance in B cells results from defective elimination of self-reactive B cells in the bone marrow or defects in peripheral tolerance mechanisms.
- Activation of CD4⁺ helper T cells specific for nucleosomal antigens that escape tolerance and help B cells to produce high-affinity pathogenic autoantibodies.
- TLR engagement by nuclear DNA and RNA contained in immune complexes may activate B lymphocytes. These TLRs function normally to sense microbial products, including nucleic acids. B cells specific for nuclear antigens may get activating signals from TLRs, resulting in increased production of antinuclear autoantibodies.
- Type I IFNs play a role in lymphocyte activation in SLE. High levels of circulating type I IFNs and a molecular signature in blood cells suggesting exposure to these cytokines has been reported in SLE patients and correlates with disease severity. Type I IFNs are antiviral cytokines

that are normally produced during innate immune responses to viruses. It may be that nucleic acids engage TLRs on DCs and stimulate the production of IFNs. In other words, self nucleic acids mimic their microbial counterparts. How interferons contribute to the development of SLE is unclear; these cytokines may activate DCs and B cells and promote helper T cell responses, all of which may stimulate the production of pathogenic autoantibodies and induce inflammation.

Environmental Factors. There are many indications that environmental factors must also be involved in the pathogenesis of SLE.

- Exposure to UV light exacerbates the disease in many individuals. UV irradiation may induce apoptosis in cells and may alter the DNA in such a way that its recognition by TLRs is enhanced. In addition, UV light may modulate the immune response, for example, by stimulating keratinocytes to produce IL-1, a cytokine known to promote inflammation.
- As mentioned earlier, the sex bias of SLE is partly attributable to actions of sex hormones and partly related to unknown genes on the X chromosome, independent of hormone effects.
- Drugs such as hydralazine, procainamide, and D-penicillamine can induce an SLE-like response in humans.

Model for the Pathogenesis of SLE. It is clear from this discussion that the immunologic abnormalities in SLE—both documented and postulated—are varied and complex. Nevertheless, an attempt can be made to synthesize results from patients and animal studies into a model of the pathogenesis of SLE (Fig. 6.25). UV irradiation and other environmental insults lead to the apoptosis of cells. Inadequate clearance of the nuclei of these cells results in a large burden of nuclear antigens. Underlying abnormalities in B and T lymphocytes are responsible for defective tolerance, because of which self-reactive lymphocytes survive and remain functional. These lymphocytes are stimulated by nuclear self antigens, and antibodies are produced against the antigens. Complexes of the antigens and antibodies bind to Fc receptors on B cells and DCs and may be internalized. The nucleic acid components engage TLRs and stimulate B cells to produce more autoantibodies. TLR stimuli also activate DCs to produce interferons and other cytokines, which further enhance the immune response and cause more apoptosis. The net result is a cycle of antigen release and immune activation that leads to the production of high-affinity autoantibodies.

Mechanisms of Tissue Injury. Different autoantibodies are the cause of most of the lesions of SLE.

- Most of the systemic lesions are caused by *immune complexes (type III hypersensitivity)*. DNA–anti-DNA complexes can be detected in the glomeruli and small blood vessels. Low levels of serum complement (secondary to consumption of complement proteins) and granular deposits of complement and immunoglobulins in the glomeruli further support the role of immune complex deposition. T-cell infiltrates are also frequently seen in the kidneys and may be involved in tissue damage. There is no evidence that ANAs, which are present in immune complexes, can penetrate intact cells. If cell nuclei are exposed, however, the ANAs can bind to them. In tissues, nuclei of damaged cells react with ANAs, lose their chromatin pattern, and become homogeneous, to produce so-called LE bodies or hematoxylin bodies. Related to this phenomenon is the *LE cell*, which is readily seen when blood from patients is agitated in vitro. The LE cell is a phagocytic leukocyte (blood neutrophil or monocyte) that has engulfed the denatured nucleus of an injured cell. The demonstration of LE cells was used in the past as a test for SLE. With new techniques for detection of ANAs, however, this test is now largely of historical interest. Sometimes, LE cells are found in pericardial or pleural effusions in patients.

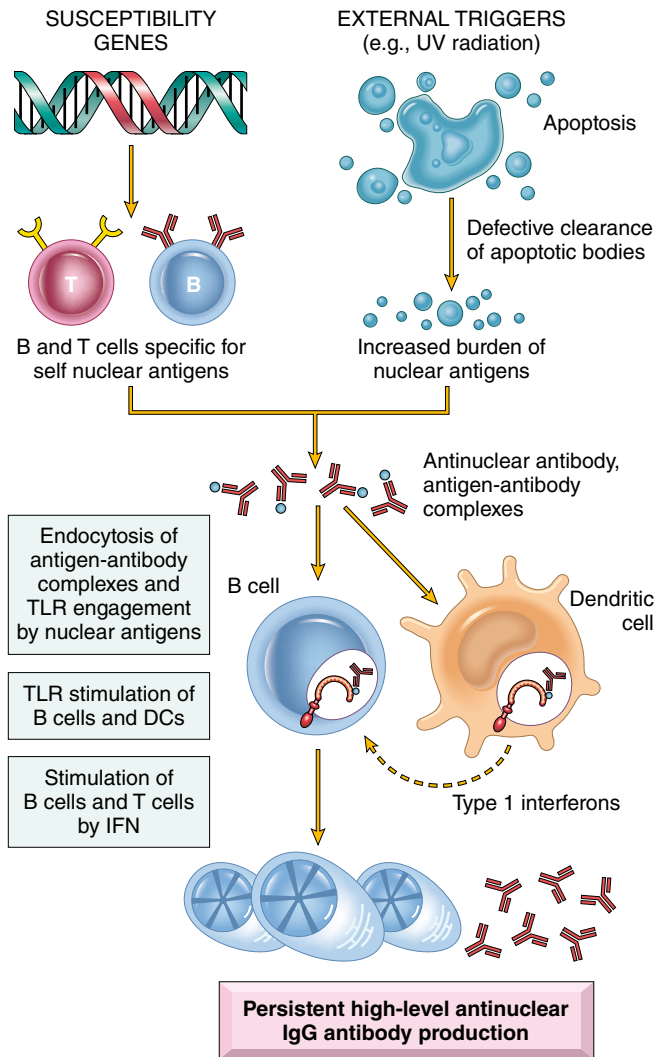


Fig. 6.25 Model for the pathogenesis of systemic lupus erythematosus. In this hypothetical model, susceptibility genes interfere with the maintenance of self tolerance, and external triggers lead to persistence of nuclear antigens. The result is an antibody response against self nuclear antigens, which is amplified by the action of nucleic acids on dendritic cells (DCs) and B cells, and the production of type 1 interferons. *IFN*, Interferon; *TLRs*, toll-like receptors; *UV*, ultraviolet.

- **Cytopenias.** Autoantibodies specific for red blood cells, white blood cells, and platelets opsonize the cells and promote their phagocytosis and destruction, resulting in depletion of these cells. These are examples of antibody-mediated (type II) hypersensitivity. The most common disorders caused by this type of autoantibody are anemia (in about 50% of SLE patients, sometimes exacerbated by chronic inflammation and iron deficiency) and thrombocytopenia (in about 10% of patients) (Chapter 14).
- **Antiphospholipid antibody syndrome.** Patients with antiphospholipid antibodies may develop venous and arterial thromboses, which may be associated with recurrent spontaneous miscarriages and focal cerebral or ocular ischemia. Primary antiphospholipid syndrome is not associated with any underlying disorder, whereas this constellation of clinical features in association with lupus is referred to as the secondary antiphospholipid antibody syndrome. The mechanisms of thrombosis are not defined, and antibodies against various serum proteins, platelets,

and endothelial cells have all been proposed as being responsible (Chapter 4).

- The *neuropsychiatric manifestations* of SLE have been attributed in part to antibodies that cross the blood-brain barrier and react with neurons or receptors for various neurotransmitters. Other immune factors, such as cytokines, may also be involved in the cognitive dysfunction and CNS abnormalities that are associated with SLE.

MORPHOLOGY

The morphologic changes in SLE are extremely variable. The frequency of individual organ involvement is shown in Table 6.11. The most characteristic lesions result from immune complex deposition in blood vessels, kidneys, connective tissue, and skin.

Blood Vessels. An acute necrotizing vasculitis involving capillaries, small arteries, and arterioles may be present in any tissue. The arteritis is characterized by fibrinoid necrosis of the vessel walls. In chronic stages, vessels undergo fibrous thickening with luminal narrowing.

Kidney. Up to 40% of SLE patients have clinically significant renal involvement mainly in the form of glomerulonephritis and tubulointerstitial nephritis. Lupus nephritis is described in Chapter 20.

Skin. Characteristic erythema affects the face along the bridge of the nose and cheeks (the “butterfly rash”) in approximately 50% of patients, and a similar rash may be seen on the extremities and trunk. Urticaria, bullae, maculopapular lesions, and ulcerations also occur. Exposure to sunlight incites or accentuates the erythema. Histologically the involved areas show vacuolar degeneration of the basal layer of the epidermis (Fig. 6.26A). In the dermis, there is variable edema and perivascular inflammation. Vasculitis with fibrinoid necrosis may be prominent. Immunofluorescence microscopy shows deposition of immunoglobulin and complement along the dermoepidermal junction (Fig. 6.26B), which may also be present in uninvolved skin. This finding is not specific for SLE and is sometimes seen in scleroderma or dermatomyositis.

Joints. Joint involvement is typically a nonerosive synovitis with little deformity, which contrasts with rheumatoid arthritis.

CNS. No clear morphologic abnormalities account for the neuropsychiatric symptoms of SLE. Noninflammatory occlusion of small vessels by intimal proliferation is sometimes noted, which may be due to endothelial damage by autoantibodies or immune complexes.

Pericarditis and Other Serosal Cavity. Inflammation of the serosal lining membranes may be acute, subacute, or chronic. During the acute phases, the mesothelial surfaces are sometimes covered with fibrinous exudate. Later they become thickened, opaque, and coated with a shaggy fibrous tissue that may lead to partial or total obliteration of the serosal cavity. Pleural and pericardial effusions may be present.

Cardiovascular System. Involvement may manifest as damage to any layer of the heart. Symptomatic or asymptomatic pericardial involvement is present in up to 50% of patients. Myocarditis, or mononuclear cell infiltration, is less common and may cause resting tachycardia and electrocardiographic abnormalities. Valvular abnormalities, primarily of the mitral and aortic valves, manifest as diffuse leaflet thickening that may be associated with dysfunction (stenosis and/or regurgitation). Valvular endocarditis (called Libman-Sacks endocarditis) was more common before the widespread use of steroids. This nonbacterial verrucous endocarditis takes the form of single or multiple 1- to 3-mm warty deposits on any heart valve, distinctively on either surface of the leaflets (Fig. 6.27). By comparison, the vegetations in infective endocarditis are considerably larger, and those in rheumatic heart disease (Chapter 12) are smaller and confined to the lines of closure of the valve leaflets.

An increasing number of SLE patients have evidence of coronary artery disease (angina, myocardial infarction) resulting from coronary atherosclerosis. This complication is particularly notable in young patients with long-standing

disease, and especially prevalent in those who have been treated with corticosteroids. The pathogenesis of accelerated coronary atherosclerosis is unclear but is probably multifactorial. Risk factors for atherosclerosis, including hypertension, obesity, and hyperlipidemia, are more frequently present in SLE patients than in the population at large. In addition, immune complexes and antiphospholipid antibodies may cause endothelial damage and promote atherosclerosis.

Spleen. Splenomegaly, capsular thickening, and follicular hyperplasia are common features. Central penicilliary arteries may show concentric intimal and smooth muscle cell hyperplasia, producing so-called onion-skin lesions.

Lungs. In addition to pleuritis and pleural effusions, which are present in almost 50% of patients, in some cases, there is chronic interstitial fibrosis and secondary pulmonary hypertension. None of these changes is specific for SLE.

Other Organs and Tissues. LE, or hematoxylin, bodies in the bone marrow or other organs are strongly indicative of SLE. Lymph nodes may be enlarged due to the presence of hyperplastic germinal centers, or may even demonstrate necrotizing lymphadenitis, usually associated with the presence of activated CTLs and macrophages. The activated T cells may be so prominent in such cases as to mimic certain features of T-cell lymphoma but are polyclonal and reactive in nature.

TABLE 6.11 Clinical and Pathologic Manifestations of Systemic Lupus Erythematosus

Clinical Manifestation	Prevalence in Patients (%) ^a
Hematologic	100
Arthritis, arthralgia, or myalgia	80–90
Skin	85
Fever	55–85
Fatigue	80–100
Weight loss	60
Renal	50–70
Neuropsychiatric	25–35
Pleuritis	45
Pericarditis	25
Gastrointestinal	20
Raynaud phenomenon	15–40
Ocular	5–15
Peripheral neuropathy	15

^aPercentages are approximate and may vary with age, ethnicity, and other factors.
Compiled with the assistance of Dr. Meenakshi Jolly, Rush Medical Center, Chicago.

Clinical Features

SLE is a multisystem disease that is highly variable in its presentation, and its diagnosis relies on a constellation of clinical, serologic, and morphologic changes (see Table 6.9). It may be acute or insidious in its onset. Often, the patient is a young female with some, but not necessarily all, of the following features: a butterfly rash over the face, fever, pain but no deformity in one or more peripheral joints (feet, ankles, knees, hips, fingers, wrists, elbows, shoulders), pleuritic chest pain, and photosensitivity. In many patients, however, the presentation of SLE is subtle and puzzling, taking forms such as a febrile illness of unknown origin, fatigue, abnormal urinary findings, or joint disease masquerading as rheumatoid arthritis or rheumatic fever. “Generic” ANAs, detected by immunofluorescence assays, are found in virtually 100% of patients, but these are not specific for SLE. A variety of clinical findings may point

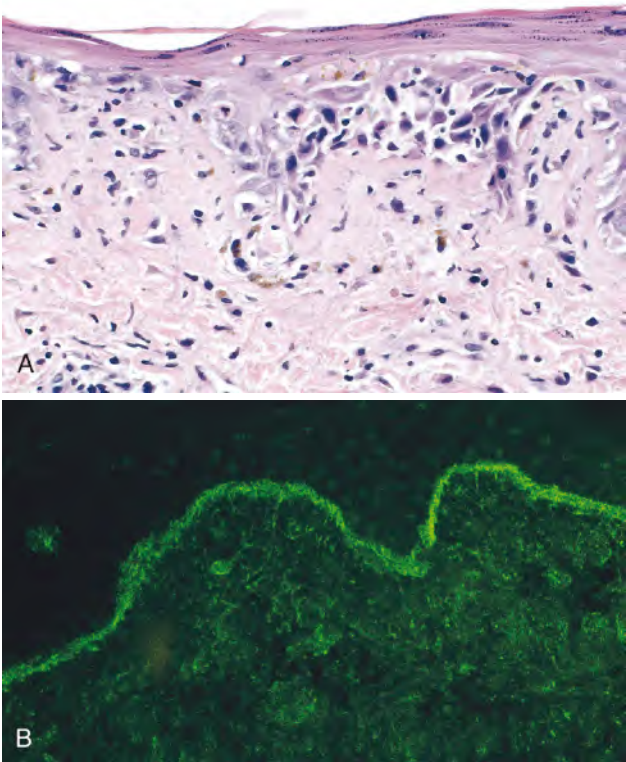


Fig. 6.26 Systemic lupus erythematosus involving the skin. (A) An H&E-stained section shows liquefactive degeneration of the basal layer of the epidermis and edema at the dermoepidermal junction. (B) An immunofluorescence micrograph stained for immunoglobulin (Ig) G reveals deposits of Ig along the dermoepidermal junction. (A, Courtesy Dr. Jag Bhawan, Boston University School of Medicine. B, Courtesy Dr. Richard Sontheimer, Department of Dermatology, University of Texas Southwestern Medical School, Dallas.)

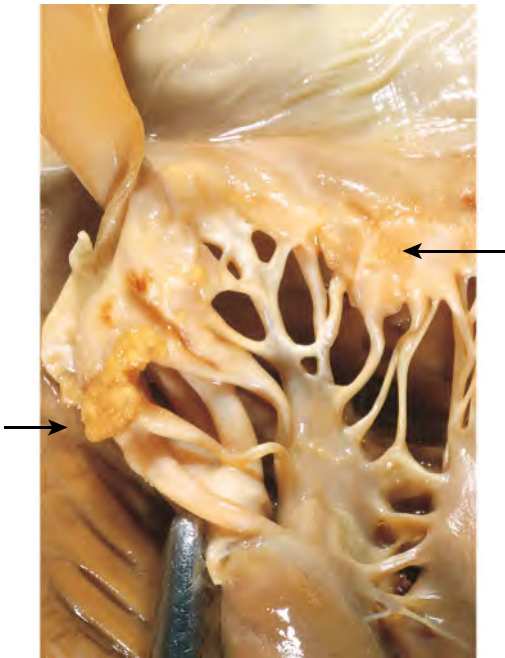


Fig. 6.27 Libman-Sacks endocarditis of the mitral valve in lupus erythematosus. The vegetations attached to the margin of the thickened valve leaflet are indicated by arrows. (Courtesy Dr. Fred Schoen, Department of Pathology, Brigham and Women’s Hospital, Boston.)

toward renal involvement, including hematuria, red cell casts, proteinuria, and in some cases nephrotic syndrome (Chapter 20). Laboratory evidence of some hematologic derangement is seen in virtually every case, and in some patients, anemia or thrombocytopenia may be the presenting manifestation as well as the dominant clinical problem. In still others, mental aberrations, including psychosis or convulsions, or coronary artery disease may be prominent. Patients with SLE are also prone to infections, presumably because of their underlying immune dysfunction and treatment with immunosuppressive drugs.

The course of the disease is variable and unpredictable. Rarely, death ensues within weeks to months. More often, with appropriate therapy, the disease follows a relapsing and remitting course spanning a period of years or even decades. During acute flareups, increased formation of immune complexes results in complement activation, often leading to hypocomplementemia. Disease flares are usually treated with corticosteroids or other immunosuppressive drugs. Even without therapy, in some patients the disease runs an indolent course with relatively mild manifestations, such as skin changes and mild hematuria, for years. The outcome has improved significantly, and an approximately 90% 5-year and 80% 10-year survival can be expected. The most frequent causes of death are kidney failure, intercurrent infections and cardiovascular diseases. Patients treated with steroids and immunosuppressive drugs incur the usual risks associated with such therapy.

As mentioned earlier, involvement of skin along with multisystem disease is fairly common in SLE. The following sections describe two syndromes in which the cutaneous involvement is the exclusive or most prominent feature.

Chronic Discoid Lupus Erythematosus

Chronic discoid lupus erythematosus is a disease in which the skin manifestations may mimic SLE, but systemic manifestations are rare. It is characterized by the presence of skin plaques showing varying degrees of edema, erythema, scaliness, follicular plugging, and skin atrophy surrounded by an elevated erythematous border. The disease usually remains confined to the skin, but up to 20% of patients develop multisystem manifestations after many years. Approximately 50% of patients test positive for generic ANAs, but antibodies to double-stranded DNA are rarely present. Immunofluorescence studies of skin biopsy specimens show deposition of immunoglobulin and C3 at the dermoepidermal junction, similar to what is seen in SLE.

Subacute Cutaneous Lupus Erythematosus

Subacute cutaneous lupus erythematosus appears to define a group of patients with features intermediate between SLE and chronic discoid lupus erythematosus. The condition presents with predominant skin involvement and photosensitivity, and is distinguished from chronic discoid lupus erythematosus by several criteria: (1) the skin rash tends to be widespread, superficial, and nonscarring (though exceptions occur); (2) most patients have mild systemic symptoms consistent with SLE; and (3) there is a strong association with antibodies to the SS-A antigen and with the *HLA-DR3* genotype.

Drug-Induced Lupus Erythematosus

An SLE-like syndrome may develop in patients receiving a variety of drugs, including hydralazine, procainamide, isoniazid, and D-penicillamine, to name only a few. Somewhat surprisingly, anti-TNF therapy, which is effective in rheumatoid arthritis and other autoimmune diseases, can also cause drug-induced lupus. Many of these drugs are associated with the development of ANAs, but most patients do not have symptoms of SLE. For example, 80% of patients receiving procainamide test positive for ANAs, but only one-third of these manifest clinical

symptoms, such as arthralgias, fever, and serositis. Although multiple organs are affected, renal and CNS involvement is uncommon. There are serologic and genetic differences from classic SLE, as well. Antibodies specific for double-stranded DNA are rare, and there is instead an extremely high frequency of antibodies specific for histones.

KEY CONCEPTS

Systemic Lupus Erythematosus

- SLE is a systemic autoimmune disease caused by autoantibodies produced against numerous self antigens and the formation of immune complexes.
- The major autoantibodies, and the ones responsible for the formation of circulating immune complexes, are directed against nuclear antigens. Other autoantibodies react with red cells, platelets, and various complexes of phospholipids with proteins.
- Disease manifestations include nephritis, skin lesions, arthritis (all caused by the deposition of immune complexes), and hematologic and neurologic abnormalities.
- The underlying cause of the breakdown in self tolerance in SLE is unknown; it may include excess or persistence of nuclear antigens (e.g., secondary to cell death caused by UV irradiation), abnormal signaling by nucleic acid-recognizing TLRs, and excessive production of type I interferons.

Rheumatoid Arthritis

Rheumatoid arthritis is a chronic inflammatory disease that primarily affects the joints, but may also involve extraarticular tissues such as the skin, blood vessels, lungs, and heart. Abundant evidence supports the autoimmune nature of the disease. Because the principal manifestations of the disease are in the joints, it is discussed in Chapter 26.

Sjögren Syndrome

Sjögren syndrome is a chronic disease characterized by dry eyes (keratoconjunctivitis sicca) and dry mouth (xerostomia) resulting from immunologically mediated destruction of the lacrimal and salivary glands. It occurs as an isolated disorder (primary form), also known as the *sicca syndrome*, or more often in association with another autoimmune disease (secondary form). Among the associated disorders, rheumatoid arthritis is the most common, but some patients have SLE, polymyositis, scleroderma, vasculitis, mixed connective tissue disease, or thyroiditis.

The characteristic decrease in tears and saliva (*sicca syndrome*) is the result of lymphocytic infiltration and fibrosis of the lacrimal and salivary glands. The infiltrate contains predominantly activated CD4⁺ helper T cells and some B cells, including plasma cells. About 75% of patients have rheumatoid factor (an antibody reactive with self IgG), whether or not coexisting rheumatoid arthritis is present. ANAs are detected in 50% to 80% of patients by immunofluorescence assay. A host of other organ-specific and non-organ-specific antibodies have also been identified. Most important, however, are antibodies directed against two ribonucleoprotein antigens, SS-A (Ro) and SS-B (La) (see [Table 6.10](#)), which can be detected in as many as 90% of patients by sensitive techniques. These antibodies are thus considered serologic markers of the disease. These autoantibodies are also present in a smaller percentage of patients with SLE and hence are not entirely specific for Sjögren syndrome.

Pathogenesis

Although the pathogenesis of Sjögren syndrome remains obscure, aberrant T-cell and B-cell activation are both implicated. The initiating trigger may be a viral infection of the salivary glands, which causes local cell death and release of tissue self antigens. In genetically susceptible individuals, CD4⁺ T cells and B cells specific for these self antigens

may have escaped tolerance and are able to react. The result is inflammation, tissue damage, and, eventually, fibrosis. However, the role of particular cytokines or T cell subsets in the development of the lesions is not established. The nature of the autoantigens recognized by these lymphocytes is also mysterious. Sjögren syndrome–like disease is seen in some patients with human T-lymphotropic virus (HTLV), HIV, and hepatitis C virus infections, but the link between these viruses and the autoimmune disorder is obscure.

As with other autoimmune diseases, Sjögren syndrome shows some association, albeit weak, with certain HLA alleles. The significance of such associations in disease causation or manifestation is not clear.

MORPHOLOGY

Lacrimal and salivary glands are the major targets of the disease, but other exocrine glands, including those lining the respiratory and gastrointestinal tracts and the vagina, may also be involved. The earliest histologic finding in involved salivary glands is periductal and perivascular lymphocytic infiltration. Eventually the lymphocytic infiltrate becomes extensive (Fig. 6.28), and lymphoid follicles with germinal centers may appear. The ductal lining epithelial cells may show hyperplasia, thus obstructing the ducts. Later there is atrophy of the acini, fibrosis, and hyalinization; still later in the course, atrophy and replacement of parenchyma with fat are seen. In some cases, the lymphoid infiltrate may be so intense as to give the appearance of a lymphoma. Indeed, these patients are at high risk for development of B-cell lymphomas.

The lack of tears leads to drying of the corneal epithelium, which becomes inflamed, eroded, and ulcerated; the oral mucosa may atrophy, with inflammatory fissuring and ulceration; and dryness and crusting of the nose may lead to ulcerations and even perforation of the nasal septum.

Clinical Features

Sjögren syndrome occurs most commonly in females between 50 and 60 years of age. As might be expected, symptoms result from inflammatory destruction of the exocrine glands. Keratoconjunctivitis produces blurring of vision, burning, and itching, and thick secretions accumulate in the conjunctival sac. Xerostomia results in difficulty in swallowing solid foods, a decrease in the ability to taste, cracks and fissures in the mouth, and dryness of the buccal mucosa. Parotid gland enlargement is present in one-half of patients; dryness of the nasal mucosa, epistaxis, recurrent bronchitis, and pneumonitis are other

symptoms. Manifestations of extraglandular disease are seen in one-third of patients and include synovitis, diffuse pulmonary fibrosis, and peripheral neuropathy. These are more common in patients with high titers of antibodies specific for SS-A. In contrast to SLE, glomerular lesions are extremely rare in Sjögren syndrome. Defects of tubular function, however, including renal tubular acidosis, uricosuria, and phosphaturia, are often seen and are associated with tubulointerstitial nephritis (Chapter 20). Over 50% of patients have another accompanying autoimmune disorder, such as rheumatoid arthritis, and these patients also have the symptoms and signs of that disorder.

As mentioned earlier, the involved glands show intense inflammatory infiltrates. In early stages of the disease, this immune infiltrate consists of a mixture of polyclonal T and B cells. However, if the reaction continues unabated, there is a strong tendency over time for individual clones within the population of B cells to gain a growth advantage, presumably because of the acquisition of somatic mutations. Emergence of a dominant B-cell clone is usually indicative of the development of a marginal zone lymphoma, a specific type of B-cell malignancy that often arises in the setting of chronic B-cell stimulation. About 5% of Sjögren patients develop lymphoma, an incidence that is 40-fold greater than normal. Certain other autoimmune disorders (e.g., Hashimoto thyroiditis) are also associated with a high risk of marginal zone lymphoma (Chapter 13), which typically arises within the organ or tissue that is the target of the autoimmune inflammation.

KEY CONCEPTS

Sjögren Syndrome

- Sjögren syndrome is an inflammatory disease that affects primarily the salivary and lacrimal glands, causing dryness of the mouth and eyes.
- The disease is believed to be caused by an autoimmune T-cell reaction against an unknown self antigen expressed in these glands, or immune reactions against the antigens of a virus that infects the tissues.

Systemic Sclerosis (Scleroderma)

Systemic sclerosis is characterized by (1) chronic inflammation thought to be the result of autoimmunity, (2) widespread damage to small blood vessels, and (3) progressive interstitial and perivascular fibrosis in the skin and multiple organs. Although the term

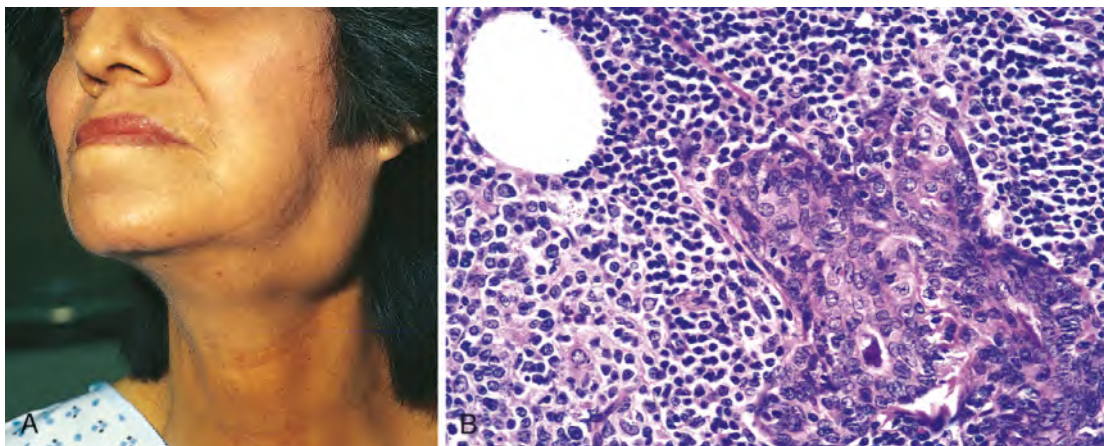


Fig. 6.28 Sjögren syndrome. (A) Enlargement of salivary glands. (B) Intense lymphocytic and plasma cell infiltration with ductal epithelial hyperplasia in a salivary gland. (A, Courtesy Dr. Richard Sontheimer, Department of Dermatology, University of Texas Southwestern Medical School, Dallas. B, Courtesy Dr. Dennis Burns, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

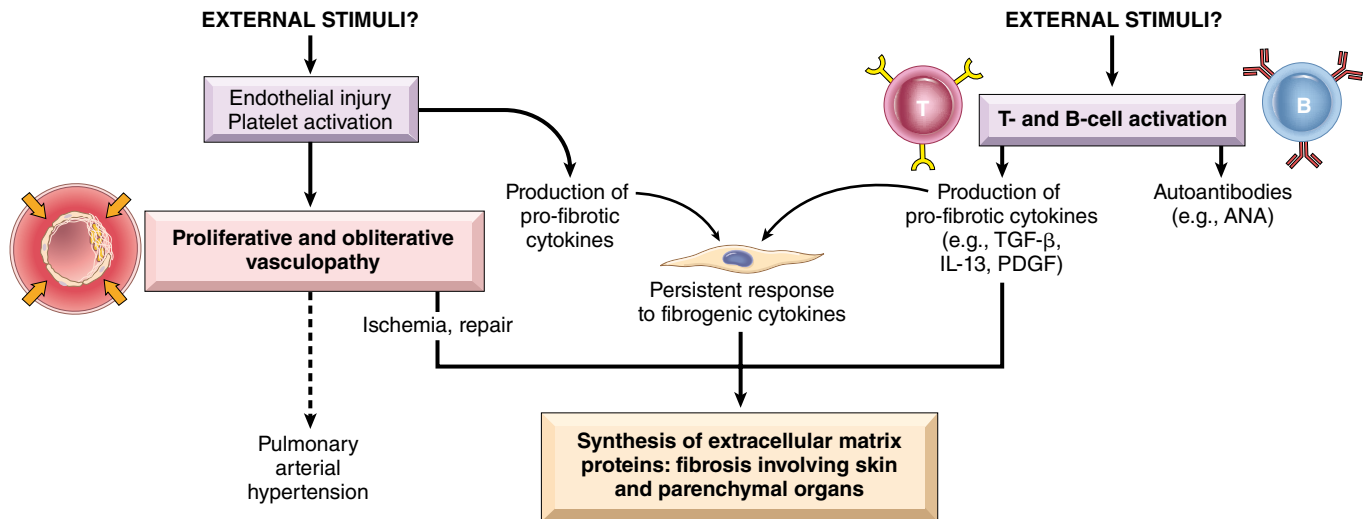


Fig. 6.29 A model for the pathogenesis of systemic sclerosis. Unknown external stimuli cause vascular abnormalities and immune activation in genetically susceptible individuals, and both contribute to the excessive fibrosis. ANA, Antinuclear antibody; IL, interleukin; PDGF, platelet derived growth factor; TGF, transforming growth factor.

scleroderma is ingrained in clinical medicine, this disease is better named *systemic sclerosis* because it is characterized by excessive fibrosis throughout the body. The skin is most commonly affected, but the gastrointestinal tract, kidneys, heart, muscles, and lungs are also frequently involved. In some patients, the disease remains confined to the skin for many years, but in the majority, it progresses to visceral involvement with death from kidney failure, cardiac failure, pulmonary insufficiency, or intestinal malabsorption. The clinical heterogeneity of systemic sclerosis has been recognized by classifying the disease into two major categories: *diffuse scleroderma*, characterized by widespread skin involvement at onset, with rapid progression and early visceral involvement; and *limited scleroderma*, in which the skin involvement is often confined to fingers, forearms, and face. Visceral involvement in the limited form occurs late; hence, the clinical course is relatively benign. Some patients with the limited disease develop a combination of calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia, called the *CREST syndrome*. Several other variants and related conditions, such as eosinophilic fasciitis, occur far less frequently and are not described here.

Pathogenesis

The cause of systemic sclerosis is not known, but the **disease likely results from three interrelated processes—autoimmune responses, vascular damage, and collagen deposition** (Fig. 6.29).

- **Autoimmunity.** It is proposed that CD4⁺ T cells responding to as yet unidentified antigen accumulate in the skin and release cytokines that activate inflammatory cells and fibroblasts. Although inflammatory infiltrates are typically sparse in the skin of patients with systemic sclerosis, activated CD4⁺ T cells can be found in many patients, and Th2 cells have been isolated from the skin. Several cytokines produced by these T cells, including TGF- β and IL-13, can stimulate the transcription of genes encoding collagen and other extracellular matrix proteins (e.g., fibronectin) in fibroblasts. Other cytokines recruit leukocytes and propagate the chronic inflammation. There is also evidence for inappropriate activation of humoral immunity, and the presence of various autoantibodies, notably ANAs, provides diagnostic and prognostic information. The role of these ANAs in the pathogenesis of the disease is unclear; it has been postulated that some of these antibodies

may stimulate fibrosis, but the evidence in support of this idea is not convincing.

- **Vascular damage.** Microvascular disease is consistently present early in the course of systemic sclerosis and may be the initial lesion. Intimal proliferation is evident in the digital arteries of patients with systemic sclerosis. Capillary dilation with leaking, as well as destruction, is also common. Nailfold capillary loops are distorted early in the course of disease, and are later obliterated. Telltale signs of endothelial activation and injury (e.g., increased levels of von Willebrand factor) and increased platelet activation (increased circulating platelet aggregates) have also been noted. However, what causes the vascular injury is not known; it could be the initiating event or the result of chronic inflammation, with mediators released by inflammatory cells inflicting damage on microvascular endothelium. Repeated cycles of endothelial injury followed by platelet aggregation lead to release of platelet and endothelial factors (e.g., PDGF, TGF- β) that trigger perivascular fibrosis. Vascular smooth muscle cells also show abnormalities, such as increased expression of adrenergic receptors. Eventually, widespread narrowing of the microvasculature leads to ischemic injury and scarring.
- **Fibrosis.** The progressive fibrosis characteristic of the disease may be the culmination of multiple abnormalities, including the accumulation of alternatively activated macrophages, actions of fibrogenic cytokines produced by infiltrating leukocytes, hyperresponsiveness of fibroblasts to these cytokines, and scarring following ischemic damage caused by the vascular lesions. There is some evidence that fibroblasts from patients with systemic sclerosis have an intrinsic abnormality that causes them to produce excessive amounts of collagen. This idea is based on studies with cultured fibroblasts, and whether, or how, this abnormality relates to the pathogenesis is unknown.

MORPHOLOGY

Virtually all organs may be involved in systemic sclerosis. Prominent changes occur in the skin, alimentary tract, musculoskeletal system, and kidney. Lesions also are often present in the blood vessels, heart, lungs, and peripheral nerves.

Skin. The majority of patients have diffuse, sclerotic atrophy of the skin, which usually begins in the fingers and distal regions of the upper extremities

and extends proximally to involve the upper arms, shoulders, neck, and face. Histologically, there are perivascular infiltrates containing CD4+ T cells and edema, together with degeneration of collagen fibers, which become eosinophilic. Capillaries and small arteries (150 to 500 μm in diameter) may show thickening of the basal lamina, endothelial cell damage, and partial occlusion. With progression of the disease, there is increasing fibrosis of the dermis, which becomes tightly bound to the subcutaneous structures. There is marked increase of compact collagen in the dermis, usually with thinning of the epidermis, loss of rete pegs, atrophy of the dermal appendages, and hyaline thickening of the walls of dermal arterioles and capillaries (Fig. 6.30B). Focal and sometimes diffuse subcutaneous calcifications may develop, especially in patients with the CREST syndrome. In advanced stages, the fingers take on a tapered clawlike appearance, joint motion is limited, and the face becomes a drawn mask. Loss of blood supply may lead to cutaneous ulcerations and to atrophic changes in the terminal phalanges (Fig. 6.30C). Sometimes the tips of the fingers undergo autoamputation.

Gastrointestinal Tract. The GI tract is affected in approximately 90% of patients. Progressive atrophy and collagenous fibrous replacement of the muscularis may develop at any level of the gut but are most severe in the esophagus. The lower two-thirds of the esophagus often develops a rubber-hose–like inflexibility. The associated dysfunction of the lower esophageal sphincter gives rise to gastroesophageal reflux and its complications, including Barrett esophagus (Chapter 17) and strictures. The mucosa is thinned and may be ulcerated, and there is excessive collagenization of the lamina propria and submucosa. Loss of villi and microvilli in the small bowel sometimes produces a malabsorption syndrome.

Musculoskeletal System. Inflammation of the synovium, associated with synovial hypertrophy and hyperplasia, is common in the early stages; fibrosis later ensues. These changes are reminiscent of rheumatoid arthritis, but joint destruction is not common in systemic sclerosis. In a small subset of patients (approximately 10%), inflammatory myositis may develop.

Kidneys. Renal abnormalities occur in over two-thirds of patients with systemic sclerosis, most prominently vascular lesions. Interlobular arteries show intimal thickening as a result of deposition of mucinous or finely collagenous material, which stains histochemically for glycoprotein and acid mucopolysaccharides. There is also concentric proliferation of intimal cells. These changes may resemble those seen in malignant hypertension, but in scleroderma the alterations are restricted to vessels 150 to 500 μm in diameter and are not always associated with hypertension. Hypertension, however, does occur in 30% of patients with scleroderma, and in 20% it takes an ominously rapid, downhill course (malignant hypertension). In hypertensive patients, vascular alterations are more pronounced and are often associated with fibrinoid necrosis of arterioles, thrombosis, and infarction. Such patients often die of kidney failure, which overall accounts for about 50% of deaths from systemic sclerosis. There are no specific glomerular changes.

Lungs. Pulmonary involvement is seen in more than 50% of individuals with systemic sclerosis and may manifest as interstitial fibrosis and pulmonary hypertension. Pulmonary vasospasm, secondary to pulmonary vascular endothelial dysfunction, is considered important in the pathogenesis of pulmonary hypertension. Pulmonary fibrosis, when present, is indistinguishable from that seen in idiopathic pulmonary fibrosis (Chapter 15).

Heart. Pericarditis with effusion, myocardial fibrosis, and thickening of intramyocardial arterioles occur in one-third of patients. Clinical impairment by myocardial involvement, however, is less common.

Clinical Features

Systemic sclerosis has a female-to-male ratio of 3:1, with a peak incidence in the 50- to 60-year age group. Although systemic sclerosis shares many features with SLE, rheumatoid arthritis (Chapter 26), and polymyositis (Chapter 27), its distinctive characteristics are the striking cutaneous changes, notably skin fibrosis (Fig. 6.30C). *Raynaud*

phenomenon, manifested as numbness and tingling of the fingers and toes caused by episodic vasoconstriction of arteries and arterioles, is seen in virtually all patients and may precede other symptoms. Gastrointestinal symptoms, such as dysphagia attributable to esophageal fibrosis and its resultant hypomotility, are present in most patients. Eventually, destruction of the esophageal wall leads to atony and dilation, especially at its lower end. Abdominal pain, intestinal obstruction, or malabsorption syndrome with weight loss and anemia due to nutritional deficiencies reflect involvement of the small intestine. Respiratory difficulties caused by the pulmonary fibrosis may result in right-sided cardiac dysfunction, and myocardial fibrosis may cause either arrhythmias or cardiac failure. Proteinuria occurs in as many as 30% of patients, but rarely is severe enough to cause nephrotic syndrome. The most ominous manifestation is malignant hypertension (Chapter 11), with the subsequent development of kidney failure, but in its absence progression of the disease may be slow. As treatment of the cardiovascular and renal complications has improved, pulmonary disease has become the major cause of death in systemic sclerosis.

Virtually all patients have ANAs that react with a variety of nuclear antigens. Two ANAs strongly associated with systemic sclerosis have been described. One of these, directed against DNA topoisomerase I (anti-Scl 70), is highly specific. Depending on the ethnic group and the assay, it may be detected in 10% to 20% of patients with diffuse systemic sclerosis. Patients who have this antibody are more likely to have pulmonary fibrosis and peripheral vascular disease. The other, an anticentromere antibody, is found in 20% to 30% of patients, who tend to have the CREST syndrome. Patients with this syndrome have relatively limited involvement of skin, often confined to fingers, forearms, and face, and calcification of the subcutaneous tissues. Involvement of the viscera, including esophageal lesions, pulmonary hypertension, and biliary cirrhosis, may not occur at all or occur late. In general, these patients live longer than those with diffuse visceral involvement at the outset.

KEY CONCEPTS

Systemic Sclerosis

- Systemic sclerosis (commonly called scleroderma) is characterized by progressive fibrosis involving the skin, gastrointestinal tract, and other tissues.
- Fibrosis may be the result of activation of fibroblasts by cytokines produced by T cells, but what triggers T-cell responses is unknown.
- Endothelial injury and microvascular disease are commonly present in the lesions of systemic sclerosis, perhaps causing chronic ischemia, but the pathogenesis of vascular injury is not known.

Inflammatory Myopathies

Inflammatory myopathies comprise an uncommon, heterogeneous group of disorders characterized by injury and inflammation of skeletal muscles that are probably immunologically mediated. These diseases are described in Chapter 27.

Mixed Connective Tissue Disease

The term **mixed connective tissue disease** is used to describe a disease with clinical features that overlap with those of SLE, systemic sclerosis, and autoimmune myositis. The disease is characterized serologically by high titers of antibodies to U1 ribonucleoprotein. Typically, mixed connective tissue disease presents with synovitis of the fingers, Raynaud phenomenon, myositis, and renal involvement, which is modest and responds well to corticosteroids, at least in the short term. It is debated whether mixed connective tissue disease is a truly distinct entity, or if different patients represent subsets of SLE, systemic sclerosis, and myositis. The disease may, over time, evolve

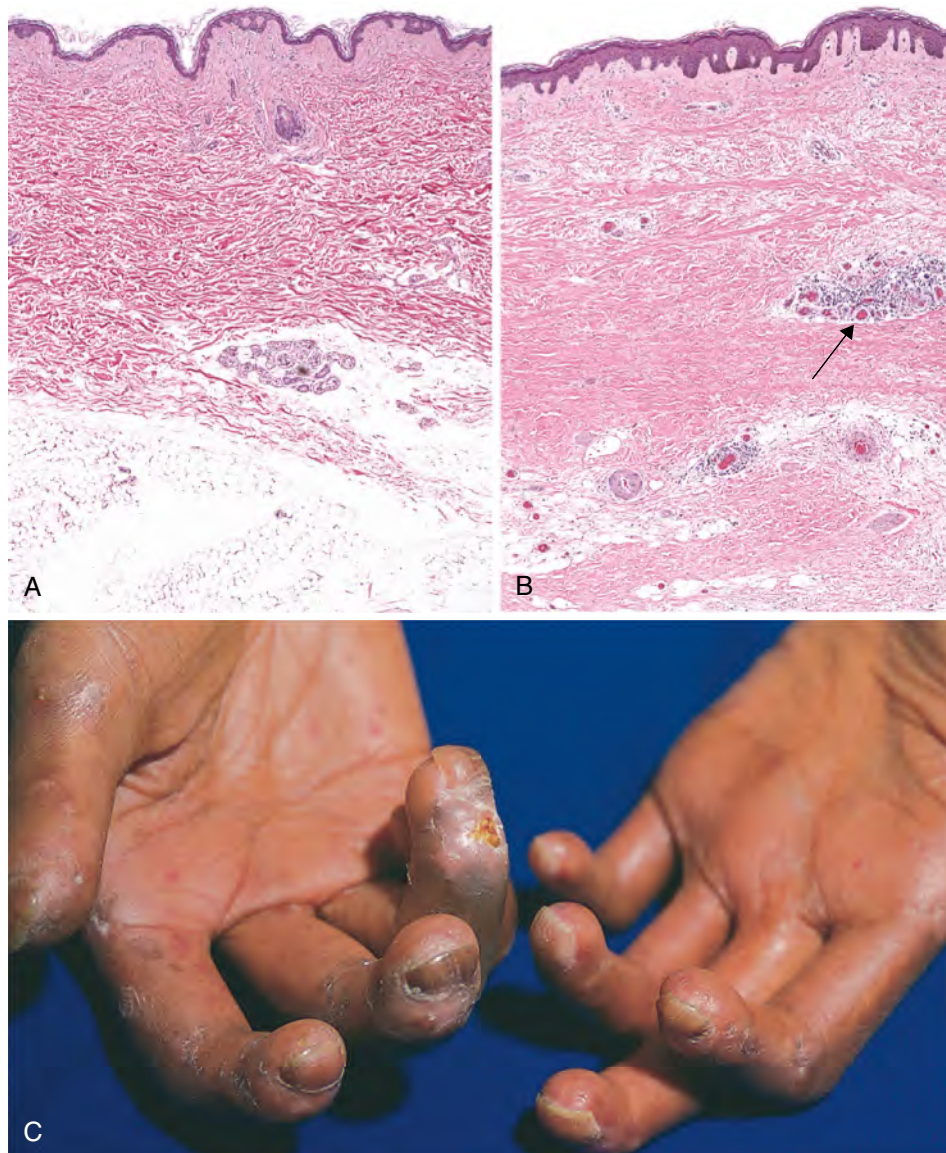


Fig. 6.30 Systemic sclerosis. (A) Normal skin. (B) Skin biopsy from a patient with systemic sclerosis. Note the extensive deposition of dense collagen in the dermis with virtual absence of appendages (e.g., hair follicles) and foci of inflammation (arrow). (C) The extensive subcutaneous fibrosis has virtually immobilized the fingers, creating a clawlike flexion deformity. Loss of blood supply has led to cutaneous ulcerations. (C, Courtesy Dr. Richard Sontheimer, Department of Dermatology, University of Texas Southwestern Medical School, Dallas.)

into classic SLE or systemic sclerosis. However, in other patients the mixture of features are maintained, and the salutary response to steroids is not universal, suggesting that there is a form of mixed connective tissue disease that is distinct from other autoimmune diseases.

Polyarteritis Nodosa and Other Vasculitides

Polyarteritis nodosa belongs to a group of diseases characterized by necrotizing inflammation of the walls of blood vessels and shows strong evidence of an immunologic pathogenic mechanism. The general term *noninfectious vasculitis* differentiates these conditions from those due to direct infection of the blood vessel wall (such as occurs in the wall of an abscess). Noninfectious vasculitis is encountered in many clinical settings. A detailed classification and description of vasculitides is presented in Chapter 11, where the immunologic mechanisms are also discussed.

IgG4-Related Disease

IgG4-related disease (IgG4-RD) is a constellation of disorders characterized by tissue infiltrates dominated by IgG4 antibody-producing plasma cells and lymphocytes (particularly T cells), fibrosis, obliterative phlebitis, and usually increased serum IgG4. IgG4-related disease has been described in virtually every organ system including the pancreas, the biliary tree, salivary glands, periorbital tissues, kidneys, lungs, lymph nodes, meninges, aorta, breast, prostate, thyroid, pericardium, and skin (Fig. 6.31). Many fibrotic disorders long viewed as confined to single organs are part of the IgG4-RD spectrum. These include some forms of Mikulicz syndrome, which affects salivary glands, Riedel thyroiditis, idiopathic retroperitoneal fibrosis, autoimmune pancreatitis, and sclerosing cholangitis, to name a few. The disease most often affects middle-aged and older males.

The pathogenesis of this condition is not understood, and although IgG4 production in lesions is a hallmark of the disease, there is no evidence that this antibody type contributes to the pathology. The IgG4

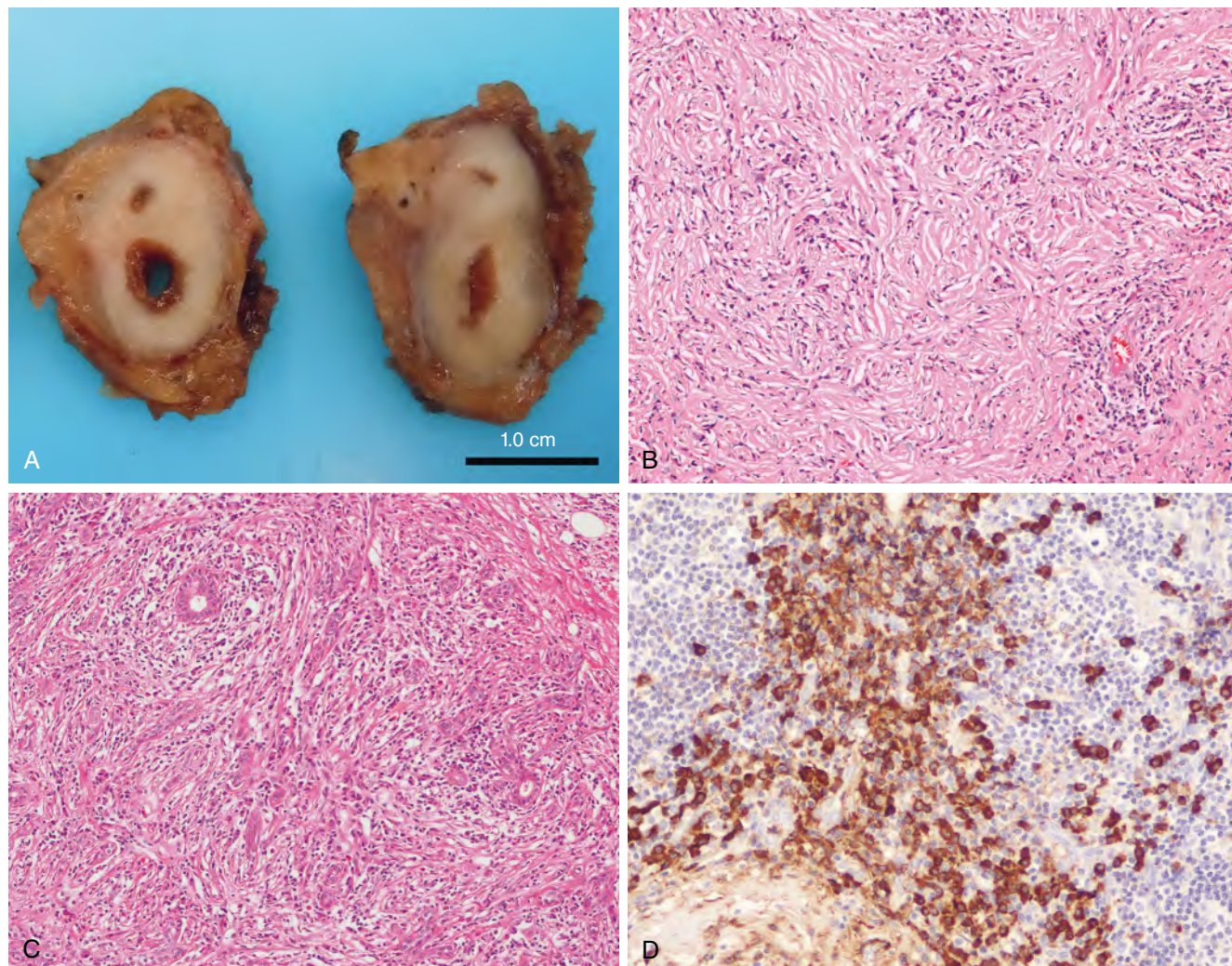


Fig. 6.31 IgG4-related disease: representative lesions. (A) Cross section of bile duct showing grayish-white area of sclerosing cholangitis. (B) Sclerotic area of the bile duct with storiform fibrosis. (C) Submandibular gland with infiltrates of lymphocytes and plasma cells and whorls of fibrosis. (D) Section of an involved lacrimal gland stained with an antibody against IgG4, showing large numbers of IgG4-producing plasma cells. (From Kamisawa T, Zen Y, Pillai S, Stone JH. IgG4-related disease. *Lancet*. 2015;385(9976):1460-1471.)

isotype does not activate complement efficiently and does not bind to leukocyte Fc receptors, and how it may induce fibrosis is unknown. Nevertheless, a key role of B cells (perhaps in antigen presentation to some T cells) is supported by clinical trials in which depletion of B cells with anti-B-cell antibodies provided clinical benefit. It is suspected that autoimmune reactions underlie the disease, but no causative auto-antigens have been identified.

REJECTION OF TISSUE TRANSPLANTS

Transplant rejection is discussed here because it involves several of the immunologic reactions that underlie immune-mediated inflammatory diseases. A major barrier to transplantation is the process of rejection, in which the recipient's immune system recognizes the graft as being foreign and attacks it.

Mechanisms of Recognition and Rejection of Allografts

Rejection is a process in which T lymphocytes and antibodies produced against graft antigens react against and destroy tissue grafts. Grafts exchanged between individuals of the same species (the usual

clinical situation) are called *allografts*, and grafts from one species to another (now being tested with genetically modified pig kidneys and hearts) are called *xenografts*. Since most clinical transplants are allografts, the discussion is focused on these.

Recognition of Graft Alloantigens by T and B Lymphocytes

The major antigenic differences between a donor and recipient that result in rejection of transplants are in HLA alleles. Because HLA genes are highly polymorphic, there are almost always some differences between individuals (except, of course, identical twins). Following transplantation, the recipient's T cells recognize donor HLA antigens from the graft (the allogeneic antigens, or alloantigens), which appear foreign to the recipient. The graft antigens are either presented directly to recipient T cells by graft APCs (direct pathway of alloantigen recognition), or the graft antigens are picked up by host APCs, processed (like any other foreign antigen), and presented to host T cells (indirect pathway of alloantigen recognition). Both lead to the activation of CD8⁺ T cells, which develop into CTLs, and CD4⁺ T cells, which become cytokine-producing effector cells, mainly Th1 cells. The direct pathway may be most

important for CTL-mediated acute rejection, and the indirect pathway may play a greater role in chronic rejection.

The frequency of T cells that can recognize the foreign antigens in a graft is much higher than the frequency of T cells specific for any microbe. For this reason, immune responses to allografts are stronger than responses to pathogens. Predictably, these strong reactions can destroy grafts rapidly, and their control requires powerful immunosuppressive agents.

B lymphocytes also recognize antigens in the graft, including HLA and other antigens that differ between donor and recipient. The activation of these B cells typically requires T cell help and leads to the production of antibodies against donor HLA molecules, called alloantibodies.

Patterns and Mechanisms of Graft Rejection

Graft rejection is classified into hyperacute, acute, and chronic based on clinical and pathologic features. This historical classification was devised by clinicians and pathologists for diagnosing rejection of kidney allografts. It has been valuable not only for diagnosis and selecting the optimal therapy but also for understanding pathogenic mechanisms because each type of rejection is mediated by a particular kind of immune response. In the following discussion, the description of the morphology of rejection is limited to kidney allografts, but similar changes are seen in other organ transplants.

- **Hyperacute rejection is mediated by preformed antibodies specific for antigens on graft endothelial cells.** The preformed antibodies may be natural IgM antibodies specific for blood group antigens or may be antibodies specific for allogeneic MHC molecules that were induced by prior exposure of the organ recipient to allogeneic cells through blood transfusions, pregnancy, or transplantation of another organ. Immediately after the graft is implanted and blood flow is restored, the antibodies bind to antigens on the graft vascular endothelium and activate the complement system, leading to endothelial injury, thrombosis, and ischemic necrosis of the graft (Fig. 6.32A). Hyperacute rejection is rare because every donor and recipient are matched for blood type, and potential recipients are tested for antibodies against the cells of the prospective donor, a test called a crossmatch.

MORPHOLOGY

In hyperacute rejection, the affected kidney rapidly becomes cyanotic, mottled, and anuric. Arterioles and arteries exhibit acute fibrinoid necrosis of their walls and narrowing or complete occlusion of their lumens by thrombi (Fig. 6.32B). Neutrophils rapidly accumulate within arterioles, glomeruli, and peritubular

capillaries. As these changes intensify and become diffuse, the glomerular capillaries also undergo thrombotic occlusion, and eventually the kidney cortex undergoes outright necrosis (infarction). Affected kidneys are nonfunctional and must be removed.

- **Acute rejection is mediated by T cells and antibodies that react against alloantigens in the graft.** It occurs within days or weeks after transplantation and is the principal cause of early graft failure. It also may appear suddenly much later after transplantation if immunosuppression is tapered or terminated. Based on the role of T cells or antibodies, acute rejection is divided into two types, although in most rejecting grafts, both patterns are present.
 - In *acute cellular rejection*, CD8⁺ CTLs may directly destroy graft cells, or CD4⁺ cells secrete cytokines and induce inflammation, which damages the graft (Fig. 6.33A). T cells may also react against graft vessels, leading to vascular damage. Current immunosuppressive therapy is designed mainly to prevent and reduce acute rejection by blocking the activation of alloreactive T cells.
 - In *acute antibody-mediated (vascular, or humoral) rejection*, antibodies bind to vascular endothelium and activate complement via the classical pathway (Fig. 6.34A). The resultant inflammation and endothelial damage cause graft failure.

MORPHOLOGY

Acute cellular (T cell–mediated) rejection may produce two different patterns of injury.

- In the tubulointerstitial pattern, there is extensive interstitial inflammation and tubular inflammation (tubulitis) associated with focal tubular injury (Fig. 6.33B). As might be expected, the inflammatory infiltrates contain activated CD4⁺ and CD8⁺ T lymphocytes.
 - The vascular pattern shows inflammation of vessels (Fig. 6.33C) and sometimes necrosis of vessel walls. The affected vessels have swollen endothelial cells, and lymphocytes are seen between the endothelium and the vessel wall, a finding termed endotheliitis or intimal arteritis. The recognition of cellular rejection is important because, in the absence of accompanying humoral rejection, most patients respond well to immunosuppressive therapy.
- Acute antibody-mediated rejection is manifested mainly by damage to glomeruli and small blood vessels. Typically, there is inflammation of glomeruli and peritubular capillaries (Fig. 6.34B), associated with deposition of complement products (e.g., C4d) due to activation of the complement system by the antibody-dependent classical pathway (Fig. 6.34C). Small vessels may also show focal thrombosis.

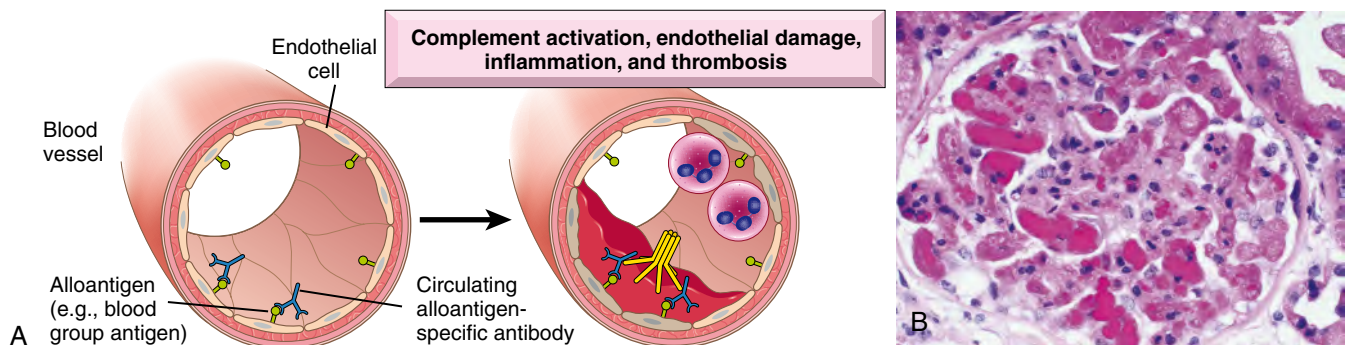


Fig. 6.32 Hyperacute rejection. (A) Deposition of antibody on endothelium and activation of complement causes thrombosis. (B) Hyperacute rejection of a kidney allograft showing platelet fibrin thrombi and severe ischemic injury in a glomerulus. (Courtesy Dr. David Howell, Department of Pathology, Duke University School of Medicine, Durham, NC.)

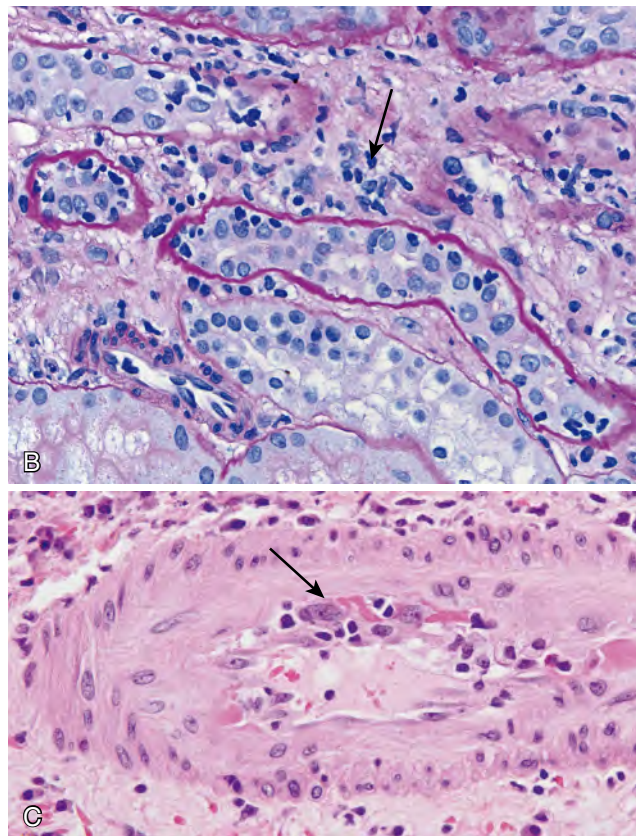
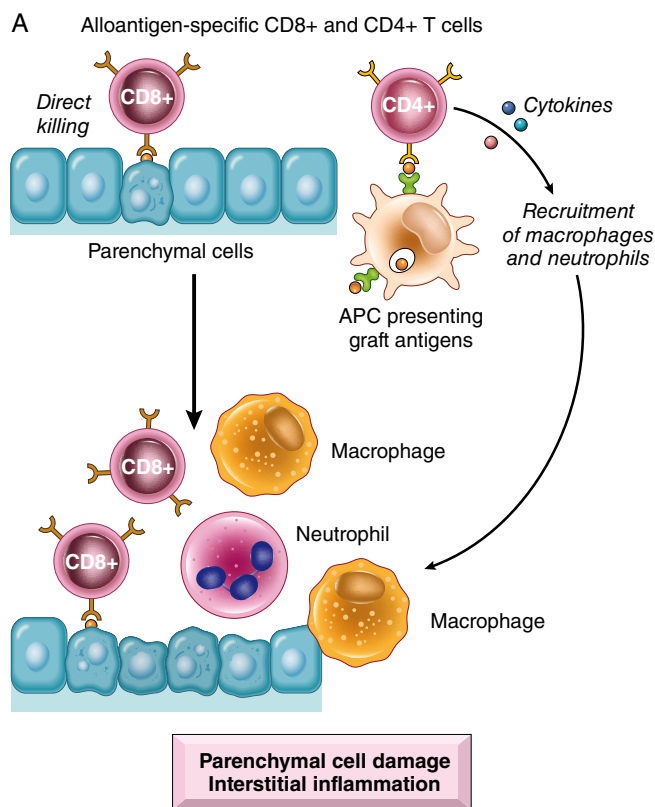


Fig. 6.33 Acute cellular rejection. (A) Destruction of graft cells by T cells. Acute T cell-mediated rejection involves direct killing of graft cells by CD8+ CTLs and inflammation caused by cytokines produced by CD4+ T cells. (B) Acute cellular rejection of a kidney graft, manifested by inflammatory cells in the interstitium and between epithelial cells of the tubules (tubulitis). Collapsed tubules are outlined by wavy basement membranes. (C) Rejection vasculitis in a kidney graft. An arteriole is shown with inflammatory cells attacking and undermining the endothelium (endotheliitis; arrow). (Courtesy Drs. Zoltan Laszik and Kuang-Yu Jen, Department of Pathology, University of California, San Francisco.)

- **Chronic rejection is an indolent form of graft damage that occurs over months or years, leading to progressive loss of graft function.** Chronic rejection manifests as interstitial fibrosis and gradual narrowing of graft blood vessels (*graft arteriosclerosis*). In both lesions, the culprits are believed to be T cells that react against graft alloantigens and secrete cytokines, which stimulate the proliferation and activities of fibroblasts and vascular smooth muscle cells in the graft (Fig. 6.35A). Alloantibodies also contribute to chronic rejection. Although treatments to prevent or curtail acute rejection have steadily improved, leading to better 1-year survival of transplants, chronic rejection is refractory to most therapies and is becoming the principal cause of graft failure.

MORPHOLOGY

Chronic rejection is dominated by vascular changes, often with intimal thickening and vascular occlusion (Fig. 6.35B). Chronically rejecting kidney grafts show glomerulopathy, with duplication of the basement membrane, likely secondary to chronic endothelial injury (Fig. 6.35C), and peritubular capillaritis with multilayering of peritubular capillary basement membranes. Interstitial fibrosis and tubular atrophy with loss of renal parenchyma may occur secondary to the vascular lesions (Fig. 6.36D). Interstitial mononuclear cell infiltrates are typically sparse.

In addition to the kidney, a variety of organs, such as the liver (Chapter 18), heart (Chapter 12), lungs, and pancreas, are also transplanted. The rejection of these grafts is described in the relevant chapters.

Methods of Increasing Graft Survival

The value of HLA matching between donor and recipient varies in different solid-organ transplants. In kidney transplants, there is substantial benefit if the most important polymorphic *HLA* alleles are matched (both alleles of *HLA-A*, *-B*, and *DR*). However, HLA matching is usually not done for transplants of liver, heart, and lungs, because other considerations, such as anatomic compatibility, severity of the underlying illness, and the need to minimize the time of organ storage, override the potential benefits of HLA matching.

Except for identical twins, immunosuppressive therapy is essential in all donor-recipient combinations. Immunosuppressive drugs in current use include steroids (which reduce inflammation), mycophenolate mofetil (which inhibits lymphocyte proliferation), and tacrolimus (FK506). Tacrolimus, like its predecessor cyclosporine, is an inhibitor of the phosphatase calcineurin, which is required for activation of a transcription factor called nuclear factor of activated T cells (NFAT). NFAT stimulates transcription of cytokine genes, in particular the gene that encodes the growth factor IL-2. Thus, calcineurin inhibitors block T cell responses. Additional drugs that are used to treat rejection include T cell-depleting antibodies, and pooled intravenous IgG (IVIG), which suppresses inflammation by unknown mechanisms. Plasmapheresis is used in cases of severe antibody-mediated rejection. Another more recent strategy for reducing antigrraft immune responses is to prevent host T cells from receiving costimulatory signals from DCs during the initial phase of sensitization. This can be accomplished by interrupting the

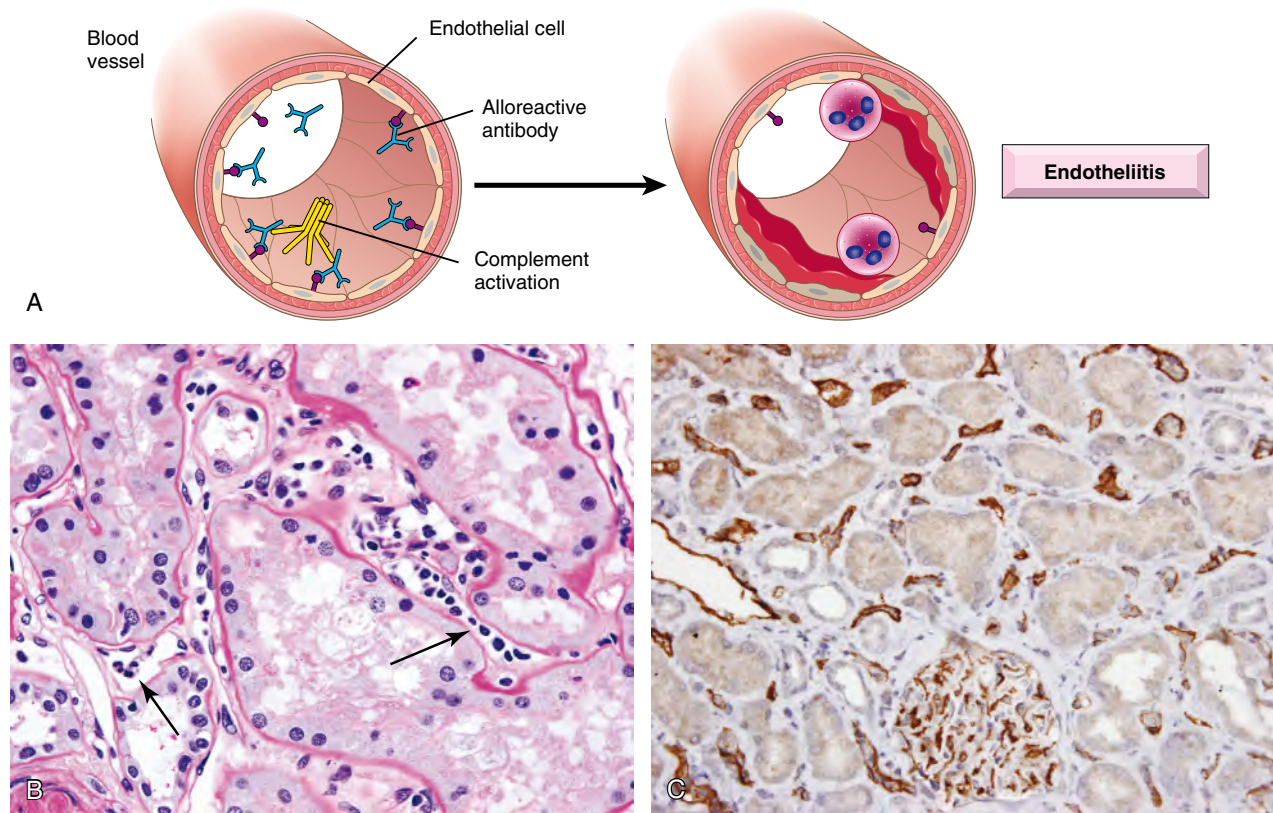


Fig. 6.34 Acute antibody-mediated (humoral) rejection. (A) Graft damage caused by antibody deposition in vessels. (B) Light micrograph showing inflammation (capillaritis) in peritubular capillaries (arrows) in a kidney graft. (C) Immunoperoxidase stain shows C4d deposition in peritubular capillaries and a glomerulus. (Courtesy Dr. Zoltan Laszik, Department of Pathology, University of California, San Francisco.)

interaction between the B7 molecules on the DCs of the graft donor with the CD28 receptors on host T cells, for example, by administration of proteins that bind to and block B7 costimulators.

Although immunosuppression prolongs graft survival, it carries its own risks. The price paid in the form of increased susceptibility to opportunistic infections and cancers is not small; in fact, infections and cancer are major barriers to disease-free survival after transplantation. Immunosuppressed individuals are susceptible to a wide variety of bacterial, viral, and fungal infections. One infectious complication unique to kidney transplant recipients is reactivation of *BK polyoma virus*. The virus establishes latent infection of epithelial cells in the lower genitourinary tract of healthy individuals, and on immunosuppression, it is reactivated, infects renal tubules, and may even cause graft failure. Immunosuppressed graft recipients are also at increased risk for developing skin cancer, EBV-induced lymphomas and human papillomavirus-induced squamous cell carcinomas, the latter two probably the result of reactivation of latent oncogenic viruses. Efforts to induce donor-specific tolerance in graft recipients have not yet been successful; this would reduce immune responses to the graft but not to microbes.

Transplantation of Hematopoietic Stem Cells

Use of HSC transplants to treat hematologic malignancies, bone marrow failure syndromes (such as aplastic anemia), and inherited bone marrow disorders (such as sickle cell anemia, thalassemia, and immunodeficiency states) is increasing each year. Transplantation of genetically “reengineered” HSCs obtained from affected patients is already used for some immunodeficiencies and inherited bone marrow disorders. Historically, HSCs were obtained from the bone marrow, but now

they are usually harvested from peripheral blood after they are mobilized from the bone marrow by administration of hematopoietic growth factors, or from the umbilical cord blood of newborn infants, a rich source of HSCs. In most of the conditions in which HSC transplantation is indicated, the recipient is irradiated or treated with chemotherapy to deplete host cells, thereby opening up stem cell niches in the microenvironment of the marrow within which the transplanted HSCs engraft and giving the transplanted cells a survival and proliferative advantage. Two problems that are unique to HSC transplantation are graft-versus-host disease (GVHD) and immunodeficiency.

GVHD occurs when immunologically competent cells or their precursors are transplanted into immunologically compromised recipients, and the transferred cells recognize alloantigens in the host and attack host tissues. It is seen most commonly in the setting of HSC transplantation but, rarely, may occur following transplantation of solid organs rich in lymphoid cells (e.g., the liver) or transfusion of unirradiated blood. When immune-compromised recipients receive HSC preparations from allogeneic donors, the immunocompetent T cells present in the donor inoculum recognize the recipient’s HLA antigens as foreign and react against them. To try to minimize GVHD, HSC transplants are done between donor and recipient who are HLA-matched using precise DNA sequencing-based methods for molecular typing of HLA alleles. There are two main categories of GVHD: acute and chronic.

- **Acute GVHD** occurs within days to weeks after allogeneic HSC transplantation. Although any organ may be affected, the major clinical manifestations result from immune-mediated injury of the epithelial cells of the skin, bile ducts, and intestines. Cutaneous involvement is characterized by lymphocyte infiltration (Fig. 6.36A)

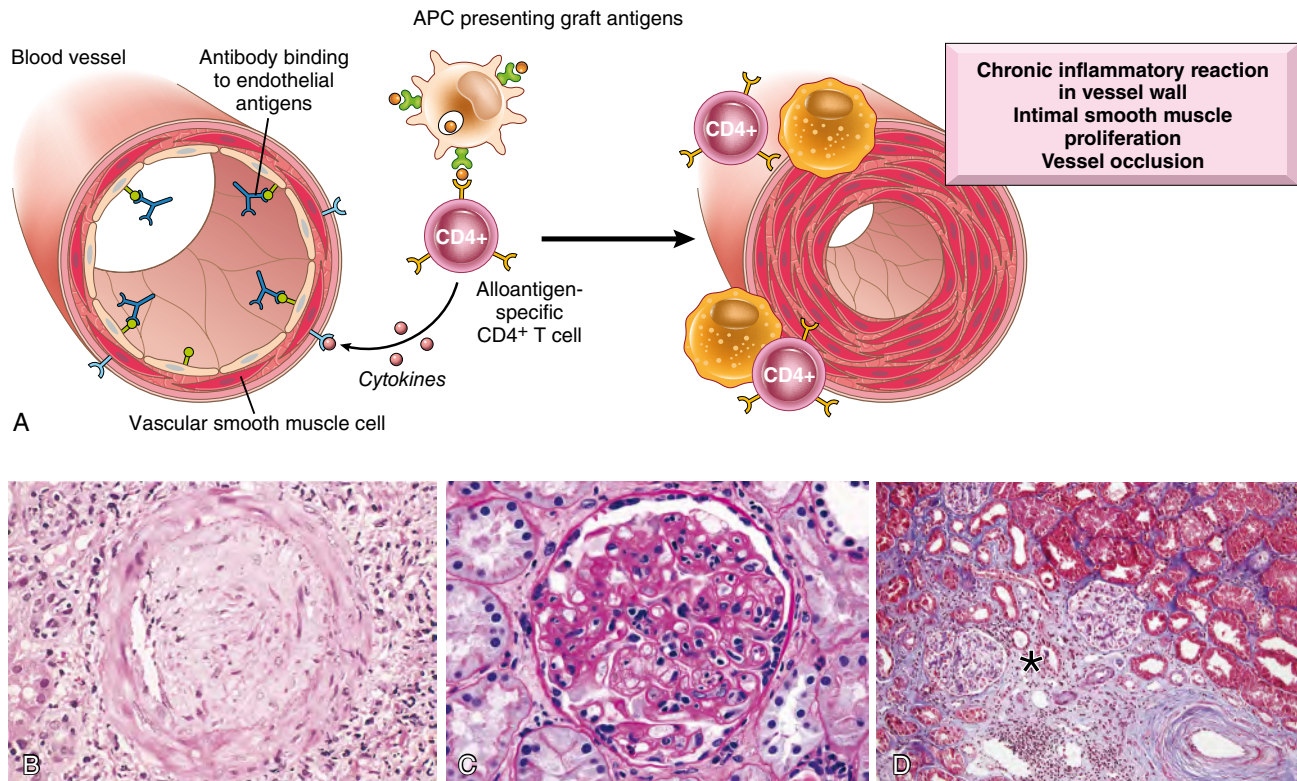


Fig. 6.35 Chronic rejection. (A) Pathogenesis of graft arteriosclerosis caused by T-cell cytokines and antibody deposition. (B) Graft arteriosclerosis in a cardiac transplant. (C) Transplant glomerulopathy, the characteristic manifestation of chronic antibody-mediated rejection in the kidney. The glomerulus shows inflammatory cells within the capillary loops (glomerulitis), accumulation of mesangial matrix, and duplication of the capillary basement membrane. (D) Interstitial fibrosis and tubular atrophy, resulting from arteriosclerosis of arteries and arterioles in a chronically rejecting kidney allograft. In this trichrome stain, the blue area (*asterisk*) shows fibrosis, contrasted with the normal kidney (*top right*). An artery showing prominent arteriosclerosis is shown (*bottom right*). APC, Antigen-presenting cell. (B, Courtesy Dr. Richard Mitchell, Department of Pathology, Brigham and Women's Hospital, Boston. C and D, Courtesy Dr. Zoltan Laszik, Department of Pathology, University of California, San Francisco.)

and apoptosis of keratinocytes (Fig. 6.36B). It manifests clinically as a rash, first appearing on the neck, ears, palms of the hands, and soles of the feet, and then becoming generalized. Destruction of small bile ducts gives rise to jaundice, and mucosal ulceration of the gut results in bloody diarrhea. Although tissue injury may be severe, the affected tissues are usually not heavily infiltrated by lymphocytes. It is believed that in addition to direct cytotoxicity by CD8⁺ T cells, considerable damage is inflicted by cytokines released by the sensitized donor T cells.

- **Chronic GVHD** may follow the acute syndrome or may occur insidiously. In this form of GVHD, there is extensive cutaneous injury with destruction of skin appendages and fibrosis of the dermis (Fig. 6.36C). The changes may resemble systemic sclerosis (discussed earlier). Chronic liver disease manifested by cholestatic jaundice is also frequent. Damage to the gastrointestinal tract may cause esophageal strictures. The immune system is affected, with involution of the thymus and depletion of lymphocytes in the lymph nodes. Not surprisingly, patients experience recurrent infections. Other patients develop manifestations of autoimmunity, postulated to result from the grafted CD4⁺ helper T cells reacting with host B cells and stimulating these cells, some of which may be capable of producing autoantibodies.

Because GVHD is mediated by T lymphocytes contained in the transplanted donor cells, depletion of donor T cells before transfusion

virtually eliminates the disease. This approach, however, is a mixed blessing: GVHD is ameliorated, but the recurrence of tumor in leukemic patients as well as the incidence of graft failures and EBV-related B-cell lymphoma increase. It seems that the T cells not only mediate GVHD but also are required for engraftment of the transplanted HSCs (perhaps by producing CSFs), suppression of EBV-infected B-cell clones, and control of leukemia cells.

Immunodeficiency is a frequent complication of HSC transplantation. The immunodeficiency may be a result of prior treatment (e.g., for leukemia), myeloablation and lymphodepletion prior to HSC transplantation, a delay in repopulation of the recipient's immune system, and attack on the host's immune cells by grafted lymphocytes. Affected individuals are profoundly immunosuppressed and are easy prey to infections. Although many different types of organisms may infect patients, infection with cytomegalovirus is particularly important. This usually results from activation of latent infection. Cytomegalovirus-induced pneumonitis can be a fatal complication.

KEY CONCEPTS

Recognition and Rejection of Transplants (Allografts)

- The rejection of solid-organ transplants is initiated mainly by host T cells that recognize the foreign HLA antigens of the graft, either directly (on APCs in the graft) or indirectly (after uptake and presentation by host APCs).

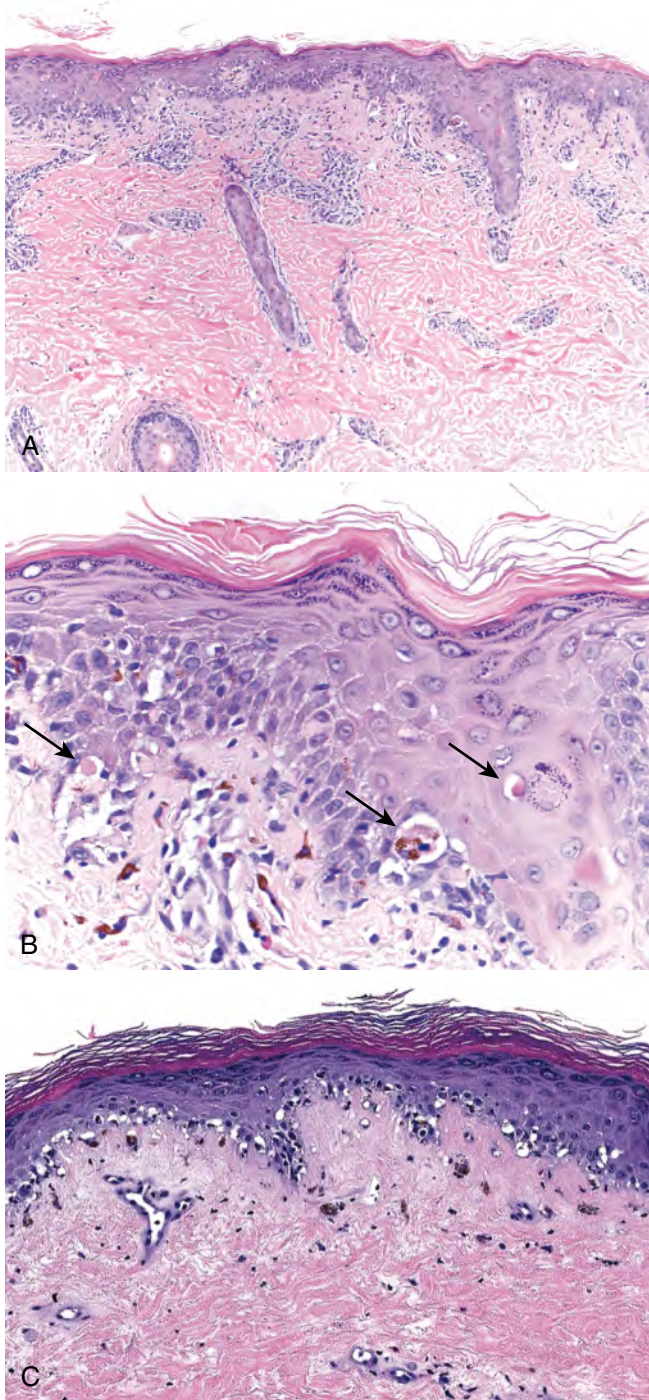


Fig. 6.36 Graft-versus-host disease (GVHD) involving the skin. Low-power (A) and high-power (B) photomicrographs are shown of a skin biopsy from a patient with acute GVHD. A sparse lymphocytic infiltrate can be seen at the dermoepidermal junction, and damage to the epithelial layer is manifested by spaces at the dermoepidermal junction (vacuolization), cells with abnormal keratin staining (dyskeratosis), apoptotic keratinocytes (arrows), and disorganization of maturation of keratinocytes from the basal layer to the surface. (C) Chronic GVHD showing a sparse lymphocytic infiltrate at the dermoepidermal junction, which has resulted in occasional damaged keratinocytes. The epidermis is thinned due to atrophy. The underlying dermis shows thick collagen bundles, indicative of sclerosis. (A and B, Courtesy Dr. Scott Granter, Department of Pathology, Brigham and Women's Hospital and Harvard Medical School, Boston. C, Courtesy Dr. Jarish Cohen, Department of Pathology, University of California, San Francisco.)

- Types and mechanisms of rejection of solid-organ grafts:
 - Hyperacute rejection. Preformed antidonor antibodies bind to graft endothelium immediately after transplantation, leading to thrombosis, ischemic damage, and rapid graft failure.
 - Acute cellular rejection. T cells destroy graft parenchyma (and vessels) by cytotoxicity and inflammatory reactions.
 - Acute humoral rejection. Antibodies damage graft vasculature.
 - Chronic rejection. Dominated by arteriosclerosis, this type is caused by T-cell activation and antibodies. The T-cells may secrete cytokines that induce proliferation of vascular smooth muscle cells, and the antibodies cause endothelial injury. The vascular lesions and T-cell reactions cause parenchymal fibrosis.
- Treatment of graft rejection relies on immunosuppressive drugs, which inhibit immune responses against the graft but also make treated patients susceptible to infections and virus-induced cancers.
- Transplantation of HSCs requires careful matching of donor and recipient and may be complicated by GVHD and immune deficiency.

IMMUNODEFICIENCY DISEASES

Immunodeficiency diseases can be divided into **primary (or congenital) immunodeficiencies**, which are caused by single-gene defects, and **secondary (or acquired) immunodeficiencies**, which may arise as complications of cancers, infections, or malnutrition, or side effects of immunosuppression, irradiation, or chemotherapy for cancer, graft rejection, and other diseases. Immunodeficiencies are manifested clinically by increased infections, which may be newly acquired or reactivation of a latent infection. The primary immunodeficiency syndromes are uncommon accidents of nature that provide valuable insights into some of the critical molecules of the human immune system. We briefly discuss the more important and best defined primary immunodeficiencies, to be followed by a more detailed description of acquired immunodeficiency syndrome (AIDS), the most devastating example of secondary immunodeficiency.

Primary Immunodeficiencies

Primary immunodeficiency diseases, sometimes called inborn errors of immunity, are caused by genetic (inherited) defects that affect the defense mechanisms of innate immunity (phagocytes, NK cells, or complement) or the humoral and/or cellular arms of adaptive immunity (mediated by B and T lymphocytes, respectively). Most of these diseases are detected in infancy, between 6 months and 2 years of life, the telltale signs being susceptibility to recurrent infections. Historically, these are considered rare diseases, but recent studies suggest the prevalence may be as high as 1 in 500 to 1 in 10,000 births in different populations. The widespread application of DNA sequencing has led to the identification of many genetic defects responsible for these diseases. Here we present selected examples of primary immunodeficiencies, beginning with defects in innate immunity and then defects in the maturation and activation of B and T lymphocytes. We conclude with immune defects associated with some systemic diseases.

Defects in Innate Immunity

Inherited defects in the early innate immune response typically affect leukocyte functions or the complement system, and all lead to increased vulnerability to infections (Table 6.12). Some of the defects whose molecular basis is defined are summarized next.

Defects in Leukocyte Function

- **Inherited defects in leukocyte adhesion.** Individuals with *leukocyte adhesion deficiency type 1* have a defect in the gene encoding the β_2 chain shared by the integrins LFA-1 and Mac-1. *Leukocyte adhesion*

TABLE 6.12 Defects in Innate Immunity

Disease	Defect
Defects in Leukocyte Function	
Leukocyte adhesion deficiency 1	Defective leukocyte adhesion because of mutations in β chain of CD11/CD18 integrins
Leukocyte adhesion deficiency 2	Defective leukocyte adhesion because of mutations in fucosyl transferase required for synthesis of sialylated oligosaccharide (receptor for selectins)
Chédiak-Higashi syndrome	Decreased leukocyte functions because of mutations affecting protein involved in lysosomal membrane traffic
Chronic granulomatous disease	Decreased oxidative burst
X-linked	Phagocyte oxidase (membrane component)
Autosomal recessive	Phagocyte oxidase (cytoplasmic components)
Defects in the Complement System	
C2, C4 deficiency	Defective classic pathway activation results in reduced resistance to infection and reduced clearance of immune complexes
C3 deficiency	Defects in all complement functions
Deficiency of complement regulatory proteins	Excessive complement activation; clinical syndromes include angioedema, paroxysmal hemoglobinuria, others

The table lists some of the more common inherited immune deficiencies affecting phagocytic leukocytes and the complement system. Modified in part from Gallin JI. Disorders of phagocytic cells. In: Gallin JI, Goldstein IM, Snyderman R, eds. *Inflammation: Basic Principles and Clinical Correlates*. 2nd ed. Raven Press; 1992:860-861.

deficiency type 2 is caused by the absence of sialyl-Lewis X, the fucose-containing ligand for E- and P-selectins, because of a mutation affecting a fucosyl transferase, an enzyme that attaches fucose moieties to protein backbones. Both conditions result in a failure of leukocyte adhesion to endothelium, preventing the cells from migrating into tissues and making patients prone to bacterial infections, which are often recurrent and frequently life threatening.

- **Inherited defects in phagolysosome function.** One such disorder is *Chédiak-Higashi syndrome*, an autosomal recessive condition characterized by defective fusion of phagosomes and lysosomes, resulting in defective phagocyte function and susceptibility to infections. The main leukocyte abnormalities are neutropenia (decreased numbers of neutrophils), defective degranulation, and delayed microbial killing. Leukocytes contain giant granules, which can be readily seen in peripheral blood smears and are thought to result from aberrant phagolysosome fusion. In addition, there are abnormalities in melanocytes (leading to albinism), cells of the nervous system (associated with nerve defects), and platelets (causing bleeding disorders). The dysfunctional gene underlying this disorder encodes a cytosolic protein called LYST, which is believed to regulate lysosomal trafficking.
- **Inherited defects in microbicidal activity.** The importance of oxygen-dependent bactericidal mechanisms is shown by the existence of a group of congenital disorders called *chronic granulomatous disease* (CGD), which are characterized by defects in bacterial killing that render patients susceptible to recurrent bacterial infection. CGD results from inherited defects in the genes encoding components of phagocyte oxidase, the phagolysosomal enzyme that generates superoxide (O_2^-). The most common variants are an X-linked defect

in one of the membrane-bound components (gp91phox) and autosomal recessive defects in the genes encoding two of the cytoplasmic components (p47phox and p67phox). The name of this disease comes from the macrophage-rich chronic inflammatory reaction that tries to control the infection when the initial neutrophil response is inadequate. This often leads to collections of activated macrophages that wall off the microbes, forming granulomas.

- **Defects in TLR signaling.** Rare defects have been described in various TLRs and their signaling molecules. Defects in TLR3, a receptor for viral RNA, result in recurrent herpes simplex encephalitis, and defects in MyD88, the adaptor protein downstream of multiple TLRs, are associated with destructive bacterial infections, especially pneumonia.
- **Cytokine defects** may be caused by mutations in genes encoding cytokines or their receptors. Acquired cytokine defects are caused by autoantibodies produced against cytokines. Defects affecting the antiviral cytokines type I IFNs are associated with viral infections, including severe forms of COVID-19. Mutations affecting the receptor for IL-12, a Th1-inducing cytokine, or for IFN- γ , the effector cytokine of Th1 cells, result in increased susceptibility to infections with intracellular bacteria such as environmental mycobacteria, which are of low virulence and do not cause disease in healthy individuals. The disorder is called *Mendelian susceptibility to mycobacterial disease*.

Deficiencies Affecting the Complement System. Hereditary deficiencies have been described for virtually all components of the complement system and several of the regulators. In addition, one disease, paroxysmal nocturnal hemoglobinuria, is caused by an acquired deficiency of complement regulatory factors.

- Deficiency of C2 is the most common complement protein deficiency. A deficiency of C2 or C4, early components of the classic pathway, may be associated with increased bacterial or viral infections. However, many patients have no clinical manifestations, presumably because the alternative complement pathway is adequate for the control of most infections. Surprisingly, in some individuals with deficiency of C2, C4, or C1q, the dominant manifestation is the autoimmune disease SLE, as discussed earlier.
- Deficiency of components of the alternative pathway (properdin and factor D) is rare. It is associated with recurrent pyogenic infections.
- The C3 component of complement is required for both the classical and alternative pathways, and hence a deficiency of this protein results in susceptibility to serious and recurrent pyogenic infections. There is also an increased incidence of immune complex-mediated glomerulonephritis. In the absence of complement, inflammation is presumably caused by Fc receptor-dependent leukocyte activation.
- The terminal components of complement (C5, 6, 7, 8, and 9) are required for the assembly of the membrane attack complex involved in the lysis of organisms. Deficiency of these late-acting components is associated with increased susceptibility to recurrent *Neisseria* (gonococcal and meningococcal) infections; *Neisseria* bacteria have thin cell walls that make them susceptible to the lytic actions of complement.
- Other patients inherit a defective form of mannose-binding lectin, the plasma protein that initiates the lectin pathway of complement. These individuals also show increased susceptibility to infections.
- A deficiency of C1 inhibitor (C1 INH) gives rise to *hereditary angioedema*. This autosomal dominant disorder is more common than complement deficiency states. The C1 inhibitor's targets are proteases, specifically C1r and C1s of the complement cascade, factor XII of the coagulation pathway, and the kallikrein system. With deficiency of C1 INH, unregulated activation of kallikrein may lead to increased production of vasoactive peptides such as bradykinin. Although the exact nature of the bioactive compound produced in hereditary angioedema is uncertain, these patients have episodes of edema affecting skin and mucosal surfaces such as the larynx and the

gastrointestinal tract. This may result in life-threatening asphyxia or nausea, vomiting, and diarrhea after minor trauma or emotional stress. Acute attacks of hereditary angioedema can be treated with C1 inhibitor concentrates prepared from human plasma.

- Deficiencies of other complement regulatory proteins are causes of paroxysmal nocturnal hemoglobinuria (Chapter 14) and are associated with chronic forms of hemolytic uremic syndrome (Chapter 20), and age-related macular degeneration (Chapter 29).

Defects in Adaptive Immunity

Defects in adaptive immunity are often subclassified based on the primary component involved (i.e., B cells or T cells or both). However, these distinctions are not clear-cut; for instance, T-cell defects almost always lead to impaired antibody synthesis, and hence isolated deficiencies of T cells are often indistinguishable clinically from combined deficiencies of T and B cells. These immunodeficiencies result from abnormalities in lymphocyte maturation or activation. The mutations responsible for many of these diseases have now been identified (Fig. 6.37).

Severe Combined Immunodeficiency. Severe combined immunodeficiency (SCID) spans a constellation of genetically distinct syndromes that all have defects in both humoral and cell-mediated immune responses. Affected infants present with prominent thrush (oral

candidiasis), extensive diaper rash, and failure to thrive. Some patients develop a morbilliform rash shortly after birth because maternal T cells are transferred across the placenta and attack the fetus, causing GVHD. Persons with SCID are extremely susceptible to recurrent, severe infections by a wide range of pathogens, including *Candida albicans*, *Pneumocystis jirovecii*, cytomegalovirus, varicella, and *Pseudomonas* and other bacteria. Without HSC transplantation, death occurs within the first year of life. Despite the common clinical manifestations, the underlying genetic defects are quite varied and in many cases are unknown. Often, the SCID defect resides in the T-cell compartment, with impaired humoral immunity being caused by the lack of T-cell help.

X-linked SCID. The most common form, accounting for 50% to 60% of cases, is X-linked, and hence SCID is more common in boys than in girls. The genetic defect in the X-linked form is a mutation in the common γ -chain (γ_c) subunit of cytokine receptors. This transmembrane protein is a signal-transducing component of the receptors for IL-2, IL-4, IL-7, IL-9, IL-11, IL-15, and IL-21. IL-7 is required for the survival and proliferation of lymphoid progenitors, particularly T-cell precursors. As a result of defective IL-7 receptor signaling, there is a profound defect in the earliest stages of lymphocyte development, especially T-cell development. T-cell numbers are greatly reduced, and although B cells may be normal in number, antibody synthesis is defective. IL-15 is important for the maturation and proliferation of NK

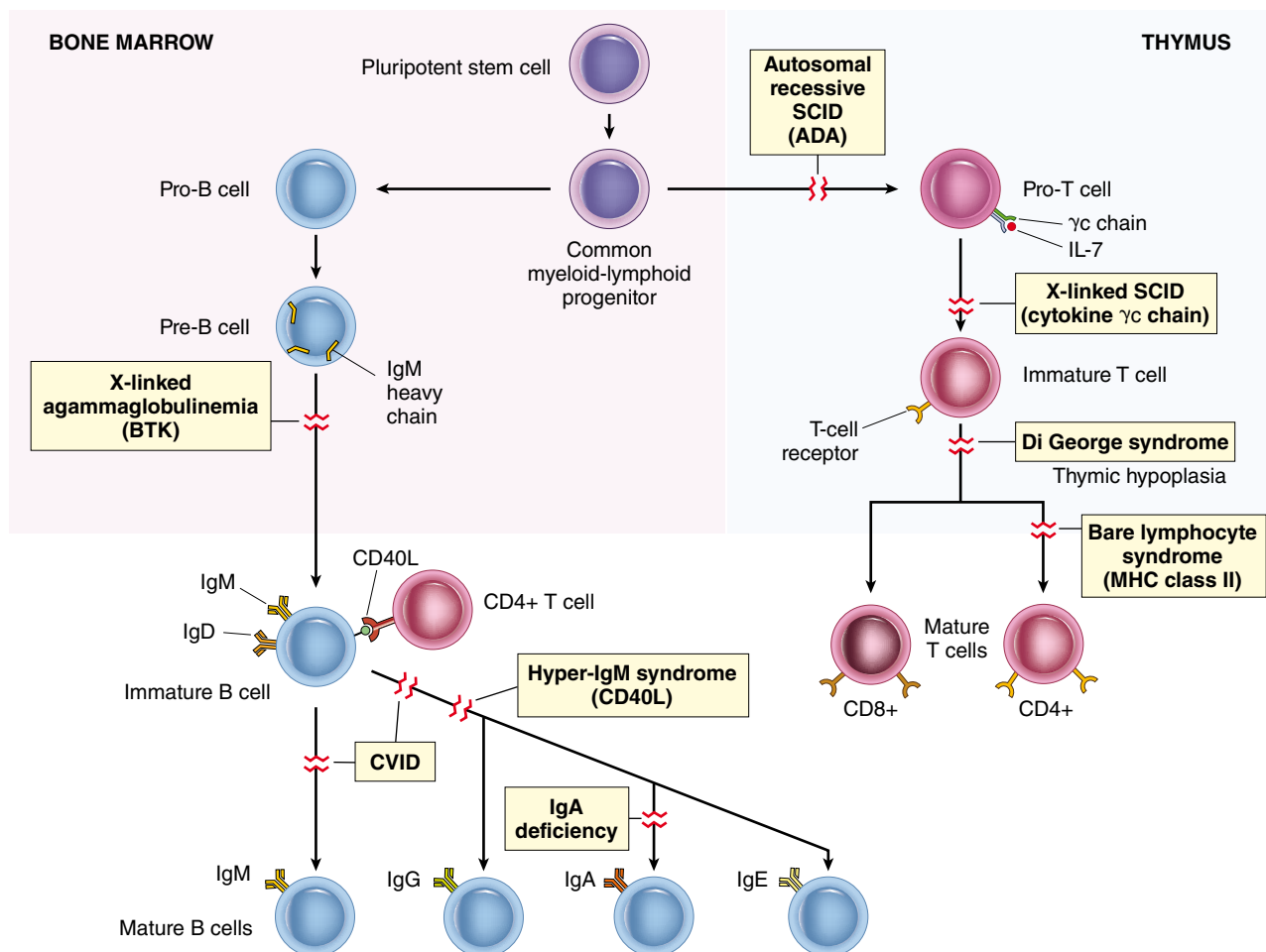


Fig. 6.37 Primary immune deficiency diseases. Shown are the principal pathways of lymphocyte development and the blocks in these pathways in selected primary immune deficiency diseases. The affected genes are indicated in parentheses for some of the disorders. ADA, Adenosine deaminase; CD40L, CD40 ligand (also known as CD154); CVID, common variable immunodeficiency; MHC, major histocompatibility complex; SCID, severe combined immunodeficiency.

cells, and because the common γ chain is a component of the receptor for IL-15, these individuals often have a deficiency of NK cells as well.

Autosomal Recessive SCID. The remaining forms of SCID are autosomal recessive disorders. The most common cause of autosomal recessive SCID is a deficiency of the enzyme adenosine deaminase (ADA). Although the mechanisms by which ADA deficiency causes SCID are not entirely clear, it has been proposed that deficiency of the enzyme leads to accumulation of deoxyadenosine and its derivatives (e.g., deoxy-ATP), which are toxic to rapidly dividing immature lymphocytes, especially those of the T-cell lineage. Hence there may be a greater reduction in the number of T lymphocytes than of B lymphocytes.

Numerous other less common causes of autosomal recessive SCID have been identified:

- Mutations in recombinase-activating genes (RAG) or other components of the antigen receptor gene recombination machinery prevent the somatic gene rearrangements that are essential for the assembly of TCR and Ig genes. This prevents the development of T and B cells.
- An intracellular tyrosine kinase called JAK3 is essential for signal transduction through cytokine receptors containing the common γ chain (which is mutated in X-linked SCID, as discussed earlier). Mutations of JAK3 therefore have the same effects as mutations in the γ -chain.
- Mutations have also been described in components of the TCR, other signaling molecules, including kinases associated with the TCR, and components of calcium channels that are required for entry of calcium and activation of many signaling pathways.

The histologic findings in SCID depend on the underlying defect. In the two most common forms (γ c mutation and ADA deficiency), the thymus is small and devoid of lymphoid cells. In X-linked SCID, the thymus contains lobules of undifferentiated epithelial cells resembling fetal thymus, whereas in SCID caused by ADA deficiency, remnants of Hassall's corpuscles can be found. In both diseases, other lymphoid tissues are hypoplastic as well, with marked depletion of T-cell areas and in some cases both T-cell and B-cell zones.

Currently, HSC transplantation is the mainstay of treatment, but X-linked SCID is the first human disease in which gene therapy has been successful. In this type of ex vivo gene therapy, a normal γ c gene is introduced using a viral vector in HSCs isolated from the patients, and the cells are then transplanted back into the same patient from which they were taken. The clinical experience is small, but some patients have shown reconstitution of their immune systems for over 10 years after therapy. Unfortunately, about 20% of patients in the initial clinical trials developed T-cell lymphoblastic leukemia, highlighting the dangers of this particular approach to gene therapy. The uncontrolled T-cell proliferation was triggered by the activation of oncogenes by the integrated virus, an effect that may have been augmented by a growth advantage conferred by the introduced γ c gene. Current trials are using new vectors with safety features built in. Patients with ADA deficiency have also been treated with HSC transplantation and, more recently, with administration of the enzyme or gene therapy involving the introduction of a normal ADA gene into T-cell precursors.

X-Linked Agammaglobulinemia (Bruton Agammaglobulinemia).

X-linked agammaglobulinemia is characterized by the failure of B-cell precursors (pre-B cells) to develop into mature B cells. It is one of the more common forms of primary immunodeficiency. During normal B-cell maturation in the bone marrow, the Ig heavy-chain genes are rearranged first, in pre-B cells. These form a complex with a "surrogate" light chain on the cell surface to form the pre-B cell receptor (pre-BCR), which delivers signals that induce rearrangement of the Ig light-chain genes and further maturation. This need for Ig-initiated signals is a

quality-control mechanism that ensures that maturation will proceed only if functional Ig proteins are expressed. X-linked agammaglobulinemia is caused by mutations in a cytoplasmic tyrosine kinase, called Bruton tyrosine kinase (BTK); the gene that encodes it is located on the long arm of the X chromosome at Xq21.22. BTK is an intracellular tyrosine kinase that is associated with the pre-BCR of immature B-cells and with the B-cell receptor (BCR) complexes that are found on mature B cells. When BTK is mutated, the pre-BCR cannot deliver the signals that are needed for proliferation of the cells and light chain rearrangement, and maturation is arrested. Because BTK is also an important transducer of BCR signals that stimulate growth and increase cell survival in benign and malignant mature B cells, inhibitors of BTK are effective therapies for several B cell malignancies (Chapter 13) and autoimmune diseases.

As an X-linked disease, this disorder is seen almost exclusively in males, but sporadic cases have been described in females, possibly caused by mutations in other genes that function in the same pathway. The disease usually does not become apparent until about 6 months of age, as maternal immunoglobulins are depleted. In most cases, recurrent bacterial infections of the respiratory tract, such as acute and chronic pharyngitis, sinusitis, otitis media, bronchitis, and pneumonia, call attention to the underlying immune defect. Almost always the causative organisms are *Haemophilus influenzae*, *Streptococcus pneumoniae*, or *Staphylococcus aureus*. These organisms are normally opsonized by antibodies and cleared by phagocytosis. Because antibodies are important for neutralizing infectious viruses, individuals with this disease are also susceptible to certain viral infections, especially those caused by enteroviruses, such as echovirus, poliovirus, and coxsackievirus. These viruses infect the gastrointestinal tract, and from here they can disseminate to the nervous system via the blood. Immunization with live poliovirus carries the risk of paralytic poliomyelitis, and echovirus can cause fatal encephalitis. For similar reasons, *Giardia lamblia*, an intestinal protozoan that is normally resisted by secreted IgA, causes persistent infections in persons with X-linked agammaglobulinemia. In general, however, most fungal, protozoal, and intracellular viral infections are handled quite well by the intact T cell-mediated immunity.

The classic form of this disease has the following characteristics:

- B cells are absent or markedly decreased in the circulation, and serum levels of all classes of immunoglobulins are depressed. Pre-B cells, which express the B-lineage marker CD19 but not membrane Ig, are found in normal numbers in the bone marrow.
- Germinal centers of lymph nodes, Peyer patches, the appendix, and tonsils are underdeveloped.
- Plasma cells are absent throughout the body.
- T cell-mediated reactions are normal.

Paradoxically, autoimmune diseases, such as arthritis and dermatomyositis, occur in as many as 30% of individuals with this disease. It is likely that these autoimmune disorders are caused by a breakdown of self tolerance resulting in autoimmunity, but chronic infections associated with the immune deficiency may play a role in inducing the inflammatory reactions. The treatment of X-linked agammaglobulinemia is replacement therapy with immunoglobulins. In the past, most patients succumbed to infection in infancy or early childhood. Prophylactic IVIG therapy allows most individuals to reach adulthood.

DiGeorge Syndrome (Thymic Hypoplasia).

DiGeorge syndrome is a T-cell deficiency that results from failure of development of the thymus. The third and fourth pharyngeal pouches, which give rise to the thymus, the parathyroids, some of the C cells of the thyroid, and the ultimobranchial body, do not develop normally. Thus, individuals with this syndrome have a variable loss of T cell-mediated immunity (resulting from hypoplasia or lack of the thymus), tetany (resulting from lack

of the parathyroids), and congenital defects of the heart and great vessels. In addition, the appearance of the mouth, ears, and facies may be abnormal. Absence of cell-mediated immunity is caused by low numbers of T lymphocytes in the blood and lymphoid tissues and poor defense against certain fungal and viral infections. The T-cell zones of lymphoid organs—paracortical areas of the lymph nodes and the periarteriolar sheaths of the spleen—are depleted. Ig levels may be normal or reduced, depending on the severity of the T-cell deficiency. The condition tends to improve with age, probably because the small amount of thymic tissue that does develop is able to support some T cell maturation.

In most (90%) of cases, DiGeorge syndrome is caused by a small germline deletion that maps to chromosome 22q11, and DiGeorge syndrome is now considered a component of the 22q11 deletion syndrome, discussed in Chapter 5. One gene in the deleted region is *TBX1*, which is required for development of the branchial arch (and hence the thymus) and the great vessels. Notably, a few cases of DiGeorge syndrome that lack 22q11 deletions have mutations in *TBX1*, strongly suggesting that its loss contributes to the observed phenotype in the classic DiGeorge syndrome.

Other Defects in Lymphocyte Maturation. Many other rare causes of immunodeficiency resulting from defective lymphocyte maturation have been documented. One of these, *bare lymphocyte syndrome*, is usually caused by mutations in transcription factors that are required for MHC class II gene expression. The lack of MHC class II molecules prevents the development and activation of CD4⁺ T cells, which are involved in cellular immunity and provide help to B cells; hence, MHC class II deficiency results in combined immunodeficiency. Other defects are caused by mutations in antigen receptor chains or signaling molecules involved in T- or B-cell maturation.

Hyper-IgM Syndrome. In this disorder, affected patients have IgM antibodies but are deficient in IgG, IgA, and IgE. Mature B cells are present, but they are incapable of Ig class switching and affinity maturation, because of a defect in CD4⁺ helper T cells or an intrinsic B-cell defect. As discussed earlier in the chapter, many of the functions of CD4⁺ helper T cells require the engagement of CD40 on B cells, macrophages, and DCs by CD40L (also called CD154) expressed on antigen-activated T cells. This interaction triggers Ig class switching and affinity maturation in B cells, and also stimulates the microbicidal functions of macrophages. Approximately 70% of individuals with hyper-IgM syndrome have the X-linked form of the disease, caused by mutations in the gene encoding CD40L located on Xq26 that interfere with CD4⁺ helper T cell function. In the remaining patients, the disease is inherited in an autosomal recessive pattern. Most of these patients have loss-of-function mutations involving either CD40 or activation-induced cytidine deaminase (AID), a DNA-editing enzyme expressed in B cells that is required for Ig class switching and affinity maturation.

The serum of persons with this syndrome contains normal or elevated levels of IgM but no IgA or IgE and extremely low levels of IgG. The numbers of B and T cells in the blood are normal. Clinically, patients present with recurrent pyogenic infections, because the level of opsonizing IgG antibodies is low, and because affinity maturation, a process necessary for production of high-affinity antibodies, is impaired. In addition, those with CD40L or CD40 mutations are susceptible to pneumonia caused by the intracellular organism *Pneumocystis jirovecii* because macrophage activation mediated by the CD40L-CD40 pathway, a key reaction of cell-mediated immunity, is also defective. Occasionally, the IgM antibodies react with blood cells, giving rise to autoimmune hemolytic anemia, thrombocytopenia, and neutropenia. In older patients, there may be a proliferation of IgM-producing plasma cells that infiltrate the mucosa of the gastrointestinal tract.

Common Variable Immunodeficiency. This relatively frequent entity encompasses a heterogeneous group of disorders in which the common feature is hypogammaglobulinemia, generally affecting all the antibody classes but sometimes only IgG. The diagnosis of common variable immunodeficiency is based on exclusion of other well-defined causes of decreased antibody production.

The disease has both familial and sporadic forms. Relatives of patients have a high incidence of selective IgA deficiency (see later), suggesting that in at least some cases, selective IgA deficiency and common variable immunodeficiency represent different expressions of a common genetic defect in antibody synthesis. In contrast to X-linked agammaglobulinemia, most individuals with common variable immunodeficiency have normal or near-normal numbers of B cells in the blood and lymphoid tissues. These B cells, however, are not able to differentiate into plasma cells.

Common variable immunodeficiency is genetically heterogeneous; mutations in over 50 genes have been implicated in its causation. Both intrinsic B-cell defects and abnormalities in helper T cell-mediated activation of B cells may account for the antibody deficiency in this disease. Examples include families in which the underlying abnormality is in a receptor for a cytokine called *BAFF* that promotes the survival and differentiation of B cells, or in a molecule called *ICOS* (inducible costimulator) that is homologous to CD28 and is involved in the development of T_H cells and in T_H cell-mediated activation of B cells. However, the known mutations explain less than 10% of cases.

The clinical manifestations of common variable immunodeficiency are caused by antibody deficiency, and hence they resemble those of X-linked agammaglobulinemia. Patients typically present with recurrent sinopulmonary pyogenic infections. In addition, about 20% of patients have recurrent herpesvirus infections. Serious enterovirus infections causing meningoencephalitis may occur. Individuals with this disorder are also prone to the development of persistent diarrhea caused by *G. lamblia*. Common variable immunodeficiency affects both sexes equally, and the onset of symptoms is later than in X-linked agammaglobulinemia, usually in childhood or adolescence. Histologically, the germinal centers in lymphoid tissues (i.e., nodes, spleen, and gut) are hyperplastic, presumably because the B cells can proliferate in response to antigen but are unable to differentiate into plasma cells.

As in X-linked agammaglobulinemia, these patients have a high frequency of autoimmune diseases (approximately 20%), including rheumatoid arthritis. The risk of lymphoid malignancy is also increased, and an increase in gastric cancer has been reported.

Selective IgA Deficiency. IgA deficiency is a common immunodeficiency caused by impaired differentiation of naive B lymphocytes to IgA-producing plasma cells. In the United States, it occurs in about 1 in 600 individuals of European descent. It is less common in people of Asian descent. Affected individuals have extremely low levels of both serum and secretory (mucosal) IgA. The molecular basis of this defect in most patients is unknown; defects in a receptor for a B cell-activating cytokine, *BAFF*, have been identified in some patients.

Most individuals with IgA deficiency are asymptomatic. Because IgA is the major antibody in mucosal secretions, mucosal defenses are weakened, and infections occur in the respiratory, gastrointestinal, and urogenital tracts. Symptomatic patients commonly present with recurrent sinopulmonary infections and diarrhea. In addition, IgA-deficient patients have a high frequency of respiratory tract allergy and a variety of autoimmune diseases, particularly SLE and rheumatoid arthritis. The basis of the increased frequency of autoimmune and allergic diseases is not known. When transfused with blood containing normal IgA, some patients develop severe, even fatal, anaphylactic reactions, because the administered IgA behaves like a foreign antigen.

X-Linked Lymphoproliferative Disease. X-linked lymphoproliferative disease is characterized by an inability to eliminate EBV, eventually leading to fulminant infectious mononucleosis and the development of B-cell tumors. In more than 60% of cases, the disease is due to mutations in the gene encoding an adapter molecule called SLAM-associated protein (SAP). SAP binds to a family of cell surface molecules involved in the activation of NK cells and T and B lymphocytes, including the signaling lymphocyte activation molecule (SLAM). Defects in SAP attenuate NK and T-cell activation and result in increased susceptibility to viral infections. SAP is also required for the development of Tfh cells, and because of this defect X-linked lymphoproliferative disease patients are unable to form germinal centers or produce high-affinity antibodies, additional abnormalities that likely contribute to susceptibility to viral infection. This immunodeficiency is most often manifested by severe EBV infection, including severe and often fatal infectious mononucleosis (Chapter 8), but not other viral infections, for reasons that are not clear.

Other Defects in Lymphocyte Activation. Many rare cases of lymphocyte activation defects have been described that affect antigen receptor signaling and various biochemical pathways. Inherited defects in Th17 responses lead to chronic mucocutaneous candidiasis and bacterial infections of the skin (a disorder called *Job syndrome*).

Primary Immune Regulatory Disorders. Recently, a new group of genetic disorders, called primary immune regulatory disorders (PIRDs), has been appreciated in which patients show dysregulation of key components of the immune system, resulting in a combination of autoimmunity, infection susceptibility, and lymphoproliferation (as well as malignancy, in some cases). These disorders are caused by a variety of mutations that amplify cytokine signaling (e.g., gain of function mutations in STAT1, STAT3, and JAK1); alter the baseline activation tone in the immune system by either reducing inhibitory signaling (e.g., CTLA-4 haploinsufficiency) or amplifying activating signaling (e.g., amplified P13 kinase delta syndrome [APDS]); or impair tolerance by reducing the function of regulatory T cells (e.g., FOXP3 deficiency and STAT5B deficiency). Typically, patients develop early-onset autoimmunity and a constellation of symptoms mentioned above. Some of the disorders have targeted therapies (e.g., blockade of upstream signaling with small molecule JAK inhibitors in STAT gain of function syndromes), but most do not and HSC transplantation for these disorders has not been generally successful so far.

Immunodeficiencies Associated With Systemic Diseases

In some inherited systemic disorders, immune deficiency is a prominent clinical problem. Two representative examples of such diseases are described next.

Wiskott-Aldrich Syndrome. Wiskott-Aldrich syndrome is an X-linked disease characterized by thrombocytopenia, eczema, and a marked vulnerability to recurrent infection, resulting in early death. The thymus is morphologically normal, at least early in the course of the disease, but there is progressive loss of T lymphocytes in the peripheral blood and in the T-cell zones (paracortical areas) of the lymph nodes, with variable defects in cellular immunity. Patients do not make antibodies to polysaccharide antigens, and the response to protein antigens is poor. IgM levels in the serum are low, but levels of IgG are usually normal. Paradoxically, the levels of IgA and IgE are often elevated. Patients are also prone to developing B-cell lymphomas. Wiskott-Aldrich syndrome is caused by mutations in a gene located at Xp11.23 that encodes *Wiskott-Aldrich syndrome protein* (WASP). WASP belongs to a family of proteins that are believed to link membrane receptors, such as antigen receptors, to cytoskeletal elements. WASP may be involved in

cytoskeleton-dependent responses, including cell migration and signal transduction. The only treatment is HSC transplantation.

Ataxia Telangiectasia. Ataxia telangiectasia is an autosomal recessive disorder characterized by abnormal gait (ataxia), vascular malformations (telangiectases), neurologic deficits, increased incidence of tumors, and immunodeficiency. The immunologic defects are of variable severity and may affect both B and T cells. The most prominent humoral immune abnormalities are defective production of isotype-switched antibodies, mainly IgA and IgG₂. The T-cell defects, which are usually less pronounced, are associated with thymic hypoplasia. Patients experience upper and lower respiratory tract bacterial infections, multiple autoimmune phenomena, and increasingly frequent cancers with advancing age. The gene responsible for this disorder is located on chromosome 11 and encodes a protein kinase called ATM (ataxia telangiectasia mutated). In response to DNA damage (double-strand breaks), ATM activates p53 by phosphorylation, which in turn can activate cell cycle checkpoints and apoptosis in the cells with damaged DNA. ATM has also been shown to contribute to the stability of DNA double-strand break complexes during V(D)J recombination. The abnormalities in DNA repair resulting from ATM deficiency may impair the generation of antigen receptors. In addition, defective DNA repair may lead to abnormalities in the DNA recombination events that are involved in Ig isotype switching. Like some other immunodeficiency syndromes, patients with ataxia telangiectasia have a markedly increased incidence of lymphoma.

KEY CONCEPTS

Primary (Inherited) Immunodeficiencies

- These diseases are caused by inherited mutations in genes involved in lymphocyte maturation or function, or in innate immunity.
- These diseases present clinically with increased susceptibility to infections in early life.
- Deficiencies in innate immunity include defects of phagocyte function, complement, and innate immune receptors.
- Some of the common disorders affecting lymphocytes and the adaptive immune response are:
 - X-SCID: failure of T-cell and B-cell maturation; mutation in the common γ chain of a cytokine receptor, leading to failure of IL-7 signaling and defective lymphopoiesis
 - Autosomal recessive SCID: failure of T-cell development, secondary defect in antibody responses; approximately 50% of cases caused by mutation in the gene encoding ADA, leading to accumulation of toxic metabolites during lymphocyte maturation and proliferation
 - X-linked agammaglobulinemia: failure of B-cell maturation, absence of antibodies; caused by mutations in the *BTK* gene, which encodes B-cell tyrosine kinase, required for maturation signals from the pre-BCRs and BCRs
 - Common variable immunodeficiency: defects in antibody production; genetically heterogeneous
 - Selective IgA deficiency: failure of IgA production; genetic overlap with common variable immunodeficiency
 - X-linked hyper-IgM syndrome: failure to produce isotype-switched high-affinity antibodies (IgG, IgA, IgE); mutation in gene encoding CD40L
 - X-linked lymphoproliferative disease: defect in a signaling molecule causing defective responses against Epstein-Barr virus and lymphoproliferation

Secondary Immunodeficiencies

Secondary (acquired) immune deficiencies may be encountered in individuals with cancer, diabetes and other metabolic diseases, malnutrition, aging, chronic infection, and in persons receiving chemotherapy

TABLE 6.13 Causes of Secondary (Acquired) Immunodeficiencies

Cause	Mechanism
Human immunodeficiency virus infection	Depletion of CD4+ helper T cells
Irradiation and chemotherapy treatments for cancer	Reduced leukocyte production
Involvement of bone marrow by cancers (metastases, leukemias)	Reduced leukocyte production
Protein-calorie malnutrition	Metabolic derangements inhibit lymphocyte maturation and function
Removal of spleen	Decreased phagocytosis of blood-borne microbes

or radiation therapy for cancer, or immunosuppressive drugs to prevent graft rejection or to treat autoimmune diseases (Table 6.13). As a group, the secondary immune deficiencies are more common than the disorders of primary genetic origin. Some of these secondary immunodeficiency states can be caused by defective lymphocyte maturation (when the bone marrow is damaged by radiation or chemotherapy or involved by tumors, such as leukemias), inadequate Ig synthesis (as in malnutrition), or lymphocyte depletion (from drugs or severe infections). The most serious secondary immunodeficiency is AIDS, which is described next.

Acquired Immunodeficiency Syndrome

AIDS is caused by the retrovirus human immunodeficiency virus (HIV) and is characterized by profound immunosuppression that leads to opportunistic infections, tumors, and neurologic manifestations. The magnitude of this modern plague is truly staggering. Although initially recognized in the United States, AIDS is a global problem. It has now been reported from most countries around the world, and the pool of HIV-infected persons in Africa and Asia is large. By 2021, HIV had infected over 80 million people worldwide, and about 40 million adults and children have died of the disease. There are about 38 million people living with HIV (1.1 million in the United States), of whom 70% are in Africa and more than 20% in Asia; the prevalence rate of infection in adults in sub-Saharan Africa is 7% overall and exceeds 25% in some countries. It is estimated that 1.5 million people were newly infected with HIV in 2021, and about 650,000 deaths were caused by AIDS.

There is some good news. Because of public health measures and improved antiviral drugs, the infection rate is decreasing, and some authorities believe it may have peaked in the late 1990s. Furthermore, antiviral therapies have resulted in fewer people dying of the disease. However, these newer treatments are not readily available in many low-resource countries, and toxic side effects remain a problem. The advent of these drugs raises its own tragic concern; because more people are living with HIV, the risk of spreading the infection will increase if vigilance is relaxed.

The enormous medical and social burden of AIDS has led to an explosion of research aimed at understanding HIV and its remarkable ability to cripple host defenses. The literature on HIV and AIDS is vast. Here we summarize the currently available data on the epidemiology, pathogenesis, and clinical features of HIV infection.

Epidemiology

Epidemiologic studies in the United States have identified five groups of adults at high risk for developing AIDS.

- *Males who have sex with males* account for more than 50% of the reported cases. This includes about 5% who are intravenous drug

users as well. Transmission of HIV in this category appears to be on the decline, but in 2016 this remained the largest affected group, constituting about 70% of new cases.

- *Heterosexual transmission*, chiefly due to contact with members of other high-risk groups (e.g., intravenous drug users), is responsible for approximately 20% of cases in the United States. Heterosexual transmission has declined modestly since 2011 in the United States. Globally, however, heterosexual transmission is by far the most common mode by which HIV is spread, and as a result, infection of females is much more frequent outside of the United States. In sub-Saharan Africa, more than half of infected individuals are female. Heterosexual spread of the virus is occurring most rapidly in female sex workers and in females in long-term marital or cohabitating relationships, particularly among adolescents.
- *Intravenous drug users* with no previous history of homosexuality are the next largest group, representing about 10% of infected individuals.
- *HIV infection of the newborn*. Children of HIV-positive females are at risk for infection in utero, at birth, or through breast milk (discussed later). Close to 2% of all AIDS cases occur in children, and it is estimated that worldwide there are 1.7 million HIV-infected individuals younger than 15 years of age.
- *Recipients of blood and blood components*, including patients with hemophilia, who received transfusions of HIV-infected blood or components (e.g., platelets, plasma) account for a small fraction of new infections. (Organs obtained from HIV-infected donors can also transmit the virus.)

It should be apparent from the preceding discussion that transmission of HIV occurs under conditions that facilitate exchange of blood or body fluids containing the virus or virus-infected cells. **The three major routes of transmission are sexual contact, parenteral inoculation, and passage of the virus from infected mothers to their newborns.**

- *Sexual transmission* is the dominant mode of infection worldwide, accounting for more than 75% of all cases of HIV transmission. Because most infected people in the United States are males who have sex with males, most sexual transmission has occurred among homosexual males. The virus is carried in the semen, and it enters the recipient's body through abrasions in rectal or oral mucosa or by direct contact with mucosal lining cells. Viral transmission occurs in two ways: (1) direct inoculation into the blood vessels breached by trauma and (2) infection of DCs or CD4+ cells within the mucosa. In addition to male-to-male and male-to-female transmission, female-to-male transmission also occurs.

Sexual transmission of HIV is enhanced by coexisting sexually transmitted diseases, especially those associated with genital ulceration, such as syphilis, chancroid, and herpes. Other sexually transmitted diseases, including gonorrhea and chlamydia, are also cofactors for HIV transmission, perhaps because the genital inflammation provides more cells that can be infected.

- *Parenteral transmission* of HIV has occurred in three groups of individuals: intravenous drug users, hemophiliacs who received factor VIII and factor IX concentrates, and random recipients of blood transfusion. Of these three, intravenous drug users constitute by far the largest group. Transmission occurs by sharing of needles, syringes, and other paraphernalia contaminated with HIV-containing blood.

Transmission of HIV by transfusion of blood or blood products, such as lyophilized factor VIII and factor IX concentrates, has been virtually eliminated. This fortunate outcome is the result of increasing use of recombinant clotting factors and three public health measures: screening of donated blood and plasma for antibody to HIV, stringent purity criteria for factor VIII and factor IX preparations, and screening of donors for anti-HIV antibody when warranted by

their history. However, an extremely small risk of acquiring AIDS through transfusion of seronegative blood persists, because a recently infected individual may be antibody negative. Currently, this risk is estimated to be 1 in more than 2 million units of blood transfused.

- As alluded to earlier, *mother-to-infant transmission* is the major cause of pediatric AIDS. Infected mothers can transmit the infection to their offspring by three routes: (1) in utero by transplacental spread, (2) during delivery through an infected birth canal, and (3) after birth by ingestion of breast milk. Of these, transmission during birth (intrapartum) and in the immediate period thereafter (peripartum) is considered to be the most common mode in the United States. The reported transmission rates vary from 7% to 49% in different parts of the world. Higher risk of transmission is associated with high maternal viral load and low CD4⁺ T-cell counts as well as chorioamnionitis. Fortunately, antiretroviral therapy (ART) given to infected pregnant females in the United States has virtually eliminated mother-to-child transmission, but it remains a major source of infection in areas where these treatments are not readily available.

There has been concern about the spread of HIV infection outside these high-risk groups. Extensive studies indicate that HIV infection cannot be transmitted by casual personal contact in the household, workplace, or school. Spread by insect bites is virtually impossible. There is an extremely small risk of transmission of HIV infection to health care workers. Seroconversion has been documented after accidental needle-stick injury or exposure of nonintact skin to infected blood in laboratory accidents. After needle-stick accidents, the risk of seroconversion is believed to be about 0.3%, and ART given within 24 to 48 hours of a needle stick can reduce the risk of infection eight-fold. By comparison, approximately 30% of those accidentally exposed to hepatitis B–infected blood become seropositive.

Etiology: Properties of HIV

HIV is a nontransforming human retrovirus belonging to the lentivirus family. Included in this group are feline immunodeficiency virus, simian immunodeficiency virus, visna virus of sheep, bovine immunodeficiency virus, and the equine infectious anemia virus.

Two genetically different but related forms of HIV, called *HIV-1* and *HIV-2*, have been isolated from patients with AIDS. HIV-1 is the most common type associated with AIDS in the United States, Europe, and Central Africa, whereas HIV-2 causes a similar disease principally in West Africa and India. Specific tests for HIV-2 are available, and blood collected for transfusion is routinely screened for both HIV-1 and HIV-2 seropositivity. The ensuing discussion relates primarily to HIV-1 and diseases caused by it, but the information is generally applicable to HIV-2 as well.

Structure of HIV. Like most retroviruses, the HIV-1 virion is spherical and contains an electron-dense, cone-shaped core surrounded by a lipid envelope derived from the host cell membrane (Fig. 6.38). The virus core contains (1) the major capsid protein p24; (2) nucleocapsid protein p7/p9; (3) two copies of viral RNA; and (4) three viral enzymes (protease, reverse transcriptase, and integrase). p24 is the most abundant viral antigen and is detected by an ELISA that is widely used to diagnose HIV infection. The viral core is surrounded by a matrix protein called p17, which lies underneath the virion envelope. Studding the viral envelope are two viral glycoproteins, gp120 and gp41, which are critical for HIV infection of cells.

The HIV-1 RNA genome contains the *LTR*, *gag*, *pol*, and *env* genes, which are typical of retroviruses (Fig. 6.39). The long terminal repeat (*LTR*) contains sequences that regulate expression of HIV genes. The products of the *gag* and *pol* genes are large precursor proteins that are cleaved by the viral protease to yield the mature proteins. In addition to

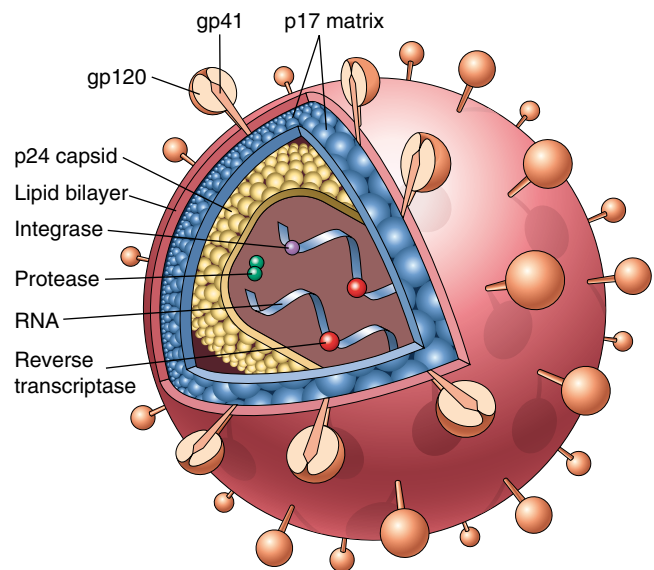


Fig. 6.38 The structure of the HIV-1 virion. The viral particle is covered by a lipid bilayer derived from the host cell and studded with viral glycoproteins gp41 and gp120.

these standard retroviral genes, HIV contains several other accessory genes, including *tat*, *rev*, *vif*, *nef*, *vpr*, and *vpu*, which regulate the synthesis and assembly of infectious viral particles and the pathogenicity of the virus. For example, the product of the *tat* (transactivator) gene causes a 1000-fold increase in the transcription of viral genes and is critical for virus replication. The functions of other accessory proteins are indicated in Fig. 6.39.

Pathogenesis of HIV Infection and AIDS

The major target of HIV infection is the immune system. The CNS may also be affected.

Profound immune deficiency, primarily affecting cell-mediated immunity, is the hallmark of AIDS. This results chiefly from infection and subsequent death of CD4⁺ T cells as well as impairment in the function of surviving helper T cells, but infection of macrophages and DCs also contributes (discussed later). HIV enters the body through mucosal tissues and blood and first infects T cells as well as DCs and macrophages. The infection becomes established in lymphoid tissues, where the virus may remain latent for long periods. Active viral replication is associated with more infection of cells and progression to AIDS. We first describe the mechanisms involved in viral entry into T cells and macrophages and the replicative cycle of the virus within cells. This is followed by a more detailed review of the interaction between HIV and its cellular targets.

Life Cycle of HIV. The life cycle of HIV consists of infection of cells, integration of the provirus into the host cell genome, activation of viral replication, and production and release of infectious virus (Fig. 6.40). The molecules and mechanisms of each of these steps are understood in considerable detail.

Infection of Cells by HIV. HIV infects cells by using the CD4 molecule as a receptor and various chemokine receptors as coreceptors (see Fig. 6.40). The requirement for CD4 binding explains the selective tropism of the virus for CD4⁺ T cells. Macrophages and DCs may be infected because of low-level expression of CD4 or because they ingest viral particles. Binding to CD4 is not sufficient for infection, however. HIV gp120 must also bind to other cell surface molecules (coreceptors) for entry into the cell. Chemokine receptors, particularly CCR5 and CXCR4, serve this role. HIV isolates can be distinguished by

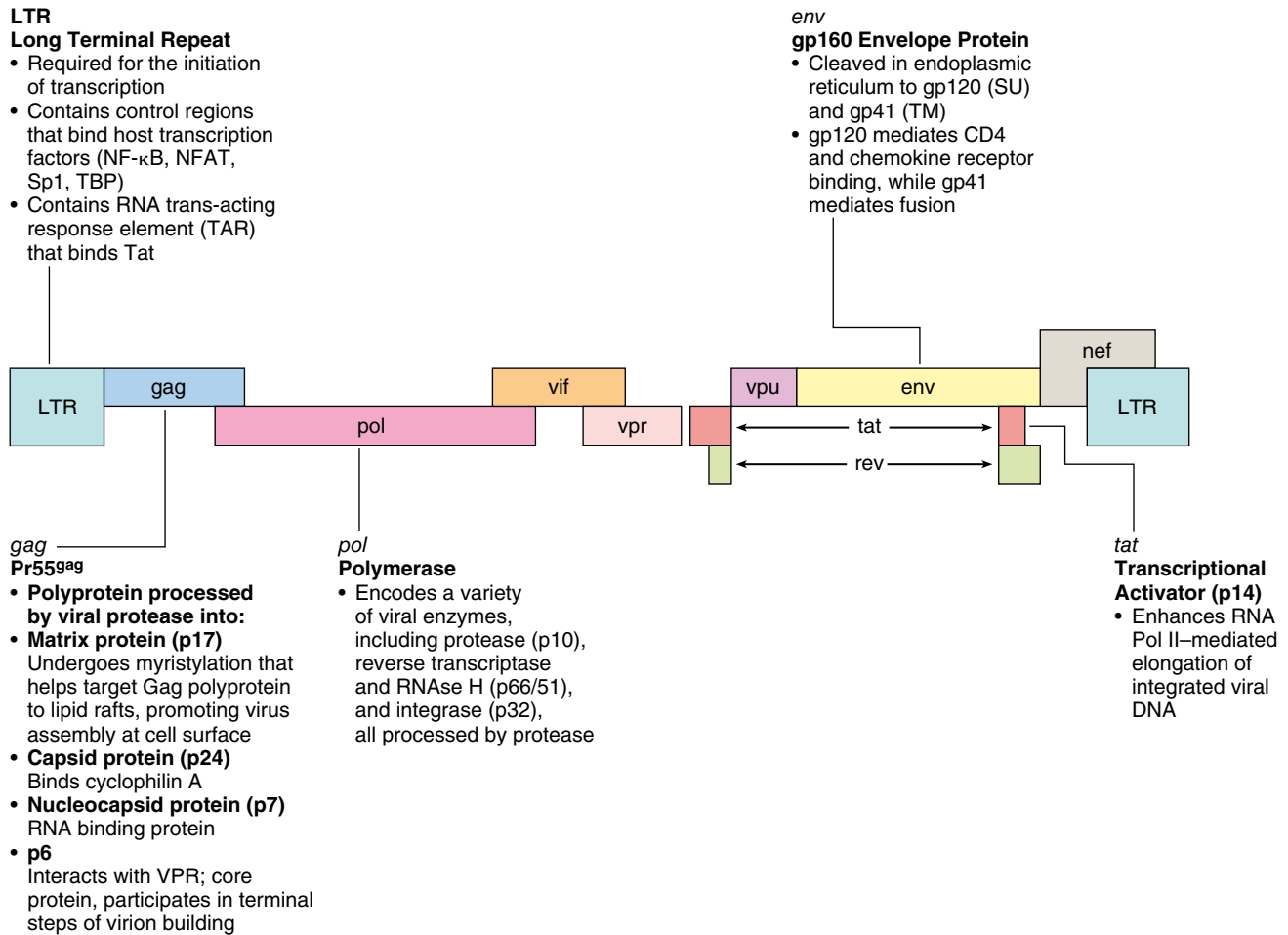


Fig. 6.39 HIV genome. The functions of selected viral genes are shown.

their use of these receptors: R5 strains use CCR5, X4 strains use CXCR4, and some strains (R5X4) use both. R5 strains preferentially infect cells of the monocyte/macrophage lineage and are thus referred to as M-tropic, whereas X4 strains are T-tropic, preferentially infecting T cells. In approximately 90% of cases, the R5 (M-tropic) type of HIV is the dominant virus found in the blood of acutely infected individuals and early in the course of infection. Over the course of infection, however, T-tropic viruses gradually accumulate; these are especially virulent because T-tropic viruses can infect many T cells and even thymic T-cell precursors and cause greater T-cell depletion and impairment.

Molecular details of the deadly handshake between HIV glycoproteins and their cell surface receptors have been elucidated and are important to understand because they may provide the basis for additional anti-HIV drugs. The HIV envelope contains two noncovalently associated glycoproteins, surface gp120 and the transmembrane protein gp41. The initial step in infection is the binding of the gp120 envelope glycoprotein to CD4, which leads to a conformational change that produces a new recognition site on gp120 for the coreceptors CCR5 or CXCR4. Binding to the coreceptors induces conformational changes in gp41 that result in the exposure of a hydrophobic region called the fusion peptide at the tip of gp41. This peptide inserts into the cell membrane of the target cells (e.g., T cells or macrophages), leading to fusion of the virus and the host cell membrane, an event that allows the virus core containing the HIV genome to enter the cell. The requirement for HIV binding to coreceptors may have important implications for the pathogenesis of AIDS. Chemokines sterically hinder HIV infection of cells in culture by occupying their

receptors, and therefore, the level of chemokines in the tissues may influence the efficiency of viral infection in vivo. Also, polymorphisms in the gene encoding CCR5 are associated with differences in susceptibility to HIV infection. About 1% of Americans of European descent inherit two mutant copies of the CCR5 gene and are resistant to infection and the development of AIDS associated with R5 HIV isolates. About 20% of individuals are heterozygous for this protective CCR5 allele; these persons are not protected from AIDS, but the onset of their disease after infection is somewhat delayed. Only rare homozygotes for the mutation have been found in people of African or East Asian ancestry.

Viral Replication. Once internalized, the RNA genome of the virus undergoes reverse transcription, leading to the synthesis of double-stranded complementary DNA (cDNA; proviral DNA) (see Fig. 6.40). In quiescent T cells, the proviral DNA may remain in the cytoplasm in a linear episomal form. In dividing T cells, the DNA circularizes, enters the nucleus, and integrates into the host genome. After integration, the provirus may be silent for months or years, a form of latent infection. Alternatively, proviral DNA may be transcribed, with the formation of complete viral particles that bud from the cell membrane. Such productive infection, when associated with extensive viral budding, leads to death of infected cells.

HIV infects memory and activated T cells but is inefficient at productively infecting naive (resting) T cells. Naive T cells contain an active form of an enzyme that introduces mutations in the HIV genome. This enzyme has the cumbersome name APOBEC3G (for apolipoprotein B mRNA-editing, enzyme-catalytic, polypeptide-like

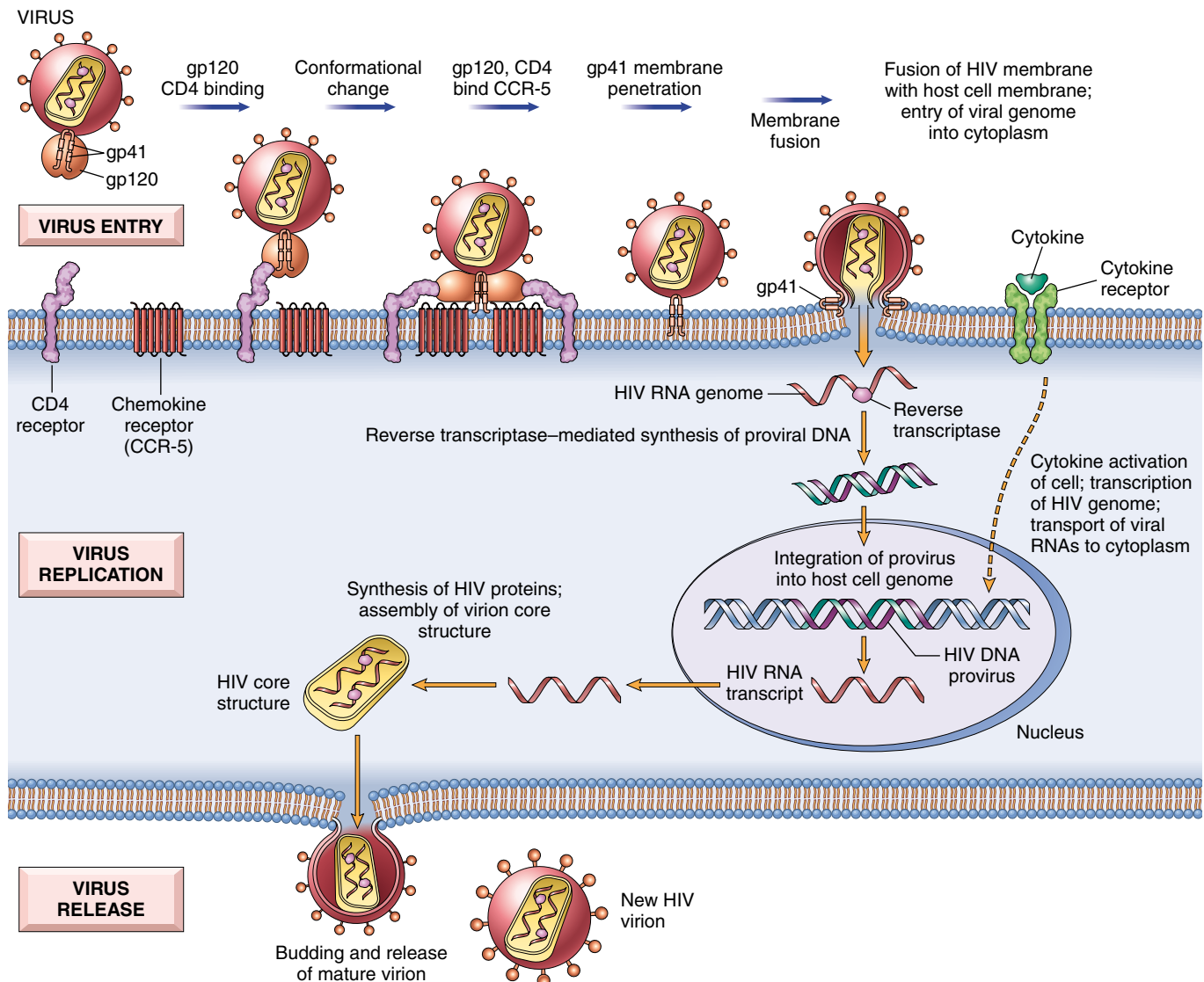


Fig. 6.40 Life cycle of HIV showing the steps from viral entry to production of infectious virions. (Modified from Wain-Hobson S. HIV. One on one meets two. *Nature*. 1996;384:117-118.)

3G). It is a cytidine deaminase that introduces cytosine-to-uracil mutations in the viral DNA. These mutations inhibit further DNA replication by mechanisms that are not fully defined. Activation of T cells converts APOBEC3G into an inactive, high-molecular-mass complex, explaining why the virus can replicate in previously activated (e.g., memory) T cells. HIV has also evolved to counteract this cellular defense mechanism; the viral protein Vif binds to APOBEC3G and promotes its degradation by cellular proteases.

Completion of the viral life cycle in latently infected cells occurs only after cell activation, and in the case of most CD4⁺ T cells, viral replication and shedding result in cell lysis. Activation of T cells by antigens or cytokines upregulates several transcription factors, including NF- κ B, that stimulate transcription of genes encoding cytokines such as IL-2 and its receptor. In resting T cells, NF- κ B is held inactive in the cytoplasm in a complex with the I κ B (inhibitor of κ B) protein. Stimulation of cells by antigen or cytokines activates cytoplasmic kinases that phosphorylate I κ B and target it for enzymatic degradation, thus releasing NF- κ B and allowing it to translocate to the nucleus. In the nucleus, NF- κ B binds to the regulatory sequences of several genes, including those of cytokines that are expressed in activated T cells. The long-terminal-repeat sequences that

flank the HIV genome also contain NF- κ B-binding sites that can drive the expression of viral RNA. Imagine now a latently infected CD4⁺ cell that encounters an environmental antigen. Induction of NF- κ B in such a cell (a physiologic response) activates the expression of HIV genes (a pathologic outcome) and leads ultimately to the production of virions and to cell lysis. Furthermore, TNF and other cytokines produced by activated macrophages also stimulate NF- κ B activity in T cells. Thus, it seems that HIV thrives when the host T cells and macrophages are physiologically activated, an act that can be described as “subversion from within.” Such activation in vivo may result from antigenic stimulation by HIV itself or by other infecting microorganisms. HIV-infected people are at increased risk for other infections, which lead to increased lymphocyte activation and production of proinflammatory cytokines. These, in turn, stimulate more HIV production, loss of additional CD4⁺ T cells, and more infection. Thus, it is easy to visualize how in individuals with AIDS a vicious cycle is set up that culminates in inexorable destruction of the immune system.

Mechanism of T-Cell Depletion in HIV Infection. Loss of infected CD4⁺ T cells is mainly because of the direct cytopathic effects of the replicating virus. In infected individuals, approximately 100 billion

new viral particles are produced every day, and 1 to 2 billion CD4⁺ T cells die each day. Because the frequency of infected cells in the circulation is very low, for many years it was suspected that the immunodeficiency is out of proportion to the level of infection and cannot be attributed to death of infected cells. In fact, many infected cells are within tissues (e.g., secondary lymphoid organs and mucosal sites), and death of these cells is a major cause of the relentless, and eventually profound, cell loss. Up to a point, the immune system can replace the dying T cells, and hence the rate of T cell loss may appear deceptively low, but as the disease progresses, renewal of CD4⁺ T cells cannot keep up with their loss. Memory T cells in tissues are highly susceptible to HIV infection and death, so loss of recall responses to previously encountered microbes is profound.

In addition to direct killing of cells by the virus, other mechanisms may contribute to the loss of T cells. These include:

- Chronic activation of uninfected cells, responding to HIV itself or to infections that are common in individuals with AIDS, leads to apoptosis of these cells by the process of activation-induced cell death. Thus, the numbers of CD4⁺ T cells that die may be considerably more than the numbers of infected cells. The molecular mechanism of this type of cell death is not known.
- Noncytopathic (abortive) HIV infection activates the inflammatory pathway and leads to a form of cell death called pyroptosis (Chapter 2). During this process, inflammatory cytokines and cellular contents are released, leading to recruitment of new cells, and increasing the numbers of cells that can be infected. This form of cell death may play a role in spread of the infection.
- HIV infects cells in secondary lymphoid organs (spleen, lymph nodes, tonsils) and may cause progressive destruction of the architecture and cellular composition of lymphoid tissues.
- Precursors of CD4⁺ T cells may be decreased, either by direct infection of thymic progenitor cells or by infection of other cells that secrete cytokines essential for CD4⁺ T-cell maturation.
- Fusion of infected and uninfected cells with formation of syncytia (giant cells) can lead to death of uninfected cells. In tissue culture, the gp120 expressed on infected cells binds to CD4 molecules on uninfected T cells, followed by cell fusion. Fused cells usually die within a few hours. This property of syncytium formation is generally confined to the T-tropic X4 type of HIV-1.
- Although marked reduction in CD4⁺ T cells, a hallmark of AIDS, can account for most of the immunodeficiency, qualitative defects in T cells have been reported even in asymptomatic HIV-infected persons. These include reduced antigen-induced T-cell proliferation, decreased Th1-type responses, defects in intracellular signaling, and many more. The loss of Th1 responses results in defective cell-mediated immunity, leading to increased susceptibility to infections by viruses and other intracellular microbes.

Low-level chronic or latent infection of T cells is an important feature of HIV infection. It is widely believed that integrated provirus, without viral gene expression (latent infection), can remain in the cells for months to years. Even with potent antiviral therapy, which practically sterilizes the peripheral blood, latent virus lurks within as many as 0.05% of CD4⁺ cells in the lymph nodes. Because most of these T cells are long-lived memory cells, with a life span of months to years, they provide a persistent reservoir of virus. Follicular helper T cells in germinal centers may also be infected, and because CD8⁺ CTLs tend to be excluded from these germinal centers, the infected follicular cells are not eliminated by the host immune response and may be another reservoir for the virus.

CD4⁺ T cells play a pivotal role in regulating both cellular and humoral immune responses. Therefore, loss of this “master regulator”

TABLE 6.14 Major Abnormalities of Immune Function in AIDS

Lymphopenia
Predominantly caused by death of the CD4 ⁺ helper T-cell subset
Decreased T-Cell Function in Vivo
Preferential loss of activated and memory T cells
Decreased delayed-type hypersensitivity
Susceptibility to opportunistic infections
Susceptibility to neoplasms
Altered T-Cell Function in Vitro
Decreased proliferative response to mitogens, alloantigens, and soluble antigens
Decreased cytotoxicity
Decreased helper function for B-cell antibody production
Decreased IL-2 and IFN- γ production
Polyclonal B-Cell Activation
Hypergammaglobulinemia and circulating immune complexes
Inability to mount de novo antibody response to new antigens
Poor responses to normal B-cell activation signals in vitro
Altered Monocyte or Macrophage Functions
Decreased chemotaxis and phagocytosis
Decreased class II HLA expression
Diminished capacity to present antigen to T cells

HLA, Human leukocyte antigen; IFN- γ , interferon- γ ; IL-2, interleukin-2; TNF, tumor necrosis factor.

has ripple effects on virtually every other component of the immune system, as summarized in Table 6.14.

HIV Infection of Non-T Cells. In addition to infection and loss of CD4⁺ T cells, **infection of macrophages is also important in the pathogenesis of HIV infection.** Macrophages express the coreceptors and low levels of CD4. Like T cells, the number of macrophages infected by HIV in the tissues greatly exceeds the number of infected blood monocytes. In certain tissues, such as the lungs and brain, 10% or more of macrophages are infected. Several aspects of HIV infection of macrophages should be emphasized:

- Although cell division is required for nuclear entry and replication of most retroviruses, HIV-1 can infect and multiply in terminally differentiated nondividing macrophages. This property of HIV-1 is dependent on the viral *vpr* gene. The Vpr protein allows nuclear targeting of the HIV preintegration complex through the nuclear pore.
- Infected macrophages contain large numbers of virus particles, often located in intracellular vesicles, but these cells release relatively small amounts of virus. Macrophages are quite resistant to the cytopathic effects of HIV, in contrast to CD4⁺ T cells. Thus, in late stages of HIV infection, when CD4⁺ T-cell numbers decline greatly, macrophages may be an important reservoir for viral persistence.
- Macrophages may act as portals of infection because in more than 90% of cases, acute HIV infection is by M-tropic strains.
- Even uninfected macrophages are reported to have unexplained functional defects that may have important consequences for host defense. These defects include impaired microbicidal activity, decreased chemotaxis, decreased secretion of IL-1, inappropriate

secretion of TNF, and poor capacity to present antigens to T cells. Also, infected monocytes may carry HIV from the blood to various parts of the body, including the nervous system, where infiltrating infected monocytes may be a source of the virus that infects resident microglial cells (described later).

Studies have documented that, in addition to macrophages, mucosal DCs and FDCs are also important targets for the initiation and maintenance of HIV infection. It is thought that mucosal DCs may ingest the virus and transport it to regional lymph nodes, where it is transmitted to CD4⁺ T cells. DCs also express a lectin-like receptor that binds HIV and displays it in an intact, infectious form to T cells, thus promoting infection of the T cells.

FDCs in the germinal centers of lymph nodes are potential reservoirs of HIV. Although some FDCs may be susceptible to HIV infection, most virus particles are found on the surface of their dendritic processes. FDCs have receptors for the Fc portion of immunoglobulins, and hence they trap HIV virions coated with anti-HIV antibodies. The antibody-coated virions localized to FDCs retain the ability to infect CD4⁺ follicular helper T cells as they traverse the intricate meshwork formed by the dendritic processes of the FDCs.

B-Cell Function in HIV Infection. Although HIV does not infect B cells, individuals with AIDS may have profound abnormalities of B-cell function. Paradoxically, early in the disease course there is polyclonal activation of B cells, resulting in germinal center B-cell hyperplasia, sometimes accompanied by autoimmune phenomena like immune thrombocytopenia. B-cell activation may result from multiple mechanisms: reactivation of or reinfection with EBV, which is a polyclonal B-cell activator; viral gp41 may stimulate B-cell growth and differentiation; and increased production of IL-6 from HIV-infected macrophages, which stimulates proliferation of B cells. Plasma cells also increase in number, leading to hypergammaglobulinemia and bone marrow plasmacytosis.

Despite the increased activation of B cells, patients with AIDS are unable to mount effective antibody responses to newly encountered pathogens. This could be due, in part, to lack of T-cell help, but antibody responses against T-independent antigens are also suppressed, and hence there may be intrinsic defects in B cells as well. Impaired humoral immunity renders these patients prey to disseminated infections caused by encapsulated bacteria, such as *S. pneumoniae* and *H. influenzae*, both of which require antibodies for effective opsonization and clearance.

Pathogenesis of Central Nervous System Involvement. The pathogenesis of neurologic manifestations deserves special mention because, in addition to the lymphoid system, the nervous system is a target of HIV infection. The clinical syndrome of CNS abnormalities is called *HIV-associated neurocognitive disorder (HAND)*. In the CNS, the virus is most often found in macrophages that carry it into the brain from peripheral sites and in microglia, the resident macrophages of the CNS. The mechanism of HIV-induced damage of the brain remains obscure. Because neurons and astrocytes are not infected by HIV and the extent of neuropathologic changes is less than might be expected from the severity of neurologic symptoms, it is believed that the neurologic deficit is caused indirectly by viral products and by soluble factors produced by infected microglia. Included among the soluble factors are cytokines, such as IL-1, TNF, and IL-6. In addition, nitric oxide induced in neuronal cells by gp41 has been implicated. Direct damage of neurons by soluble HIV gp120 has also been postulated.

Natural History of HIV Infection

The virus typically enters through mucosal epithelia. The subsequent pathologic and clinical manifestations of the infection can be divided into several phases: (1) an acute retroviral syndrome; (2) a middle, chronic phase, in which most individuals are asymptomatic; and (3) clinical AIDS (Figs. 6.41 and 6.42).

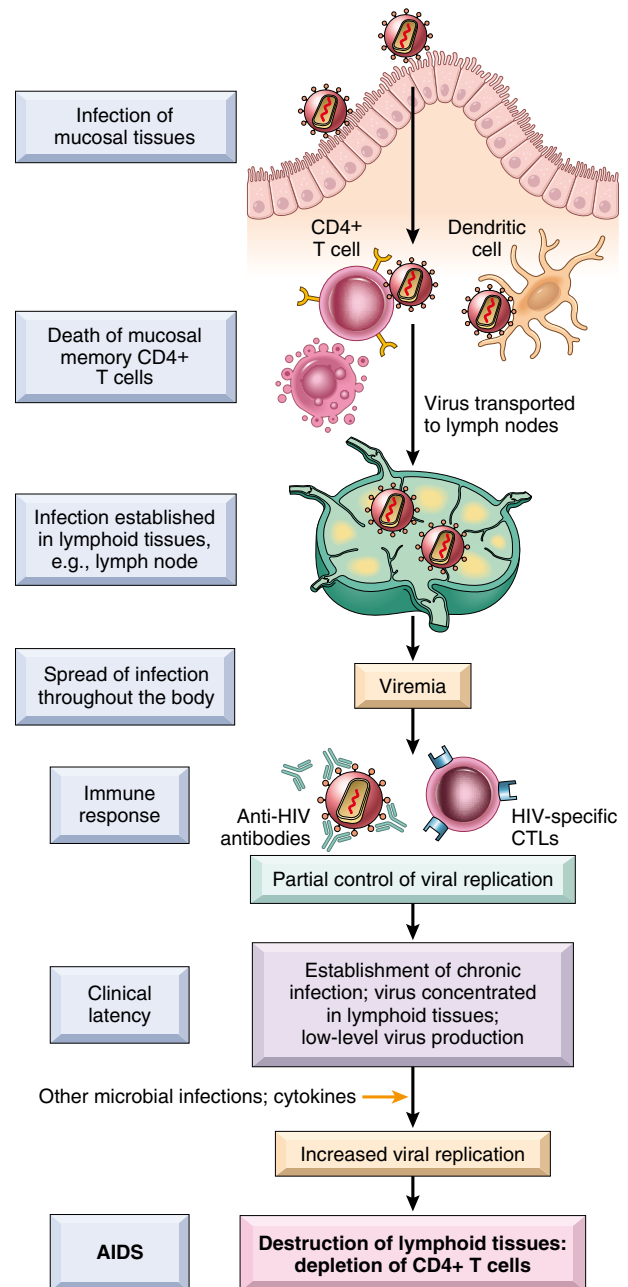


Fig. 6.41 Pathogenesis of HIV-1 infection. The initial infection starts in mucosal tissues, involving mainly memory CD4⁺ T cells and dendritic cells, and spreads to lymph nodes. Viral replication leads to viremia and widespread seeding of lymphoid tissue. This corresponds to the early acute phase of HIV infection. The viremia is controlled by the host immune response, and the patient then enters a phase of clinical latency. During this phase, viral replication in both T cells and macrophages continues unabated, but there is some immune containment of virus (not shown). There continues a gradual erosion of CD4⁺ cells, and ultimately CD4⁺ T-cell numbers decline, and the patient develops clinical symptoms of full-blown AIDS. CTL, Cytotoxic T lymphocyte.

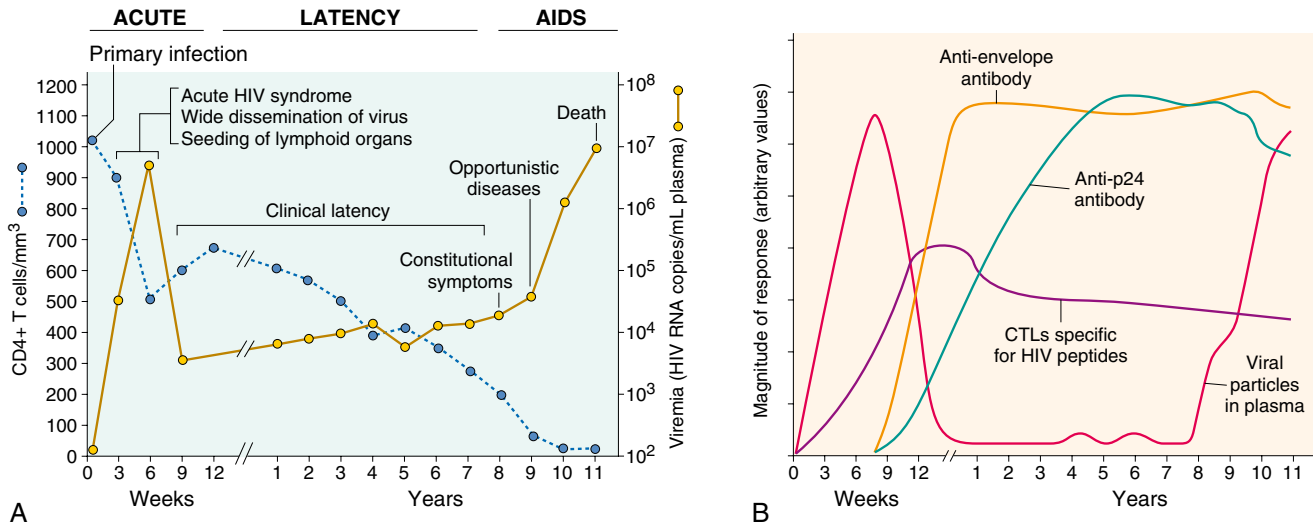


Fig. 6.42 Clinical course of HIV infection. (A) During the early period after primary infection, there is dissemination of virus, development of an immune response to HIV, and often an acute viral syndrome. During the period of clinical latency, viral replication continues, and the CD4+ T-cell count gradually decreases until it reaches a critical level below which there is a substantial risk of AIDS-associated diseases. (B) Immune response to HIV infection. A cytotoxic T lymphocyte (CTL) response to HIV is detectable by 2 to 3 weeks after the initial infection, and it peaks by 9 to 12 weeks. Marked expansion of virus-specific CD8+ T-cell clones occurs during this time, and up to 10% of a patient's CTLs may be HIV specific at 12 weeks. The humoral immune response to HIV peaks at about 12 weeks. (A, Modified from Fauci AS, Lane HC. Human immunodeficiency virus disease: AIDS and related conditions. In: Harrison TR, Fauci AS, eds. *Harrison's Principles of Internal Medicine*. 14th ed. McGraw-Hill; 1997:1791.)

Primary Infection, Virus Dissemination, and the Acute Retroviral Syndrome. Acute (early) infection is characterized by infection of memory CD4+ T cells in mucosal lymphoid tissues and death of many infected cells. Because the mucosal tissues are the largest reservoir of memory T cells in the body, and memory T cells are susceptible to HIV infection, this local loss results in considerable depletion of lymphocytes. Few infected cells are detectable in the blood and other tissues. Predisposing factors for mucosal infection are damage to the epithelium and defects in mucosal barrier functions.

Mucosal infection is followed by dissemination of the virus and the development of host immune responses. DCs in epithelia at sites of virus entry capture the virus and then migrate into the lymph nodes. Once in lymphoid tissues, DCs may pass HIV on to CD4+ T cells through direct cell-to-cell contact. Within days after the first exposure to HIV, viral replication can be detected in the lymph nodes. This replication leads to viremia, during which high numbers of HIV particles are present in the patient's blood. The virus disseminates throughout the body and infects helper T cells and macrophages in peripheral lymphoid tissues.

As the HIV infection spreads, the individual mounts antiviral humoral and cell-mediated immune responses. These responses lead to seroconversion (usually within 3 to 7 weeks of presumed exposure) and the development of virus-specific CD8+ cytotoxic T cells. The immune responses partially control the infection and viral production, as reflected by a drop in viremia to low but detectable levels by about 12 weeks after the primary exposure.

The *acute retroviral syndrome* is the clinical presentation of the initial spread of the virus and the host response. It is estimated that 40% to 90% of individuals who acquire a primary infection develop this syndrome. This typically occurs 3 to 6 weeks after infection and resolves spontaneously in 2 to 4 weeks. This phase is marked by a non-specific self-limited acute illness with flu-like symptoms, including sore

throat, myalgias, fever, weight loss, and fatigue, sometimes accompanied by rash, cervical adenopathy, diarrhea, and vomiting.

The level of HIV-1 RNA in the blood (viral load) is a useful marker of HIV disease progression and is of clinical value in the management of people with HIV infection. The viral load at the end of the acute phase reflects the equilibrium reached between the virus and the host response, and in a given patient it may remain fairly stable for several years. This level of steady-state viremia, called the viral set point, is a predictor of the rate of decline of CD4+ T cells, and, therefore, progression of HIV disease. In one study, only 8% of patients with a viral load of less than 4350 copies of viral mRNA per microliter of blood progressed to clinical AIDS in 5 years, whereas 62% of those with a viral load of greater than 36,270 copies developed AIDS in the same period.

Because the loss of immune containment is associated with declining CD4+ T-cell counts, the Centers for Disease Control and Prevention (CDC) classification stratifies HIV infection into three categories based on CD4+ cell counts: CD4+ cells greater than or equal to 500 cells/ μ L, 200 to 499 cells/ μ L, and fewer than 200 cells/ μ L (Table 6.15). For clinical management, the blood CD4+ T-cell count is perhaps the most reliable short-term indicator of disease progression. For this reason, the CD4+ cell count (rather than viral load) is the primary clinical measurement used to determine when to start ART.

Chronic Infection: Phase of Clinical Latency. In the chronic phase of the disease, lymph nodes and the spleen are sites of continuous HIV replication and cell destruction (Fig. 6.41). During this period of the disease, few or no clinical manifestations of the HIV infection are present. Therefore, this phase of HIV disease is called the *clinical latency* period. Although most peripheral blood T cells do not harbor the virus, destruction of CD4+ T cells within lymphoid tissues continues and the number of CD4+ T cells in the blood steadily declines. More

TABLE 6.15 CDC Classification Categories of HIV Infection

Clinical Categories	CD4+ T-Cell Categories		
	1 ≥500 cells/μL	2 200–499 cells/μL	3 <200 cells/μL
A. Asymptomatic, acute (primary) HIV, or persistent generalized lymphadenopathy	A1	A2	A3
B. Symptomatic, not A or C conditions	B1	B2	B3
C. AIDS indicator conditions: including opportunistic infections, neurologic disease, and tumors			

Data from Centers for Disease Control and Prevention (CDC). 1993 revised classification system and expanded surveillance definition for AIDS among adolescents and adults. *MMWR*. 1992;41(RR-17):1.

than 90% of the body's approximately 10^{11} T cells are normally found in secondary lymphoid organs, and it is estimated that HIV destroys up to 2×10^9 CD4+ T cells every day. Early in the course of the disease, the T cells can be replaced almost as quickly as they are destroyed. At this stage, up to 10% of CD4+ T cells in lymphoid organs may be infected, but the frequency of infected CD4+ T cells in the blood may be less than 0.1% of the circulating CD4+ T cells. Eventually, over a period of years, the slowly amplifying cycle of virus infection, T-cell death, and new infection leads to a steady decline in the number of CD4+ T cells in the lymphoid tissues and the circulation.

Concomitant with this loss of CD4+ T cells, host defenses wane, and the proportion of the surviving CD4+ cells infected with HIV increases, as does the viral burden per CD4+ cell. As expected, HIV RNA levels increase as the host begins to lose the battle with the virus. How HIV escapes immune control is not entirely clear, but several mechanisms have been proposed. These include destruction of the CD4+ T cells that are critical for effective immunity, antigenic variation, and down-modulation of MHC class I molecules on infected cells so that viral antigens are not recognized by CD8+ CTLs. During this period, the virus may evolve and switch from relying primarily on CCR5 to enter its target cells to relying on either CXCR4 or both CCR5 and CXCR4. This coreceptor switch is associated with more rapid decline in CD4+ T-cell counts, presumably because of greater infection of T cells.

In this chronic phase of infection, patients are either asymptomatic or develop minor opportunistic infections, such as oral candidiasis (thrush), vaginal candidiasis, herpes zoster, and perhaps tuberculosis (the latter being particularly common in resource-poor regions such as sub-Saharan Africa). Autoimmune thrombocytopenia may also be noted (Chapter 14).

AIDS. The final phase is progression to AIDS, characterized by a breakdown of host defense, a dramatic increase in viral load, and severe, life-threatening clinical disease. The typical patient presents with long-lasting fever (>1 month), fatigue, weight loss, diarrhea, and generalized lymph node enlargement. After a variable period, serious opportunistic infections, secondary neoplasms, or clinical neurologic disease (grouped under the rubric AIDS indicator diseases, discussed later) emerge, and the patient is said to have developed AIDS.

Without treatment, most patients with HIV infection progress to AIDS after a chronic phase lasting from 7 to 10 years. Exceptions to this typical course are rapid progressors and long-term nonprogressors. In *rapid progressors*, the middle, chronic phase is telescoped to 2 to 3 years after primary infection. Conversely, other infected individuals show slow disease progression; about 1% remain asymptomatic for 10 years or more, with stable CD4+ T-cell counts and low viral loads, even without ART. These individuals are divided into *long-term nonprogressors* (~5000 copies of viral RNA per mL of blood) and *elite controllers* (usually <50 viral RNA copies per mL). Individuals with such uncommon clinical courses have attracted great attention in the hope that studying them may shed light on

host and viral factors that influence disease progression. Studies thus far indicate that this group is heterogeneous with respect to the variables that influence the course of the disease. In most cases, the viral isolates do not show qualitative abnormalities, suggesting that the course of the disease cannot be attributed to a “wimpy” virus. In all cases, there is evidence of a vigorous anti-HIV immune response, but the immune correlates of protection are still unknown. Some of these individuals have high levels of HIV-specific CD4+ and CD8+ T-cell responses, and these levels are maintained over the course of infection. The inheritance of particular HLA alleles seems to correlate with slow disease progression, perhaps reflecting the ability to mount potent antiviral T-cell responses.

Clinical Features of AIDS

The salient clinical features of the acute early and chronic middle phases of HIV infection were described earlier; here we summarize the clinical manifestations of the terminal phase, AIDS. At the outset, it should be noted that the clinical manifestations and opportunistic infections associated with HIV infection vary in different parts of the world. Also, the course of the disease has been greatly modified by antiretroviral drug therapy, and many devastating complications that were once common are now infrequent.

In the United States, the typical adult patient with AIDS presents with fever, weight loss, diarrhea, generalized lymphadenopathy, multiple opportunistic infections, neurologic disease, and, in many cases, secondary neoplasms. The infections and neoplasms listed in Table 6.16 are included in the surveillance definition of AIDS.

Opportunistic Infections. Opportunistic infections account for most deaths in untreated patients with AIDS. These infections are caused by low-virulence environmental microbes or by reactivation of latent infections, which are normally kept in check by a healthy immune system but are not completely eradicated because the infectious agents have evolved to coexist with their hosts. The frequency of infections varies in different regions of the world, and has been markedly reduced by antiretroviral therapy (ART), which relies on a combination of three or four drugs that block different steps of the HIV life cycle. The following description of opportunistic infections reflects the clinical experience before widespread use of antiviral drugs.

- Approximately 15% to 30% of untreated HIV-infected people develop pneumonia caused by the fungus *Pneumocystis jirovecii* (reactivation of a prior latent infection) during the course of the disease. Before the advent of ART, this infection was the presenting feature in about 20% of cases, but the incidence is much less in patients who respond to ART.
- Many patients present with other opportunistic infections. Among the most common pathogens are *Candida*, cytomegalovirus, atypical and typical mycobacteria, *Cryptococcus neoformans*, *Toxoplasma gondii*, *Cryptosporidium*, herpes simplex virus, papovaviruses, and *Histoplasma capsulatum*.

TABLE 6.16 AIDS-Defining Opportunistic Infections and Neoplasms Found in Patients With HIV Infection

Infections
Protozoal and Helminthic Infections
<i>Cryptosporidium</i> or <i>Cystoisospora</i> (enteritis)
<i>Pneumocystis</i> (pneumonia or disseminated infection)
<i>Toxoplasma</i> (pneumonia or CNS infection)
Fungal Infections
<i>Candida</i> (esophageal, tracheal, or pulmonary)
<i>Cryptococcus</i> (CNS infection)
<i>Coccidioides</i> (disseminated)
<i>Histoplasma</i> (disseminated)
Bacterial Infections
<i>Mycobacterium</i> ("atypical," e.g., <i>Mycobacterium avium-intracellulare</i> , disseminated or extrapulmonary; <i>Mycobacterium tuberculosis</i> , pulmonary or extrapulmonary)
<i>Nocardia</i> (pneumonia, meningitis, disseminated)
<i>Salmonella</i> infections, disseminated
Viral Infections
Cytomegalovirus (pulmonary, intestinal, retinitis, or CNS infections)
Herpes simplex virus (localized or disseminated infection)
Varicella-zoster virus (localized or disseminated infection)
Progressive multifocal leukoencephalopathy
Neoplasms
Kaposi sarcoma
Primary lymphoma of brain
Invasive cancer of the uterine cervix

CNS, Central nervous system.

- **Candidiasis** is the most common fungal infection in patients with AIDS, and infection of the oral cavity, vagina, and esophagus are its most common clinical manifestations. In asymptomatic HIV-infected individuals, oral candidiasis is a sign of immunologic decompensation and often heralds the transition to AIDS. Invasive candidiasis is infrequent in patients with AIDS, and it usually occurs when there is drug-induced neutropenia or use of indwelling catheters.
- **Cytomegalovirus (CMV)** may cause disseminated disease or may be localized to the eye and gastrointestinal tract. At one time, CMV chorioretinitis was seen in approximately 25% of patients, but its incidence has decreased dramatically due to ART. Cytomegalovirus retinitis occurs almost exclusively in patients with CD4⁺ T cell counts less than 50 cells/ μ L. Gastrointestinal disease, seen in 5% to 10% of cases, manifests as esophagitis and colitis, the latter associated with multiple mucosal ulcerations.
- Disseminated bacterial infection with *atypical mycobacteria* (mainly *Mycobacterium avium-intracellulare*) also occurs late, in the setting of severe immunosuppression. Coincident with the AIDS epidemic, the incidence of tuberculosis has risen dramatically. Worldwide, almost one-third of all deaths in AIDS patients are attributable to tuberculosis, but this complication remains uncommon in the United States. Patients with AIDS are at risk for reactivation of latent pulmonary disease as well as primary infection. In contrast to infection with atypical mycobacteria, which is a late complication, *M. tuberculosis* infection manifests early in the course of AIDS. As with tuberculosis in other settings, the infection may be confined to lungs or may involve multiple organs. The extent of infection depends on

the degree of immunosuppression; dissemination is more common in patients with very low CD4⁺ T-cell counts. Most worrisome are reports indicating that a growing number of isolates are resistant to multiple antimycobacterial drugs.

- A variety of opportunistic infections target the CNS. *Cryptococcosis* occurs in about 10% of AIDS patients. As in other settings with immunosuppression, meningitis is the major clinical manifestation of cryptococcosis. *Toxoplasma gondii*, another frequent invader of the CNS in AIDS, causes encephalitis and is responsible for 50% of all mass lesions in the CNS. *JC virus*, a human papovavirus, is another important cause of CNS infections in HIV-infected patients. It causes progressive multifocal leukoencephalopathy (Chapter 28).
- *Herpes simplex virus infection* is manifested by mucocutaneous ulcerations involving the mouth, esophagus, external genitalia, and perianal region.
- *Persistent diarrhea*, which is common in untreated patients with advanced AIDS, is often caused by infections with protozoans such as *Cryptosporidium*, *Cystoisospora belli*, or microsporidia. These patients have chronic, profuse, watery diarrhea with massive fluid loss. Diarrhea may also result from infection with enteric bacteria, such as *Salmonella* and *Shigella*, as well as *M. avium-intracellulare*.

Tumors. Patients with AIDS have a high incidence of certain tumors, especially Kaposi sarcoma (KS), B-cell lymphoma, cervical cancer in females, and anal cancer in males. It is estimated that 25% to 40% of untreated HIV-infected individuals will eventually develop a malignancy. A common feature of these tumors is that they are caused by oncogenic DNA viruses, specifically human herpesvirus-8 (KS), EBV (B-cell lymphoma), and human papillomavirus (cervical and anal carcinoma). These viruses may establish latent infections even in healthy people, but they are usually kept in check by a competent immune system. The risk of malignancy in AIDS patients is increased mainly because of failure to contain the oncogenic viruses and decreased immunity against the tumors.

Kaposi Sarcoma. KS, a vascular tumor that is otherwise rare in the United States, is the most common neoplasm in patients with AIDS. The morphology of KS and its occurrence in patients not infected with HIV are discussed in Chapter 11. At the onset of the AIDS epidemic, up to 30% of infected homosexual or bisexual males had KS, but in recent years, with use of ART there has been a dramatic decline in its incidence. In contrast, in areas of sub-Saharan Africa where HIV infection is both frequent and largely untreated, KS is one of the most common tumors.

The lesions of KS are characterized by the proliferation of spindle-shaped cells that express markers of both endothelial cells (vascular or lymphatic) and smooth muscle cells. There is also a profusion of slit-like vascular spaces, suggesting that the lesions may arise from primitive mesenchymal precursors of vascular channels. In addition, KS lesions contain numerous chronic inflammatory cells (lymphocytes and macrophages). Many of the features of KS suggest that it is not a malignant tumor (despite its ominous name). For instance, spindle cells in many KS lesions are polyclonal or oligoclonal, although more advanced lesions occasionally show monoclonality. The current model of KS pathogenesis is that the spindle cells produce proinflammatory and angiogenic factors, which recruit the inflammatory and neovascular components of the lesion, and the latter components supply signals that aid in spindle cell survival and growth.

There is compelling evidence that KS is caused by *human herpesvirus 8* (HHV8), also called KS herpesvirus (KSHV). Exactly how HHV8 infection leads to KS is still unclear. Like other herpesviruses, HHV8 establishes latent infection, during which several proteins are produced with potential roles in stimulating spindle cell proliferation and

preventing apoptosis. These include a viral homologue of cyclin D and several inhibitors of p53. However, HHV8 infection, although necessary for KS development, is not sufficient, and additional cofactors are needed. In the AIDS-related form, that cofactor is clearly HIV (the relevant cofactors for HIV-negative KS remain unknown). HIV-mediated immune suppression may aid in widespread dissemination of HHV8 in the host.

HHV8 infection is not restricted to endothelial cells. The virus is related phylogenetically to the lymphotropic subfamily of herpesviruses (γ -herpesvirus); in keeping with this, its genome is found in B cells of infected subjects. In fact, HHV8 infection is also linked to rare B-cell lymphomas in AIDS patients (called primary effusion lymphoma) and to multicentric Castleman disease, a B-cell lymphoproliferative disorder.

Clinically, AIDS-associated KS is quite different from the sporadic form (Chapter 11). In HIV-infected individuals, the tumor is usually widespread, affecting the skin, mucous membranes, gastrointestinal tract, lymph nodes, and lungs. These tumors also tend to be more aggressive than classic KS.

Lymphomas. Lymphoma occurs at a markedly increased rate in individuals with AIDS, making it one of several AIDS-defining conditions. Roughly 5% of AIDS patients present with lymphoma, and approximately another 5% develop lymphoma during their subsequent course, virtually all of which originate from transformed B cells. The risk of aggressive B-cell lymphoma is about 100-fold higher in HIV-infected persons than in uninfected people, and the risk of Hodgkin lymphoma is about 30-fold higher. With the advent of effective ART, the incidence of lymphoma has fallen substantially in some HIV-infected populations. The pathogenesis of B-cell tumors in HIV-infected individuals appears to involve at least two mechanisms (Fig. 6.43).

- *B-cell proliferation associated with germinal center reactions in the setting of HIV infection.* As mentioned, even in the face of effective ART, the overall rate of lymphoma in the HIV-infected population remains elevated, even in those with normal CD4⁺ T-cell counts. Most of the lymphomas that arise in patients with preserved CD4 T-cell counts are not associated with EBV or HHV8. What then explains the increased risk of lymphoma? The answer is not known, but it may be related to the profound germinal center B-cell hyperplasia that oc-

curs early in HIV infection. Recall that in germinal centers, B cells undergo class switching and somatic hypermutation in their immunoglobulin genes. Both processes are associated with breaks in DNA and are error-prone, sometimes leading to translocation of oncogenes. Of note, B cell tumors that arise outside of the setting of full-blown AIDS in HIV-infected individuals, such as Burkitt lymphoma and diffuse large B-cell lymphoma, are often associated with translocations of oncogenes into Ig gene loci. The striking germinal center B-cell hyperplasia that occurs early in HIV infection may contribute to lymphomagenesis by increasing the number of B cells that are at risk for acquiring lymphoma-initiating events.

- *Infection of B cells with oncogenic viruses such as EBV and HHV8, leading to unchecked proliferation of the cells.* B cells infected with these oncogenic viruses are normally eliminated by an intact immune system, particularly cytotoxic T cells, but with the T-cell deficiency associated with AIDS, this control is lost, and patients are at high risk of developing aggressive B-cell lymphomas.

By adulthood, most normal individuals are infected with EBV. Once immunity is established, EBV persists in such individuals as a latent infection in approximately 1 in 100,000 B cells. Activation of such cells, by antigen or by cytokines, reawakens an EBV-encoded program of gene expression that drives B-cell proliferation. Patients with AIDS have high levels of several cytokines, some of which, including IL-6, are growth factors for B cells. These patients are also chronically infected with pathogens that may activate B-cells. In the absence of T-cell immunity, these activated, EBV-infected clones proliferate and eventually acquire additional somatic mutations, leading to their outgrowth as full-blown EBV-positive B-cell lymphomas. The tumors often occur in extranodal sites, such as the CNS, but also the gut, the orbit, the lungs, and elsewhere. The primary effusion lymphomas mentioned earlier present as malignant effusions and are remarkable in that the tumor cells are typically coinfecting by both EBV and HHV8, an unusual example of cooperativity between two oncogenic viruses.

Several other EBV-related proliferations also merit mention. Hodgkin lymphoma, an unusual B cell tumor associated with a pronounced tissue inflammatory response (Chapter 13), occurs at increased frequency in HIV-infected individuals. In virtually all instances of HIV-associated Hodgkin lymphoma, the characteristic tumor cells (Reed-Sternberg cells) are infected with EBV. Many (but not all) HIV patients with Hodgkin lymphoma have low CD4 counts at the time of disease presentation. EBV infection is also responsible for oral hairy leukoplakia (white projections on the tongue), which results from EBV-driven squamous cell proliferation of the oral mucosa (Chapter 16).

Other Tumors. In addition to KS and lymphoma, patients with AIDS are at increased risk of human papilloma virus (HPV)-associated carcinomas of the uterine cervix and the anus. HPV is intimately associated with squamous cell carcinoma of the cervix and its precursor, squamous intraepithelial lesion (Chapters 7 and 22). HPV-associated squamous intraepithelial lesion is 10 times more common in HIV-infected females than in uninfected females attending family planning clinics and is most prevalent in HIV-infected females with CD4 counts of less than 500 cells/ μ L, suggesting that the risk is attributable to diminished immune surveillance. The rate of progression from squamous intraepithelial lesion to overt cervical carcinoma also appears to be accelerated in HIV-infected females. Hence, gynecologic examination is an important part of a routine workup for HIV-infected females.

Central Nervous System Disease. Involvement of the CNS is a common manifestation of AIDS. Ninety percent of patients demonstrate some form of neurologic involvement at autopsy, and 40% to 60% have

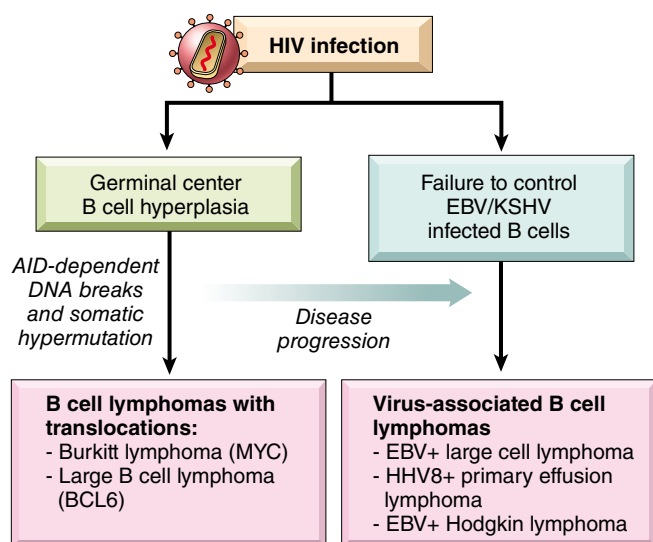


Fig. 6.43 A model for the pathogenesis of B-cell lymphomas in HIV infection. HIV infection results in several changes that may cooperate to produce B-cell lymphomas. AID, Activation-induced cytidine deaminase; EBV, Epstein-Barr virus; KSHV, Kaposi sarcoma-associated herpesvirus; HHV8, human herpesvirus 8.

clinically apparent neurologic dysfunction. Importantly, in some patients, neurologic manifestations may be the sole or earliest presenting feature of HIV infection. In addition to opportunistic infections and neoplasms, several HIV-related neuropathologic changes occur. These include a self-limited meningoencephalitis occurring at the time of seroconversion, aseptic meningitis, vacuolar myelopathy, peripheral neuropathies, and, most commonly, a progressive encephalopathy designated clinically as HIV-associated neurocognitive disorder (Chapter 28). It is believed to result from a combination of HIV infection of microglia and an immune response in the CNS.

Effect of Antiretroviral Drug Therapy. Fortunately for those affected, ART with drugs that target the viral reverse transcriptase, protease, and integrase have altered the clinical course of HIV infection quite dramatically. These drugs are given in combination to reduce the emergence of drug-resistant mutants. Over 25 antiretroviral drugs from six distinct drug classes have been developed for the management of HIV infection. When a combination of at least three effective drugs is used appropriately, HIV viral load is reduced to below the level of detection (<50 copies RNA/mL) and remains there indefinitely (while the patient is on therapy). Even when a drug-resistant virus breaks through, there are several second- and third-line options to combat the virus. Once the virus is suppressed, the progressive loss of CD4⁺ T cells is halted, and over a period of several years the peripheral blood CD4⁺ T-cell count slowly increases and often returns to normal. With the use of ART, in the United States the annual death rate from AIDS has decreased from its peak of 18 per 100,000 people in 1996 to less than 2 per 100,000. Many AIDS-associated disorders, such as *P. jirovecii* infection and KS, are very uncommon now. ART also has reduced the transmission of the virus, especially from infected mothers to newborns. However, because of the reduced mortality, more people are living with HIV, and the risk of HIV spread in the population will inevitably increase if vigilance is relaxed. Indeed, there is compelling evidence that even patients who have virtually undetectable viral loads for years on ART develop active infection if they stop drug treatment.

Despite these dramatic improvements, several new complications associated with HIV infection and its treatment have emerged. Some patients with advanced disease who are given ART have a paradoxical deterioration in their clinical condition as the immune system recovers. This occurs despite increasing CD4⁺ T-cell counts and decreasing viral load. This disorder, called *immune reconstitution inflammatory syndrome*, is not understood but is postulated to be a poorly regulated host response to the high antigenic burden of persistent microbes. Perhaps a more important complication of long-term ART pertains to adverse side effects of the drugs. These include lipodystrophy (loss of facial fat), lipid accumulation (excess fat deposition centrally), elevated lipids, insulin resistance, peripheral neuropathy, and potentially deleterious effects on cardiovascular, renal, and hepatic function. Because CD4⁺ T-cell counts are normalized, non-AIDS morbidity is far more common than classic AIDS-related morbidity in long-term ART-treated patients. Major causes of morbidity are cancer, and accelerated cardiovascular, kidney, and liver disease. The mechanism for these non-AIDS-related complications is not known, but persistent inflammation and immune dysfunction may play a role.

MORPHOLOGY

The anatomic changes in the tissues are neither specific nor diagnostic. Common pathologic features of AIDS include opportunistic infections, KS, and B-cell lymphomas. Most of these lesions are discussed elsewhere because they also occur in individuals who do not have HIV infection. Lesions in the CNS are described in Chapter 28.

Biopsy specimens from enlarged lymph nodes in the early stages of HIV infection reveal a marked hyperplasia of B-cell follicles. The follicles are enlarged and often take on unusual, serpiginous shapes. The mantle zones that surround the follicles are attenuated, and the germinal centers impinge on interfollicular T-cell areas. This hyperplasia of B cells is the morphologic reflection of the polyclonal B-cell activation that is seen in HIV-infected individuals.

With disease progression, the frenzy of B-cell proliferation subsides and gives way to a pattern of lymphoid involution. The lymph nodes are depleted of lymphocytes, and the organized network of FDCs is disrupted. The germinal centers may even become hyalinized. These small, atrophic, “burnt-out” lymph nodes may harbor numerous opportunistic pathogens, often within macrophages. Because of profound immunosuppression, inflammatory responses to infections, both in the lymph nodes and at extranodal sites, may be sparse or atypical. For example, mycobacteria may not evoke granuloma formation because CD4⁺ cells are deficient. In the empty-looking lymph nodes and in other organs, the presence of infectious agents may not be readily apparent without special stains. As might be expected, lymphoid involution is not confined to the nodes; in later stages of AIDS, the spleen and thymus also are converted to “wastelands” that are virtually devoid of lymphocytes.

Despite the advances in our understanding of HIV infection, the long-term prognosis of HIV-positive patients remains guarded. With ART, the mortality rate has declined in the United States, but treated patients still carry viral DNA in their lymphoid tissues, and a cure remains elusive. Although a considerable effort has been mounted to develop a vaccine, many hurdles remain to be crossed before vaccine-based prophylaxis becomes a reality. Molecular analyses have revealed an alarming degree of variation in viral isolates from patients; this renders the task of producing a vaccine extremely difficult. Recent efforts have focused on producing broadly neutralizing antibodies against relatively invariant portions of HIV proteins. The goal of developing an effective vaccine is complicated by the fact that the correlates of immune protection are not fully understood. At present, therefore, prevention, public health measures, and antiretroviral drugs remain the mainstays in the fight against AIDS.

KEY CONCEPTS

Pathogenesis and Course of HIV Infection and AIDS

- Virus entry into cells: requires CD4 and coreceptors, which are receptors for chemokines; involves binding of viral gp120 and fusion with the cell, mediated by viral gp41 protein; main cellular targets are CD4⁺ helper T cells; macrophages and DCs may also be infected
- Viral replication: provirus genome integrates into host cell DNA; viral gene expression is triggered by stimuli that activate infected cells (e.g., infectious microbes, cytokines produced during normal immune responses)
- Progression of infection: acute infection of mucosal T cells and DCs; viremia with dissemination of virus; latent infection of cells in lymphoid tissue; continuing viral replication and progressive loss of CD4⁺ T cells
- Mechanisms of immune deficiency
 - Loss of CD4⁺ T cells: T-cell death during viral replication and budding (similar to other cytopathic infections); apoptosis as a result of chronic stimulation; decreased thymic output; functional defects
 - Defective macrophage and DC functions
 - Destruction of architecture of lymphoid tissues (late)
- *Progression of disease.* HIV infection progresses through phases
 - *Acute HIV infection.* Manifestations of acute viral illness
 - *Chronic (latent) phase.* Dissemination of virus, host immune response, progressive destruction of immune cells
 - *AIDS.* Severe immune deficiency

- **Clinical features.** Full-blown AIDS manifests with several complications, mostly resulting from immune deficiency
 - Opportunistic infections
 - Tumors, especially tumors caused by oncogenic viruses
 - Neurologic complications of unknown pathogenesis
- Antiretroviral therapy has greatly decreased the incidence of opportunistic infections and tumors but also has numerous complications.

AMYLOIDOSIS

Amyloidosis is a generic term for disorders in which proteins with the propensity to aggregate form fibrils and deposit in extracellular tissues, causing tissue damage and functional compromise. These abnormal deposits are produced by the aggregation of proteins that are often misfolded (but are soluble in their normal folded configuration), forming oligomers and then large aggregates that coalesce into insoluble fibrils (Fig. 6.44). The deposits bind a wide variety of proteoglycans and glycosaminoglycans, including heparan sulfate and dermatan sulfate, and plasma proteins, notably serum amyloid P-component. The presence of abundant charged sugar groups in these adsorbed proteins gives the deposits staining characteristics that were thought to resemble starch (amylose). Therefore, the deposits were called *amyloid*, a name that is firmly entrenched despite the realization that the deposits are unrelated to starch.

Amyloid is deposited in the extracellular space in various tissues and organs in several clinical settings. With progressive accumulation, it encroaches on and impairs the function of adjacent cells. Because amyloid deposition appears insidiously and sometimes mysteriously, its clinical recognition depends on morphologic identification of this distinctive substance in appropriate biopsy specimens. With the light microscope and hematoxylin and eosin stains, amyloid appears as an amorphous, eosinophilic, hyaline, extracellular substance. To differentiate amyloid from other hyaline materials (e.g., collagen, fibrin), a variety of histochemical techniques, described later, are employed. The most widely used is the Congo red stain, which under ordinary light imparts a pink or red color to tissue deposits, but far more striking and specific is its characteristic birefringence when observed by polarizing microscopy (see later).

Properties of Amyloid Proteins

Even though all amyloid deposits have a similar microscopic appearance, amyloid is not a single chemical entity. In fact, more than 40 different proteins can aggregate to form amyloid. There are three major and several minor biochemical forms, which are deposited by different pathogenic mechanisms. Therefore, amyloidosis is not one disease, but rather a group of diseases having in common the deposition of similar-appearing proteins. At the heart of the morphologic similarity is the uniform physical organization of amyloid, which we consider first.

Physical Nature of Amyloid

By electron microscopy, all types of amyloid, regardless of clinical setting or chemical composition, consist of continuous, nonbranching fibrils with a diameter of approximately 8 to 10 nm with a characteristic conformation. Each fibril consists of stacks of protofilaments with β -pleated structure that are bound to one another by their side-chains. This conformation is responsible for the distinctive Congo red staining of amyloid.

Chemical Nature of Amyloid

Approximately 95% of the amyloid consists of fibril proteins, with the remaining 5% being serum amyloid P and other glycoproteins. The three most common forms of amyloid are:

- **Amyloid light chain (AL) protein** is made up of complete immunoglobulin light chains, the amino-terminal fragments of light chains, or both. Most of the AL proteins analyzed are composed of λ light chains

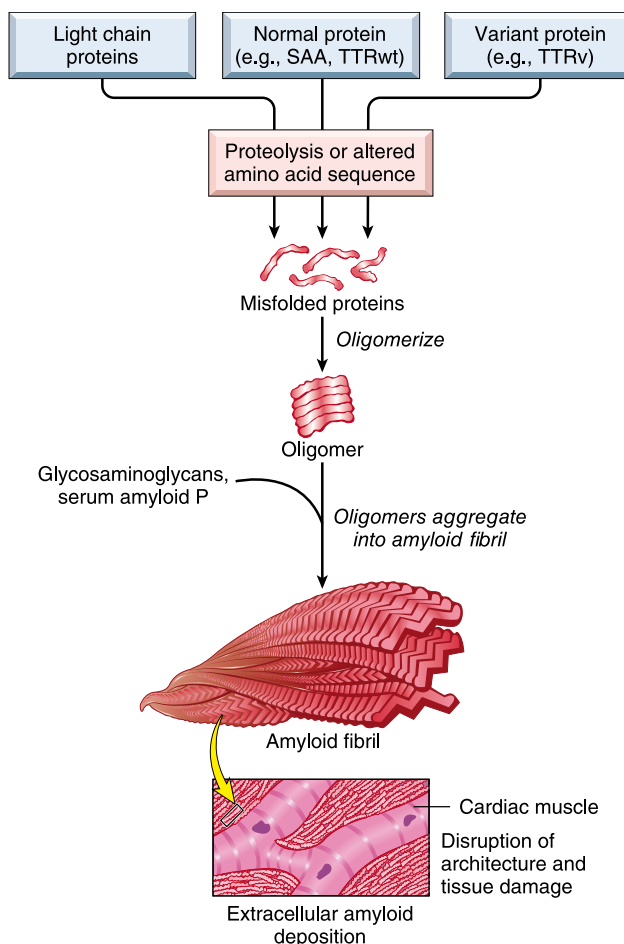


Fig. 6.44 Mechanism of amyloid deposition. Misfolded proteins (monomers) associate to form oligomers and then aggregate and deposit in tissues. SAA, Serum amyloid A protein; TTRv, variant transthyretin; TTRwt, wildtype transthyretin. (With the advice of Dr. Vaishali Sanchurawala, Boston Medical Center.)

or their fragments, but κ chains may also form amyloid. The amyloid fibril protein of the AL type is produced from free Ig light chains secreted by a monoclonal population of plasma cells, and its deposition is associated with certain plasma cell tumors (Chapter 13).

- **Transthyretin amyloid (ATTR)** is composed of transthyretin (TTR), a plasma protein that binds and transports thyroxine and retinol. The most common form is acquired, in which normal (wild-type) transthyretin aggregates to produce amyloid deposits in the heart usually in males older than 70 years (ATTRwt). Variant forms of transthyretin produce amyloid deposits mostly in the heart, peripheral nerves, and other tissues (ATTRv); these disorders are described in more detail later.
- **Amyloid-associated (AA) protein** is a form of amyloid derived from a protein made by the liver. The AA protein found in the fibrils is created by proteolysis of a larger precursor called SAA (serum amyloid-associated) protein that is synthesized in the liver and circulates in the blood bound to high-density lipoproteins. The production of SAA protein is increased in inflammatory states as part of the acute-phase response; therefore, this form of amyloidosis is associated with chronic inflammation.

AL amyloidosis is the most common form of systemic amyloidosis, ATTRwt being next in incidence. ATTR-wt is becoming increasingly prevalent likely because of an aging population. As mentioned, multiple

other biochemically distinct proteins can also deposit as amyloid in a variety of clinical settings. In these other forms of amyloid, the proteins most often involved are:

- β -amyloid ($A\beta$) protein constitutes the core of cerebral plaques found in Alzheimer disease as well as the amyloid deposited in walls of cerebral blood vessels in individuals with this disease. The $A\beta$ protein is derived by proteolysis from a much larger transmembrane glycoprotein called amyloid precursor protein. This form of amyloid is discussed in Chapter 28.
- β_2 -microglobulin, a component of MHC class I molecules and a normal plasma protein, has been identified as the major component of a form of amyloid ($A\beta_2m$) that deposits in or around the joints or soft tissues of patients on long-term hemodialysis.
- In a minority of cases of prion disease in the CNS, the misfolded prion proteins aggregate in the extracellular space and acquire the structural and staining characteristics of amyloid.

Pathogenesis and Classification of Amyloidosis

Amyloidosis results from abnormal folding of proteins, which become insoluble, aggregate, and deposit as fibrils in extracellular tissues. Normally, misfolded proteins are degraded intracellularly in proteasomes, or extracellularly by macrophages. It appears that in amyloidosis, these quality-control mechanisms fail, leading to accumulation of a misfolded protein outside cells. The proteins that form amyloid fall into two general categories (Fig. 6.45): (1) normal proteins that have an inherent tendency to fold improperly, associate and form fibrils, and do so when they are produced in increased amounts or when their degradation is

impaired, and (2) variant proteins that are prone to misfolding and subsequent aggregation. The mechanisms of deposition of different types of amyloid are discussed here along with classification. Deposits may form at the site of production of the amyloidogenic protein (localized amyloidosis) or away from the site (systemic amyloidosis).

Organ dysfunction in amyloidosis is caused mainly by the abnormal tissue architecture resulting from the deposits. In addition, some amyloidogenic proteins may activate signaling pathways that lead to increased production of ROS and defective calcium homeostasis, which may contribute to cellular dysfunction.

Amyloidosis occurs in diverse clinical settings (Table 6.17). In current classifications, the three most common forms are named according to the biochemical nature of the deposited amyloid protein. Several other clinical forms occur less frequently.

AL Amyloidosis

Amyloid in this category (previously called primary amyloidosis) is usually systemic in distribution and is of the AL type. With approximately 2000 to 3000 new cases every year in the United States, this is the most common form of amyloidosis. In all cases, the disorder is caused by a clonal proliferation of plasma cells that synthesize an Ig light chain that is prone to form amyloid due to its intrinsic physicochemical properties. Best defined is the occurrence of systemic amyloidosis in 5% to 15% of individuals with multiple myeloma, a plasma-cell tumor characterized by multiple osteolytic lesions throughout the skeletal system (Chapter 13). The malignant plasma cells synthesize abnormal amounts of a single Ig, producing an M (myeloma) protein “spike” on serum electrophoresis. In

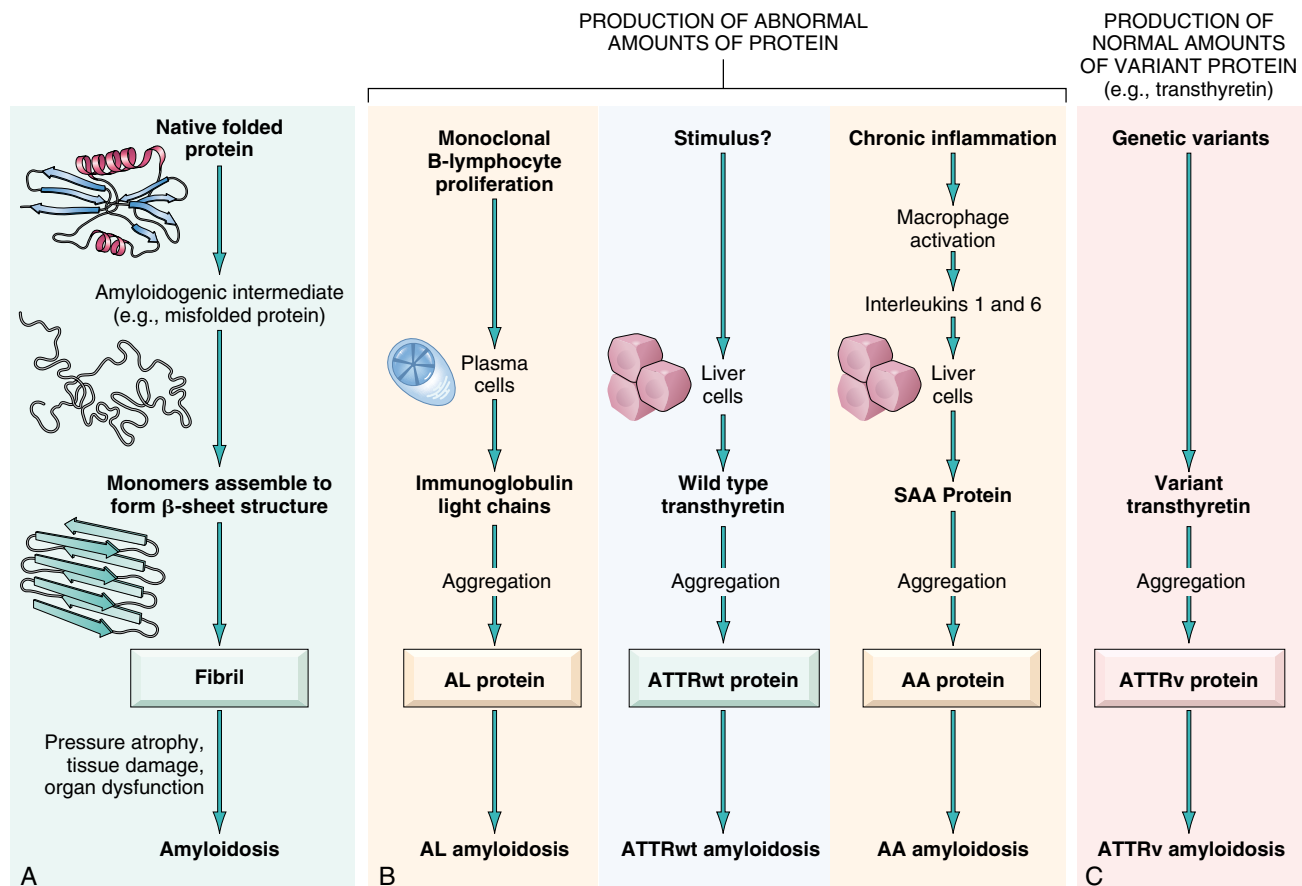


Fig. 6.45 Pathogenesis of amyloidosis. (A) General mechanism of formation of amyloid fibrils. (B) Formation of amyloid from excessive production of proteins prone to misfolding. (C) Formation of amyloid from a variant protein. AA, Amyloid A; AL, amyloid light chain; ATTR, amyloid transthyretin; SAA, serum amyloid A.

TABLE 6.17 Classification of Amyloidosis

Clinicopathologic Category	Associated Diseases	Major Fibril Protein	Precursor Protein
Systemic (Generalized) Amyloidosis			
AL (amyloid light chain) amyloidosis	Multiple myeloma and other monoclonal plasma cell proliferations	AL	Immunoglobulin light chains, chiefly λ type
ATTR (transthyretin) amyloidosis	Aging (males >70 years old)	ATTRwt	Transthyretin
AA (amyloid A) amyloidosis	Chronic inflammatory conditions (e.g., rheumatoid arthritis, Crohn disease)	AA	SAA
Hereditary Amyloidosis			
Familial Mediterranean fever		AA	SAA
Familial amyloidotic neuropathies (several types); cardiac amyloidosis		ATTRv	Transthyretin (variant)
Localized Amyloidosis			
Senile cerebral	Alzheimer disease	A β	APP
Endocrine	Type 2 diabetes Medullary carcinoma of thyroid Islets of Langerhans	ACal A β APP	Calcitonin Islet amyloid peptide
Isolated atrial amyloidosis		AANF	Atrial natriuretic factor

addition to the synthesis of whole Ig molecules, the malignant plasma cells often secrete free, unpaired κ or λ light chains (referred to as Bence-Jones protein). These are often found in the serum, and due to their small molecular size, Bence-Jones proteins are excreted and concentrated in the urine. In primary amyloidosis, the free light chains are not only present in serum and urine but are also deposited in tissues as amyloid. It should be noted, however, that most myeloma patients who have free light chains in serum and urine do not develop amyloidosis. Clearly, not all free light chains are equally likely to produce amyloid, and it is believed that the amyloidogenic potential of any particular light chain is largely determined by its specific amino acid sequence.

Most persons with AL amyloid do not have classic multiple myeloma or any other overt B-cell neoplasm; such cases have been traditionally classified as primary amyloidosis, because their clinical features derive from the effects of amyloid deposition without any other associated disease. In virtually all such cases, however, monoclonal immunoglobulins or free light chains, or both, can be found in the serum or urine. Most of these patients also have a modest increase in the number of plasma cells in the bone marrow, which presumably secrete the precursors of AL protein. Thus, these patients have an underlying monoclonal proliferation of Ig-producing cells (*monoclonal gammopathy*) in which production of an abnormal protein, rather than presence of tumor masses, is the dominant manifestation.

ATTR (Transthyretin) Amyloidosis

This form is characterized by deposition of TTR amyloid fibrils in various tissues. Genetic mutation or aging of the protein can lead to pathological misfolding, eventually forming aggregates that develop into insoluble amyloid fibrils. As mentioned earlier, wildtype TTR aggregates to produce amyloid deposits (ATTRwt) in the heart but can be found in connective tissues (e.g., ligaments), usually in males older than 70 years. ATTRwt is the second most common form of systemic amyloidosis and is increasing in prevalence because of an aging population.

ATTRv is an autosomal dominant condition. Remarkably, specific variant TTR polypeptides tend to form amyloid in different organs. Thus, in some families, deposits are seen primarily in peripheral and autonomic nerves (familial amyloid polyneuropathy), and in other families, cardiac deposits predominate (familial cardiomyopathy, also known as ATTRv-CM). One variant of the TTR gene that leads to

cardiac amyloidosis is carried by approximately 4% of the African-American population in the United States; restrictive cardiomyopathy has been identified in both homozygous and heterozygous patients. The origin of the allele in this population is in West African regions where much of the African-American population of the United States originated. Penetrance of this variant differs substantially among individuals, leading to severe disease in some and no disease in others.

AA Amyloidosis

The amyloid deposits in this pattern are systemic in distribution and are composed of AA protein. This category was previously referred to as secondary amyloidosis because it is secondary to an associated inflammatory condition. At one time, tuberculosis, bronchiectasis, and chronic osteomyelitis were the most important underlying conditions, but with the advent of effective antibiotic therapy, the importance of these conditions has diminished. More commonly now, reactive systemic amyloidosis complicates rheumatoid arthritis, other connective tissue disorders such as ankylosing spondylitis, and inflammatory bowel disease (both Crohn disease and ulcerative colitis). Among these, the most frequently associated condition is rheumatoid arthritis. Amyloidosis is reported to occur in approximately 3% of patients with rheumatoid arthritis and is clinically significant in one-half of those affected. Heroin users who inject the drug subcutaneously also have a high occurrence rate of generalized AA amyloidosis. The chronic skin infections associated with “skin-popping” of narcotics seem to be responsible for the amyloidosis. AA amyloidosis may also occur in association with solid tumors, the most common being renal cell carcinoma and Hodgkin lymphoma. Overall, the prevalence of this form of amyloidosis has been decreasing, presumably because the underlying inflammatory disorders are now treated effectively.

In this form of amyloidosis, SAA synthesis by liver cells is stimulated by cytokines such as IL-6 and IL-1 that are produced during inflammation; thus, long-standing inflammation leads to a sustained elevation of SAA levels. SAA is normally degraded to soluble end products by the action of macrophage-derived enzymes, so increased production of SAA by itself is not sufficient for the deposition of amyloid. It is possible that individuals who develop amyloidosis have an enzyme defect that results in incomplete breakdown of SAA, thus generating insoluble AA molecules. Alternatively, a genetically determined structural abnormality in the SAA molecule may render it resistant to degradation by macrophages.

Other Types of Amyloidosis

Hereditary Amyloidosis. A variety of familial forms of amyloidosis have been described. Most of them are rare and occur in limited geographic areas. The most common and best studied is an autosomal recessive condition called *familial Mediterranean fever* (FMF). This is an autoinflammatory syndrome associated with excessive production of the cytokine IL-1. It is characterized clinically by attacks of fever accompanied by inflammation of serosal surfaces, including peritoneum, pleura, and synovial membrane. The gene for familial Mediterranean fever encodes a protein called pyrin (for its relation to fever), which is one of a complex of proteins that regulate inflammatory reactions via activation of the inflammasome (Chapter 3). This disorder is encountered largely in individuals of Armenian, Sephardic Jewish, and Arabic origins. It is sometimes associated with widespread amyloidosis. The amyloid fibril proteins are made up of AA proteins, reflecting the increased production of SAA in response to excessive IL-1 secretion.

Another familial form of amyloidosis, ATTRv, was discussed earlier.

Hemodialysis-Associated Amyloidosis. Patients on long-term hemodialysis for kidney failure can develop amyloidosis as a result of deposition of β_2 -microglobulin. This protein is present in high concentrations in the serum of persons with kidney disease, and in the past it was retained in the circulation because it could not be filtered through dialysis membranes. Patients usually present with symptoms related to β_2 -microglobulin deposition in joints, muscle, tendons, or ligaments; one relatively common presentation is as carpal tunnel syndrome. With new dialysis filters, the incidence of this complication has decreased substantially.

Localized Amyloidosis. Sometimes, amyloid deposits are limited to a single organ or tissue without involvement of any other site in the body. The deposits may produce grossly detectable nodular masses or be evident only on microscopic examination. Nodular deposits of amyloid are most often encountered in the lung, larynx, skin, urinary bladder, tongue, and the region around the eye. Frequently, there are infiltrates of lymphocytes and plasma cells in the periphery of these amyloid masses. At least in some cases, the amyloid consists of AL protein and may therefore represent a localized form of plasma cell–derived amyloid.

Endocrine Amyloid. Microscopic deposits of localized amyloid may be found in certain endocrine tumors, such as medullary thyroid carcinoma, pancreatic islet tumor, and pheochromocytoma, and in the islets of Langerhans in individuals with type 2 diabetes mellitus. In these settings, the amyloidogenic proteins seem to be derived either from polypeptide hormones (e.g., in medullary carcinoma) or from unique proteins (e.g., islet amyloid polypeptide). In medullary carcinoma of the thyroid, the presence of amyloid is a helpful diagnostic feature.

MORPHOLOGY

There are no consistent or distinctive patterns of organ or tissue distribution of amyloid deposits in any of the categories cited. Nonetheless, a few generalizations can be made. In amyloidosis secondary to chronic inflammatory disorders, the kidneys, liver, spleen, lymph nodes, adrenals, and thyroid, as well as many other tissues, are typically affected. Although amyloidosis associated with plasma cell proliferations cannot reliably be distinguished from the secondary form by its organ distribution, it more often involves the heart, gastrointestinal tract, respiratory tract, peripheral nerves, skin, and tongue. The localization of amyloid deposits in the hereditary syndromes is varied. In familial Mediterranean fever, the amyloidosis may be widespread, involving the kid-

neys, blood vessels, spleen, respiratory tract, and (rarely) liver. The localization of amyloid in the remaining hereditary syndromes can be inferred from the designation of these entities.

Whatever the clinical disorder, the amyloid may or may not be apparent on macroscopic examination. When amyloid accumulates in larger amounts, the organ is frequently enlarged and the tissue appears gray with a waxy, firm consistency. Histologically, the amyloid deposition is always extracellular and begins between cells, often closely adjacent to basement membranes (Fig. 6.46A). As the amyloid accumulates, it encroaches on the cells, in time surrounding and destroying them. In the form associated with plasma cell proliferation, perivascular and vascular deposits are common.

The histologic diagnosis of amyloid is based almost entirely on its staining characteristics. The most common staining technique uses the dye Congo red, which under ordinary light imparts a pink or red color to amyloid deposits. Under polarized light, the Congo red–stained amyloid shows so-called apple-green birefringence (Fig. 6.46B). This reaction is shared by all forms of amyloid and is caused by the cross- β -pleated configuration of amyloid fibrils. Confirmation can be obtained by electron microscopy, which reveals amorphous nonoriented thin fibrils. AA, AL, and amyloid TTR types of amyloid can also be distinguished by specific immunohistochemical staining, and all types can be identified by mass spectroscopy, which is available in some centers.

The pattern of organ involvement in different clinical forms of amyloidosis is variable.

Kidney. Amyloidosis of the kidney is the most common and potentially the most serious form of organ involvement. Grossly, the kidneys may be of normal size and color, or, in advanced cases, they may be shrunken because of ischemia caused by vascular narrowing induced by the deposition of amyloid within arterial and arteriolar walls.

Histologically, the amyloid is deposited primarily in the glomeruli, but the interstitial peritubular tissue, arteries, and arterioles are also affected. The glomerular deposits first appear as subtle thickenings of the mesangial matrix, accompanied usually by uneven widening of the basement membranes of the glomerular capillaries. In time, the mesangial depositions and the deposits along the basement membranes cause capillary narrowing and distortion of the glomerular vascular tuft. With progression of the glomerular amyloidosis, the capillary lumens are obliterated, and the obsolescent glomerulus is flooded by confluent masses or interlacing broad ribbons of amyloid (Fig. 6.46C).

Spleen. Amyloidosis of the spleen may be inapparent grossly or may cause moderate to marked splenomegaly (up to 800 g). For completely mysterious reasons, one of two patterns of deposition is seen. In one, the deposits are largely limited to the splenic follicles, producing tapioca-like granules on gross inspection, designated *sago spleen*. In the other pattern, the amyloid involves the walls of the splenic sinuses and connective tissue framework in the red pulp. Fusion of the early deposits gives rise to large, maplike areas of amyloidosis, creating what has been designated *lardaceous spleen*.

Liver. The deposits may be inapparent grossly or may cause moderate to marked hepatomegaly. Amyloid appears first in the space of Disse and then progressively encroaches on adjacent hepatic parenchymal cells and sinusoids (see Fig. 6.46A). In time, deformity, pressure atrophy, and disappearance of hepatocytes occur, causing total replacement of large areas of liver parenchyma. Vascular involvement is frequent. Even with extensive involvement, liver function is usually preserved.

Heart. Amyloidosis of the heart (Chapter 12) may occur in any form of systemic amyloidosis. It is also the major organ involved in senile systemic amyloidosis. The heart may be enlarged and firm, but more often it shows no significant changes on gross inspection. The deposits begin as focal subendocardial accumulations and within the myocardium between the muscle fibers. Expansion of these myocardial deposits eventually causes pressure atrophy of myocardial fibers. When the amyloid deposits are subendocardial, the conduction system may be damaged, accounting for the electrocardiographic abnormalities noted in some patients.

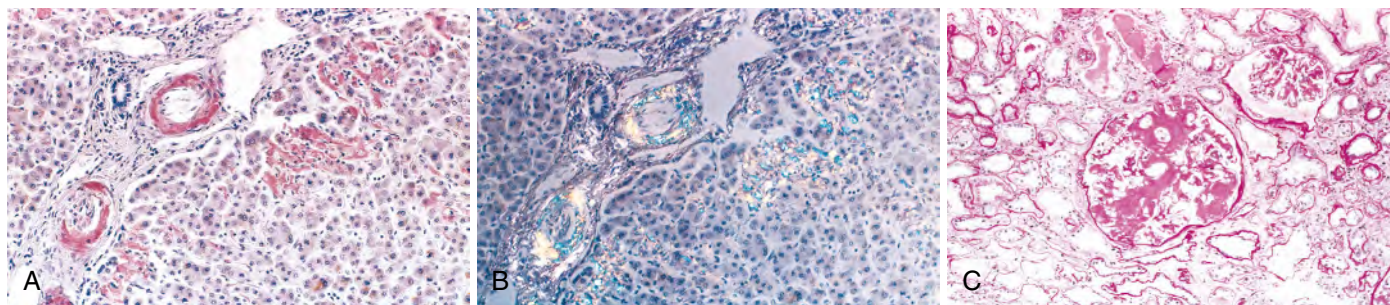


Fig. 6.46 Amyloidosis. (A) A section of the liver stained with Congo red reveals pink-red deposits of amyloid in the walls of blood vessels and along sinusoids. (B) Note the yellow-green birefringence of the deposits when observed by polarizing microscope. (C) Amyloidosis of the kidney. The glomerular architecture is almost totally obliterated by the massive accumulation of amyloid. (A and B, Courtesy Dr. Trace Worrell and Sandy Hinton, Department of Pathology, University of Texas Southwestern Medical School, Dallas.)

Other Organs. Nodular deposits in the tongue may cause macroglossia, giving rise to the designation *tumor-forming amyloid of the tongue*. The respiratory tract may be involved focally or diffusely from the larynx down to the smallest bronchioles. As mentioned earlier, a distinct form of amyloid is found in the brains of patients with Alzheimer disease. It may be present in so-called **plaques** as well as blood vessels (Chapter 28). Amyloidosis of peripheral and autonomic nerves is a feature of several familial amyloidotic neuropathies. Depositions of amyloid in patients on long-term hemodialysis are most prominent in the carpal ligament of the wrist, resulting in compression of the median nerve (carpal tunnel syndrome). These patients may also have extensive amyloid deposition in the joints.

Clinical Features

Amyloidosis may be an unsuspected histologic finding, having produced no clinical manifestations, or it may cause serious clinical problems and even death. The symptoms depend on the magnitude of the deposits and on the sites or organs affected. At first, clinical manifestations are often nonspecific, such as weakness, weight loss, lightheadedness, or syncope. Somewhat more specific findings appear later and most often relate to renal, cardiac, and gastrointestinal involvement.

Renal involvement gives rise to proteinuria that may be severe enough to cause the nephrotic syndrome (Chapter 20). Progressive obliteration of glomeruli in advanced cases ultimately leads to kidney failure and uremia. Kidney failure is a common cause of death. **Cardiac amyloidosis** may present as an insidious congestive heart failure. The most serious aspects of cardiac amyloidosis are conduction disturbances and arrhythmias, which may prove fatal. Occasionally, cardiac amyloidosis produces a restrictive pattern of cardiomyopathy and masquerades as chronic constrictive pericarditis (Chapter 12). **Gastrointestinal amyloidosis** may be entirely asymptomatic, or it may present in a variety of ways. Amyloidosis of the tongue may cause sufficient enlargement and inelasticity to hamper speech and swallowing. Depositions in the stomach and intestine may lead to malabsorption, diarrhea, and disturbances in digestion. **Vascular amyloidosis** causes vascular fragility that may lead to bleeding, sometimes massive, that can occur spontaneously or following seemingly trivial trauma. Additionally, in some cases AL amyloid binds and inactivates factor X, a critical coagulation factor, leading to a life-threatening bleeding disorder.

The diagnosis of amyloidosis depends on the histologic demonstration of amyloid deposits in tissues. The most common sites biopsied are the kidney, when renal manifestations are present, or rectal

or gingival tissues in patients suspected of having systemic amyloidosis. Examination of abdominal fat aspirates stained with Congo red can also be used for the diagnosis of systemic amyloidosis. The test is quite specific, but its sensitivity is low. In suspected cases of AL amyloidosis, serum and urine protein electrophoresis and immunoelectrophoresis should be performed. Bone marrow aspirates in such cases often show monoclonal plasmacytosis, even in the absence of overt multiple myeloma. Scintigraphy with radiolabeled serum amyloid P component is a rapid and specific test since it binds to the amyloid deposits and reveals their presence. It also gives a measure of the extent of amyloidosis and can be used to follow patients undergoing treatment.

The prognosis for individuals with generalized amyloidosis is poor. Those with AL amyloidosis have an overall median survival of 2 years after diagnosis, and the prognosis is even poorer with myeloma-associated AL amyloidosis. Nevertheless, many new, more effective treatments have recently been deployed for plasma cell tumors and offer hope. The outlook for individuals with reactive systemic amyloidosis is somewhat better and depends to some extent on the control of the underlying condition. Resorption of amyloid after treatment of the associated condition has been reported, but this is a rare occurrence. New therapeutic strategies aimed at correcting protein misfolding and inhibiting fibrillogenesis are being developed.



KEY CONCEPTS

Amyloidosis

- Amyloidosis is a disorder characterized by the extracellular deposits of proteins that are prone to aggregate and form insoluble fibrils.
- The deposition of these proteins may result from: excessive production of proteins that are prone to aggregation; mutations that produce proteins that cannot fold properly and tend to aggregate; and defective or incomplete proteolytic degradation of extracellular proteins.
- Amyloidosis may be localized or systemic. It is seen in association with a variety of primary disorders, including monoclonal B-cell proliferations (in which the amyloid deposits consist of immunoglobulin light chains); chronic inflammatory diseases such as rheumatoid arthritis (deposits of amyloid A protein, derived from an acute-phase protein produced in inflammation); familial conditions in which the amyloid deposits consist of mutants of normal proteins (e.g., transthyretin in familial amyloid polyneuropathies); Alzheimer disease (amyloid β protein).
- Amyloid deposits cause tissue injury without evoking an inflammatory response and impair normal function by causing pressure on cells and tissues.

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